

THE INTEGRATION LAYER • THE FACTORY FLOOR • INDUSTRY 4.0

THE FACTORY, END TO END

A COLLECTED TECHNICAL LIBRARY ON HOW A MODERN FACTORY WORKS



THE COMPLETE COLLECTED EDITION

The Whole Plant, collected into one volume

From the bolt and the sensor on the floor — through the controllers, networks, motors, pneumatics and software that run the line — all the way up to the data on the cloud. Every issue, in one English edition.

The Integration Layer, Issues 01–19, with Appendix & Device Index • **The Factory Floor** — four field guides • **Industry 4.0** — two data-journey editions.

26 issues & guides • **273 pages** • 270+ glossary terms • 110+ tool & product families.

Tesla is used throughout as the running example; methods, equipment, figures and numbers are representative of an automotive-class plant. Internal Tesla systems are proprietary. Illustrations are original; standard codes and strength values are real, example figures are illustrative.

Contents

This volume collects the complete English run of three companion series into one bound edition. **Part One** is the full nineteen-issue **Integration Layer** — control, networks, motion, sensing, logistics, planning, software and measurement — reproduced with its own cover, contents and two reference volumes. **Part Two** gathers the **Factory Floor** field guides on fastening and planning. **Part Three** follows a single reading from the machine to the cloud. Each constituent issue keeps its own internal page numbers in the footer; the figures below are positions within this collected edition.



Part One — The Integration Layer

AUTOMATION • SOFTWARE • PLANNING

I	The Complete Collected Edition — Issues 01–19, Appendix & Reference, and the Device & Catalog-Link Index	3
	01 Tesla MOS • 02 Siemens • 03 Protocols • 04 Rockwell & CIP • 05 Rockwell portfolio • 06 Beckhoff • 07 Motors & drives • 08 Logistics • 09 Sensors • 10 Conveyors • 11 Atlas Copco • 12 Festo • 13 Emerson • 14 Material handling • 15 Production planning • 16 Tesla software stack • 17 KPIs & OEE • 18 Demand • 19 Excel • App. Appendix • Idx. Device index	



Part Two — The Factory Floor

FASTENING & PLANNING FIELD GUIDES

II-1	The Bolt is a Spring — bolted-joint engineering & tightening (The Fastening Issue)	205
II-2	The Language of Fasteners — types, technical specs & the standards system (The Standards Issue)	217
II-3	The Fastener Catalog — every type illustrated, and where to find the real thing (The Catalog Issue)	228
II-4	Planning the Machine — MPS, MRP & MRP II, handed to MOS (The Planning Issue)	234



Part Three — Industry 4.0

THE DATA JOURNEY • FLOOR TO CLOUD

III-1	The Factory Cloud — how a factory's machines send their data to the internet (Plain-English Edition)	245
III-2	From Sensor to Cloud — one reading traced end to end, with real values (Technical Deep-Dive)	264

About this collection. Built from the individual issues and reference documents in English; nothing here supersedes the sourcing and scope notes inside each issue. The Persian/bilingual editions and unrelated material have been omitted so this volume reads as a single English library. Company, brand and standard names are trademarks of their respective owners, referenced for identification only.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION



THE COMPLETE COLLECTED EDITION

Volume One Issues 01-19

Nineteen issues on how a modern factory really works — control, networks, motion, sensing, logistics, planning, software, measurement, demand, and the spreadsheet — collected in production order into one volume, with a full appendix and a device & catalog-link index.

19 issues • 2 reference volumes • 202 pages

273 terminology entries • 114 tool & product families • 146 devices indexed.

Tesla as the running example; methods, equipment, and figures are representative of an automotive-class plant.
Internal Tesla systems are proprietary.

CONTENTS

The Complete Collected Edition

Nineteen issues of *The Integration Layer*, in production order, followed by two reference volumes. The colour chip matches each issue's theme; the page number is its start within this bound edition.

#	Title	Page
01	 Tesla MOS — The Operating System of the Factory	3
02	 The Siemens Stack Behind a Gigafactory	11
03	 The Languages of the Factory Floor (protocols)	17
04	 The Rockwell Stack & the Language of CIP	26
05	 The Rockwell & Allen-Bradley Portfolio	35
06	 The Beckhoff Way — PC-based Control & EtherCAT	45
07	 Motors, Drives & the Art of Controlled Motion	54
08	 Logistics, Inventory & the Art of Not Holding Stock	62
09	 The Sensors That Watch Every Car Get Built	71
10	 The Lines That Move the Line (conveyors)	80
11	 Atlas Copco — Torque, Air & the Perfect Joint	90
12	 Festo — Pneumatics, Motion & the Grip	99
13	 Emerson — One Company, a Dozen Names	108
14	 Material Handling, Flow & Inventory Control	117
15	 Production Planning — Plan, Schedule, MRP & Heijunka	128
16	 Tesla's Real Software Stack — Warp, MES & MOS	137
17	 KPIs, OEE & Measuring a Plant	145
18	 Demand — Orders, Forecasting & What Feeds Them	154
19	 Excel for Inventory & Production Planning	163
App.	 Appendix & Reference — Summary, Terminology, Tools, Links	171
Idx.	 Device & Catalog-Link Index	191

Issues 01–19 each retain their own internal page numbering in the footer; the numbers above are positions within this collected edition. Built from the individual issues and the two reference documents; nothing here supersedes the sourcing and scope notes inside each issue.

THE FACTORY FLOOR

A TECHNICAL MAGAZINE ON MANUFACTURING SOFTWARE & SYSTEMS · ISSUE
01

COVER STORY · THE SOFTWARE THAT BUILDS THE CAR

Inside Tesla's Operating System for the Factory

How a piece of software called MOS — built entirely in-house — choreographs the machines, captures every bolt, and turns a car factory into something that behaves like a living program.

BUILT IN-HOUSE

Why Tesla wrote its own factory OS instead of buying SAP or Oracle.

THE VERTICAL OS

Treating factory, product, and cloud as one breathing organism.

ONE BOLT, FOREVER

How a single torque reading becomes a permanent part of a car's record.

FACTORY MODE

Why a Tesla can change every few hours, not every two years.

The Operating System That Builds the Car

THE FACTORY FLOOR · Feature · An explainer grounded in public reporting

Walk onto the floor of a Tesla Gigafactory and the first thing you notice is motion — robots swinging body panels, conveyors gliding, sparks arcing in welding cells. What you cannot see is the thing actually running the show. Beneath the choreography sits a layer of software that schedules the machines, records what they do, and decides whether a half-built car is allowed to move to the next station. Tesla calls it MOS — the Manufacturing Operating System.

It is, in the most literal sense, the operating system of the factory. Just as a computer's OS manages the hardware, schedules tasks, and arbitrates resources so that applications can run, MOS manages the equipment, sequences the operations, and arbitrates the flow of vehicles so that production can run. And like a good OS, it is mostly invisible until something goes wrong.

According to reporting by CleanTechnica after a rare tour of Tesla's Fremont plant, the system that coordinates the entire intricate dance was built completely in-house and has grown to support nearly all of the company's manufacturing equipment. That single design decision — to write the software rather than buy it — is the thread that runs through everything interesting about MOS.

"The custom-built system lets Tesla fine-tune its equipment and processes — the upshot of all that internal complexity is flexibility."

What MOS actually does

Strip away the mystique and MOS performs a recognisable set of jobs that any manufacturing execution system would perform — it just performs them on Tesla's own terms. As CNBC reported in 2018, MOS is used to track where a vehicle is in the process of manufacturing, repair, and testing. In practice that means four things at once.

THE FOUR JOBS OF A FACTORY OS

- **Orchestrate** — tell each station and robot what to do, and in what order, for the specific vehicle in front of it.
- **Capture** — record the result of every meaningful operation: torques, welds, tests, part serial numbers.
- **Bind** — attach every one of those results to the individual car's identity, building a complete as-built history.
- **Gate** — refuse to let a vehicle advance, or ship, until every required step has been done and has passed.

That fourth job is the one operators feel most directly. A car does not roll forward because a timer expired; it rolls forward because MOS is satisfied that everything required at this station actually happened. The software is not just a ledger — it holds a gate.

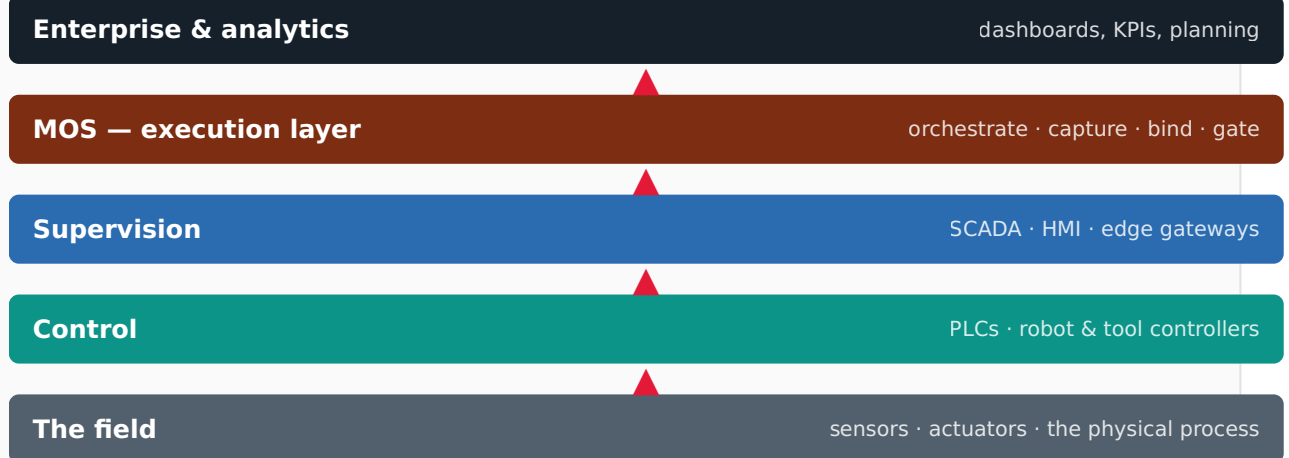


Figure 1. Where MOS sits. Data is born at the bottom and climbs; MOS occupies the execution layer where results are bound to a specific vehicle. *(Conceptual, based on standard ISA-95 architecture.)*

Why Write Your Own?

Here is the part that makes industry veterans raise an eyebrow. Most large manufacturers do not build their own factory software. They buy it. As a long-time enterprise-tech investor told CNBC, it is uncommon for big manufacturers to roll their own ERP-class software; most license it from vendors like SAP and Oracle and customise heavily on top. Tesla went the other way.

The cost of that choice is real. Every change has to be designed, written, and validated internally; there is no vendor to call, no off-the-shelf module to switch on. But the reporting consistently lands on the same conclusion: the payoff is speed. Because Tesla owns the code, it can change the line as fast as it can change a program.

THE TRADE-OFF, IN ONE LINE

Buy: someone else maintains it. Build: you can change it tonight.

A bought system means waiting on a vendor's roadmap and a chain of approvals to change how the floor behaves. A built system means the factory's behaviour is yours to rewrite — at the price of having to maintain every line of it yourself.

A widely cited illustration captures the spirit: a traditional automaker that wants to change how a single icon looks on an instrument cluster might contact a supplier on the other side of the world and wait through a long approval cycle. A vertically integrated software shop can simply change the code. The same logic, Tesla's defenders argue, applies on the factory floor — not just in the car.

A traditional line might take **two years to validate a change. Tesla measures the same loop in **hours**.**

"Factory mode" and the agile habit

That hours-not-years figure comes from interviews with Joe Justice, who worked on Tesla's agile approach. The pace of innovation, in this framing, is measured as the speed at which a change can be deployed to the product and reach customers — and at Tesla that cycle is described in hours, not the roughly two-year validation cadence of a traditional, test-heavy manufacturer. The enabling trick is that the car and the factory are both architected for modularity, so a change in one place does not cascade into dependencies everywhere else. A testing suite sometimes referred to as "factory mode" allows for incremental change without breaking the whole.

DISAMBIGUATION

MOS is not "Tesla OS"

It is easy to confuse two different Tesla software stacks. **Tesla OS** (or the vehicle firmware) is the software inside the car — the brain behind the touchscreen, Autopilot, and over-the-air updates. **MOS** is the software inside the *factory* that builds the car. This article is about the second one. Both reflect the same instinct: own the stack.

How the World Learned MOS Existed

For a system so central, MOS spent years almost entirely out of public view. What dragged its name into headlines was a lawsuit. In 2018, Tesla sued a former employee, alleging he had accessed the company's MOS software and exfiltrated several gigabytes of data to outside parties. The former employee disputed the charges. The legal merits are not the point here — what matters for our story is what the episode revealed: that a single internal platform sat close enough to the heart of production to be worth fighting over in court.

CASE FILE · 2018

The lawsuit that named the system

The dispute put three letters — M, O, S — into mainstream technology coverage for the first time. In trying to explain what had allegedly been breached, journalists pieced together, from job listings and former-employee accounts, the first public sketch of how Tesla runs its factories in software. A system designed to be invisible became, briefly, the most discussed piece of software in the company.

That same reporting surfaced a more mundane but revealing detail: MOS does not stand alone. It is one of several systems Tesla has built and run in parallel.

THE WIDER SOFTWARE ESTATE

- **MOS** — the manufacturing operating system tracking materials and vehicle/battery production.
- **MES** — a manufacturing execution system reported as supporting Model S and X production.
- **Executive Factory dashboard** — surfacing equipment effectiveness, line data, and high-level production numbers to executives.
- **Garage / Garage Portal** — tracking which firmware updates vehicles have or may need.

A former employee noted the flip side of running several disparate systems: data — repair notes and the like — can sometimes slip between them, occasionally causing duplicate work. It is a useful reminder that even a sophisticated, in-house software estate is a work in progress, not a finished cathedral. ■

1

in-house OS, not bought off the shelf

~GBs

data at issue in the 2018 case

Hours

Tesla's change-deploy loop (per interviews)

4+

parallel software systems on the floor

The Factory as One Program

Step back far enough and MOS stops looking like a piece of factory software and starts looking like one organ of something larger. A 2026 essay framed Tesla's real innovation not as the car itself but as an unmatched level of systems integration — coining the idea of a "Vertical OS." In this view, the factory, the product, and the cloud are not three separate domains stitched together but one single, breathing organism, with a high-velocity feedback loop running between them.

Hardware & Physics

the car, the machines, the line

Software & Firmware

Tesla OS, MOS, the code

Manufacturing Ops

the people and process on the floor

Figure 2. The three pillars of the "Vertical OS." The argument: value comes from the **intersection** — the feedback loop between them — not from any pillar alone. (After the framing in a 2026 essay.)

MOS is what makes the manufacturing pillar legible to the other two. When fleet data from cars on the road suggests a change, the factory has to be able to absorb it quickly; when a new component or chip needs validating, the floor has to be able to test it fast. The questions that animate this culture — how fast can we get fleet data to the floor, how fast can we validate a new part — only have good answers if the factory itself runs on software you control.

The car is the most visible output. The deeper product is the **integration — and MOS is where it touches the floor.**

First principles, on the floor

This is also why Tesla is described as hiring "first-principles" thinkers for these roles — people who look at a crowded factory and design a genuinely new solution from spatial constraints, rather than only optimising the existing line. The "Unboxed" assembly concept is the headline example of that mindset applied to the line itself. Software like MOS is the substrate that makes such radical reconfiguration survivable: if the rules of the factory live in code you own, you can rewrite them.

WHY IT MATTERS TO A TECHNICIAN

Owning the OS changes the job

On a floor that runs on bought software, your job is to operate within a vendor's system. On a floor that runs on MOS, the system can change under you in days — new checks, new gates, new data captured. Understanding that the factory is a *program*, not a fixed machine, is the mental model that keeps you ahead of it.

The Quiet Layer That Decides Everything

MOS will never be the star of a Tesla keynote. It has no horsepower figure, no 0-to-60 time, no screen the customer ever touches. But it is, in a real sense, the most consequential software the company runs — because it is the layer that decides, thousands of times a day, whether a car is being built correctly. Every torque reading, every weld, every test either satisfies MOS or does not, and the line moves accordingly.

The lesson that survives all the specifics is the one Tesla keeps demonstrating: when you own the operating system of your factory, the factory becomes something you can edit. That is the whole bet — complexity and maintenance burden traded for the ability to change the line as quickly as you change your mind. Whether that bet pays off is, in the end, written in the same place everything else is: in the data MOS quietly captures, one bolt at a time. ■

FIELD GLOSSARY

The terms in this issue, in plain language

- **MOS (Manufacturing Operating System)** — Tesla's in-house software that runs the factory: orchestrating equipment, capturing results, binding them to each vehicle, and gating progress. Its exact internal design is not public.
- **MES (Manufacturing Execution System)** — the general class of software that manages and records shop-floor production; MOS is Tesla's MES-class platform.
- **ERP (Enterprise Resource Planning)** — business-wide planning software (e.g. SAP, Oracle); the kind of thing most manufacturers buy rather than build.
- **Tesla OS / vehicle firmware** — the software *inside the car*; distinct from MOS, which is the software that *builds* the car.
- **As-built record / genealogy** — the complete, linked history of how one specific vehicle was made, assembled from every captured result.
- **Factory mode** — a testing approach that allows incremental change to the line without breaking the whole.
- **Vertical OS** — the idea of treating factory, product, and cloud as one integrated system with a fast feedback loop.

EDITOR'S NOTE · ON SOURCING & SCOPE

Tesla's MOS is proprietary, and the company does not publish its internal design. This article is an explainer assembled from public reporting — including CNBC's 2018 account of the MOS lawsuit, a CleanTechnica/Evannex factory tour, interviews on Tesla's agile "factory mode," and a 2026 essay on Tesla's "Vertical OS" — together with the standard architecture of manufacturing systems. Specific internal figures are not disclosed by Tesla; diagrams here are conceptual. Nothing in this issue should be read as confirmed internal Tesla documentation.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 02

COVER STORY · THE SILENT BACKBONE OF THE LINE

The Siemens Stack Behind a Gigafactory

From the S7-1500 controller to SINAMICS drives, SCALANCE networks, and the TIA Portal that ties them together — a tour of the automation hardware that quietly runs a modern car plant.

THE BRAIN

Why the S7-1500 is the default controller of automotive lines.

ONE NETWORK

PROFINET and SCALANCE as the nervous system of the floor.

MOTION, MASTERED

How SINAMICS drives and SIMOTICS motors move a robot's axes.

TOTALLY INTEGRATED

How TIA Portal turns a pile of devices into one system.

The Stack Behind the Line

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

A car factory looks like a triumph of mechanical engineering — robots, presses, conveyors, sparks. But mechanical motion is only the surface. Underneath every clamp and weld is a layer of industrial automation hardware deciding, thousands of times a second, what moves and when. In plants of the Gigafactory class, a very large share of that hardware carries one name: Siemens. This issue is a guided tour of that stack — what each family does, and how the pieces lock together.

The reason a single vendor's catalog can run so much of a floor is not brand loyalty. It is integration. Siemens does not sell a controller here and a drive there; it sells a portfolio designed so the controller, the drive, the motor, the network, and the engineering software all speak the same language. That coherence — branded "Totally Integrated Automation," or TIA — is the thread of this entire issue.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact bill of materials inside its factories, and the specific part numbers on any given fixture are proprietary. This article describes the **standard Siemens automation portfolio for a plant of this class** — the same families (SIMATIC S7-1500, SINAMICS, TIA Portal, and so on) that are extensively documented for automotive manufacturing. Treat models and figures as representative real-world examples, drawn from Siemens' public product documentation, not as confirmed internal Tesla equipment lists.

To make the tour concrete, we will follow the same organising idea Siemens itself uses: the automation pyramid. Data and commands flow between five layers, and Siemens has a product family living on each one.

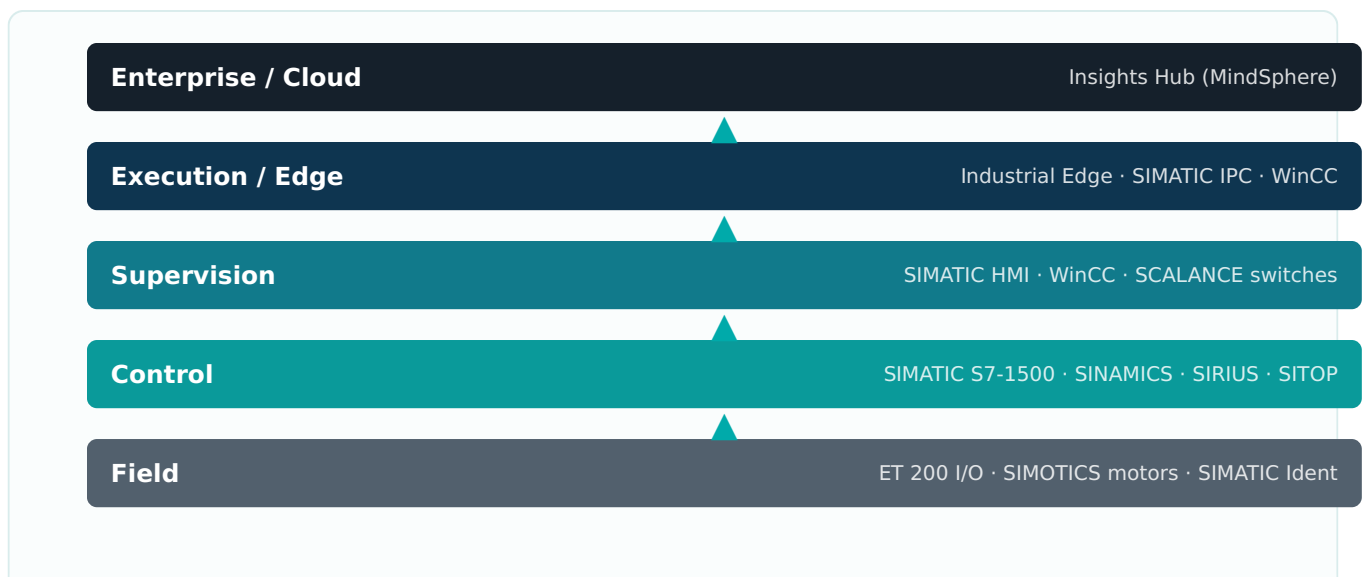


Figure 1. The Siemens portfolio, mapped to the automation pyramid. Each layer has a dedicated product family.
(Conceptual, based on standard ISA-95 architecture.)

THE CONTROLLER · SIMATIC S7-1500

The Brain on the DIN Rail

If a cell has a brain, it is the PLC — the programmable logic controller. And in demanding automotive automation, the default brain is the SIMATIC S7-1500: a modular, high-performance controller that executes the cell's logic in real time and, crucially, deterministically. It does not merely run fast; it runs *predictably*, reacting within a known and bounded time. That guarantee is the whole reason a PLC, rather than an ordinary PC, is trusted to run machinery.

An S7-1500 station is a row of modules clipped onto a rail and joined by a high-speed backplane: a power supply, the CPU itself (with a small diagnostic display on its face), then signal, technology, and communication modules. The CPU range scales from modest to formidable, and the spread is dramatic.

SIMATIC S7-1500 — CPU range

SIMATIC · order prefix 6ES7

From CPU 1511 (entry, ~60 ns bit time) **to CPU 1518** (flagship, ~1 ns bit time — roughly 60× faster). Fail-safe (**F**) variants run safety logic in the same CPU; technology (**T**) variants add motion control; **MFP** variants even run C/C++. PROFINET and an OPC UA server are built in; programming is in TIA Portal using LAD, FBD, SCL, and GRAPH.

It does not merely run fast. It runs *predictably* — and predictability is what you can build a safety case on.

How a PLC actually "thinks": the scan cycle

The S7-1500 does not watch its inputs continuously. It repeats a loop — the scan cycle — many times per second: read all inputs into a memory snapshot, execute the program top to bottom over that fixed snapshot, then write the outputs. Because the loop is cyclic and bounded, the controller's response time is something an engineer can rely on. It is the heartbeat beneath every motion on the line.

WHY IT WINS IN AUTOMOTIVE

One family, cabinet to machine-edge

The same S7-1500 logic can run as a central cabinet controller, or in the compact ET 200SP form right beside the machine, or as a software controller on an industrial PC. Add fail-safe and motion variants and a single, coherent family covers nearly every control job on the floor — which is exactly why it dominates.

How a Robot Moves

Every precise movement on the line — a robot swinging a body panel, an axis indexing a fixture — traces back to a drive commanding a motor. Siemens splits this into two families that are designed to work as one: SINAMICS, the drives, and SIMOTICS, the motors. Together with a motion-capable controller they form a chain that runs from a line of code all the way to a spinning shaft.

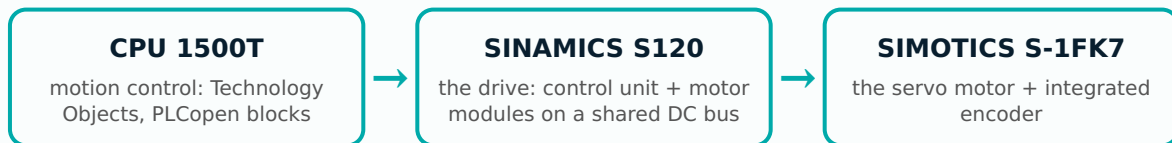


Figure 2. Siemens' integrated drive chain, linked over PROFINET with the PROFIdrive profile and DRIVE-CLiQ. The encoder closes the loop, so motion is precise and synchronised. *(Representative configuration.)*

The drives come in tiers. For high-performance, multi-axis work — the kind a robot or a coordinated line needs — the **SINAMICS S120** is the flagship: a modular system where a control unit drives several motor modules sharing a common DC bus. That shared bus is a quiet piece of elegance: when one axis brakes, its energy can feed another axis that is accelerating, instead of being burned off as heat.

SINAMICS — the drive tiers

SINAMICS · order prefix 6SL3

S120 — high-performance multi-axis servo (robots, lines). **S210** — highly dynamic single-axis servo, fast to commission. **V90** — economical servo for basic axes. **G120** family — general-purpose frequency converters for pumps, fans, and conveyors. The S120 pairs with SIMOTICS S-1FK7 servo motors over DRIVE-CLiQ, which carries the motor's "electronic nameplate" so the drive configures itself.

When one axis brakes, its energy can feed another that is accelerating. The factory *recycles its own motion*.

The motors that do the work

On the motor side, the **SIMOTICS S-1FK7** is the workhorse servo for high-performance motion — compact, rugged, with an integrated encoder that tells the drive exactly where the shaft is. For machine-tool spindles there is the asynchronous 1PH8; for general pumps and fans, plain induction motors driven by G120 converters. The point is the matching: each motor is designed to be driven by a specific SINAMICS drive, so the whole chain behaves as a single engineered system rather than a bag of parts.

One Network to Carry It All

A pyramid only works because its layers are connected, and on a Siemens floor the connective tissue is PROFINET — industrial Ethernet engineered for the deterministic, real-time traffic that motion control demands. Every S7-1500 CPU has a PROFINET port built in. The physical network — the switches, the rings, the firewalls — is the SCALANCE family.

SCALANCE — the network hardware

SCALANCE · industrial Ethernet

XB / XC-200 — managed switches for a cell. **XR / XM-400** — rack-mount backbone switches with Layer-3 routing. **W** — industrial WLAN for AGVs and moving parts. **SC-600 / S** — industrial firewalls and VPN that segment and defend cells. The whole estate is built to carry control traffic and IT traffic on one infrastructure without the control traffic ever waiting.

Above the wire sits the second bridge: **OPC UA**, built into every S7-1500, which carries *structured* data — a value with its units, limits, and meaning attached — up toward the IT systems. Between PROFINET below and OPC UA above, a Siemens floor has a clean path from a single sensor all the way to the cloud.

THE SUPPORTING CAST

The families you don't notice — until they fail

- **ET 200** — distributed I/O out at the machine, so wiring shrinks to one network cable per cell.
- **SIRIUS** — the contactors, overload relays, and safety relays that actually switch and protect the motors.
- **SITOP** — the clean 24 V DC power that every controller and sensor quietly depends on.
- **SIMATIC Ident** — the RFID and code readers that give each vehicle its identity for tracking.

The thing that ties it together: TIA Portal

Hardware is only half the story. The reason a Siemens floor behaves as one system is **TIA Portal** — the Totally Integrated Automation engineering environment. In a single project, an engineer configures the PLC (STEP 7), the HMI (WinCC), the drives (Startdrive), and the safety logic, all sharing one tag database. A name defined once is available everywhere. That shared environment, more than any single device, is the product Siemens is really selling.

A name defined once is available everywhere. The real product isn't a box — it's the *integration*.

The Coherence Is the Point

Walk the stack from bottom to top — SIMOTICS motor, SINAMICS drive, S7-1500 controller, PROFINET and SCALANCE network, WinCC and IPC, TIA Portal over all of it — and a pattern emerges. No single Siemens product is magic. The S7-1500 is an excellent PLC, but excellent PLCs exist elsewhere. What is hard to replicate is the coherence: a portfolio in which every layer was designed to connect to the ones above and below it, engineered from one tool, speaking one set of protocols.

That is why a plant of this class can run so much of its automation on one vendor's catalog. The bet is the same one integration always makes — you trade the freedom to mix best-of-breed parts for a system that simply fits together. On a floor where a controller, a drive, a motor, and a network must move in microsecond lockstep, that fit is not a luxury. It is the product. ■

FIELD GLOSSARY

The Siemens vocabulary in this issue

- **SIMATIC S7-1500** — Siemens' flagship modular PLC; the controller that runs a cell or line in real time.
- **ET 200** — the distributed I/O family, mounted near the machine and linked over PROFINET.
- **SINAMICS** — the drive family (S120, S210, V90, G120) that controls motor speed, torque, and position.
- **SIMOTICS** — the electric-motor family, matched to SINAMICS drives.
- **PROFINET** — Siemens' industrial-Ethernet network for real-time control traffic.
- **SCALANCE** — the network hardware: switches, routers, wireless, and firewalls.
- **SIRIUS** — low-voltage controls: contactors, overload relays, safety relays.
- **SITOP** — the power-supply family delivering stable 24 V DC.
- **SIMATIC Ident** — RFID and optical code readers for traceability.
- **TIA Portal** — the integrated engineering software that configures the whole stack from one project.
- **DRIVE-CLiQ** — the digital interface linking Siemens drives, motors, and encoders.
- **OPC UA** — a secure, structured-data standard, built into the S7-1500, bridging the floor to IT.

WHERE TO SEE THE REAL CATALOG

Every family in this issue has an official Siemens product page with photos, datasheets, and an order-number selector. Start at the **SIMATIC controllers overview** ([siemens.com/global/en/products/automation/systems/industrial/plc.html](https://www.siemens.com/global/en/products/automation/systems/industrial/plc.html)) and the **SINAMICS drives hub** ([siemens.com/global/en/products/drives/sinamics.html](https://www.siemens.com/global/en/products/drives/sinamics.html)); the online Industry Mall (mall.industry.siemens.com) lists exact order numbers.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 03

COVER STORY · HOW THE MACHINES TALK

The Languages of the Factory Floor

PROFINET, IO-Link, OPC UA, MQTT — the communication protocols that connect a PLC to every sensor, drive, and robot on the line, and let a single torque reading travel from a bolt to the cloud.

REAL-TIME OR BUST

Why control networks must be deterministic, not just fast.

THE LAST METRE

How IO-Link gives a humble sensor a voice.

THE FIELDBUS WARS

PROFINET vs EtherNet/IP vs EtherCAT — and why it matters.

BRIDGING TO IT

OPC UA and MQTT carry the floor's data to the cloud.

How the Machines Talk

THE INTEGRATION LAYER · Feature · A protocol tour grounded in public standards

A PLC by itself controls nothing. It is a brain in a box until it can talk — to the sensor that reports a part is present, to the drive that turns a motor, to the robot that swings an arm, to the system upstairs that records what happened. On a Gigafactory-class line, that conversation never stops, and it is carried by a stack of communication protocols, each engineered for a very specific kind of talking. This issue is a field guide to those languages.

The first thing to understand is that factory communication is not one network but a layered set of them, each with different demands. A safety stop must be heard in milliseconds with absolute certainty. A torque result must arrive intact and bound to the right car. A dashboard upstairs can tolerate a one-second delay. No single protocol is best at all three, so a real plant runs several, stitched together — and the art is in how they connect.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact protocols, topologies, or part numbers wired into its factories. This article describes the **standard industrial-communication stack for a plant of this class** — protocols (PROFINET, IO-Link, OPC UA, MQTT, and peers) that are openly documented and ubiquitous in automotive automation. Treat examples as representative, drawn from public standards and vendor documentation, not as confirmed internal Tesla configuration.

The one idea that explains everything: determinism

Before any protocol names, one concept governs the whole field. An office network is judged on average speed; if an email arrives 200 milliseconds late, no one notices. A control network is judged on *determinism* — the guarantee that a message arrives within a known, bounded time, every single time. A motion command that arrives late, or at an unpredictable moment, can ruin a process or trigger a fault. This need for guaranteed timing, more than raw bandwidth, is what separates industrial protocols from ordinary IT.

An office network optimises for average speed. A control network optimises for the worst case.

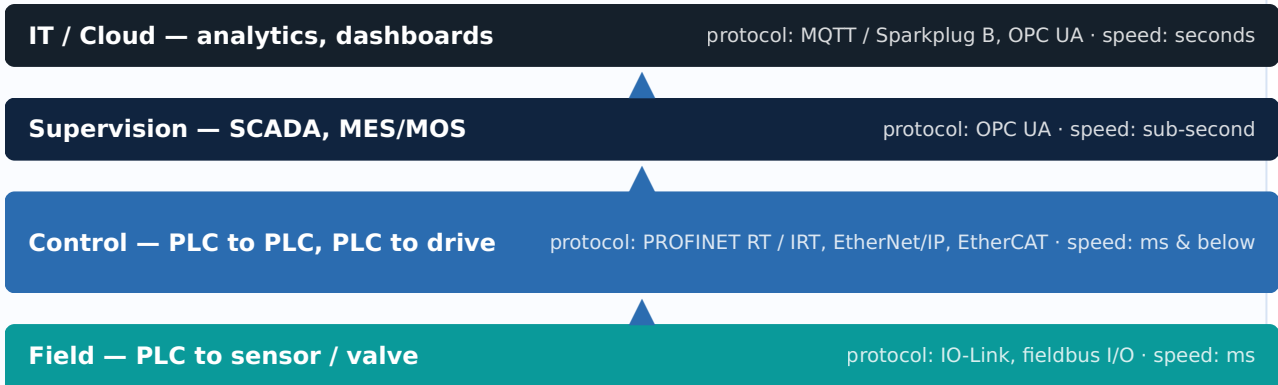


Figure 1. The protocol stack, by layer. Different jobs, different protocols — the faster and more deterministic, the lower in the stack. (Conceptual, based on standard architecture.)

THE BACKBONE · PROFINET

The Real-Time Workhorse

On a Siemens-centric automotive floor, the dominant control network is PROFINET — industrial Ethernet re-engineered for the deterministic traffic that control demands. It runs on standard Ethernet hardware, which lets it share infrastructure and tools with the IT world, but it adds the timing guarantees that ordinary Ethernet lacks. Every S7-1500 CPU has a PROFINET port built in, and the distributed I/O, drives, and HMIs all speak it.

PROFINET is not one speed but three, layered by how strict the timing must be. Most communication uses the standard real-time channel; the most demanding, like coordinated motion, uses an isochronous channel synchronised to the microsecond.

PROFINET

Industrial Ethernet · the control backbone

Three classes: **TCP/IP** (non-time-critical config and diagnostics), **RT (Real-Time)** for standard cyclic control, and **IRT (Isochronous Real-Time)** for tightly synchronised motion. Supports line, star, and ring topologies; rings give redundancy so a single cable break does not stop the line.

TYPE	TIMING	CARRIES
Industrial Ethernet	RT / IRT, µs-class	Control, motion, I/O

PROFINET also has an older sibling still found across many existing lines: **PROFIBUS**, a reliable serial fieldbus that predates the move to Ethernet. New installations favour PROFINET, but a technician will meet both, and gateways let them coexist.

TWO PROFILES RIDING ON TOP

PROFIsafe and PROFIdrive

PROFINET is a carrier; specialised *profiles* ride on it. **PROFIsafe** carries safety-critical signals (an emergency stop, a light curtain) over the very same cable as normal data, certified so a fault leads to a safe state — no separate safety wiring needed. **PROFIdrive** defines how a controller and a drive talk, so the CPU, drive, and motor stay in deterministic lockstep during motion.

Not Everyone Speaks the Same Tongue

PROFINET dominates Siemens-built lines, but it is not the only industrial-Ethernet protocol, and a large plant with equipment from many vendors will have islands speaking different languages. The three big control protocols all run on Ethernet hardware yet are mutually incompatible at the wire level — a genuine modern echo of the old "fieldbus wars."

Protocol	Home turf	Strength	Typical use
PROFINET	Siemens ecosystem	Deterministic RT/IRT, ubiquitous in Europe/automotive	Cell & line control, motion, I/O
EtherNet/IP	Rockwell / Allen-Bradley	Strong in North America, CIP messaging	Plants on Rockwell controllers
EtherCAT	Beckhoff ecosystem	Ultra-low latency, tight multi-axis sync	High-performance motion control
Modbus TCP	Vendor-neutral	Dead simple, near-universal support	Simple devices, legacy & 3rd-party gear

The practical consequence on the floor is the **gateway** — a device that translates between protocols so an island speaking one language can hand data to an island speaking another. Much of the quiet engineering in a large plant is exactly this: making incompatible networks cooperate. When data seems to "stop" at a boundary, a technician learns to look first at the gateway.

Three protocols, one cable type, zero compatibility. The gateway is where they shake hands.

µs sync precision for motion (IRT/EtherCAT)	3 major incompatible Ethernet protocols	1989 Modbus's ancestor era — still in use	1 cable safety + data together via PROFIsafe
---	---	---	--

Giving a Sensor a Voice

All the protocols so far connect controllers and major devices. But at the very edge — the last metre, between an I/O module and an individual sensor or valve — lives a different and increasingly important standard: IO-Link. It is not a fieldbus; it is a simple point-to-point digital link that replaces the old "one wire, one on/off signal" connection with a two-way digital conversation.

The difference is profound. A traditional sensor can only say "yes" or "no" — part present or not. An IO-Link sensor can report its exact measured value, its health, its temperature, its

settings, and warn that it is about to fail. It can also be reconfigured from the controller without anyone touching it. When a sensor is swapped, the IO-Link master can push the old settings to the new device automatically.

IO-Link

IEC 61131-9 · the last-metre link

Point-to-point digital communication between an IO-Link master (which sits on the fieldbus) and a device. Carries **process data, diagnostics, and parameters** over a standard 3-wire sensor cable. Enables automatic device replacement, remote configuration, and rich health data — the foundation of predictive maintenance at the sensor level.

TYPE	CARRIES	WINS AT
Point-to-point digital	Value + health + config	The last metre

WHY IT MATTERS TO A TECHNICIAN

From "replace and recalibrate" to "plug and play"

On an IO-Link cell, a failed sensor is not a calibration ordeal. The master recognises the replacement and pushes the stored parameters automatically. And because each sensor streams its own health, the system can flag a sensor drifting toward failure *before* it causes a single bad part — turning a breakdown into a scheduled swap.

Carrying the Data Upstairs

Deterministic control protocols are perfect for the floor and wrong for the climb to IT. The systems upstairs — MES/MOS, dashboards, the cloud — do not need microsecond timing; they need data that is structured, secure, and easy for software to consume. Two protocols own this bridge, and they are increasingly used together.

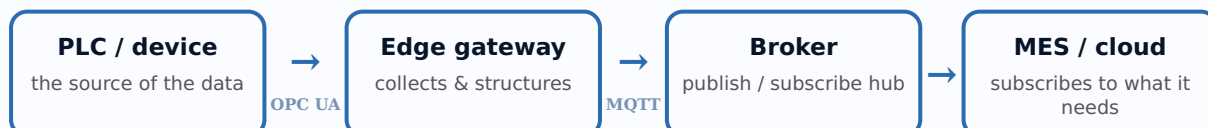


Figure 2. The path from floor to cloud. OPC UA carries structured data off the device; MQTT moves it efficiently to any consumer. (Representative data path.)

OPC UA (Open Platform Communications Unified Architecture) is the standard for carrying *structured* data. A value does not arrive as a bare number; it arrives with its units, its limits, its timestamp, and its meaning attached — so the receiving software knows it is a "torque result in N·m," not just "46.2." Crucially for modern security, OPC UA has authentication and encryption built in, and a server is embedded in every S7-1500.

OPC UA			IEC 62541 · the structured bridge
Platform-independent, secure, self-describing. Data carries its own model — units, type, limits, meaning. Built-in authentication and encryption. The de-facto standard for OT-to-IT data exchange, embedded in S7-1500 controllers and a cornerstone of Industry 4.0.			
TYPE	CARRIES	WINS AT	
Client/server & pub/sub	Structured, secure data	OT → IT integration	

MQTT takes a different, beautifully simple approach: *publish and subscribe*. Devices publish messages to named topics on a central broker; any system that wants the data subscribes to those topics. The publisher never needs to know who is listening. This decoupling is ideal for moving telemetry from thousands of points to many consumers over unreliable or constrained links. Its industrial profile, **Sparkplug B**, adds a defined topic structure, device birth/death awareness, and efficient binary payloads.

The publisher never needs to know who is listening. That decoupling is the whole trick.

OPC UA VS MQTT — NOT A RIVALRY

They do different jobs, and often team up

OPC UA excels at *describing* data richly; MQTT excels at *moving* it efficiently. A common modern pattern uses OPC UA to pull structured data off the equipment and MQTT/Sparkplug to distribute it to

the broker and onward. Together they form the backbone of the "Unified Namespace" idea — a single, live, structured picture of everything happening in the operation.

The Right Language for the Job

Trace a single torque reading on its journey — born at a sensor, judged by a PLC, carried to a dashboard — and you cross four or five protocols, each handing off to the next. IO-Link gives the sensor its voice; PROFINET carries the result deterministically across the cell; a gateway bridges to another island if needed; OPC UA structures the data; MQTT moves it to the cloud. No single language does it all, and that is not a flaw. It is the design.

The lesson that survives the alphabet soup is the one that makes a good technician: each protocol exists because a specific job demanded it. Determinism low, structure and reach high. When something breaks — a value missing on a screen, a cell that will not sync — the question is never just "is the network down?" but "*which* language failed, and at which handoff?" Understanding the stack is what turns a mystery into a diagnosis. ■

FIELD GLOSSARY

The protocols in this issue, in plain language

- **Determinism** — the guarantee that a message arrives within a known, bounded time, every time; the defining requirement of control networks.
- **Fieldbus** — the general family of digital industrial networks connecting field devices to controllers.
- **PROFINET** — Siemens' industrial-Ethernet control network, with real-time (RT) and isochronous (IRT) classes.
- **PROFIBUS** — the older serial fieldbus that PROFINET succeeds; still common on existing lines.
- **PROFIsafe** — a profile carrying safety signals over the same network as normal data.
- **PROFIdrive** — a profile defining controller-to-drive communication for synchronised motion.
- **EtherNet/IP** — the Rockwell-world industrial-Ethernet protocol, built on CIP.
- **EtherCAT** — an ultra-low-latency Ethernet protocol favoured for high-performance motion.
- **Modbus TCP** — a simple, near-universal protocol for basic and legacy devices.
- **IO-Link** — a point-to-point digital link giving individual sensors and valves a two-way voice (value, health, config).
- **OPC UA** — a secure, structured, self-describing standard bridging the floor to IT.
- **MQTT / Sparkplug B** — a lightweight publish/subscribe protocol (and its industrial profile) for moving telemetry to the cloud.
- **Gateway** — a device that translates between two incompatible protocols.
- **Unified Namespace** — a single, live, structured picture of an operation, often built on MQTT/OPC UA.

FURTHER READING · THE STANDARDS BODIES

These protocols are governed by open standards organisations whose documentation is public: PROFINET, PROFIBUS, PROFIsafe and PROFIdrive by **PI (PROFIBUS & PROFINET International)**; IO-Link under **IEC 61131-9**; OPC UA by the **OPC Foundation** (IEC 62541); MQTT by **OASIS**, with Sparkplug B under the **Eclipse Foundation**; EtherNet/IP by **ODVA**; EtherCAT by the **EtherCAT Technology Group**. Vendor product pages (e.g. siemens.com) document how each is implemented in real controllers.

THE INTEGRATION **LAYER**

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 04

COVER STORY · THE AMERICAN STANDARD ON THE FLOOR

The Rockwell Stack & the Language of CIP

ControlLogix, CompactLogix, GuardLogix — Allen-Bradley's Logix controllers, and the EtherNet/IP and CIP protocols that connect them to every drive, I/O block, and robot on the line.

THE LOGIX FAMILY

ControlLogix vs CompactLogix — one control engine, two scales.

THE RESILIENT RING

Device Level Ring: surviving a cut cable without stopping.

ONE PROTOCOL, ONE STACK

How EtherNet/IP and CIP carry control, motion, and safety.

SAFETY IN THE SAME CHASSIS

GuardLogix folds SIL 3 safety into the controller.

The Other Half of the Floor

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

In an earlier issue we toured the Siemens stack — the dominant automation portfolio on European-style lines. But walk a large plant in North America, or any floor assembled from American-built equipment, and you meet a different giant with an entirely different accent: Allen-Bradley, the brand of Rockwell Automation. Its controllers, its network, and its engineering software form a parallel universe to the Siemens world — equally complete, equally coherent, and built on one unifying idea of its own.

Where Siemens organises everything under "Totally Integrated Automation," Rockwell organises everything under what it calls the **Integrated Architecture** — and at the technical heart of that architecture sits a single protocol that runs through the entire portfolio. Understanding Rockwell means understanding that protocol. This issue is a field guide to the hardware and, above all, to the language that ties it together.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact controllers, networks, or part numbers wired into its factories, and the floor of any given plant mixes vendors. This article describes the **standard Allen-Bradley / Rockwell portfolio for a plant of this class** — the Logix controllers, EtherNet/IP network, and Studio 5000 software that are openly documented and ubiquitous in North American automotive automation. Treat models and figures as representative, drawn from Rockwell's public product documentation, not as confirmed internal Tesla equipment.

The one idea: the Logix control engine

Rockwell's modern coherence rests on a single re-architected control engine shared across its controllers. Whether the hardware is a large rack-mounted ControlLogix or a compact DIN-rail CompactLogix, the same Logix engine runs inside, programmed in the same Studio 5000 software, using the same tag-based logic. Learn one Logix controller and you have largely learned them all — the direct equivalent of the coherence that makes the Siemens portfolio so sticky.

**One control engine, one programming tool, one protocol.
Rockwell's whole bet is coherence.**

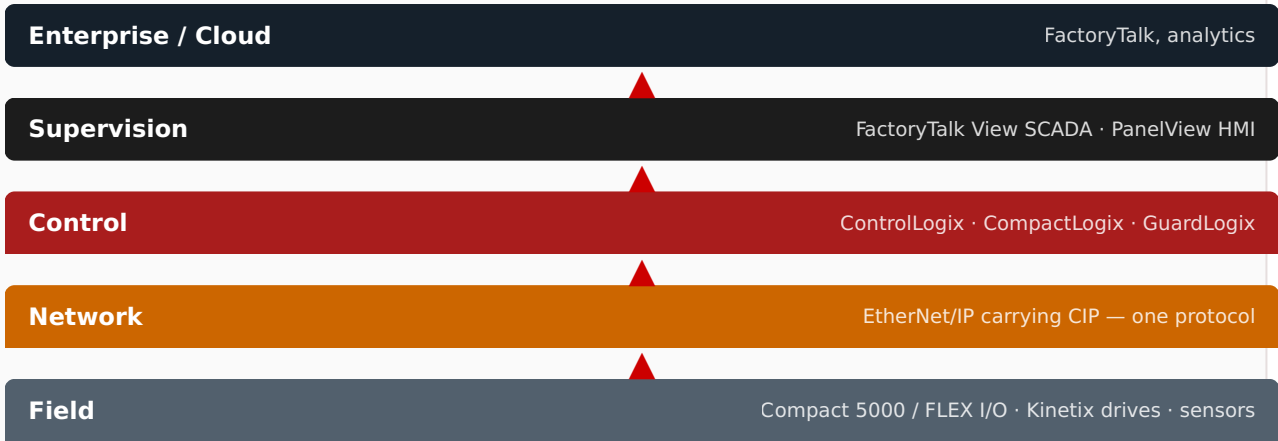


Figure 1. The Rockwell Integrated Architecture, by layer. One protocol — CIP over EtherNet/IP — runs from field to enterprise. (Conceptual, based on public documentation.)

THE CONTROLLERS · THE LOGIX FAMILY

One Engine, Two Scales

The Logix family splits cleanly by size, but not by concept. At the top sits **ControlLogix**, a chassis-based controller for large, distributed, high-performance plants; below it, **CompactLogix**, a DIN-rail controller for compact and stand-alone machines. Both run the same engine and the same code — the difference is capacity and architecture, not language.

ControlLogix 5580Allen-Bradley · 1756 · chassis-based

The flagship. A modular, chassis-based controller (catalog 1756-L8x) offering the memory, speed, and capacity for large process and motion applications. Excels in **distributed network environments**, supports controller redundancy, and comes in process, extreme-temperature, and no-stored-energy variants. Centralised authentication, access control, and digitally-signed firmware. Programmed in Studio 5000 Logix Designer.

FORM	SCALE	NETWORK
Rack / chassis (1756)	Large, distributed	EtherNet/IP

CompactLogix 5380 & 5480Allen-Bradley · 5069 · DIN-rail

The compact workhorse. Uses the same re-architected engine as the 5580, on the Compact 5000 I/O platform, with **built-in dual EtherNet/IP ports** supporting linear and Device Level Ring topologies, and integrated motion over EtherNet/IP for up to 32 axes. The **5480** adds a hybrid twist: a Logix control engine running alongside an embedded Windows 10 IoT environment for edge analytics. Supports up to 31 local Compact I/O modules.

FORM	SCALE	MOTION
DIN-rail (5069)	Small to medium	Up to 32 axes

There is also the older but still widespread **MicroLogix** and **Micro800** lines for the smallest tasks, and legacy **SLC 500** and **PLC-5** systems that a technician will still encounter on older lines — with Rockwell positioning the CompactLogix 5380 as the modern migration target.

THE SIEMENS PARALLEL

Reading Rockwell if you know Siemens

The mapping is almost one-to-one. **ControlLogix** ≈ **S7-1500** (the high-end controller); **CompactLogix** ≈ **S7-1200 / ET 200SP CPU** (the compact controller); **Studio 5000** ≈ **TIA Portal** (the one engineering tool); **EtherNet/IP** ≈ **PROFINET** (the house network); **Kinetix** ≈ **SINAMICS** (the motion drives). Different accents, same grammar.

One Protocol to Carry Everything

Here is the concept that defines the entire Rockwell world. Most vendors use different protocols for different jobs — one for I/O, another for motion, another for safety. Rockwell's defining idea is the opposite: a single application protocol, the **Common Industrial Protocol (CIP)**, carries all of them. Control messaging, motion, and safety are all just different "profiles" of the same CIP language.

CIP is the *application* layer — the meaning of the messages. To travel over a wire it needs a transport, and the dominant one is **EtherNet/IP** (where "IP" stands for *Industrial Protocol*, not Internet Protocol). EtherNet/IP is simply CIP carried over standard Ethernet and TCP/IP — which is exactly why it integrates so naturally with IT networks and is so strong in North America.

EtherNet/IP + CIP

ODVA · the Rockwell house network

CIP is the application protocol carrying control, motion, and safety as profiles; **EtherNet/IP** carries CIP over standard Ethernet and TCP/IP. One network and one protocol family for the whole stack — I/O, drives, HMI, and safety all speak CIP. Strong IT integration, broad third-party support, governed by the open **ODVA** standards body.

TRANSPORT

Standard Ethernet / TCP-IP

APP PROTOCOL

CIP (one for all jobs)

GOVERNED BY

ODVA

Most vendors switch languages for each job. Rockwell speaks **one language — CIP — for all of them.**

The CIP family — same language, specialised dialects

Profile	What it carries	The job
CIP (core)	Standard control messaging & I/O	PLC-to-I/O, PLC-to-PLC, general control
CIP Motion	Synchronised drive/axis commands	Integrated motion over EtherNet/IP (Kinetix)
CIP Safety	Safety-rated signals (SIL 3 / PLe)	E-stops, guards — on the same network
CIP Sync	Precise time distribution	Timestamping & coordinating devices

The payoff mirrors what PROFI-safe and PROFI-drive do for Siemens, but with even tighter unity: because motion and safety are *profiles of the same protocol* rather than separate protocols, a single EtherNet/IP network can carry ordinary control, microsecond-coordinated motion, and certified safety traffic all at once.

1

protocol (CIP) for control, motion & safety

32

axes of integrated motion (CompactLogix 5380)

SIL 3

safety rating via CIP Safety / GuardLogix

2

built-in EtherNet/IP ports for ring topology

The Ring That Survives a Cut

A control network's nightmare is a single broken cable taking down a line. Rockwell's answer, built into the Logix controllers' dual Ethernet ports, is the **Device Level Ring (DLR)** — a topology where devices are wired in a closed loop rather than a simple chain. In normal operation traffic flows around the ring; if a cable is cut or a device fails, the ring instantly reconfigures and traffic flows the other way. The line keeps running.



Figure 2. A Device Level Ring. Traffic normally flows one way around the loop; a break triggers instant reconfiguration so no single cut stops the line. *(Representative topology.)*

WHY IT MATTERS TO A TECHNICIAN

A cut cable becomes a warning, not a shutdown

On a DLR, the first cable fault does not stop production — the ring heals and keeps running, but it logs the fault and flags it. That turns an emergency into a scheduled repair: you fix the break at the next planned stop, before a *second* fault on the now-open ring can actually halt the line. Knowing whether you are on a ring or a chain changes how you read a network fault entirely.

Drives, Guardians, and I/O

Around the Logix controller sits the rest of the Integrated Architecture — and, true to the philosophy, all of it speaks CIP over EtherNet/IP. Three families do the heavy lifting.

Kinetix

Allen-Bradley · servo drives

The motion family. Kinetix servo drives (e.g. Kinetix 5700 for high-axis-count machines) deliver integrated motion over EtherNet/IP using CIP Motion — the controller commands axes over the same network that carries everything else, no separate motion bus. The Rockwell counterpart to Siemens SINAMICS.

CARRIES MOTION VIA
CIP Motion / EtherNet/IP

SIEMENS PARALLEL
SINAMICS

GuardLogix & Compact GuardLogix

Allen-Bradley · safety controllers

Safety in the same chassis. GuardLogix 5580 and Compact GuardLogix 5380 run safety logic alongside standard control in one controller, reaching up to **SIL 3 / PLe**. A "safety partner" co-processor handles the higher integrity levels; safety signals travel as CIP Safety over the same EtherNet/IP network. The direct counterpart to a Siemens fail-safe (F) CPU.

RATING
up to SIL 3 / PLe

CARRIES SAFETY VIA
CIP Safety

SIEMENS PARALLEL
S7-1500F

Compact 5000 & FLEX I/O

Allen-Bradley · distributed I/O

The senses. The Compact 5000 I/O platform (catalog 5069) provides local and distributed digital, analog, safety, and specialty modules that clip side-by-side on a DIN rail; FLEX 5000 extends I/O out to the machine over EtherNet/IP. The Rockwell counterpart to the Siemens ET 200 family.

PLATFORM
Compact 5000 (5069)

NETWORK
EtherNet/IP

SIEMENS PARALLEL
ET 200

The software and visualisation tier

Holding it together is **Studio 5000 Logix Designer**, the single environment for programming every Logix controller, configuring motion, and writing safety logic — Rockwell's TIA-Portal equivalent. Above the controllers, the **FactoryTalk** suite provides visualisation and analytics: **FactoryTalk View** for SCADA and the **PanelView** family for panel HMIs at the machine.

THE ROCKWELL STACK AT A GLANCE

Every layer, one protocol

- **Control** — ControlLogix (large), CompactLogix (compact), GuardLogix (safety).
- **Motion** — Kinetix servo drives, via CIP Motion.
- **I/O** — Compact 5000 and FLEX 5000.

- **Network** — EtherNet/IP carrying CIP, with Device Level Ring for resilience.
- **HMI / SCADA** — PanelView, FactoryTalk View.
- **Software** — Studio 5000, FactoryTalk.

Two Worlds, One Lesson

Set the Rockwell stack beside the Siemens stack and the symmetry is striking. Both are complete portfolios spanning every layer; both are unified by a single engineering tool and a single network philosophy; both trade the freedom to mix vendors for the reliability of a system that fits together. The accents differ — chassis-based ControlLogix and CIP over EtherNet/IP on one side, modular S7-1500 and PROFINET on the other — but the underlying grammar is the same.

What makes Rockwell distinct is the purity of its one-protocol idea: that control, motion, and safety are not separate networks to be bridged but profiles of a single language, CIP, riding one EtherNet/IP wire. For a technician, the lesson that survives is portability of understanding. Learn how a layered, protocol-unified architecture works on one vendor's floor, and you can walk onto the other's and read it — because beneath the brand names, every modern factory is solving the same problem the same way. ■

FIELD GLOSSARY

The Rockwell vocabulary in this issue

- **Allen-Bradley** — Rockwell Automation's brand for its industrial control hardware.
- **Integrated Architecture** — Rockwell's name for its unified portfolio, the parallel to Siemens TIA.
- **Logix engine** — the common control engine shared across all modern Logix controllers.
- **ControlLogix (1756)** — the large, chassis-based flagship controller.
- **CompactLogix (5069)** — the compact, DIN-rail controller; the 5480 adds embedded Windows.
- **GuardLogix** — the safety controller running safety logic in the same chassis, up to SIL 3 / PLe.
- **Studio 5000 Logix Designer** — the single engineering software for all Logix controllers.
- **CIP (Common Industrial Protocol)** — the one application protocol carrying control, motion, and safety as profiles.
- **EtherNet/IP** — CIP carried over standard Ethernet; "IP" means Industrial Protocol.
- **CIP Motion / CIP Safety / CIP Sync** — the motion, safety, and time-sync profiles of CIP.
- **Device Level Ring (DLR)** — a ring topology that survives a single cable break without stopping.
- **Kinetix** — the servo-drive family for integrated motion over EtherNet/IP.
- **Compact 5000 / FLEX I/O** — the distributed I/O families.
- **FactoryTalk / PanelView** — the SCADA/analytics suite and the panel-HMI family.
- **ODVA** — the open standards body governing CIP and EtherNet/IP.

WHERE TO SEE THE REAL CATALOG

Every family in this issue has an official Rockwell Automation product page with specs and selectors. Start at the programmable controllers hub (rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html); the ControlLogix 5580 and CompactLogix 5380 pages carry catalog numbers, and EtherNet/IP & CIP are documented by ODVA (odva.org).

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 05

SPECIAL ISSUE · THE FULL PORTFOLIO

The Rockwell & Allen-Bradley Portfolio

A family-by-family tour of the Allen-Bradley catalog behind a Gigafactory-class line — controllers, I/O, drives, motion, networks, safety, HMI, components, and the software that unifies them all.

CONTROL TO COMPONENT

Every layer of the stack, one Integrated Architecture.

STRATIX & ETHERNET/IP

The Cisco-built switches and the CIP network.

POWERFLEX & KINETIX

The VFD and servo families that move the line.

STUDIO 5000

The one environment that programs the whole catalog.

One Catalog, Floor to Enterprise

THE INTEGRATION LAYER · Portfolio guide · Grounded in public product documentation

An earlier issue made the case for why Rockwell's world hangs together: one control engine, one network protocol, one engineering tool. This issue is the companion reference — a guided walk through the actual catalog, family by family, the way you would meet it on the floor. From the controller in the cabinet down to the push-button an operator presses and the contactor that switches a motor, almost every layer of a Gigafactory-class line can be filled from a single Allen-Bradley catalog.

That breadth is the point of the **Integrated Architecture** — Rockwell's name for a portfolio designed so the controller, the drive, the I/O, the network, the safety system, and the HMI all connect natively and are configured from one place. This guide is organised by the layers of the automation pyramid; each section profiles the families that live there.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact equipment, part numbers, or vendors in its factories, and any real floor mixes brands. This is a **portfolio reference for the standard Allen-Bradley / Rockwell catalog** as used in automotive-class automation — drawn from Rockwell's public product documentation. Model names and figures are representative, not confirmed internal Tesla equipment. Catalog families evolve; check the official product pages (listed at the end) for current details.

Enterprise / Information

FactoryTalk suite · analytics · Studio 5000

Visualization

PanelView HMI · FactoryTalk View SCADA

Control & Safety

ControlLogix · CompactLogix · GuardLogix · Micro800

Motion & Power

Kinetix servo · PowerFlex VFDs · servo motors

Network & Field

Stratix switches · Compact 5000 / FLEX / POINT I/O · sensors & components

Figure 1. The Allen-Bradley portfolio mapped to the automation pyramid — the structure of this issue.
(Conceptual, based on public documentation.)

In this issue

1. **Controllers** — the Logix & Micro families
2. **Distributed I/O** — Compact 5000, FLEX, POINT
3. **Drives (VFDs)** — the PowerFlex family
4. **Motion** — Kinetix drives & servo motors
5. **Networking** — Stratix & EtherNet/IP

6. **Safety** — GuardLogix & Guardmaster
7. **Visualization** — PanelView & FactoryTalk
8. **Components & sensors** — the field devices

9. **Software** — Studio 5000 & FactoryTalk
10. **Portfolio at a glance** & the Siemens parallel

PART 1 · CONTROL

The Controllers

Every cell needs a brain, and Rockwell's brains share one re-architected **Logix control engine** — the same engine whether the hardware is a large rack or a compact rail-mount unit, all programmed in one tool. The family scales by size and capability, not by language.

ControlLogix 5580

Allen-Bradley · 1756 · flagship

Large, chassis-based control. The top of the line (catalog 1756-L8x): the memory, speed, and capacity for large, distributed process and motion applications, with controller redundancy and process / extreme-temperature / no-stored-energy variants. The reference platform for plant-wide control.

FORM

Rack (1756)

SCALE

Large / distributed

NETWORK

EtherNet/IP

CompactLogix 5380 & 5480

Allen-Bradley · 5069 · compact

Mid-range, DIN-rail control. Same engine as the 5580, on the Compact 5000 I/O platform, with built-in dual EtherNet/IP ports, integrated motion for up to 32 axes, and ring topologies. The **5480** adds an embedded Windows 10 IoT environment beside the Logix engine for edge computing.

FORM

DIN-rail (5069)

SCALE

Small to medium

MOTION

Up to 32 axes

Micro800 & MicroLogix

Allen-Bradley · micro & nano control

Small, cost-conscious control. The Micro800 family (e.g. Micro820, Micro850, Micro870) and the legacy MicroLogix line handle stand-alone machines and simpler tasks where a full Logix controller is more than the job needs — programmed in Connected Components Workbench.

FORM

Compact micro

SCALE

Small / stand-alone

SOFTWARE

Connected Components Workbench

STILL ON THE FLOOR

The legacy you will still meet

Many running lines still use **SLC 500** and **PLC-5** controllers from earlier generations. They remain supported and in service; Rockwell positions the CompactLogix 5380 and ControlLogix 5580 as the modern migration targets when these are finally retired.

Distributed I/O

Controllers rarely wire every sensor directly; they use distributed I/O mounted near the machine and linked back over EtherNet/IP. Rockwell offers three I/O platforms, scaled to how much wiring density and ruggedness a spot on the floor demands.

Compact 5000 I/O

Allen-Bradley · 5069

The modern flagship I/O. The platform shared with CompactLogix 5380 controllers: digital, analog, safety, and specialty modules that clip side-by-side on a DIN rail. Used both as local controller I/O and, with an EtherNet/IP adapter, as distributed I/O out on the line.

CATALOG

5069

USE

Local & distributed

MODULES

Digital/analog/safety

FLEX 5000 & POINT I/O

Allen-Bradley · distributed I/O

Distributed and granular. FLEX 5000 extends high-performance I/O (including safety) out to the machine over EtherNet/IP; **POINT I/O** offers fine-grained, point-by-point modularity so you add exactly the channels a location needs. Both cut home-run wiring to a single network drop per node.

FLEX 5000

Rugged distributed

POINT I/O

Granular modular

NETWORK

EtherNet/IP

WHY DISTRIBUTED WINS

One cable instead of a hundred

Placing an I/O block beside a cell means a single EtherNet/IP cable runs back to the controller, instead of dozens of individual sensor wires crossing the plant. Less wiring, cleaner cabinets, faster commissioning, and far easier diagnostics — the same logic that drives every modern distributed-I/O design.

Drives: The PowerFlex Family

When a motor needs its speed or torque controlled — a pump, a fan, a conveyor, a spindle — the device that does it is a variable frequency drive (VFD). Rockwell's VFD family is **PowerFlex**, spanning compact machine drives to high-performance, information-rich drives, all designed to integrate with Logix over EtherNet/IP.

PowerFlex 520-Series

Allen-Bradley · compact VFD

Compact, full-featured machine drives. The 520-series (e.g. PowerFlex 525) packs a modular design across roughly 0.4–22 kW with **embedded EtherNet/IP, safety, and USB programming** — the everyday workhorse for conveyors, pumps, and fans on a machine.

POWER

~0.4–22 kW

NETWORK

Embedded EtherNet/IP

EXTRAS

Safety + USB

PowerFlex 753 / 755 / 755T

Allen-Bradley · high-performance VFD

High-performance, information drives. The 750-series scales to large motors with rich control and optional embedded safety. The **755T** adds **TotalFORCE** adaptive field-oriented control of velocity, torque, and position. A dual-port EtherNet/IP option adds Device Level Ring; crucially, these VFDs can mix with Kinetix servo drives on the *same* integrated-motion network.

CLASS

High performance

TECH

TotalFORCE (755T)

NETWORK

EtherNet/IP + DLR option

A VFD **programmed like a servo** — same software, same network, fewer surprises.

PART 4 · MOTION

Motion: Kinetix & Servo Motors

Precise positioning — a robot axis, an indexer, a coordinated multi-axis machine — is the job of servo drives and matched servo motors. Rockwell's servo family is **Kinetix**, delivering integrated motion over EtherNet/IP using CIP Motion, so axes are commanded over the same network that carries everything else.

Kinetix 5500 / 5700

Allen-Bradley · servo drives

Integrated multi-axis servo. The Kinetix 5500 (with VPL servo motors) offers fast loop closure and a shared AC/DC bus for scalable single- or multi-axis systems; the **5700** targets higher axis counts and bigger machines. Both connect over EtherNet/IP and are configured in Studio 5000 alongside the controller.

5500

Compact multi-axis

5700

High axis count

MOTION VIA

CIP Motion

Kinetix 7000 & servo motors

Allen-Bradley · high-power servo & motors

High power and the motors themselves. The Kinetix 7000 handles high-power, single-axis applications with built-in SIL3 Safe Torque Off. On the motor side, families such as **VPL/MPL** (low-inertia, compact, high-torque servo motors) are matched to Kinetix drives — often over a single cable carrying power and feedback together.

KINETIX 7000

High-power, SIL3 STO

MOTORS

VPL / MPL servo

CABLING

Single-cable option

THE ONE-NETWORK PAYOFF

VFDs and servos on the same wire

Because both PowerFlex VFDs and Kinetix servo drives speak integrated motion over EtherNet/IP, a single network can carry simple conveyor speed control and microsecond-coordinated multi-axis motion at once — all programmed in the same Studio 5000 project. That unity is the practical heart of the Integrated Architecture.

Networking: Stratix & EtherNet/IP

The connective tissue of the whole portfolio is **EtherNet/IP** — the CIP protocol carried over standard Ethernet (the "IP" is *Industrial Protocol*). The physical network that carries it is the **Stratix** switch family, built jointly with Cisco so the plant network speaks the same language as enterprise IT.

Stratix Industrial Switches

Allen-Bradley · powered by Cisco

The plant network backbone. Managed industrial Ethernet switches (e.g. the Stratix 5700 Layer-2 managed switch) built on Cisco IOS, configured and diagnosed from within the Integrated Architecture. They bridge IT and OT, support Device Level Ring for resilient topologies, and scale from compact machine switches to backbone units.

BUILT WITH
Cisco IOS

TOPOLOGY
Star / linear / DLR

ROLE
OT-IT bridge

ONE PROTOCOL, MANY PROFILES

CIP carries control, motion, and safety

The defining Rockwell idea: a single application protocol, **CIP**, carries everything as profiles — **CIP Motion** for drives, **CIP Safety** for safety signals, **CIP Sync** for time. So one EtherNet/IP network, switched by Stratix, can run ordinary control, coordinated motion, and certified safety all at once. (Full protocol detail is in Issue 04.)

Safety: GuardLogix & Guardmaster

A plant full of robots and presses needs a certified safety layer, and Rockwell folds it into the same architecture rather than bolting on a separate system. The result spans from the safety controller down to the safety relays, switches, and light curtains on the machine.

GuardLogix & Compact GuardLogix

Allen-Bradley · safety controllers

Safety in the same chassis. GuardLogix 5580 and Compact GuardLogix 5380 run safety logic alongside standard control in one controller, up to **SIL 3 / PLe**, using a "safety partner" co-processor for the highest integrity. Safety travels as CIP Safety over the same EtherNet/IP network as everything else.

RATING
up to SIL 3 / PLe

CARRIES VIA
CIP Safety

PROGRAMMED IN
Studio 5000

Guardmaster Safety Devices

Allen-Bradley · safety field devices

The devices that watch the machine. The Guardmaster line and related families provide safety relays, light curtains, safety interlock and gate switches, emergency-stop devices, and safety I/O — the field hardware that detects a hazard and commands a safe stop, reporting into the GuardLogix controller.

INCLUDES

Relays, curtains, switches

FUNCTION

Detect & safe-stop

REPORTS TO

GuardLogix

PART 7 · VISUALIZATION

Visualization: PanelView & FactoryTalk

An operator needs a window into the machine, and a supervisor needs a view of the whole line. Rockwell covers both: the **PanelView** hardware at the machine, and the **FactoryTalk View** software above it.

PanelView 5000 & 800

Allen-Bradley · operator terminals (HMI)

The face of the machine. The PanelView 5000 graphic terminals are designed to pair tightly with Logix controllers — tags are shared, reducing duplicate engineering — while the PanelView 800 covers smaller, cost-sensitive applications. Touch terminals across a range of sizes for status, alarms, and operator control.

PV 5000

Logix-integrated HMI

PV 800

Compact / economical

TYPE

Graphic touch terminals

FactoryTalk View

Rockwell · HMI / SCADA software

From one machine to the whole site. FactoryTalk View ME (Machine Edition) drives PanelView terminals and single-station HMIs; FactoryTalk View SE (Site Edition) is the distributed SCADA for line- and plant-wide visualization, alarming, and historical data — the supervisory layer above the controllers.

VIEW ME

Machine / terminal HMI

VIEW SE

Site-wide SCADA

LAYER

Supervision

PART 8 · FIELD DEVICES

Components & Sensors

Beneath the controllers and networks lies the layer an operator actually touches and a machine actually senses with — the push-buttons, switches, contactors, and sensors. Allen-Bradley's component catalog is vast, and it is one of the reasons a plant can standardize on a single brand right down to the last device.

Category	Allen-Bradley families (examples)	What they do
Push-buttons & signaling	Bulletin 800F / 800T, tower lights	Operator command & status indication
Contactors & motor control	Bulletin 100 contactors, motor starters	Switching and protecting motors
Circuit & motor protection		Overload & short-circuit protection

	Bulletin 140 motor-protection circuit breakers	
Sensors	Photoelectric, inductive proximity, sensors	Detect presence, position, distance
Safety field devices	Guardmaster relays, interlocks, light curtains	Detect hazards, command safe stop
Relays & signal	Bulletin 700 control & safety relays	Switching control circuits

WHY IT MATTERS TO A TECHNICIAN

Standardizing to the last device

When the contactor, the push-button, the proximity sensor, and the controller all come from one catalog, spare-parts inventory simplifies, documentation is consistent, and a technician learns one set of conventions. The depth of the component catalog is exactly what lets a plant commit to a single architecture end to end.

PART 9 · SOFTWARE

Software: Studio 5000 & FactoryTalk

Hardware breadth would be chaos without one place to configure it. The software is what turns the catalog into a system — and it is, more than any single box, the product Rockwell is really selling.

Studio 5000 Logix Designer

Rockwell · engineering software

The one engineering environment. The single application for programming and configuring every Logix controller, plus motion, drives, and safety logic — using tag-based, common instructions across discrete, process, batch, motion, and safety. The direct counterpart to Siemens TIA Portal.

PROGRAMS

All Logix controllers

COVERS

Motion, drives, safety

PARALLEL

TIA Portal

FactoryTalk Suite

Rockwell · operations & information software

Above the controllers. A family covering visualization (View), historical data (Historian), analytics, asset management, and more — the operations and information layer that turns control data into dashboards, trends, and business insight across a site.

INCLUDES

View, Historian, analytics

LAYER

Operations / information

CONNECTS

Floor → enterprise

The catalog is the hardware. The **system is the software that makes it one thing.**

The Portfolio at a Glance

Laid out side by side, the pattern of the Integrated Architecture is clear: a complete, layered catalog in which every family was designed to connect to the ones above and below it, all carried on one network and built from one tool. Here is the whole tour on one page — with its Siemens equivalents, for anyone fluent in that world.

Layer	Allen-Bradley / Rockwell	Siemens parallel
Large control	ControlLogix 5580 (1756)	SIMATIC S7-1500
Compact control	CompactLogix 5380 / 5480 (5069)	S7-1200 / ET 200SP CPU
Micro control	Micro800 / MicroLogix	S7-1200 / LOGO!
Safety control	GuardLogix / Compact GuardLogix	S7-1500F
Distributed I/O	Compact 5000 · FLEX 5000 · POINT I/O	ET 200
VFD drives	PowerFlex 520 / 750-series	SINAMICS G120
Servo drives	Kinetix 5500 / 5700 / 7000	SINAMICS S120 / S210
Network	EtherNet/IP + CIP · Stratix switches	PROFINET · SCALANCE
HMI	PanelView 5000 / 800	SIMATIC HMI
SCADA / software	FactoryTalk View · Studio 5000	WinCC · TIA Portal

No single family in this catalog is irreplaceable; competitors make excellent controllers, drives, and switches too. What is hard to replicate is the *coherence* — a portfolio where the controller, the drive, the I/O, the network, the safety system, and the HMI all speak CIP over EtherNet/IP and are engineered from one Studio 5000 project. For a Gigafactory-class line, that fit is the reason a plant commits to one catalog from the cabinet to the contactor. ■

FIELD GLOSSARY

The portfolio vocabulary, in brief

- **Integrated Architecture** — Rockwell's name for its unified, connect-everything portfolio.
- **Logix engine** — the shared control engine inside all modern Logix controllers.
- **ControlLogix / CompactLogix / GuardLogix / Micro800** — the large, compact, safety, and micro controllers.
- **Compact 5000 / FLEX 5000 / POINT I/O** — the distributed I/O platforms.
- **PowerFlex** — the VFD (variable frequency drive) family.
- **Kinetix** — the servo-drive family for integrated motion over EtherNet/IP.
- **Stratix** — the Cisco-built industrial Ethernet switch family.
- **EtherNet/IP & CIP** — the network transport and the one application protocol it carries.
- **Guardmaster** — the safety field-device line (relays, curtains, interlocks).
- **PanelView** — the operator-terminal (HMI) family.
- **FactoryTalk** — the visualization, historian, and analytics software suite.
- **Studio 5000** — the single engineering environment for the whole catalog.

WHERE TO SEE THE REAL CATALOG

Every family here has an official Rockwell product page with specs and selectors. Start at the Allen-Bradley products hub (rockwellautomation.com/en-us/products/hardware/allen-bradley.html); the programmable controllers, PowerFlex drives, Kinetix motion, and Stratix networking pages carry catalog numbers; EtherNet/IP and CIP are documented by ODVA (odva.org).

THE INTEGRATION LAYER · Issue 05 · A technical magazine · Sources: Rockwell Automation & ODVA public documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 06

COVER STORY · WHEN THE PC BECOMES THE PLC

The Beckhoff Way: One PC, One Bus, One Software

PC-based control and EtherCAT — the radical idea that an industrial PC running TwinCAT can replace a cabinet full of dedicated controllers, all synchronised over the fieldbus Beckhoff invented.

THE PC IS THE CONTROLLER

Why Beckhoff puts a computer, not a PLC, at the centre.

TWINCAT DOES IT ALL

PLC, motion, CNC, vision, safety — one software, one PC.

ETHERCAT

The bus Beckhoff invented — microsecond-fast, nanosecond-synced.

FLYING MAGNETS

XTS and XPlanar: motion products no one else has.

When the PC Becomes the PLC

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

The first two vendors in this series — Siemens and Rockwell — disagree about many things, but they agree on one: the controller is a dedicated box, a PLC, purpose-built to run machine logic. Beckhoff built its entire company on rejecting that premise. Its founding idea, dating to the 1980s, is that a standard industrial PC is powerful enough to be the controller — that the PLC, the motion controller, the CNC, and more are not separate hardware but *software tasks* running on one computer. Beckhoff calls it PC-based control, and it makes the company the contrarian of the automation world.

If that idea is the engine, then EtherCAT is the driveshaft. Having decided the PC is the brain, Beckhoff needed a way for that brain to reach every motor, sensor, and valve with hard real-time precision — so in 2003 it invented its own industrial-Ethernet fieldbus. Together, PC-based control and EtherCAT define a stack that looks genuinely different from its competitors': flatter, more software-centric, and unusually fast.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact equipment, part numbers, or vendors in its factories, and any real floor mixes brands. This is a **portfolio reference for the standard Beckhoff catalog** as used in high-performance automation — drawn from Beckhoff's public product documentation. Model names and figures are representative, not confirmed internal Tesla equipment. Beckhoff is especially associated with high-dynamic motion and packaging-style machinery; treat the examples as illustrative of the platform, not as a claim about any specific line.

The shape of a PC-based stack

The practical consequence of the philosophy is a flatter architecture. Where a traditional design stacks a PLC, then a separate motion controller, then a separate safety controller, Beckhoff collapses those into TwinCAT tasks on one Industrial PC, and fans everything out over a single EtherCAT network. The diagram below is deliberately drawn to contrast with the classic pyramid of earlier issues.

The Industrial PC + TwinCAT 3

PLC + Motion + CNC + Vision + Safety + Robotics — all as software tasks on one PC

EtherCAT**EtherCAT network**

one real-time bus to everything — I/O, drives, safety

Field

EtherCAT Terminals · servo drives & motors · XTS / XPlanar · sensors

Figure 1. The Beckhoff architecture is flatter than the classic pyramid: control layers become software tasks on one PC, unified by EtherCAT. (Conceptual, based on public documentation.)

In this issue

1. **The Industrial PC** — the controller, reimagined
2. **EtherCAT** — the bus Beckhoff invented
3. **EtherCAT I/O** — terminals & boxes
4. **Drive technology** — AX, AMP & motors
5. **Mechatronics** — XTS & XPlanar
6. **TwinCAT 3** — the software that does everything
7. **Safety** — TwinSAFE
8. **Portfolio at a glance** & the parallels

PART 1 · THE CONTROLLER

The Industrial PC, Reimagined

In the Beckhoff world there is no "PLC product" to buy — you buy an **Industrial PC** and the control runs on it as software. The hardware ranges from server-class machines down to devices small enough to clip beside a sensor, but all of them run the same TwinCAT runtime, so the choice is purely about how much computing power the job needs and where it has to live.

C60xx Ultra-Compact IPC

Beckhoff · cabinet Industrial PC

The high-performance brain. The ultra-compact C60xx series (e.g. C6030, and the C6032 with extra PCIe slots) packs quad-core, 3.6 GHz Core™ i7-class processors into a small cabinet-mount housing — enough to run demanding PLC, motion, vision, and analytics tasks simultaneously in TwinCAT.

CLASS

Ultra-compact IPC

CPU

up to Core i7 quad

RUNS

TwinCAT 3

Embedded PC (CX series)

Beckhoff · DIN-rail controller

The DIN-rail computer. The CX family are modular embedded PCs that mount on a DIN rail and connect directly to EtherCAT or Bus Terminals — the closest Beckhoff comes to a "PLC form factor," but it is still a full PC running TwinCAT, scalable from ARM to Intel CPUs.

FORM

DIN-rail embedded PC

I/O

Direct terminal bus

CPU

ARM to Intel

Control on the machine, and the operator's screen. The C7015 is an IP65/67 ultra-compact PC for direct mounting on a machine, with EtherCAT P onboard. Beckhoff also builds Panel PCs and Control Panels — combining a computer and a multi-touch display — so the same PC platform provides both control and the HMI.

C7015

IP65/67 on-machine

PANEL PC

PC + touch display

HMI VIA

TwinCAT HMI

There is no PLC to buy. You buy a **computer — and the control is software.**

PART 2 · THE NETWORK

EtherCAT: The Bus Beckhoff Invented

If one technology defines Beckhoff, it is EtherCAT — the real-time industrial-Ethernet system it developed and then opened to the industry (now governed by the EtherCAT Technology Group). Its trick is "processing on the fly": instead of each device receiving a packet, reading it, and forwarding it, an EtherCAT frame passes *through* every node, which reads its data and inserts its reply as the frame streams by. The result is famous low latency — one frame can service hundreds of devices.

EtherCAT

EtherCAT Technology Group · the real-time bus

Processing on the fly. A single frame streams through all nodes, each reading and writing as it passes — giving microsecond cycle times. **Distributed Clocks** synchronise every device to within nanoseconds, the key to coordinated multi-axis motion. Variants: **EtherCAT G** (gigabit, for high-bandwidth devices like cameras) and **EtherCAT P** (power and communication in a single cable).

CYCLE

Microsecond-class

SYNC

Distributed Clocks, ns

VARIANTS

EtherCAT G · EtherCAT P

THE DETERMINISM ANGLE

Why "on the fly" matters

Recall the governing idea of control networks (Issue 03): determinism, not raw speed. EtherCAT's pass-through design makes its timing both extremely fast *and* extremely predictable, while its Distributed Clocks give every axis a shared nanosecond-accurate sense of time. That combination is precisely what high-dynamic, tightly coordinated motion demands — and it is why a single EtherCAT line can run an entire machine.

2003

 year Beckhoff introduced
EtherCAT

μs

cycle times across the bus

ns

 axis sync via Distributed
Clocks

1 cable

 power + data with EtherCAT
P

EtherCAT I/O: Terminals & Boxes

Beckhoff's I/O is granular to an unusual degree: rather than fixed multi-channel cards, it offers thousands of slim terminals, often just one or a few channels wide, so a designer assembles exactly the I/O a machine needs, channel by channel. They snap together on a rail and speak EtherCAT directly.

EtherCAT Terminals (EL series)

Beckhoff · in-cabinet I/O

One slice per function. The EL terminals are slim, DIN-rail EtherCAT I/O modules — digital, analog, measurement, communication, and specialty types — assembled side by side into exactly the mix a machine needs. Specialised terminals add power measurement, overcurrent protection, and condition monitoring right in the I/O. (The earlier KL Bus Terminals serve other fieldbuses.)

FORM

Slim DIN-rail terminals

BUS

EtherCAT (native)

RANGE

Digital/analog/measure

EtherCAT Box (EP / EPP series)

Beckhoff · on-machine I/O (IP67)

I/O out on the machine, no cabinet. The EtherCAT Box modules are fully sealed IP67 I/O for direct mounting on the machine, connecting sensors and actuators in the field. The EPP (EtherCAT P) versions carry power and communication on one cable — enabling decentralised, cabinet-free machine designs.

PROTECTION

IP67 sealed

MOUNT

On the machine

EPP

EtherCAT P single-cable

Drives & Motors

Because Beckhoff is, at heart, a motion company, its drive technology is where the EtherCAT advantage shows most clearly. The motion controller is a TwinCAT task on the PC; it commands the drives over EtherCAT with new setpoints as often as every 62.5 microseconds — against the roughly one-millisecond cycle of more conventional systems.

AX8000 Multi-Axis Servo System

Beckhoff · high-performance servo

One of the fastest drives on the market. The modular AX8000 accepts new setpoints every **62.5 µs** over EtherCAT, with microsecond current scans and nanosecond axis synchronisation via Distributed Clocks. Newer axis modules use silicon-carbide (SiC) semiconductors for more output in the same compact footprint.

TYPE

Multi-axis servo

SETPOINT

every 62.5 µs

TECH

SiC axis modules

AMP8000 Distributed Servo System

Beckhoff · drive-in-motor

The drive moves into the motor. AMP8000 integrates the servo drive directly into a compact servomotor, with power and communication delivered in one cable via EtherCAT P. The result removes drives from the control cabinet entirely — a decentralised approach that shrinks or eliminates the cabinet. The AMI series integrates drive, motor, and I/O together.

CONCEPT

Drive built into motor

CABLING

EtherCAT P, one cable

BENEFIT

Cabinet-free design

Servomotors & linear motors

Beckhoff · AM, AL, AA families

The motors themselves. Rotary servomotors (AM series), linear servomotors (AL80xx) for direct linear motion without mechanical conversion, and tubular motors (e.g. AA2518, delivering high force to replace spindles). TwinCAT Motion Designer helps engineers match motor, gearbox, cable, and drive into an optimal combination.

ROTARY

AM servomotors

LINEAR

AL80xx

TUBULAR

AA series

New setpoints every 62.5 microseconds, versus the usual one millisecond. The bus is the advantage.

PART 5 · MECHATRONICS

Flying Magnets: XTS & XPlanar

Here is where Beckhoff offers something its rivals simply do not have in the same form: motion *systems* that rethink how products move through a machine. Both turn the conveyor and the indexing table into a software-defined, magnetically driven surface — and both are controlled as ordinary axes in TwinCAT.

XTS — eXtended Transport System

Beckhoff · linear transport

Independently controlled movers on a track. XTS is a modular track with magnetically driven **movers** that can each be positioned independently and moved at will — replacing rigid mechanical conveyors and indexers. Movers can group, space out, and re-sequence on the fly, enabling flexible assembly and handling. Hygienic and washdown (IP69K) versions exist for food and pharma.

FORM

Modular track + movers

CONTROL

each mover an axis

USE

Flexible transport

XPlanar — Planar Motor System

Beckhoff · 2D levitating motion

Flying magnets with six degrees of freedom. XPlanar floats magnetic movers above a tiled motor surface, moving them freely in two dimensions — and even tilting and lifting them — with no contact and no wear. It enables non-linear, lot-size-1 process flows for adaptive packaging, material handling, and inspection. The integrated AMI servo drive concept underpins the tiles.

MOTION

2D, contact-free

FREEDOM

6 degrees

USE

Adaptive, lot-size-1

WHY IT MATTERS

The machine becomes programmable in space

A traditional conveyor moves everything at one speed in one direction. With XTS and XPlanar, the *path* of every product is software — movers accelerate, pause, regroup, and re-route under program control. For a manufacturer chasing flexibility and frequent product change, that turns mechanical re-tooling into a code change, the same instinct that drives the whole PC-based philosophy.

PART 6 • SOFTWARE

TwinCAT 3: One Software for Everything

Everything in the Beckhoff world converges on one piece of software. **TwinCAT 3** is both the engineering environment and the real-time runtime — and, critically, it is built inside Microsoft Visual Studio, so PLC code, C++, and MATLAB/Simulink models live in one project. On the running PC, TwinCAT turns the standard processor into a real-time controller, executing all the disciplines below as parallel tasks.

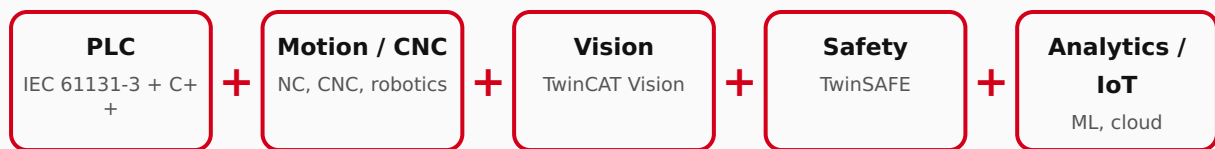


Figure 2. TwinCAT 3 runs every automation discipline as parallel real-time tasks on one Industrial PC — the software embodiment of PC-based control. (*Representative module set.*)

This is the payoff of the whole philosophy. Because control is software, adding a capability — machine vision, a robot kinematic, a machine-learning model, condition monitoring — means adding a TwinCAT module, not another box on the rail. **TwinCAT Vision** brings real-time image processing into standard PLC languages; **TwinCAT Motion / CNC** handles coordinated motion and machining; **TwinCAT Analytics** and **Cloud Engineering** reach to IoT and the cloud.

THE ENGINEERING ANGLE

One project, one language base, one debugger

Building on Visual Studio means an engineer programs the PLC, writes C++ algorithms, configures motion, and integrates simulation models in a single environment, with one set of tools and one debugger. It is Beckhoff's answer to TIA Portal and Studio 5000 — but with the boundary between "PLC programming" and "software development" deliberately erased.

PART 7 • SAFETY

Safety: TwinSAFE

T rue to the platform, safety is not a separate system but another layer riding on EtherCAT and configured in TwinCAT. **TwinSAFE** provides safety I/O terminals and a safety logic that

travels over the standard EtherCAT network, so emergency stops, light curtains, and guards integrate into the same architecture as everything else — with the diagnostics and data access that brings.

TwinSAFE

Beckhoff · functional safety

Safety over the same bus. A scalable, modular safety system using EtherCAT: TwinSAFE I/O terminals capture safety signals, and safety logic runs without a separate safety controller, expanding design flexibility, diagnostics, and data insight. Safe motion functions integrate with the drives. The counterpart to PROFIsafe and CIP Safety in earlier issues.

CARRIED OVER EtherCAT	FORM Safety I/O + logic	CONFIGURED IN TwinCAT
--------------------------	----------------------------	--------------------------

REFERENCE

The Portfolio at a Glance

Set beside the Siemens and Rockwell catalogs, the Beckhoff portfolio is the same idea reached from the opposite direction. The others build dedicated hardware and unify it with software and a network; Beckhoff starts from software on a PC and a single fast bus, and adds only the hardware the physics demands. The families map cleanly onto their rivals' — with two products, XTS and XPlanar, that have no direct equivalent.

Function	Beckhoff	Siemens / Rockwell parallel
Controller	Industrial PC (C60xx) + TwinCAT	S7-1500 / ControlLogix
Compact controller	Embedded PC (CX series)	S7-1200 / CompactLogix
On-machine controller	C7015	(distributed CPU variants)
Network	EtherCAT (+ G, + P)	PROFINET / EtherNet/IP
Distributed I/O	EtherCAT Terminals (EL) · EtherCAT Box	ET 200 / Compact 5000
Servo drives	AX8000 · AMP8000 distributed	SINAMICS S120 / Kinetix
Safety	TwinSAFE	PROFIsafe / CIP Safety
HMI	Panel PC + TwinCAT HMI	SIMATIC HMI / PanelView
Software	TwinCAT 3 (in Visual Studio)	TIA Portal / Studio 5000
Mechatronics	XTS · XPlanar	— (no direct equivalent)

No single Beckhoff product is irreplaceable; the contrarian bet is the architecture itself. By making the PC the controller and EtherCAT the one bus, Beckhoff trades the reassuring familiarity of a dedicated PLC for raw performance, software flexibility, and a strikingly flat system — and then offers motion products, in XTS and XPlanar, that only this software-and-magnets worldview could produce. For a line that prizes high-dynamic motion and frequent change, that is a genuinely different answer to the same question the whole series has been asking. ■

FIELD GLOSSARY

The Beckhoff vocabulary in this issue

- **PC-based control** — Beckhoff's founding philosophy: a standard industrial PC *is* the controller, running control as software.
- **EtherCAT** — the real-time industrial-Ethernet bus Beckhoff invented (2003); "processing on the fly" gives microsecond cycles.
- **Distributed Clocks** — EtherCAT's mechanism for synchronising all devices to within nanoseconds.
- **EtherCAT G / EtherCAT P** — the gigabit variant, and the variant carrying power + data on one cable.
- **Industrial PC (C60xx / CX / C7015)** — the cabinet, DIN-rail embedded, and on-machine computers that run TwinCAT.
- **EtherCAT Terminals (EL) / EtherCAT Box** — the slim in-cabinet I/O and the IP67 on-machine I/O.
- **AX8000 / AMP8000** — the multi-axis servo system, and the distributed drive-in-motor system.
- **XTS** — the eXtended Transport System: independently controlled magnetic movers on a track.
- **XPlanar** — the planar motor system: magnetic movers levitating and moving freely in 2D.
- **TwinCAT 3** — the engineering environment and real-time runtime, built in Visual Studio, running PLC, motion, vision, and more.
- **TwinSAFE** — Beckhoff's EtherCAT-based functional-safety system.

WHERE TO SEE THE REAL CATALOG

Every family here has an official Beckhoff product page with specs and configurators. Start at the Beckhoff products hub (beckhoff.com/en-en/products/); the Industrial PC, EtherCAT Terminals, drive technology, XTS/XPlanar, and TwinCAT 3 sections carry full detail. EtherCAT itself is governed by the EtherCAT Technology Group (ethercat.org).

THE INTEGRATION LAYER · Issue 06 · A technical magazine · Sources: Beckhoff & EtherCAT Technology Group public documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 07

COVER STORY · WHERE ELECTRICITY BECOMES MOTION

Motors, Drives & the Art of Controlled Motion

How a VFD spins a pump, how a servo positions a robot to the micron, and why the difference between them — open loop versus closed loop — explains almost everything on the factory floor.

THE VFD

Rectify, smooth, invert — how a drive makes adjustable AC.

THE SERVO SYSTEM

Drive, motor, and encoder in a microsecond feedback loop.

OPEN VS CLOSED LOOP

The one distinction that separates a VFD from a servo.

INDUCTION VS PM

Why the motor type decides the drive type.

Where Electricity Becomes Motion

THE INTEGRATION LAYER · Feature · A cross-vendor explainer grounded in engineering fundamentals

Every other system in this magazine exists to command this one moment: the instant electricity becomes motion. A controller can reason, a network can carry, a sensor can report — but nothing physically moves until current flows into a motor and a shaft turns. The pair that governs that moment is the **drive** and the **motor**, and understanding them is understanding the muscle of the factory. This issue steps away from any single brand to explain the technology itself.

Two families dominate the floor, and they answer two different questions. A **variable frequency drive (VFD)** answers "how fast?" — it controls the *speed* of an ordinary AC motor turning a pump, fan, or conveyor. A **servo system** answers "exactly where, and exactly when?" — it controls the precise *position* of a specialised motor moving a robot axis or an indexing table. The line between them runs through one idea: feedback.

EDITOR'S NOTE · ON SCOPE

This is a fundamentals issue, not a brand tour. The principles here — rectifier-bus-inverter, V/Hz, open versus closed loop, induction versus permanent-magnet — are universal and govern every vendor's products, including the Siemens SINAMICS, Rockwell PowerFlex/Kinetix, and Beckhoff AX/AMP families covered in earlier issues. Examples are illustrative of the technology, not claims about any specific factory. Figures are conceptual.

The whole field on one axis: open loop vs closed loop

Before the hardware, the governing distinction. A VFD typically runs **open loop**: it sends the motor a commanded speed and trusts it to follow, with no continuous check. A servo runs **closed loop**: an encoder constantly reports the motor's actual position back to the drive, which compares it to the command and corrects the error thousands of times a second. That single difference — whether the system watches what the motor actually did — cascades into everything: cost, precision, response, and the type of motor each one drives.



Figure 1a. Open loop (a typical VFD): command goes in, motor spins, nothing checks the result. Simple, robust, economical.



Figure 1b. Closed loop (a servo): the encoder's feedback closes the loop, so the drive corrects any error continuously. Precise, dynamic, more costly.

In this issue

1. **The electric motor** — how a shaft turns
2. **The VFD** — making adjustable AC
3. **V/Hz & vector control** — how a VFD thinks
4. **The servo system** — the feedback loop
5. **Induction vs permanent-magnet** — the motor decides
6. **Selecting & surviving** — specs, heat, and faults
7. **The vendor landscape** & glossary

PART 1 · THE MACHINE

The Electric Motor: How a Shaft Turns

At its core every industrial motor does the same thing: it turns electrical energy into rotating mechanical energy using magnetism. A current in the stationary windings (the **stator**) creates a rotating magnetic field; that field drags the rotating part (the **rotor**) around with it, and the rotor's shaft does the work. The families differ in *how* the rotor is made to follow the field — and that single design choice ripples all the way out to which drive you must use.

The AC Induction Motor

the workhorse · squirrel-cage rotor

The factory's most common motor. The rotating stator field *induces* a current in a simple cage-shaped rotor, which creates its own field and follows along — always slightly behind, a gap called **slip**. Rugged, cheap, and needs no magnets or rotor connections. Ideal for pumps, fans, and conveyors. Paired with a VFD for speed control.

ROTOR

Cage (induced)

TRAIT

Has slip

DRIVEN BY

VFD

The Permanent-Magnet (PM) Servo Motor

the precision machine

The motor built for control. The rotor carries **rare-earth permanent magnets**, so it locks exactly in step with the stator field — a synchronous motor with no slip. This precise, predictable behaviour, plus low inertia for fast acceleration, is what makes accurate positioning possible. Almost always paired with a servo drive and an encoder.

ROTOR

Permanent magnets

TRAIT

Synchronous, no slip

DRIVEN BY

Servo drive

THE KEY INSIGHT

The motor and the drive are a matched pair

You cannot freely mix a motor and a drive. A cage-rotor induction motor is built for a VFD's open-loop, frequency-based control; a permanent-magnet servo motor is built for a servo drive's closed-loop, feedback-based control. Choosing the motor type effectively chooses the drive type — which is why every vendor sells them as engineered families (SINAMICS+SIMOTICS, Kinetix+its motors, AX8000+its servomotors).

PART 2 • THE DRIVE

The VFD: Making Adjustable AC

An AC motor's speed is set by the frequency of its supply — but the grid delivers a fixed frequency (50 or 60 Hz). To vary the speed you must manufacture AC at whatever frequency you want, and that is precisely what a variable frequency drive does. It performs a clever three-stage trick on the incoming power.



Figure 2. The three stages inside every VFD: rectify the grid's AC to DC, smooth it on the DC bus, then synthesise fresh AC at any frequency. *(Conceptual.)*

The magic is in stage three. Using fast-switching power transistors (IGBTs), the inverter chops the DC bus into pulses that, on average, look like a smooth sine wave at the frequency the drive chooses — a technique called pulse-width modulation (PWM). Want the motor slower? Synthesise a lower frequency. The same stages, incidentally, are why the shared DC bus of a multi-axis system (Issue 02) can recycle braking energy: it all meets on that middle stage.

The grid gives you **one frequency. A VFD manufactures **any frequency** you ask for.**

A NOTE FOR TECHNICIANS

Why a VFD saves so much energy

On a pump or fan, the power needed drops with roughly the *cube* of speed — so running at 80% speed instead of full-throttle-plus-a-valve can cut energy dramatically. That cube law is the single biggest reason VFDs are everywhere: slowing the motor to match the actual demand, rather than throttling a full-speed motor, is enormous savings.

How a VFD Thinks: V/Hz to Vector

Not all VFD control is equally clever, and the method matters when the load gets demanding. Drives offer a ladder of control strategies, from simple and robust to precise and responsive — and knowing which a machine needs is a core selection decision.

Control method	How it works	Best for
V/Hz (scalar)	Keeps voltage proportional to frequency by a fixed rule; open loop, no model of the motor	Pumps, fans, simple conveyors — cheap and robust
Sensorless vector	Builds a mathematical model of the motor to control torque and speed precisely — still no encoder	Demanding loads needing good torque at low speed
Closed-loop vector	Adds an encoder for full feedback — approaching servo-like control of an induction motor	High-precision speed/torque without a true servo

Two more concepts a technician meets constantly. **Field weakening:** a motor can run *above* its rated speed if you accept less torque — running at less than rated volts-per-hertz above base speed (so a motor can deliver, say, 80% torque at 125% speed for the same power). And the **cube law** from Part 2, which is why variable-torque (pump/fan) and constant-torque (conveyor) loads are configured differently in the drive.

PART 4 · PRECISION MOTION

The Servo System: The Feedback Loop

A servo is not a single device but a *system* — four parts working as one tight loop: a controller issuing commands, a servo drive, a servo motor, and a feedback device. What makes it a servo, rather than just a fast VFD, is that loop: the constant, high-speed comparison of commanded position against actual position.

The Closed-Loop Servo			controller · drive · motor · encoder
Command, compare, correct — thousands of times a second. The drive takes a command (from a PLC or motion controller) and compares it to feedback from the motor's encoder or resolver. The instant there is a discrepancy — the motor is a hair off where it should be — the drive responds to the error. Servo speed-response bandwidths can reach the kHz range, versus the hundreds of Hz typical of a VFD.			
LOOP Closed (feedback)	FEEDBACK Encoder / resolver	BANDWIDTH up to kHz	

The three things a servo controls, in order of sophistication, are position, velocity, and torque — usually as nested loops, the position loop wrapping the velocity loop wrapping the torque (current) loop. This is why a servo can hold a robot arm dead still against a force, snap it to a new position in milliseconds, and know its location to a fraction of a degree the entire time.

VFD VS SERVO — THE HONEST SUMMARY

It is about the job, not "better or worse"

A servo is not a "better VFD" — it is the right tool for a different job. Need to spin a fan or pump efficiently? A VFD is simpler, cheaper, and ideal. Need to position a robot axis to the micron and snap between points? Only a closed-loop servo will do. Many machines use both: VFDs on the conveyors and pumps, servos on the precise axes — often, today, on the same network.

kHz

servo response bandwidth

<20ms

servo 0→2000 rpm, no load

~cube

how pump power scales with speed

2

loops: open (VFD) vs closed (servo)

PART 5 · THE MATCHED PAIR

Induction vs Permanent-Magnet

We can now draw the whole picture together, because the motor type and the drive type are two ends of one decision. The table below is the heart of this issue — the map a technician carries in their head when they look at any axis on the floor and ask, "what is moving this, and why?"

	AC Induction + VFD	PM Servo + Servo Drive
Rotor	Squirrel cage (induced current)	Rare-earth permanent magnets
Speed behaviour	Asynchronous — lags by slip	Synchronous — locked to field
Control loop	Usually open loop	Always closed loop (encoder)
Controls	Speed (and torque, with vector)	Position, velocity, and torque
Precision	Good speed control	Micron-level positioning
Inertia / response	Higher inertia, slower	Low inertia, very fast
Cost	Lower	Higher
Typical job	Pumps, fans, conveyors, spindles	Robots, indexing, CNC, packaging

Choose the motor and you have **chosen the drive. They are two halves of one decision.**

PART 6 · IN PRACTICE

Selecting & Surviving: Specs, Heat & Faults

Knowing the theory is half the job; living with the hardware is the other half. A few practical realities define day-to-day work with motors and drives on the floor.

What the nameplate and the spec sheet really tell you

A motor is rated by power (kW or hp), base speed (rpm), voltage, and — critically — torque. The distinction that trips people up is **constant-torque vs variable-torque** loads: a conveyor needs the same torque at any speed (constant torque), while a pump or fan needs far less torque at low speed (variable torque, the cube law). The drive must be configured for the right one, or it will be either undersized or wildly inefficient.

THE THINGS THAT ACTUALLY GO WRONG

A technician's short list

- **Heat** — the enemy of every drive. Drives are derated at high ambient temperature; a hot cabinet shortens their life. Cooling and clean filters matter more than people expect.
- **Harmonics & EMC** — a VFD's fast switching injects electrical noise back onto the supply and out along the motor cable; shielded cable, grounding, and line reactors exist to tame it.
- **Encoder faults** — on a servo, the feedback device is the single point the whole loop depends on. A flaky encoder shows up as erratic motion or position errors before anything mechanical is actually wrong.
- **Bearing currents** — PWM drives can induce small currents through motor bearings; over time this pits them. It is why drive-rated motors and proper grounding exist.

The recurring lesson, echoing earlier issues: when motion is wrong, the cause is often electrical or in the feedback, not mechanical. A servo axis hunting or drifting points to the encoder or tuning long before it points to a worn gear. Read the drive's fault log first.

REFERENCE

The Landscape, and the Vocabulary

Every vendor in this series sells the same two technologies under their own names, engineered into matched families. Set side by side, the pattern from the whole magazine repeats one last time: the same fundamental physics, dressed in different brands. Here is where the motors and drives of the earlier issues fit on the map this issue has drawn.

Technology	Siemens	Rockwell / Allen-Bradley	Beckhoff
VFD (induction)	SINAMICS G120	PowerFlex 520 / 750	(general drives)
Servo drive	SINAMICS S120 / S210	Kinetix 5500 / 5700	AX8000 / AMP8000
Servo motor	SIMOTICS S-1FK7	Kinetix VPL / MPL motors	AM / AL servomotors
Induction motor	SIMOTICS GP / SD	(MP-series & others)	(catalog motors)

No motor and no drive is magic on its own; the engineering is in the matching, and the understanding is in the loop. Master the one distinction this issue is built on — open loop for speed, closed loop for position — and you can walk up to any moving axis in any factory, on any vendor's hardware, and know what is turning that shaft and why. That is the muscle of the factory, and now you can read it. ■

FIELD GLOSSARY

The motion vocabulary in this issue

- **VFD (Variable Frequency Drive)** — a drive that controls an AC motor's speed by synthesising AC at an adjustable frequency; also called an inverter or AC drive.
- **Servo drive / servo system** — a closed-loop system (controller + drive + motor + feedback) for precise position, velocity, and torque control.
- **Open loop / closed loop** — whether the system continuously checks the motor's actual output (closed) or not (open); the defining VFD-vs-servo distinction.
- **Stator / rotor** — the stationary windings that create the rotating field, and the rotating part that follows it.
- **Induction motor** — an AC motor whose cage rotor follows the field by induced current, always lagging by slip.
- **Permanent-magnet (PM) motor** — a synchronous motor with magnets in the rotor, locking it to the field; the basis of servo motors.
- **Slip** — the small speed gap by which an induction rotor lags the stator field.
- **Rectifier / DC bus / inverter** — the three stages inside a VFD: AC→DC, smoothing, DC→adjustable AC.
- **PWM (Pulse-Width Modulation)** — the technique of switching the DC bus into pulses that average to a sine wave at the desired frequency.
- **V/Hz & vector control** — the ladder of VFD control methods, from a simple voltage/frequency rule to a full motor model.
- **Field weakening** — running a motor above base speed by accepting reduced torque.
- **Encoder / resolver** — the feedback device reporting a servo motor's actual position.
- **Constant- vs variable-torque load** — loads needing the same torque at all speeds (conveyor) vs much less at low speed (pump/fan, the cube law).

FURTHER READING

The motor and drive families named here are detailed on each vendor's product pages — Siemens SINAMICS / SIMOTICS ([siemens.com](https://www.siemens.com)), Rockwell PowerFlex / Kinetix ([rockwellautomation.com](https://www.rockwellautomation.com)), and Beckhoff drive technology ([beckhoff.com](https://www.beckhoff.com)) — and were covered in Issues 02, 05, and 06 of this series. The fundamentals here (V/Hz, vector control, PWM, slip, field weakening) are standard motor-control engineering, documented widely in drive manufacturers' technical literature.

THE INTEGRATION LAYER · Issue 07 · A technical magazine · Sources: motor-control engineering fundamentals & vendor technical documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 08

COVER STORY · THE FACTORY THAT REFUSES TO STORE THINGS

Logistics, Inventory & the Art of Not Holding Stock

How a Gigafactory keeps parts flowing just-in-time, tracks every component to the bin, and runs a warehouse it would rather not need — the quiet discipline of material flow behind every car built.

JUST-IN-TIME

Why the cheapest part to store is the one you never store.

THE WMS

The software that knows where everything is, to the bin.

THE TWO-BIN KANBAN

The elegant trick that reorders parts automatically.

VERTICAL INTEGRATION

Making it in-house to shorten the supply chain.

The Factory That Refuses to Store Things

THE INTEGRATION LAYER · Feature · An explainer grounded in public reporting & logistics fundamentals

Inventory is a paradox. A factory cannot build cars without parts on hand — yet every part sitting in a warehouse is money frozen, space consumed, and risk accumulating. A bolt in a bin earns nothing; it only costs. The entire discipline of inventory control is the art of resolving that paradox: having exactly the right part, in exactly the right place, at exactly the moment it is needed — and not one minute, or one bin, sooner. This issue is about how a Gigafactory-class operation does that.

The governing philosophy, borrowed and sharpened from Toyota, is **just-in-time (JIT)**: components arrive precisely when they are needed, so the amount of inventory that must be stored is minimised. Reporting consistently describes Tesla as running a JIT system in this Toyota lineage — ordering parts and materials only as required, which reduces storage cost and ensures parts are there when the line calls for them. The aim is a factory where material flows through, rather than piles up.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the internal design of its inventory and warehouse systems — like MOS (Issue 01), the specifics are proprietary. This article is an explainer assembled from public reporting on Tesla's supply chain (its JIT approach, two-bin kanban for small parts, vertical integration, and Gigafactory siting) together with the **standard, universal principles** of logistics and inventory control (WMS, ABC analysis, cycle counting, JIT/JIS, warehouse automation). Treat numbers and mechanisms as representative of how a plant of this class operates, not as confirmed internal Tesla configuration. Diagrams are conceptual.

The one idea: inventory is a cost to be minimised, not a comfort to be hoarded

Traditional manufacturing kept large buffers "just in case." Lean manufacturing inverts that: inventory hides problems and ties up cash, so you drive it down deliberately and fix the problems that buffers used to mask. The whole toolkit in this issue — kanban, the WMS, cycle counting, vertical integration — exists to let a factory hold *less* stock without ever starving the line.

A bolt in a bin earns nothing. The cheapest part to store is the one you never store.

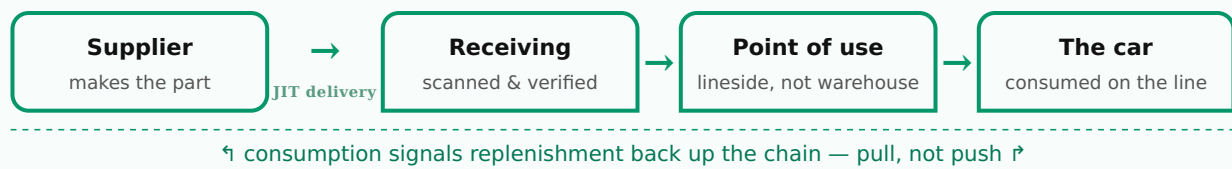


Figure 1. The JIT ideal: parts flow from supplier to the point of use with minimal storage in between, and *consumption itself triggers the next delivery. (Conceptual.)*

In this issue

1. **Just-in-time & pull** — the core philosophy
2. **Kanban** — the two-bin reorder trick
3. **The WMS** — software that knows where everything is
4. **Tracking & counting** — barcodes, RFID, cycle counts
5. **ABC & the math of stock** — what to watch
6. **Vertical integration** — shortening the chain
7. **Warehouse automation** & the glossary

PART 1 · THE PHILOSOPHY

Just-in-Time & the Pull System

The deepest idea in modern inventory control is the difference between *push* and *pull*. A push system builds to a forecast: make a lot, store it, hope it sells or gets used. A pull system makes nothing until downstream demand asks for it — each step is triggered by the consumption of the step after it. JIT is a pull system, and it changes a factory from a place that stockpiles into a place that flows.

PUSH VS PULL

The signal travels backwards

In a pull system, the line consuming a part is what *authorises* the next part to be delivered — the signal travels upstream, opposite to the parts. Nothing is made or moved "just in case"; everything moves "just in time," because something downstream actually used the previous one. Figure 1's backward arrow is the whole idea in one line.

The benefits are large: less cash frozen in stock, less warehouse space, less risk of parts becoming obsolete or damaged while waiting, and faster detection of quality problems (a defect can't hide in a mountain of buffer stock). But JIT is demanding — it relies on dependable suppliers, reliable logistics, and tight coordination, because there is little buffer to absorb a disruption.

THE HONEST TRADE-OFF

Lean is efficient, but it is exposed

The same thinness that makes JIT efficient makes it fragile. When global logistics seized up in recent years, lean factories with little buffer felt it first — which is exactly why the conversation has shifted toward resilience, and toward vertical integration (Part 6) as a way to shorten and control the chain rather than just thin it. Lean and resilient are a balance, not a free lunch.

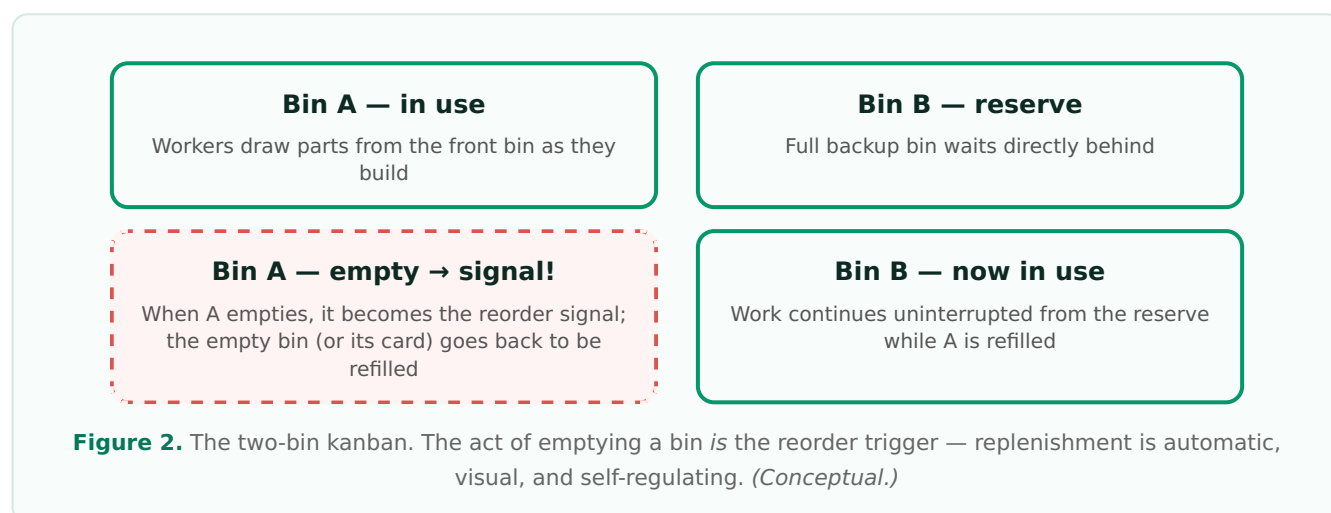
JIT vs JIS — a useful refinement

In car assembly you will meet a stricter cousin: **just-in-sequence (JIS)**. Not only do parts arrive just in time — they arrive in the exact *order* the vehicles will need them, matched to the sequence of cars coming down the line. For variant-rich products (different seats, trims, colours per car), JIS means the right specific part for *this* car arrives at the station exactly as that car does.

PART 2 · THE MECHANISM

Kanban: The Two-Bin Trick

JIT is the philosophy; **kanban** is the elegant mechanism that makes it run on the floor, especially for the thousands of small, inexpensive parts a car needs. Kanban (Japanese for "signboard") is simply a signal that authorises replenishment — and its most beautiful form needs no computer at all.



Reporting describes exactly this at Tesla for small, inexpensive parts: a kanban system that prompts inventory in coordination with run rate and replenishment time, using a two-bucket arrangement where the quantity in each bucket corresponds to the number of parts used during the time it takes to replenish. The genius is its self-regulation — the number of parts in a bin is sized so that the reserve bin lasts exactly as long as it takes to refill the first.

The act of emptying the bin **is** the order. No forecast, no paperwork — just a signal.

WHY TECHNICIANS & PLANNERS LOVE IT

Self-correcting and visual

A two-bin kanban automatically adjusts to how fast a line is actually running: build faster, empty bins faster, reorder faster. It needs no central computer to function, it is visible at a glance (an empty bin is unmissable), and it cannot easily over-order — you only ever refill what was actually used. It is lean inventory control distilled to its simplest physical form.

The WMS: Knowing Where Everything Is

For everything kanban cards don't cover — and to give management a real-time picture — a modern operation runs a **Warehouse Management System (WMS)**. If MOS (Issue 01) is the operating system of the *production line*, the WMS is the operating system of the *material world* around it: it knows what stock exists, exactly where it sits, and orchestrates every move.

Warehouse Management System (WMS)

the material operating system

What it does. Tracks every item by **location** (down to the rack, shelf, and bin), directs **receiving, put-away, picking, and shipping**, optimises travel paths for pickers or robots, and maintains a live, accurate stock count. It integrates upward with the ERP (which plans materials) and sideways with MES/MOS (which consumes them), so the picture stays consistent end to end.

TRACKS Every item, by location	DIRECTS Receive/put-away/pick/ship	INTEGRATES ERP + MES/MOS
-----------------------------------	---------------------------------------	-----------------------------

The reporting on Tesla emphasises exactly this layer: a sophisticated supply-chain system that tracks and monitors the flow of materials from suppliers to the factories, with real-time visibility and AI/analytics used to predict demand and optimise inventory. Real-time visibility is the recurring phrase — the ability to know, at any instant, what is where.

THE THREE SOFTWARE LAYERS

ERP, WMS, MES — who does what

- **ERP** (enterprise planning) decides *what to order and make* — the business-level plan, demand, and procurement.
- **WMS** (warehouse) decides *where everything is and how it moves* through storage and handling.
- **MES / MOS** (execution) *consumes* the materials on the line and records production (Issue 01).

Tracking & Counting: How Stock Stays Honest

A WMS is only as good as the data feeding it — and the enemy of inventory control is the gap between what the system *thinks* is on hand and what *actually* is. Two technologies keep the record honest in real time, and one discipline audits it continuously.

Method	How it works	Best for
Barcode / DataMatrix	A scan at each move updates the record; cheap, universal, needs line-of-sight	Most items; the workhorse of tracking
RFID	Radio tags read without line-of-sight, many at once (recall SIMATIC Ident, Issue 02)	Pallets, containers, fast bulk reads, gates
Cycle counting	Counting a small subset of stock every day, rotating through all of it — instead of one big annual count	Continuously verifying accuracy without stopping

The shift from the old annual "stop everything and count the warehouse" to continuous **cycle counting** is the inventory-control equivalent of SPC (Issue... the quality issue): you sample continuously and catch drift early, rather than discovering a year's worth of error all at once. High-value items get counted more often than low-value ones — which leads directly to the next idea.

WHY IT MATTERS TO A TECHNICIAN

The phantom-stock problem

The most common inventory failure isn't theft — it's the record drifting from reality through un-scanned moves, miscounts, and damage. When the WMS says a part is in bin 12 and it isn't, the line stops with a "phantom" shortage. Disciplined scanning at every move, plus cycle counting, is what keeps the digital twin of the warehouse matching the physical one.

PART 5 · THE MATH OF STOCK

ABC Analysis & the Numbers That Matter

Not every part deserves equal attention. A factory may stock tens of thousands of distinct items, and treating a multi-thousand-dollar battery module like a ten-cent washer would waste effort on one and risk the other. **ABC analysis** sorts inventory by value and focus, so control effort goes where the money is.

Class	Roughly	Control approach
A items	~20% of items, ~80% of value	Tight control, frequent counts, low buffer, close supplier links
B items	moderate value	Moderate control, periodic review
C items	~50% of items, ~5% of value	Simple control — this is where two-bin kanban shines

A few numbers recur whenever planners discuss stock. **Reorder point**: the stock level that triggers a new order, set so the remaining stock lasts through the supplier's **lead time**. **Safety stock**: a deliberate small buffer against variability — the one piece of inventory lean accepts on purpose. **Inventory turns**: how many times a year stock is fully used and replaced — higher turns mean leaner, faster-flowing inventory, a number Tesla-style operations push hard.

A·B·C value tiers of inventory focus	ROP reorder point = lead-time demand + safety	Turns higher = leaner, faster flow	2-bin simplest control, ideal for C items
--	---	--	---

PART 6 · THE STRATEGY

Vertical Integration: Shortening the Chain

There is a more radical answer to supply risk than managing inventory cleverly: make more of the supply chain yourself. **Vertical integration** — producing components in-house rather

than buying them — is a defining feature of the Tesla model, and it reshapes the inventory problem from the outside in.

Reporting frames it consistently: vertical integration gives the business more control over critical components, reduces reliance on third parties, and makes the company more resilient to interruptions — visible in moves like in-house battery cell manufacturing and the Gigafactory network that consolidates many production stages under one roof. The logic, in the words of one supply-chain analysis: if a capability is central to product economics and repeatedly exposed to external risk, the case for pulling it inward gets stronger.

HOW IT CHANGES INVENTORY

A shorter chain is an easier chain

- **Co-location** — making cells beside where cars are built (the Gigafactory idea) collapses transport and storage between stages to almost nothing.
- **Strategic siting** — placing Gigafactories near their markets (e.g. Shanghai for China) cuts finished-goods logistics and delay.
- **Control** — owning a stage means you schedule it to feed the line directly, rather than negotiating deliveries with an outside supplier.

THE CATCH

More control, more burden

Vertical integration is easier to admire than to execute. The broader the enterprise boundary, the more complexity management must absorb: factory ramping, material strategy, and global logistics all become internal problems. And consolidating stages under one roof amplifies the impact of any internal bottleneck — if your own upstream stage stumbles, there is no outside supplier to fall back on. Control and exposure rise together.

Don't just manage the supply chain — **absorb it. Then live with the complexity that brings.**

PART 7 • THE WAREHOUSE ITSELF

Warehouse Automation, and the Whole Picture

Where inventory must be stored and moved, the modern answer is to automate the handling — the same robotics-and-software instinct that runs the assembly line, applied to material flow. The technologies range from vehicles that fetch parts to systems that store and retrieve them without a human walking an aisle.

AGVs & AMRs — automated guided vehicles and autonomous mobile robots that ferry parts from storage to lineside (often over the industrial wireless of Issue 03). **AS/RS** — automated storage and retrieval systems: dense racking served by robotic cranes/shuttles, storing and fetching totes with no aisle walking. **Conveyors & sortation** for high-volume flow. All directed by the WMS, all reporting their moves back in real time.

TRANSPORT	STORAGE	DIRECTED BY
AGV / AMR	AS/RS	WMS

Pull the whole issue together and a single pattern emerges, the same one that runs through this entire magazine: *information makes the physical efficient*. JIT and kanban minimise what must be held; the WMS and real-time tracking know precisely what is held and where; ABC and cycle counting focus attention and keep the record true; vertical integration shortens the chain that has to be coordinated at all; and automation moves what remains. Each layer is the supply chain's version of the control-and-feedback idea that governs the factory floor.

The destination is a factory that behaves less like a warehouse and more like a river — parts entering and flowing to the point of use with as little pooling as possible, every item visible the whole way, consumption itself pulling the next delivery forward. A bolt should arrive, get fitted, and be gone. That is the quiet discipline of material flow, and it is as much a part of building a car as any robot on the line. ■

FIELD GLOSSARY

The logistics vocabulary in this issue

- **Just-in-Time (JIT)** — supplying parts precisely when needed to minimise stored inventory.
- **Just-in-Sequence (JIS)** — JIT plus arriving in the exact order the line will use them.
- **Push vs pull** — building to forecast (push) versus building only when downstream demand signals (pull).
- **Kanban** — a visual signal authorising replenishment; the two-bin form reorders automatically when a bin empties.
- **WMS (Warehouse Management System)** — software tracking every item by location and directing all warehouse moves.
- **ERP** — enterprise planning software deciding what to order and make; sits above the WMS.
- **Cycle counting** — continuously counting small subsets of stock to keep records accurate, instead of one annual count.
- **ABC analysis** — sorting items by value so control effort focuses on the high-value few.
- **Reorder point / lead time / safety stock** — the level that triggers an order, the time a supplier takes, and the deliberate buffer against variability.
- **Inventory turns** — how many times a year stock is used and replaced; higher is leaner.
- **Vertical integration** — making components in-house to control and shorten the supply chain.
- **AGV / AMR / AS/RS** — automated guided vehicles, autonomous mobile robots, and automated storage/retrieval systems.

FURTHER READING & SOURCING

This issue draws on public reporting about Tesla's supply chain — its JIT approach and two-bin kanban for small parts, its vertical integration and Gigafactory siting, and its use of real-time visibility and analytics — from outlets including supply-chain trade press and Tesla owners' accounts, combined with standard logistics and inventory-control engineering (JIT/JIS, kanban, WMS, ABC, cycle counting). Specific internal figures and system designs are not disclosed by Tesla; mechanisms and numbers here are representative, and diagrams are conceptual.

THE INTEGRATION LAYER · Issue 08 · A technical magazine · Sources: public supply-chain reporting & logistics fundamentals

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 09

COVER STORY · THE SENSES OF THE FACTORY

The Sensors That Watch Every Car Get Built

Inductive, photoelectric, vision, force, pressure — the thousands of sensors on a Gigafactory line that detect a part, measure a bolt, and turn the physical world into the data every other system depends on.

PRESENCE & POSITION

Inductive, capacitive, photoelectric — is the part there, and where?

MEASURING FORCE

The torque and load cells behind a safe bolt.

THE MACHINE EYE

Vision systems that read a VIN and inspect a weld.

THE SMART SENSOR

How IO-Link gives a sensor a voice and a memory.

The Senses of the Factory

THE INTEGRATION LAYER · Feature · A cross-vendor explainer grounded in sensing fundamentals

Every system in this magazine — the controller that reasons, the network that carries, the drive that moves, the MOS that records — is blind without one humble layer beneath it all. A PLC cannot judge a bolt it cannot measure; a robot cannot grasp a part it cannot find; a quality system cannot trace a weld it never saw. The sensors are the factory's sense organs, and they are where the physical world first becomes the data everything else runs on. This issue is a field guide to those senses.

A modern car line carries thousands of them, but they answer a surprisingly small set of questions. *Is the part there? Exactly where is it? What does it look like — is it correct? How hard am I pushing, and how tight is that bolt? How hot, how fast, how full?* Group the sensors by the question they answer, and the bewildering catalog organises itself into a handful of families.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact sensors, part numbers, or vendors on its lines. This is a cross-vendor explainer of the **standard sensing technologies universal to automotive assembly** — inductive, capacitive, photoelectric, ultrasonic, magnetic, vision, force/torque, pressure, temperature, and position sensing — drawn from sensor-engineering fundamentals and vendor documentation. It connects to earlier issues: the torque transducer of the MOS data-journey (Issue 01), SIMATIC Ident readers (Issue 02), and IO-Link smart sensors (Issue 03). Examples are representative of a plant of this class, not confirmed Tesla equipment. Diagrams are conceptual.

The one idea: a sensor converts a physical fact into an electrical signal

Strip every sensor to its essence and it does one thing: it detects a physical input — distance, presence, light, force, temperature, pressure — and converts it into an electrical signal a controller can read. That signal, refined and digitised, is the first link in the data chain that climbs all the way to the dashboard (Issue 01). Everything else in this issue is variation on *how* a sensor performs that one conversion, and *which* physical fact it is tuned to catch.

A PLC can only judge what a sensor can [measure](#). The data chain begins here, or not at all.

The question: "what is true, physically, right now?"

a part's presence, position, shape, force, or state

The sensor: converts that fact to an electrical signal

inductive · photoelectric · vision · force · pressure · temperature

Signal conditioning & digitising

amplify · filter · 4–20 mA / IO-Link / digital

The PLC reads it — and the data journey begins

(Issue 01: from here to the dashboard)

Figure 1. Every sensor sits at the bottom of the data chain: a physical fact becomes a signal, which becomes data. *(Conceptual.)*

In this issue

1. **Presence & proximity** — inductive, capacitive, magnetic
2. **Photoelectric & distance** — sensing with light
3. **Machine vision** — the factory's eye
4. **Force, torque & pressure** — measuring effort
5. **Position & the rest** — encoders, temperature, flow
6. **The smart sensor** — IO-Link & diagnostics
7. **Choosing & surviving** & the glossary

PART 1 · PRESENCE & PROXIMITY

"Is the Part There?"

The most common question on any line is also the simplest: is something present, yes or no? Proximity sensors answer it without touching the object — no wear, no contact, millions of cycles. They come in three main flavours, distinguished by the physics they exploit and, crucially, by what material they can "see."

Inductive Proximity Sensor

detects metal · the floor's workhorse

The most common industrial sensor. Generates an electromagnetic field; when a **metal** object enters it, eddy currents are induced and the sensor trips. Rugged, fast (response times under 1 ms), immune to dust and moisture, no contact. The default for confirming a metal part is seated in a jig, positioning welding fixtures, and counting. Detects metal only.

DETECTS

Metal objects

SPEED

<1 ms response

USE

Part-seated, fixtures

Capacitive Proximity Sensor

detects almost anything

When it isn't metal. Detects a change in **capacitance** as any object — metal, plastic, glass, liquid, powder — enters its field. Ideal for sensing non-metallic parts, or fill levels through a container wall. More sensitive to humidity and temperature, so it needs careful setup; typical range up to ~40 mm.

DETECTS

Metal + non-metal

RANGE

up to ~40 mm

USE

Plastics, levels

Magnetic (Hall / Reed) Sensor

detects a magnetic field

For position and cylinders. Hall-effect and reed-switch sensors detect a magnetic field, typically from a magnet on the moving part. Simple, reliable, and work through non-magnetic walls — the standard for sensing piston position inside a pneumatic/hydraulic cylinder, and for motor speed and limit sensing.

DETECTS

Magnetic field

USE

Cylinder position

TRAIT

Sees through walls

THE SELECTION RULE

Material decides the sensor

The first question in picking a proximity sensor is *what is the target made of?* Metal → inductive (rugged, cheap, fast). Anything else, or a level behind a wall → capacitive. A magnet on a moving part, especially inside a cylinder → magnetic. Get the material right and most of the decision is made.

PART 2 · SENSING WITH LIGHT

Photoelectric & Distance Sensors

When you need more range than proximity sensors offer, or must detect objects regardless of material, you reach for light. Photoelectric sensors ("photo eyes") use an LED or laser beam and come in three modes that trade range against reliability — a distinction every technician must know to aim and troubleshoot them.

Mode	How it works	Trade-off
Through-beam	Separate emitter and receiver face each other; object breaks the beam	Longest range, most reliable — but two units to wire & align
Retro-reflective	Emitter + receiver in one unit; a reflector returns the beam	One unit + a reflector; medium range
Diffuse	One unit; detects light bounced back off the object itself	Simplest to mount; range depends on object colour/finish

Beyond simple presence, **laser distance sensors** measure exact distance — by triangulation or time-of-flight — and are used for precise positioning in machining centres, with durable glass optics and metal housings to survive coolant and lubricant. They quickly determine a correct clamping position with high precision. Photo eyes are immune to background moisture, temperature, and sound, but are vulnerable to stray ambient light — the first thing to suspect when one misbehaves.

Through-beam for range and reliability. Diffuse for simplicity. The **mode** is the decision.

PART 3 · THE FACTORY'S EYE

Machine Vision

A proximity sensor knows something is there; a vision system knows *what* it is, whether it is *correct*, and *where* exactly it sits. Equipped with a camera and image-processing software, machine vision is the factory's eye — and one of its most important quality-control tools, verifying each step is right.

Machine Vision System

camera + lighting + processing

Seeing, reading, and judging. A vision system perceives and inspects objects, verifies positions, identifies parts, and reads codes. On a car line it reads the **VIN barcode/DataMatrix** that anchors traceability (Issue 01), inspects welds and seals, checks that the right component and colour are present, and guides robots to a precise grasp point. Lighting is half the battle — controlled illumination is what makes a feature reliably visible.

READS VIN, codes, features	INSPECTS Welds, presence, color	GUIDES Robot grasp points
-------------------------------	------------------------------------	------------------------------

VISION VS A SIMPLE SENSOR

One eye replaces many photo eyes

Where a handful of proximity and photoelectric sensors each answer one yes/no question, a single vision system can answer dozens at once — presence, orientation, dimension, colour, code, defect — from one image. That richness is why vision is central to modern quality control, and why it integrates tightly with the controller (recall TwinCAT Vision, Issue 06): the image becomes data on the same network as everything else.

<1ms proximity sensor response time	3 photoelectric modes (beam/retro/diffuse)	2-D DataMatrix codes vision reads for the VIN	IP69K washdown rating for harsh-zone sensors
--	---	--	---

PART 4 · MEASURING EFFORT

Force, Torque & Pressure

Detection and vision tell you *what* and *where*; another family tells you *how hard*. These are measurement sensors — they don't just trip a yes/no signal, they report a precise analog value, the kind that becomes an SPC data point (Issue 01) and the difference between a safe joint and a recall.

Torque Transducer (the nutrunner sensor)

strain gauges · the safety-critical bolt

The star of Issue 01. Inside a powered nutrunner, strain gauges on a precision shaft deform microscopically as the tool tightens, changing resistance in proportion to torque — while an angle encoder counts rotation. Together they let the controller judge a fastening OK/NOK against a torque-and-angle window. This is the sensor whose 46.2 N·m reading we followed all the way to the dashboard.

ELEMENT

Strain gauges

MEASURES

Torque + angle

STAKES

Safety-critical joints

Load Cell & Force Sensor

press force · assembly verification

How hard is it pushing? A load cell (also strain-gauge based) measures force — the tonnage of a press, the insertion force of pressing a bearing or pin into place. A force-vs-distance curve during an assembly press is itself a quality signature: the right part, correctly seated, produces the right curve.

MEASURES

Force / load

USE

Presses, press-fits

SIGNAL

Force-distance curve

Pressure Sensor

pneumatics · process · 4–20 mA

Watching the air and fluids. Pressure transmitters monitor the compressed air that drives pneumatic clamps and tools, plus coolant and process fluids. Often output the classic **4–20 mA** signal (Issue 01) — where 0 mA means a broken wire. A clamp that won't hold or a tool that won't fire often traces to an air-pressure sensor reading low.

MEASURES

Pressure

OUTPUT

often 4–20 mA

USE

Pneumatics, fluids

PART 5 · POSITION & THE REST

Position, Temperature & Flow

A few more families round out the factory's senses — the ones that measure exactly how far, how hot, and how much is flowing. Together with the detection and force families, they cover essentially every physical fact a line needs to know.

Sensor	Measures	Where on the line
Encoder (rotary/linear)	Exact position, angle, or speed of a moving axis	Every servo motor (Issue 07); conveyors, robots
LVDT / linear gauge	Precise linear displacement, to microns	Dimensional checks, part gauging
Ultrasonic sensor	Distance to any material via sound time-of-flight	Clear/shiny parts, AGV obstacle detection, levels
Temperature (RTD/thermocouple)	Temperature of a process, motor, or fluid	Paint, e-coat, battery, motor monitoring
Flow sensor	Rate of liquid or gas flow	Coolant, adhesive dispensing, fluids

Two worth a closer note. The **encoder** is the most ubiquitous measurement sensor of all — every servo axis in Issue 07 depends on one to close its loop, reporting actual position thousands of times a second. And **ultrasonic** sensors solve the cases that defeat light: they

detect any material regardless of colour or transparency, which is why they handle clear glass, shiny metal, and AGV obstacle avoidance where a photo eye would fail.

Detection, force, position, state — together, the sensors cover every physical fact a line must know.

PART 6 · THE SMART SENSOR

When a Sensor Gets a Voice

A traditional sensor can only say one thing, one way: a single on/off wire, or a single analog value. The modern shift — introduced in Issue 03 — is the **smart sensor**, which holds a two-way digital conversation with the controller over **IO-Link**. This changes what a sensor is, from a switch into a small, self-describing device.

The IO-Link Smart Sensor

IEC 61131-9 · the last metre, revisited

Value, health, identity, and memory. Over a standard 3-wire cable, an IO-Link sensor reports not just "part present" but its exact measured value, its internal health and temperature, and a warning that it is drifting toward failure. It can be **reconfigured remotely**, and when one is swapped, the IO-Link master pushes the old settings to the new device automatically — turning a calibration ordeal into plug-and-play.

CARRIES

Value + health + config

ENABLES

Auto-replacement

BUS

IO-Link

WHY IT MATTERS TO A TECHNICIAN

The sensor that warns you before it fails

A smart sensor streaming its own health is the foundation of **predictive maintenance** at the edge: instead of discovering a dead sensor when the line stops, the system flags one degrading *before* it produces a bad part. And because the IO-Link master stores each sensor's parameters, replacing a failed unit no longer means manual recalibration — the configuration travels with the port, not the part.

PART 7 · IN PRACTICE

Choosing & Surviving on the Line

Knowing the families is the theory; specifying and maintaining sensors is the daily craft. A handful of practical realities decide whether a sensor lasts for years or fails every month.

THE SPEC-SHEET CHECKLIST

What actually matters when you choose

- **Target material** — the first filter: metal (inductive), anything (capacitive/photoelectric), magnet (magnetic).
- **Sensing range** — and remember many sensors don't "start" at zero; they detect best at a set standoff distance.
- **Response time** — high-speed lines need <1 ms to keep up with fast-moving parts.
- **IP rating & housing** — washdown or weld-spatter zones need IP68/IP69K and stainless or metal housings.
- **Connector & voltage** — M8/M12 connectors for quick swap-out; 24 V DC is the industrial standard.

And the failure modes a technician learns to expect: **misalignment** (a knocked photo eye no longer hits its reflector), **contamination** (weld spatter, coolant, or dust blinding an optical face), **ambient-light interference** (the first suspect for an erratic photo eye), and **cable/connector damage** in high-flex or high-heat spots. The recurring lesson of this whole magazine applies in miniature here: when a cell misbehaves, the sensor and its wiring are very often the cause — check the simplest sense organ before the complex brain.

Step back and the picture is complete. The controllers, networks, drives, and software of the earlier issues are all, in the end, in service of this lowest layer — because none of them can act on a fact no sensor caught. The sensors are where the factory touches reality: a part detected, a bolt measured, a weld inspected, a code read. Get the senses right, and everything above them has truth to work with. Get them wrong, and the smartest controller in the world is reasoning about a world it cannot actually see. ■

FIELD GLOSSARY

The sensing vocabulary in this issue

- **Proximity sensor** — detects presence without contact; inductive, capacitive, or magnetic.
- **Inductive sensor** — detects metal via an electromagnetic field; the rugged floor workhorse.
- **Capacitive sensor** — detects almost any material (incl. levels) via capacitance change.
- **Magnetic (Hall / reed) sensor** — detects a magnetic field; standard for cylinder position.
- **Photoelectric sensor** — detects objects with a light beam; through-beam, retro-reflective, or diffuse.
- **Laser distance sensor** — measures precise distance by triangulation or time-of-flight.
- **Machine vision** — a camera + processing system that reads codes, inspects, and guides.
- **Torque transducer** — strain-gauge sensor measuring fastening torque (and, with an encoder, angle).
- **Load cell** — a strain-gauge sensor measuring force or load.
- **Pressure sensor** — measures pneumatic or fluid pressure, often as a 4–20 mA signal.
- **Encoder** — reports exact position, angle, or speed; closes every servo loop.
- **LVDT** — a precise linear-displacement sensor for dimensional gauging.
- **Ultrasonic sensor** — measures distance to any material via sound time-of-flight.
- **RTD / thermocouple** — temperature sensors for processes, motors, and fluids.
- **IO-Link / smart sensor** — a two-way digital link giving a sensor value, health, and remote config.
- **IP rating** — ingress protection (e.g. IP69K) defining dust/water resistance.

FURTHER READING & SOURCING

This issue draws on standard industrial-sensing engineering — proximity (inductive/capacitive/magnetic), photoelectric, ultrasonic, vision, force/torque, pressure, position, and temperature sensing — from sensor-manufacturer and automation technical literature, and connects to Issues 01 (the torque transducer and data journey), 02 (SIMATIC Ident readers), 03 (IO-Link and 4–20 mA), 06 (TwinCAT Vision), and 07 (encoders on servo axes). Tesla's specific sensor selection is proprietary; the technologies and figures here are representative of automotive-class lines, and diagrams are conceptual.

THE INTEGRATION LAYER · Issue 09 · A technical magazine · Sources: industrial-sensing fundamentals & vendor documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 10

COVER STORY · HOW THE WORK FLOWS THROUGH THE PLANT

The Lines That Move the Line

Conveyors, skillets, overhead carriers and AGVs — the material-transport backbone that carries a car body from bare steel to finished vehicle, and the quiet shift from fixed track to flexible flow.

PACED vs ACCUMULATING

The one distinction that defines every conveyor.

OVERHEAD POWER & FREE

The automotive workhorse that hangs from the ceiling.

THE SKILLET & THE SKID

The platforms that carry a car through assembly.

THE AGV FUTURE

Why fixed track is giving way to mobile robots.

The Backbone of Movement

THE INTEGRATION LAYER · Feature · An explainer grounded in material-handling fundamentals

Henry Ford's single greatest contribution was not a car — it was an idea about movement: bring the work to the worker, at a steady pace, and a factory's output transforms. Every assembly plant since runs on a descendant of that idea, and the machinery that physically embodies it is the conveyor system. It is the circulatory system of the factory: while controllers reason and robots weld, the conveyors are what actually carry a car body, station by station, from a sheet of steel to a finished vehicle. This issue is a guide to that backbone of movement.

A modern car plant is really three conveyor worlds stitched together, because the body changes form as it travels. In the **body shop**, bare metal is carried on rugged skids while robots weld it into a shell. In the **paint shop**, that shell rides overhead carriers through dip tanks and ovens. In **general assembly**, the painted body sits on a platform — a skillet — that carries it past hundreds of workers fitting it out. Each world uses a different conveyor because each asks something different of it.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact conveyors, layouts, or vendors in its factories. This is a cross-vendor explainer of the **standard material-transport technologies universal to automotive assembly** — belt/roller/chain/slat conveyors, skids and skillets, overhead power-and-free and EMS, transfer mechanisms, and the move toward AGVs — drawn from material-handling engineering and conveyor-vendor documentation, plus public reporting on Tesla's "Unboxed" manufacturing direction. Examples are representative of a plant of this class, not confirmed Tesla equipment. Diagrams are conceptual.

The one idea: paced flow, and how much it bends

Beneath all the variety sits a single organising question, inherited straight from Ford: should the line move as one synchronised body, or should each unit be free to stop and wait? That choice — **synchronous** (paced) versus **asynchronous** (accumulating) flow — is the deepest distinction in conveyance, and it cascades into which conveyor a designer picks for each part of the plant. Everything in this issue is, in some sense, a variation on how rigidly the work is paced.

Bring the work to the worker, at a steady pace. Every conveyor since is a variation on [Ford's one idea](#).



Figure 1. A car body's journey crosses three conveyor worlds, each suited to what the body needs there.
(Conceptual.)

In this issue

1. **Synchronous vs asynchronous** — the core distinction
2. **Floor-level conveyors** — belt, roller, chain, slat
3. **Skids & skillets** — carrying the car itself
4. **Overhead systems** — power & free and EMS
5. **Transfer & vertical** — shuttles, lifts, turntables
6. **Drive, tracking & control** — what runs the conveyor
7. **The flexible future** — AGVs & "Unboxed"

PART 1 · THE CORE DISTINCTION

Synchronous vs Asynchronous Flow

Before any hardware, the governing choice. In a **synchronous** (continuous or indexing) line, every unit is coupled to the same pace — the whole line moves together, and a single speed control matches all units to the desired takt time (Issue 01). In an **asynchronous** (accumulating) line, units are free to stop, queue, and release independently, so one station can pause without halting its neighbours.

Synchronous — paced

All units locked to one speed; the line indexes or flows as a single body. Simple, predictable, perfectly matched to takt — but a stop anywhere stops everything.

Asynchronous — accumulating

Units stop, queue, and release independently via stops along the track. A station can pause without halting neighbours; buffers absorb hiccups — at the cost of more complexity.

This is the conveyor world's version of the rigid-versus-flexible tension that runs through this whole magazine. A paced line is wonderfully efficient when everything runs smoothly; an accumulating line trades some of that simplicity for resilience, letting work pool between stations so a brief problem at one does not starve or block the others.

THE KEY CONCEPT · ACCUMULATION

Buffers, in physical form

Accumulation is simply letting units queue on the conveyor itself — a physical buffer between stations. It is the material-transport cousin of safety stock in inventory (Issue 08): a deliberate cushion against variability. Power & free and EMS systems excel at it because each carrier can be independently stopped and released; a single coupled belt cannot accumulate at all.

The Conveyors Underfoot

The simplest conveyors move things along the floor, and a plant uses several types depending on load, distance, and whether power is even needed. These carry everything from small parts in totes to the heavy platforms beneath a car.

Belt Conveyor

continuous belt · parts & totes

The everyday mover. A continuous belt over pulleys carries parts, totes, and packages smoothly and quietly. Gentle on the load and ideal for small or irregular items that rollers would not handle. The workhorse for parts feeding, kitting, and packaging flows around the main line.

CARRIES

Parts, totes, packages

FLOW

Continuous

TRAIT

Gentle, quiet

Roller Conveyor

gravity or powered rollers

For flat-bottomed loads. Rollers move boxes, trays, and pallets — by gravity on a slope, or powered for controlled movement. Powered roller zones are a natural place to **accumulate**: zones can stop and release independently, queuing cartons without crushing them. Common in logistics and parts supply.

CARRIES

Boxes, pallets

POWER

Gravity or driven

TRAIT

Easy accumulation

Chain & Slat Conveyor

heavy loads · mainline duty

For the heavy work. Chain conveyors drive heavy loads and handle inclines and declines economically; **slat conveyors** use linked slats (steel, stainless, or composite) to form a moving floor strong enough for car-sized loads — from tens of kilograms up to 20+ metric tons — in continuous or step-by-step modes. The backbone of many mainlines.

CARRIES

Heavy, large loads

MODES

Continuous / step

USE

Mainline floors

PART 3 · CARRYING THE CAR ITSELF

Skids & Skillets

The car body cannot simply sit on a belt — it needs a carrier that holds it precisely and, in assembly, at the right working height. Two purpose-built platforms dominate automotive mainlines, each named for the part of the plant it rules.

The Skid

body-in-white · the body shop & paint

The body shop's pallet. A skid is a rugged steel fixture with two longitudinal runners that cradles a bare body shell, carrying it through welding and (often) paint. It holds the body rigidly and at a known position so robots can weld and handle it repeatably. Skids cycle back to the start of the line empty, ready for the next body.

CARRIES

Bare body shell

WHERE

Body shop, paint

TRAIT

Rigid, repeatable

The Skillet

general assembly · trim & final line

The platform workers ride with. Derived from the skid and pallet systems, the skillet is the mainline carrier in general assembly for trim and final operations. Its defining feature is an **onboard scissor lift** that raises or lowers the body to each operator's ideal working height — an ergonomic centrepiece. Often powered **inductively** (contactless, through an air gap), so there are no wear-prone power contacts even through lifts and transfers.

CARRIES

Painted body + workers

FEATURE

Onboard scissor lift

POWER

Often inductive

A STATION WORTH NAMING

The "body marriage"

One transfer station in every car plant has its own name: the **body marriage** (or "decking") station, where the painted upper body is lowered and joined to the powertrain/chassis module coming up from below. It is a precise, conveyor-choreographed meeting of two flows — the moment a car truly becomes a car — and a showcase of how conveyors are synchronised to bring two sub-assemblies together at exactly the right instant.

PART 4 · OVERHEAD SYSTEMS

Power & Free and the Electric Monorail

Lifting the work off the floor frees the space beneath it — for workers, for access, for other flows — which is why overhead conveyors are an automotive staple, especially in body and paint. Two families dominate, and the difference between them is exactly the synchronous-versus-asynchronous distinction made physical.

Power & Free (P&F)

overhead · the automotive workhorse

Two tracks, one clever trick. P&F runs two parallel tracks: a powered chain above, and a "free" track below carrying the trolleys. Carriers are pushed by the chain but can be **disengaged** and held by simple **pneumatic stops** anywhere on the line, then released when ready — giving true accumulation. Long the first choice in automotive for its durability and relatively few motors and moving parts; the pneumatic stops relocate easily as needs change.

TRACKS

Powered + free

STOPS

Pneumatic, relocatable

TRAIT

Rugged, accumulates

Electrified Monorail System (EMS)

overhead · independently powered carriers

Every carrier its own vehicle. EMS uses motorised trolleys, each independently powered from an electrified rail and individually controlled. Because every carrier drives itself, the system supports **dynamic routing**, per-carrier speeds, accumulation, and decoupling at any station — the "smart" overhead system, ideal for variant-rich, flexible production. More electrically complex than P&F, but far more flexible.

DRIVE

Per-carrier motor

CONTROL

Individual, routable

TRAIT

Flexible, smart

System	How it paces	Strength	Trade-off
--------	--------------	----------	-----------

Synchronous belt/ slat	All units one speed	Simple, matched to takt	One stop halts all
Power & Free	Chain push + pneumatic stops	Rugged, few motors, accumulates	Less flexible routing
EMS	Each carrier self-driven	Dynamic routing, very flexible	More electrical complexity
AGV (Part 7)	No track at all	Reroutable in software	Throughput & coordination limits

A belt paces everything as one. An EMS carrier is *its own vehicle*. That is accumulation, made of motors.

PART 5 • TRANSFER & VERTICAL

Shuttles, Lifts & Turntables

A line is never one straight conveyor; bodies must change direction, change height, and jump between conveyors. A family of transfer mechanisms does this connective work — the junctions and elevators of the conveyor network.

Mechanism	What it does	Typical use
Shuttle / transfer car	Moves a carrier sideways to a station or another line, then back	Linking parallel lines; feeding cells
Fork / lift transfer	Lifts a body off one conveyor and sets it on another	Body shop to paint; level changes
Vertical conveyor / lift	Raises or lowers loads between floors or levels	Multi-level plants; buffer storage
Turntable	Rotates a body or carrier to a new orientation	Reorienting for access or direction change

These mechanisms are where conveyor systems get genuinely three-dimensional. A modern plant routes bodies up to a mezzanine for buffering, across to a parallel line via a shuttle, and back down for the next process — all choreographed so a body arrives at each step in the right orientation, at the right height, at the right moment. The transfers are the verbs that connect the conveyors' nouns.

PART 6 • WHAT RUNS THE CONVEYOR

Drive, Tracking & Control

A conveyor is not just steel — it is one of the largest consumers of the motors, drives, networks, and sensors covered in earlier issues. Pull back the housing and the whole magazine is present in the thing that simply "moves the work."

The Drive: Gearmotors & VFDs

motion, applied to a belt

Where Issue 07 meets the floor. Most conveyors are driven by **garmotors** — an induction motor plus a gearbox — whose speed is set by a **VFD** (Issue 07). Conveyor duty is a classic constant-torque load, and a VFD lets the line ramp smoothly, hold an exact pace to match takt, and recover energy on declines. Big systems coordinate many drives over the industrial network (Issue 03).

MOTOR

Garmotor (induction)

SPEED BY

VFD (Issue 07)

LOAD

Constant-torque

Tracking & Identification

knowing which body is where

Where Issues 02 & 09 meet the line. Every carrier and body carries an identity the line must track — via **RFID** tags on skids/skillets (recall SIMATIC Ident, Issue 02) and barcode/vision reads (Issue 09). This is what lets MOS (Issue 01) bind each operation to the right VIN, and what tells a station which variant is arriving so it fits the correct parts.

TAGS

RFID on carriers

READS

Barcode / vision

FEEDS

MOS / VIN binding

SENSORS & SAFETY ON THE CONVEYOR

The senses that keep it moving — and safe

Photoelectric and proximity sensors (Issue 09) detect carrier presence and position for accumulation control and precise stops. And because conveyors move heavy loads through spaces people share, they are wrapped in safety (Issues 04/06): emergency-stop pull-cords along the line, light curtains and guards at transfers, and safety-rated stops — all carried over the same networks as everything else. A conveyor is a safety system as much as a transport one.

PART 7 · THE FLEXIBLE FUTURE

AGVs and the "Unboxed" Idea

For a century, the conveyor's defining trait was also its limitation: it is fixed. Bolted into the floor, a conveyor line is fast and reliable but hard to change — rerouting it means rebuilding it. The current revolution in material transport is the move from fixed track to **mobile robots**, and Tesla sits near the center of that shift.

AGVs & AMRs

conveyance without a conveyor

The track becomes software. Automated guided vehicles and autonomous mobile robots (Issue 08) carry bodies and modules with *no fixed track at all* — their route is a program, rerouted in software rather than rebuilt in steel. They navigate the floor (often over the industrial wireless of Issue 03), bringing flexibility a bolted-down conveyor can never match. The trade-off is throughput and traffic coordination at the highest volumes.

TRACK

None — software route

STRENGTH

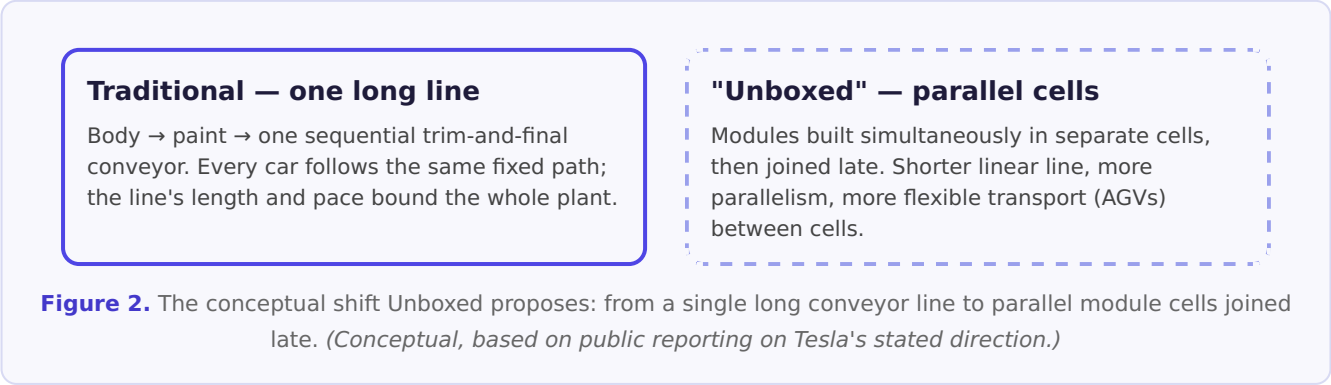
Reroutable, flexible

TRADE-OFF

Throughput / traffic

This is the backdrop to Tesla's most-discussed manufacturing idea: the **"Unboxed" process**. A traditional plant builds the car as one body moving down one long sequential line — body, then paint, then a single trim-and-final conveyor. Unboxed proposes building large modules —

front, rear, floor, sides — *in parallel* in separate cells, painting the pieces, and bringing them together late. The aim is to compress the long linear conveyor line, work on many modules at once, and lean heavily on flexible transport between parallel cells.



THE HONEST TRADE-OFF
Flexibility versus the proven line
The fixed conveyor line earned its century of dominance for good reasons: at high volume it is fast, simple, and brutally reliable. Mobile, parallel approaches promise flexibility and a smaller footprint, but trade away some of that raw throughput and add the hard problem of coordinating many independent vehicles. The future on most floors is not "conveyors or AGVs" but a blend — fixed track where volume and pace demand it, mobile robots where flexibility wins. As with every choice in this magazine, it is a balance, not a verdict.

1913 Ford's moving line — the idea's origin	3 conveyor worlds: body, paint, assembly	20t+ slat-conveyor load capacity	0 fixed track an AGV needs
---	--	--	--------------------------------------

REFERENCE

The Whole Backbone, on One Page

Trace a body from bare steel to finished car and you ride nearly every system in this issue: a skid through the body shop, an overhead carrier through paint, a skillet through assembly with its scissor lift rising to meet each worker, a body-marriage station joining it to its chassis, and — increasingly — an AGV ferrying a module between parallel cells. Each was chosen for one reason: what that stretch of the journey asks of the way the work moves.

Conveyor type	Carries	Where on the line	Paces by
Belt	Parts, totes, packages	Parts feed, kitting, packaging	Continuous
Roller	Boxes, pallets	Logistics, parts supply	Gravity / accumulating
Chain / slat	Heavy, car-sized loads	Mainline floors	Continuous or step
Skid	Bare body shell	Body shop, paint	With the line
Skillet	Painted body + workers	General assembly (trim/final)	With the line, height-adjustable

Power & Free	Bodies, sub-assemblies	Body, paint, overhead routes	Chain + pneumatic stops (accumulates)
EMS	Bodies, carriers	Flexible overhead routes	Per-carrier motor (routable)
AGV / AMR	Modules, bodies, parts	Flexible, parallel-cell transport	Software route (no track)

The arc of the whole story is the same one bending through every issue of this magazine: from rigid and fast toward flexible and smart. The conveyor began as Ford's iron embodiment of paced flow, and it is now learning to let go of the track entirely. Whether a plant moves its work on a century-proven slat line or a swarm of mobile robots, the goal has never changed since 1913 — bring the work to where it needs to be, exactly when it needs to be there. The machinery just keeps getting cleverer about how. ■

FIELD GLOSSARY

The conveyance vocabulary in this issue

- **Synchronous (paced) line** — all units locked to one speed, moving as a single body matched to takt.
- **Asynchronous (accumulating) line** — units that stop, queue, and release independently.
- **Accumulation** — letting units queue on the conveyor as a physical buffer between stations.
- **Takt time** — the line's target pace, the drumbeat the conveyor is set to (Issue 01).
- **Belt / roller / chain / slat conveyor** — the floor-level families, by load and duty.
- **Skid** — the rigid steel fixture carrying a bare body through body shop and paint.
- **Skillet** — the general-assembly mainline carrier with an onboard scissor lift; often inductively powered.
- **Body marriage (decking)** — the station where the upper body is joined to the chassis/powertrain.
- **Power & Free (P&F)** — a two-track overhead conveyor whose carriers disengage and accumulate via pneumatic stops.
- **EMS (Electrified Monorail System)** — an overhead system of individually powered, individually controlled carriers.
- **Shuttle / fork transfer / turntable** — mechanisms that move carriers sideways, between conveyors, or rotate them.
- **Gearmotor** — an induction motor plus gearbox, the typical conveyor drive (speed set by a VFD).
- **AGV / AMR** — automated guided vehicles and autonomous mobile robots: conveyance with no fixed track.
- **"Unboxed"** — Tesla's stated direction of building modules in parallel cells and joining them late, shrinking the linear line.

FURTHER READING & SOURCING

This issue draws on standard material-handling engineering — belt/roller/chain/slat conveyors, skids and skillets, overhead power-and-free and EMS, transfer mechanisms — from conveyor-vendor and automotive-assembly technical literature, plus public reporting on Tesla's "Unboxed" manufacturing direction and its use of AGVs. It connects to Issue 01 (takt & VIN binding), 02 (SIMATIC Ident/RFID), 03 (industrial networks & wireless), 04/06 (safety), 07 (motors, gearmotors & VFDs), 08 (AGV/AMR & flow), and 09 (sensors). Tesla's specific conveyor selection and layouts are proprietary; technologies and figures here are representative of automotive-class plants, and diagrams are conceptual.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 11

COVER STORY · THE COMPANY THAT TIGHTENS THE BOLT

Atlas Copco: Torque, Air & the Perfect Joint

The smart nutrunners that fasten a safety-critical bolt to spec, the oil-free compressors that power the plant's air, and the controllers that prove every joint — a tour of the tools behind the tightening.

THE SMART NUTRUNNER

How a Tensor tool turns a bolt to an exact torque and angle.

PROVING THE JOINT

Power Focus controllers and the traceable fastening.

THE PLANT'S LUNGS

Oil-free, Class 0 compressors that power every tool.

WHERE 46.2 N·m IS BORN

The tool behind Issue 01's data journey.

The Company That Tightens the Bolt

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

In the very first issue of this magazine, we followed a single number — 46.2 N·m at 58 degrees — on its journey from a torque sensor up through every layer of the factory to the manager's dashboard. We left one question unasked: what actually *tightened* that bolt, and how did it know it had done so correctly? The answer, on a great many car lines, is a tool from Atlas Copco. This issue is about the company behind the fastening — and behind the compressed air that powers the whole plant.

Atlas Copco occupies an unusual place in the factory. It does not build the controller, the robot, or the conveyor; it builds the things that *act on the product* and the *air that drives them*. Its industrial portfolio splits into two great halves: **assembly tools** — above all, the smart nutrunners that fasten every critical joint — and **compressed air and vacuum**, the utility that powers pneumatic tools, clamps, and processes across the plant. This issue tours both.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact tools, compressors, or vendors in its factories. This is a tour of the **standard Atlas Copco industrial portfolio for a plant of this class** — the Tensor tightening tools, Power Focus controllers, and oil-free compressors and vacuum that are openly documented and ubiquitous in automotive assembly. Models and figures are representative, drawn from Atlas Copco's public product documentation, not confirmed internal Tesla equipment. The 46.2 N·m example is the illustrative data point from Issue 01.

The one idea: a fastening is a measured event, not a muscle

The thread of this whole issue is a shift in what it means to tighten a bolt. A hand wrench or a dumb impact gun applies force and hopes. A modern Atlas Copco tool treats every fastening as a *measured, recorded, and verified event*: it controls the torque and the angle precisely, judges the result against a window, refuses to pass a bad joint, and reports the outcome for traceability. The tool is not a muscle — it is an instrument. That is the difference between hoping a bolt is tight and *proving* it.

A dumb tool applies force and hopes. A smart tool measures, judges, and proves.

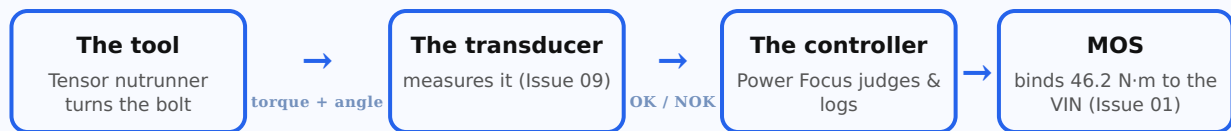


Figure 1. Where Atlas Copco sits in the data journey: the tool and its controller are the *origin* of the 46.2 N·m that Issue 01 followed to the dashboard. (*Conceptual.*)

In this issue

1. **The fastening problem** — why torque alone isn't enough
2. **The Tensor tools** — the smart electric nutrunners
3. **Strategies** — TurboTight, pulse, and the IxB idea
4. **The controllers** — Power Focus & traceability
5. **Pneumatic & low-torque** — the rest of the tools
6. **Compressed air** — the oil-free compressors
7. **Vacuum & the whole picture** & glossary

PART 1 · THE FASTENING PROBLEM

Why Torque Alone Isn't Enough

A bolt's job is to clamp two parts together with a precise force — the **clamp load** — and that is what actually holds a car together. But you cannot measure clamp load directly while tightening; you can only measure **torque** (turning effort) and **angle** (rotation). The whole science of controlled tightening is inferring the invisible clamp load from those two measurable things — and it is harder than it sounds, because friction steals a large and variable share of the torque.

THE CORE PROBLEM

Most of your torque fights friction, not the joint

Of the torque a tool applies, a large fraction is consumed overcoming friction under the bolt head and in the threads — only the remainder actually stretches the bolt to create clamp load. Because friction varies bolt to bolt, torque alone is an unreliable proxy for clamp load. This is why advanced tools also watch the **angle** of rotation: torque-plus-angle monitoring catches joints that hit the right torque for the wrong reason (a cross-thread, a soft spot), which a torque-only tool would happily pass.

This is the technical reason a "safety-critical" joint — a seat bracket, a steering component, a brake — demands a **transducerised** tool: one with a built-in torque transducer (Issue 09) and angle measurement, controlled so tightly that it can guarantee, and document, that each joint was formed correctly. A safety recall can hinge on the data a single tool recorded. That is the standard Atlas Copco's tools are built to meet.

The Tensor Tool Family

Atlas Copco's flagship line of intelligent assembly tools is **Tensor** — electric nutrunners built for quality- and safety-critical tightening, with a built-in transducer and a wealth of control and error-proofing options. They are positioned as the high-accuracy alternative to traditional pneumatic tools, with better accuracy and far lower noise and vibration. The family spans cordless and cabled, pistol-grip and angle-head, to reach every joint on a car.

Tensor STB / SRB (cordless)

Atlas Copco · battery nutrunner

Cordless, for safety-critical joints in tight spots. Ergonomic pistol-grip and angle Tensor STB nutrunners are built for safety-critical work in cramped areas, partnered wirelessly (Industrial Radio Communication) with a Power Focus controller for torque monitoring and full **joint traceability**. The **SRB** is the evolution, supporting every tightening strategy from direct drive to advanced low-reaction methods. One tool covers many applications in its torque range.

TYPE	FOR	LINKS TO
Cordless nutrunner	Safety-critical joints	Power Focus (wireless)

Tensor ES / STR (cabled)

Atlas Copco · corded nutrunner

Cabled, for tough environments and high throughput. The straight cable nutrunner **Tensor ES** is built for tough working environments, its durable design and built-in torque transducer suiting quality-critical industrial work — an accurate, low-noise, low-vibration alternative to pneumatic tools. The **Tensor STR** pushes productivity with faster tightening speeds and strong ergonomics, in both straight and angle forms.

TYPE	TRAIT	ELEMENT
Cabled nutrunner	Durable, fast, accurate	Built-in transducer

Tensor SB / fixtured spindles

Atlas Copco · handheld & automated

From handheld to fully automated. The pistol cordless **Tensor SB** ensures superior joint quality with torque-and-angle monitoring and rich error-proofing. For automated stations, fixtured tightening spindles and multi-spindle systems mount on robots or fixtures to tighten several bolts at once — the same Tensor intelligence, applied without a human hand.

HANDHELD	AUTOMATED	BOTH
Tensor SB	Fixtured spindles	Torque + angle monitored

PART 3 · TIGHTENING STRATEGIES

TurboTight, Pulse & the Tool That Is Its Own Controller

Having a precise tool is only half the story; *how* it drives the bolt — the tightening strategy — decides speed, ergonomics, and quality. Atlas Copco's tools support a ladder of strategies, and one architectural idea that quietly changes the whole station.

Strategy	How it works	Why it's used
Direct-drive (two-stage)	Spin down fast, then tighten slowly to the target torque/angle	The classic, accurate baseline for critical joints
TurboTight	A fast, low-reaction strategy that tightens in a quick, smooth move	Speed plus low reaction force into the operator's wrist
Tensor Pulse / TBP	Delivers torque in pulses, cutting reaction force to the hand	One-handed, error-proof work with low operator strain

Beyond strategy, the deeper shift is **where the intelligence lives**. Traditionally a tool is dumb and a separate controller box holds the brains. The **Tensor IxB** family inverts that: it puts the **controller inside the tool**, so the tool integrates directly with station accessories and the production system with no intermediate controller box. Features like **TrueAngle** compensation guard against the operator moving mid-tightening. It is the same instinct as Beckhoff's PC-based control (Issue 06) or the CompactLogix 5480 (Issue 05) — collapse a layer by moving the brain into the device.

Put the controller **inside the tool — and the controller box on the wall disappears.**

PART 4 · PROVING THE JOINT

Controllers & Traceability

For tools that are not self-contained, the brain is the controller — and the controller is what turns a tightening into *data*. This is the bridge from Atlas Copco's hardware to the data journey of Issue 01: the controller measures, judges OK/NOK, drives error-proofing, and hands the result up to the plant's systems.

Power Focus & MTF6000 controllers		Atlas Copco · tightening controllers
The brain that proves the joint. Power Focus controllers drive Tensor tools, set the tightening program, judge each result against its torque/angle window, and store the outcome for full joint traceability . The MTF6000 is positioned as an Industry 4.0 controller for low-torque fixtured (QMC) and handheld driver families. Controllers connect over the plant network (Issue 03) so results reach SCADA and MOS.		
DRIVES Tensor tools	JUDGES OK / NOK per joint	PROVIDES Full traceability

ERROR-PROOFING IN PRACTICE

The tool that won't let you make a mistake

- **Refuses bad joints** — a tightening outside the torque/angle window is flagged NOK and the station won't release the car (the interlock of Issue 01).
- **Counts the bolts** — the controller knows how many fastenings a station requires and won't pass until all are done OK.
- **Guides the operator** — socket trays, position detection, and lights direct which bolt, in which order.
- **Records everything** — every torque, angle, and result is logged and bound to the VIN, enabling a recall in minutes.

2

things measured: torque +
angle

OK/NOK

verdict on every single joint

46.2

N·m — the joint from Issue
01

100%

of critical joints traced to the
VIN

PART 5 · THE REST OF THE TOOLS

Pneumatic & Low-Torque Tools

Not every joint is safety-critical, and not every tool is a high-end electric nutrunner. Atlas Copco's range extends to air-powered tools for heavy or simple work and precise screwdrivers for the smallest fasteners — the broad base beneath the Tensor flagships.

Pneumatic Nutrunners (LTV, RTP)

Atlas Copco · air tools

Air-driven, for speed and heavy duty. The angle nutrunner **LTV** range combines high speed and accuracy with ergonomic design; the slim, lightweight **RTP** pneumatic nutrunner reaches heavy-duty bolts in cramped spaces with high repeatability and lower noise and vibration than impact wrenches. Air tools remain common where their power-to-weight and simplicity win.

POWER

Compressed air

LTV

Fast angle nutrunner

RTP

Slim, heavy-duty

Low-Torque Screwdrivers

Atlas Copco · small fasteners

For the small, delicate joints. A range of accurate, ergonomic electric and pneumatic screwdrivers handles smaller screw sizes (up to ~M6) — electronics, trim, and interior fasteners — with shut-off clutches for precision, time after time. Many are lubrication-free. The low-torque counterpart to the Tensor nutrunners.

FOR

Screws up to ~M6

USE

Electronics, trim

TRAIT

Precise, ergonomic

PART 6 · THE PLANT'S LUNGS

Compressed Air: The Oil-Free Compressors

Atlas Copco's other great business is the utility that quietly powers a huge share of the factory: compressed air. Pneumatic tools, clamps, actuators, blow-off, and many processes all

run on it — and in a car plant, especially in paint and battery areas, the air must be scrupulously clean. That demand is why **oil-free** compressors matter.

ZR / ZT Oil-Free Rotary Compressors

Atlas Copco · oil-free, Class 0

100% oil-free air for critical processes. The ZR (screw) and ZT (tooth) ranges deliver **Class 0** certified oil-free air — the highest purity, with zero risk of oil contaminating a paint finish or a battery process. Available with integrated dryers (Full Feature), **Variable Speed Drive (VSD+)**, and energy-recovery systems. Atlas Copco pioneered VSD in compressed air; matching motor speed to demand can cut compressor energy by up to ~35%.

AIR

100% oil-free, Class 0

DRIVE

VSD+ (variable speed)

USE

Paint, battery, critical

GA Oil-Injected & ZH Centrifugal

Atlas Copco · general & high-flow air

For general air and the largest demands. The **GA** oil-injected screw range is the workhorse for general compressed air where Class 0 purity isn't required; the **ZH** centrifugal (turbo) compressors serve the very highest flows. Modern controllers (e.g. Optimizer-class) coordinate multiple machines and tie into factory energy-management systems for plant-wide efficiency.

GA

Oil-injected, general air

ZH

Centrifugal, high flow

CONTROL

Multi-machine, energy-aware

WHY ENERGY DOMINATES THE DECISION

Over 80% of a compressor's lifetime cost is electricity

Compressed air is often the single largest electrical load in a plant — its generation can be a large share of the electricity bill, and over a compressor's life, energy dwarfs its purchase price. That economics is why VSD, energy recovery (capturing the compression heat), and smart multi-machine control are not luxuries but the core of the decision. A leaky air line is, quite literally, money escaping.

PART 7 · VACUUM & THE WHOLE PICTURE

Vacuum, and the Tools Behind the Build

The portfolio closes with the inverse of compressed air: **vacuum**. Industrial vacuum pumps power robotic suction grippers that lift body panels, glass, and parts; hold workpieces on fixtures; and serve processes from degassing to leak testing. Atlas Copco's vacuum range — including oil-sealed and dry pumps — rounds out a company whose products bracket the physical act of building a car, from the air that drives a tool to the suction that lifts a windshield.

Industrial Vacuum Pumps

Atlas Copco · vacuum technique

The pull that lifts and holds. Vacuum pumps (oil-sealed rotary-vane for deep vacuum, plus dry/claw and screw types for clean or continuous duty) feed the suction grippers on robots that handle panels and glass, clamp parts to fixtures, and support processes like leak testing and degassing. The quiet counterpart to compressed air on the same plant.

DRIVES

Robot suction grippers

HOLDS

Parts on fixtures

TYPES

Oil-sealed & dry

Step back, and Atlas Copco's place in the whole series comes clear. The earlier issues built the factory's nervous system — controllers, networks, drives, motors, sensors, conveyors. Atlas Copco builds the *hands and lungs*: the tools that physically act on the car, and the air and vacuum that animate them. Its smart nutrunner is where Issue 01's 46.2 N·m is actually born; its transducer is the sensor of Issue 09; its controller feeds the data journey; its compressors breathe through the pneumatics that appear in nearly every other issue. It is, in a sense, where this magazine's abstractions finally touch the steel.

And the lesson it embodies is the one this whole series keeps circling: precision is not a feeling, it is a measurement. A bolt is not "tight enough" because it felt tight — it is correct because a tool measured the torque and the angle, judged them against a window, refused the joint if they failed, and wrote the proof to the car's permanent record. Multiply that discipline by the thousands of joints in a car, and you have the difference between a vehicle you hope is safe and one you can *prove* is. That proof begins in the tool. ■

FIELD GLOSSARY

The Atlas Copco vocabulary in this issue

- **Nutrunner** — a powered tool for tightening threaded fasteners to a controlled torque (and angle).
- **Clamp load** — the actual clamping force a tightened bolt produces; the real goal, which torque only approximates.
- **Torque & angle monitoring** — measuring both turning effort and rotation to verify a joint, catching faults torque alone misses.
- **Transducerised tool** — a tool with a built-in torque transducer (Issue 09) for measured, verified tightening.
- **Tensor** — Atlas Copco's flagship family of intelligent electric nutrunners.
- **Tensor IxB** — the Tensor line with the controller built into the tool, removing the separate controller box.
- **TurboTight / Tensor Pulse / TBP** — low-reaction tightening strategies for speed and operator ergonomics.
- **TrueAngle** — angle compensation that guards against the operator moving during tightening.
- **Power Focus / MTF6000** — Atlas Copco tightening controllers that program, judge, and log fastenings.
- **Error-proofing** — features that prevent and detect mistakes (refusing bad joints, counting bolts, guiding the operator).
- **Joint traceability** — recording every fastening result and binding it to the vehicle's VIN.
- **Oil-free / Class 0** — compressed air certified to contain no oil — essential for paint and battery processes.
- **VSD (Variable Speed Drive)** — matching a compressor's motor speed to air demand to save energy (Issue 07).
- **ZR/ZT, GA, ZH** — oil-free rotary, oil-injected, and centrifugal compressor ranges.
- **Industrial vacuum** — pumps that power suction grippers and hold parts; the inverse of compressed air.

WHERE TO SEE THE REAL CATALOG

Every family here has an official Atlas Copco product page with specs and selectors. For tools, start at the assembly-tools and Tensor hub (atlascopco.com/en-us/itba); for air and vacuum, the compressors and vacuum sections (atlascopco.com/en-us/compressors). This issue connects to Issue 01 (the 46.2 N·m data journey & VIN binding), 03 (the plant network carrying results), 05/06 (the controller-in-the-device idea), 07 (VSD & motors), and 09 (the torque transducer). Tesla's specific tooling is proprietary; models and figures here are representative of automotive-class plants.

THE INTEGRATION LAYER · Issue 11 · A technical magazine · Source: Atlas Copco public product documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 12

COVER STORY · THE COMPANY THAT MAKES AIR DO WORK

Festo: Pneumatics, Motion & the Grip

The cylinders, valve terminals, grippers and electric axes that move, clamp, and lift on every line — and the steady migration from air to electric that is reshaping how a factory makes things move.

THE CYLINDER

How compressed air becomes a precise, repeatable push.

AIR vs ELECTRIC

Why motion is migrating from pneumatic to servo.

THE VALVE TERMINAL

One smart manifold commanding dozens of motions.

THE GRIP & THE LIFT

Grippers and vacuum that handle every part.

The Company That Makes Air Do Work

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

The previous issue ended with compressed air leaving the compressor — clean, dry, and pressurised, ready to do something. This issue is about what it does. If Atlas Copco makes the plant's air, Festo is the company that puts that air to work: turning still pressure into motion, clamping, lifting, and gripping, thousands of times a minute across a car line. And increasingly, it is the company quietly replacing that air with electricity. Festo sits exactly at the hinge between two eras of factory motion.

For most of a century, "automation motion" meant pneumatics — a cylinder pushed by compressed air. It is cheap, simple, forceful, and forgiving, which is why air cylinders still number in the thousands on any line. But pneumatics is hard to position precisely and wastes energy, so the industry is steadily shifting demanding motions to **electric actuators**. Festo is unusual in leading *both* worlds — the definitive pneumatics house and a serious electric-automation supplier — which makes it the perfect lens on that transition.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact components or vendors in its factories. This is a tour of the **standard Festo automation portfolio for a plant of this class** — cylinders, valve terminals, the CPX/CPX-AP platform, CMMT servo drives, electric actuators, grippers, and vacuum — openly documented and ubiquitous in automotive assembly and machine building. Models and figures are representative, drawn from Festo's public product documentation, not confirmed internal Tesla equipment. This issue is the natural companion to Issue 11 (Atlas Copco): there, air is made; here, air does work.

The one idea: motion is a choice between air and electrons

The thread of this issue is a single engineering decision repeated thousands of times across a plant: should this motion be **pneumatic** or **electric**? Air is simplest and cheapest for a blunt push between two end stops; electric wins when you need precise position, soft control, energy efficiency, and data. Festo's portfolio is organised around giving an engineer both, and tools to choose — and the centre of gravity is slowly, steadily moving toward electric.

For a century, motion meant [air](#). Now, joint by joint, it is becoming [electrons](#).

Control & engineering

Festo Automation Suite · CPX controllers

The valve terminal / I/O

CPX-AP · VTUG / VTSA / MPA / VTEM — commands the actuators

The actuators

pneumatic cylinders · electric axes (CMMT) · grippers · vacuum

The work

push · clamp · position · grip · lift

Figure 1. The Festo stack: a controller and a smart valve terminal command a layer of actuators — pneumatic and, increasingly, electric — that do the physical work. (*Conceptual.*)

In this issue

1. **Pneumatic basics** — how a cylinder works
2. **The cylinders** — the Festo actuator families
3. **Valves & valve terminals** — commanding the motion
4. **CPX & digital pneumatics** — the smart manifold & VTEM
5. **Air vs electric** — the great migration
6. **Electric motion** — CMMT drives & axes
7. **Grippers, vacuum & air prep** & glossary

PART 1 · PNEUMATIC BASICS

How a Cylinder Turns Air Into a Push

The fundamental unit of pneumatic motion is the cylinder: a piston in a tube. Admit compressed air to one side and the piston is pushed to the other end; admit it to the other side and it returns. That is the entire principle — a simple, powerful, binary push between two end stops. Its very simplicity is why pneumatics conquered the factory: a cylinder is cheap, light, forceful for its size, tolerant of overload, and safe in wet or hazardous areas where electricity is awkward.

THE STRENGTH AND THE WEAKNESS

Brilliant between two points, awkward in between

A cylinder is superb at moving fully *out* or fully *in* — clamping, ejecting, stamping, stopping. What it is *not* good at is stopping precisely *partway*, or moving gently: air is compressible, so a pneumatic piston is springy and hard to position to a fraction of a millimetre. This single limitation — great at end stops, poor at controlled intermediate positions — is the seed of the whole air-versus-electric story this issue builds toward.

A pneumatic motion also needs supporting cast: a **valve** to direct the air (Part 3), **fittings and tubing** to carry it, **cushioning** to absorb the impact at each end (Festo's self-adjusting PPS cushions do this automatically), and **cylinder switches** (Issue 09's magnetic sensors) to confirm the piston actually reached its end position. A single clamp is a small system.

The Cylinder Families

Festo's pneumatic-drive catalog is vast, but it organises by the shape of motion a job needs — straight push, guided push, rodless travel, rotation, or grip. A handful of families cover the overwhelming majority of a car line's pneumatic motions.

ISO & Compact Cylinders (DSBC, DNC, ADVU)

Festo · piston-rod cylinders

The standard push. Piston-rod cylinders to ISO standards (DSBC, DNC) and compact forms (ADVU, DSNU) deliver a straight linear push with standardised dimensions and mounting, so they are interchangeable across vendors. Featuring self-adjusting **PPS cushioning** for smooth, precise end-of-stroke operation. The everyday workhorse for clamping, pushing, and ejecting.

MOTION

Straight linear push

STANDARD

ISO interchangeable

FEATURE

Self-adjusting cushions

Guided Drives & Mini Slides (DGSL)

Festo · guided linear motion

A push that can't twist. When a load must move precisely without rotating or deflecting — handling, pick-and-place, tooling — guided drives and mini slides like the **DGSL** add integrated guide rails so the motion is rigid and repeatable. The pneumatic precursor to a small electric slide.

MOTION

Guided linear

TRAIT

No twist / deflection

USE

Pick-and-place, tooling

Rodless Cylinders & Rotary (DGC, swivel)

Festo · long-travel & rotation

Long strokes and turns. Rodless cylinders (**DGC**) carry a load along the cylinder body itself, saving space on long travels; swivel/rotary drives and semi-rotary actuators produce controlled rotation for flipping, indexing, and orienting parts. Plus specialty stopper, tandem, and multi-position cylinders for niche motions.

RODLESS

Long-travel (DGC)

ROTARY

Swivel / semi-rotary

SPECIALTY

Stopper, tandem

PART 3 · COMMANDING THE MOTION

Valves & Valve Terminals

A cylinder cannot move until something tells the air where to go — that is the valve's job. A solenoid valve switches air to one side of a cylinder or the other on an electrical signal from the controller. But a line has hundreds of cylinders, and wiring a separate valve to each would be chaos. The elegant solution, and a Festo signature, is the **valve terminal**: a manifold that packs many valves into one block with a single electrical and pneumatic connection.

Dozens of valves, one smart block. A valve terminal mounts many solenoid valves on a shared manifold with one electrical interface and one air supply — commanding dozens of cylinders from a single networked node. Families scale from compact (VTUG) to high-flow modular (VTSA, MPA). The terminal connects to the plant network (Issue 03) through the CPX electrical platform, so the controller addresses each valve by software.

FUNCTION

Many valves, one block

CONNECTION

One electrical + air

NETWORK

via CPX (Issue 03)

WHY THE TERMINAL WINS

One cable instead of a hundred — again

The logic is exactly that of distributed I/O (Issue 05) and the conveyor cabling (Issue 10): consolidate. Instead of running power and tubing to every individual valve, a valve terminal needs one network drop and one air feed, then distributes both internally. Fewer connections, faster build, cleaner diagnostics — and each valve becomes a software-addressable point, not a hand-wired device.

250µs

CPX-AP cycle time, real-time

80

modules in one CPX-AP system

1

air feed for a whole valve bank

4+

fieldbuses CPX speaks (Issue 03)

PART 4 · THE SMART MANIFOLD

CPX, CPX-AP & Digital Pneumatics

The valve terminal is where Festo's pneumatics meets the digital factory, and the bridge is the **CPX** platform — an electrical backplane that turns a valve bank into an intelligent, networked node. It is the piece that lets a century-old technology (the air cylinder) speak the language of Industry 4.0.

CPX / CPX-AP Automation Platform

The intelligence layer for pneumatics. The robust CPX backplane connects valves, digital and analog I/O, proportional regulators, sensors, and **IO-Link** devices (Issue 03), and links to the controller over the major fieldbuses. The newer **CPX-AP** automation platform is a backplane-based system reaching ~250 µs cycle times and up to 80 freely-combined modules, letting electric and pneumatic automation mix in one architecture.

CARRIES

Valves, I/O, IO-Link

NETWORK

multi-fieldbus

CPX-AP

~250 µs, 80 modules

The Motion Terminal: valves controlled by apps

Festo's **Motion Terminal VTEM** is the boldest idea in the catalog: the first valve terminal whose valve *functions* are defined by **software apps** rather than fixed hardware. A combination of pneumatics, sensors, electronics, and software lets one physical valve take on many different behaviours — flow control, soft-stop, pressure regulation, leak detection — changed by selecting an app. It is "digitised pneumatics": the same instinct as software-defined everything elsewhere in this series, applied to air.

PART 5 · THE GREAT MIGRATION

Air vs Electric: The Defining Choice

Now the central decision this whole issue circles. For every motion on a line, an engineer weighs pneumatic against electric — and the balance has been tilting toward electric for years. Neither is "better"; they answer different needs, and knowing which to pick is core automation craft.

Pneumatic — air

Cheap, simple, forceful, light, robust, safe in wet/hazardous zones. Brilliant for blunt moves between two end stops. But springy (hard to position partway), noisy, and energy-inefficient — compressed air is costly to make (Issue 11).

Electric — servo

Precise positioning anywhere, soft and programmable motion, quiet, energy-efficient, and rich with data (position, force, diagnostics). But costlier per axis and more complex. Wins wherever precision, control, or data matter.

Need	Reach for	Why
Clamp / eject / stop	Pneumatic cylinder	Blunt force between end stops; cheap and robust
Precise positioning	Electric axis	Stops anywhere, repeatably, to fractions of a mm
Energy efficiency	Electric	No continuous air consumption or leak losses
Process data / flexibility	Electric	Reports position & force; reprogrammed in software
Lowest cost / simplicity	Pneumatic	Fewer parts, forgiving, decades of familiarity

Air for the blunt push. Electrons for the precise, quiet, data-rich move. The line is learning the difference.

Festo's answer to the migration is to make electric *as easy as* pneumatic. Its **Simplified Motion Series** of electric actuators are deliberately designed to be as simple to set up as a pneumatic cylinder — a drop-in upgrade path that lets a plant convert motion to electric without the usual servo-engineering burden. That bridge is exactly where much of the industry now sits.

Electric Axes & CMMT Servo Drives

When a motion graduates from air to electric, Festo offers a full electric-automation stack — actuators, motors, and the servo drives that control them — echoing the motor-and-drive fundamentals of Issue 07, now in Festo's catalog. The defining trait is multi-protocol flexibility.

CMMT-AS / CMMT-ST Servo Drives

Festo · multiprotocol servo drives

The electric brain, speaking every language. The CMMT servo drives control Festo's electric axes and motors — and are **multiprotocol**, working over EtherNet/IP, EtherCAT, PROFINET, and Modbus TCP (all from Issue 03). That single trait lowers the learning curve and inventory for builders working with any major PLC brand — one drive fits a Siemens, Rockwell, or Beckhoff line. Commissioned in the Festo Automation Suite.

CONTROLS

Festo electric axes

PROTOCOLS

EIP/EtherCAT/PROFINET/Modbus

SETUP

Festo Automation Suite

Electric Axes & Actuators (EGC, mini slides)

Festo · electromechanical motion

Precise linear and rotary motion. Electric cylinders, linear axes (**EGC** toothed-belt and ball-screw), slides, rotary actuators, and stoppers provide programmable positioning — any position, any speed, repeatably — for handling, assembly, and gantries. Paired with EMMS/servo motors and a CMMT drive into a complete electric axis configured with one engineering tool.

FORMS

Axes, slides, rotary

CONTROL

Programmable position

USE

Handling, gantries

MULTI-AXIS & ROBOTICS

From one axis to a handling system

Combine electric axes on two or three orthogonal directions and you have a **multi-axis positioning system** — a Cartesian gantry or planar handler that places a part anywhere in a work area, all from one Festo configuration. Festo also offers compact handling and industrial-robot solutions, so a builder can assemble pick-and-place and assembly motion from the same catalog — the actuator equivalent of buying a whole portfolio from one house (Issues 02/05).

Grippers, Vacuum & Air Preparation

The portfolio closes with the parts that actually touch the workpiece — grippers and vacuum — and the unglamorous but vital units that condition the air before any of it runs. These are where Festo meets the part directly.

Grippers (pneumatic & electric)

Festo · gripping technology

The hand at the end of the arm. Festo's grippers — parallel, angular, radial, three-point — grip, hold, position, and orient a part or tool, in both pneumatic and electric forms. Mounted on robots and handlers, they are the interface between the motion system and the workpiece. Electric grippers add controllable force and position feedback, useful for delicate or varied parts.

TYPES

Parallel/angular/radial

DRIVE

Pneumatic or electric

JOB

Grip, hold, orient

Vacuum Technology

Festo · suction handling

Lifting by suction. Vacuum generators (often compact venturi types that make vacuum from compressed air), suction cups, and vacuum grippers lift flat, smooth, or delicate parts — sheet metal, glass, trim — that mechanical fingers handle poorly. The handling counterpart to the industrial vacuum of Issue 11, integrated directly into the motion system.

MAKES VACUUM

Venturi generators

LIFTS

Sheet, glass, trim

TOOLS

Cups & vacuum grippers

Air Preparation (FRL: MS, LFR)

Festo · service units

Conditioning the air before it works. Before compressed air reaches a valve or cylinder it must be filtered, regulated to the right pressure, and (sometimes) lubricated — the job of **FRL service units** (Filter-Regulator-Lubricator), such as the MS and LFR series. They protect every downstream component; dirty or wrong-pressure air is a leading cause of pneumatic faults. Where Issue 11's clean air is tuned for each machine.

F-R-L

Filter, regulate, lubricate

PROTECTS

All downstream parts

SERIES

MS, LFR

REFERENCE

The Whole Portfolio, and the Migration

Trace a single handling motion — a gripper descending to lift a panel — and you touch the whole Festo catalog: a valve terminal commands the air or a CMMT drive commands an axis; a cylinder or electric slide moves the gripper down; a vacuum cup or gripper takes the part; cylinder switches and sensors confirm each step; and FRL units fed by Issue 11's compressor conditioned the air that made it all move. Festo's distinction is spanning that entire chain in *both* pneumatic and electric form.

Function	Pneumatic (Festo)	Electric (Festo)
Straight push	ISO/compact cylinder (DSBC, ADVU)	Electric cylinder (Simplified Motion)
Guided / slide	Mini slide (DGSL)	Electric slide / axis (EGC)
Long travel	Rodless cylinder (DGC)	Toothed-belt axis (EGC)
Rotation	Swivel / semi-rotary drive	Electric rotary actuator
Command layer	Valve terminal (VTUG/VTSA/MPA)	CMMT servo drive
Gripping	Pneumatic gripper	Electric gripper (force feedback)

The arc, once more, is the one bending through every issue: from simple and forceful toward precise, quiet, and data-rich. Pneumatics will never vanish — for a blunt, cheap, robust push it remains unbeatable, and a car line will always have thousands of air cylinders clamping and ejecting. But the demanding, precise, measured motions are migrating to electric, and Festo built its catalog to carry a plant smoothly across that line rather than forcing a choice. Air made the modern factory move; electrons are teaching it to move *exactly*. The grip at the end of the arm is where both meet the car. ■

FIELD GLOSSARY

The Festo vocabulary in this issue

- **Pneumatic cylinder** — a piston in a tube pushed by compressed air; the basic unit of pneumatic motion.
- **ISO / compact cylinder (DSBC, ADVU)** — standardised piston-rod cylinders for a straight push.
- **Guided drive / mini slide (DGSL)** — a linear motion with guide rails so it cannot twist.
- **Rodless cylinder (DGC)** — a cylinder whose load rides along the body, for long travels.
- **Solenoid valve** — an electrically switched valve that directs air to a cylinder.
- **Valve terminal (VTUG/VTSA/MPA/CPV)** — a manifold packing many valves into one networked block.
- **CPX / CPX-AP** — Festo's electrical backplane and automation platform linking valves, I/O, and IO-Link to the network.
- **Motion Terminal (VTEM)** — a valve terminal whose valve functions are defined by software apps ("digital pneumatics").
- **FRL / service unit (MS, LFR)** — Filter-Regulator-Lubricator that conditions air before use.
- **PPS cushioning** — Festo's self-adjusting end-of-stroke cushioning for smooth cylinder stops.
- **Simplified Motion Series** — electric actuators designed to be as easy to set up as a pneumatic cylinder.
- **CMMT-AS / -ST** — Festo's multiprotocol servo drives (EtherNet/IP, EtherCAT, PROFINET, Modbus TCP).
- **Electric axis (EGC)** — an electromechanical linear actuator for programmable positioning.
- **Gripper** — the pneumatic or electric "hand" that grips and orients a part.
- **Vacuum generator / gripper** — venturi-based suction handling for flat or delicate parts.

WHERE TO SEE THE REAL CATALOG

Every family here has an official Festo product page with specs and configurators. Start at the industrial-automation products hub (festo.com/us/en/c/products/industrial-automation-id_pim1/); the pneumatic drives, valve terminals, CPX-AP, CMMT drives, electric actuators, grippers, and vacuum sections carry full detail, and the Festo Automation Suite configures electric systems. This issue is the companion to Issue 11 (Atlas Copco — making the air), and connects to Issue 03 (fieldbuses & IO-Link), 05/10 (consolidated cabling), 07 (servo motors & drives), and 09 (cylinder switches & sensors). Tesla's specific components are proprietary; models and figures here are representative of automotive-class plants.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 13

COVER STORY · THE HOUSE OF MANY BRANDS

Emerson: One Company, a Dozen Names

AVENTICS pneumatics, ASCO valves, Rosemount measurement, PACSystems control, NI test — the federation of famous brands under one roof, and the "floor-to-cloud" idea that ties them together.

A FEDERATION OF BRANDS

Why Emerson is a portfolio of names you already know.

AVENTICS & ASCO

The pneumatics and valves that move and switch the line.

MEASURE EVERYTHING

Rosemount instruments and the data they create.

FLOOR TO CLOUD

PACSystems control and the edge-to-cloud bridge.

The House of Many Names

THE INTEGRATION LAYER · Feature · A portfolio tour grounded in public product documentation

Every vendor in this series so far has been, more or less, one company with one name on the product. Emerson is different, and understanding it means understanding that difference first. Emerson is not a single brand — it is a *federation* of famous automation names, assembled over decades, each a leader in its niche and each still carrying its own identity. When you meet AVENTICS, ASCO, Rosemount, TESCO, TopWorx, PACSystems, Movicon, Branson, or NI on a factory floor, you are meeting Emerson. This issue is a tour of that federation, and of the idea that unites it.

That structure makes Emerson unusually broad. Where Festo (Issue 12) is the pneumatics-and-motion house and Atlas Copco (Issue 11) the tools-and-air house, Emerson spans **discrete automation** (pneumatics, valves, control), **measurement and instrumentation** (the sensors that quantify a process), and even **test and measurement** (through its NI business). It reaches from the air cylinder on an assembly fixture to the pressure transmitter on a process line to the test rig that validates a finished module.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish the exact components or vendors in its factories. This is a tour of the **standard Emerson industrial portfolio for a plant of this class** — AVENTICS pneumatics, ASCO valves, Rosemount measurement, PACSystems control, and the NI test platform — openly documented and widely used in automotive, packaging, and process automation. Models and figures are representative, drawn from Emerson's public product documentation, not confirmed internal Tesla equipment. Brand names (AVENTICS, ASCO, Rosemount, etc.) are Emerson brands referenced descriptively.

The one idea: floor to cloud

If a single phrase unites Emerson's sprawl, it is "**floor to cloud**" — the idea of unlocking the data trapped in islands of automation and connecting it all the way from the machine on the floor up to cloud analytics. It is Emerson's framing of the same theme that runs through this whole magazine: a factory is only as good as the data it can see, and value comes from connecting the layers. Emerson's pitch is that owning brands at *every* layer — from the sensor to the controller to the edge software — lets it carry data across that whole climb.

A federation of famous names, under one roof, joined by one idea: floor to cloud.



Figure 1. Emerson's brands mapped to the layers they serve — from acting and measuring on the floor, through control and edge, up to the cloud. *(Conceptual.)*

In this issue

1. **The federation** — how Emerson is structured

2. **AVENTICS pneumatics** — valves & the Series XV

3. **ASCO & fluid control** — the solenoid valve

4. **Rosemount measurement** — quantifying the process
5. **PACSystems control** — the controllers & edge

6. **NI test & Branson** — validating & joining

7. **The whole federation** & glossary

PART 1 · HOW EMERSON IS STRUCTURED

A Federation of Famous Names

To navigate Emerson you navigate its brands, because each is a recognised leader that Emerson acquired and kept. The names matter: a technician asking "do we run Emerson?" is really asking about a dozen distinct product lines. Here is the map — the major industrial brands and what each one owns.

Brand	Domain	What it makes
AVENTICS	Pneumatics	Valves, cylinders, actuators, vacuum — factory motion
ASCO	Fluid control	Solenoid valves for air, gas, water, and fluids
Rosemount	Measurement	Pressure, temperature, level, flow instruments
PACSystems	Control	PLCs, edge controllers, industrial PCs
TESCOM	Pressure control	High-precision pressure regulators
TopWorx	Position sensing	Valve position monitors & switches
Movicon / PACEdge	Software	SCADA/HMI and edge software
Branson	Joining	Ultrasonic & laser welding of plastics/metals
NI (National Instruments)	Test & measurement	LabVIEW, automated test platforms

WHY A FEDERATION WORKS DIFFERENTLY

Best-of-breed parts, one corporate roof

Where Siemens or Rockwell sell a tightly co-engineered *single* portfolio (Issues 02/05), Emerson's model is closer to the opposite: keep each acquired brand's strength and identity, and unite them at the level of strategy, software, and the "floor-to-cloud" data story rather than forcing one electrical standard. The upside is genuine best-in-class depth in each niche; the work is making the pieces interoperate — which is exactly why open protocols (Issue 03) matter so much to Emerson.

PART 2 · PNEUMATICS

AVENTICS: The Motion Brand

Emerson's pneumatics brand is **AVENTICS**, and its role on a car line mirrors Festo's (Issue 12): turning compressed air into motion — valves, cylinders, actuators, and vacuum. Its flagship recent launch shows where the field is heading: high flow, compact size, and openness to every network.

AVENTICS Series XV Valves

Emerson · next-gen valve platform

High flow, compact, interoperable. The Series XV pneumatic valve platform packs higher flow into a compact package — the XV03 delivers up to ~350 NL/min and the XV05 up to ~880-1000 NL/min — with interoperable sizes and universal threads for global use. Crucially, it speaks **PROFINET, EtherNet/IP, EtherCAT, Modbus TCP, and IO-Link** (all Issue 03), so it drops into any major PLC environment.

FLOW

up to ~880+ NL/min

NETWORKS

all major fieldbuses

TRAIT

Compact, interoperable

AVENTICS AV Valve System & SPRA actuator

Emerson · modular valves & electric

Modular air, and a step to electric. The Advanced Valve System **AV** is a modular valve manifold that serves many applications even in tight spaces — AVENTICS's answer to the valve terminal (Issue 12). And mirroring the air-to-electric migration, the **Series SPRA** electric rod-style linear actuator brings improved load capacity, accuracy, and reliability — pneumatics and electric motion from one brand.

AV

Modular valve manifold

SPRA

Electric rod actuator

SPANS

Air + electric

THE FRONTIER · PNEUMATICS MEETS OPC UA

The first valve system with built-in OPC UA

Emerson introduced what it describes as the first pneumatic valve system with integrated **OPC UA** (Issue 03) — a "digital twin" approach that lets the valve manifold expose its data in a structured, vendor-neutral way and solve interoperability across the floor. It is the same instinct as Festo's digital pneumatics (Issue 12): make the humble air valve a first-class, self-describing data citizen on the floor-to-cloud path.

ASCO: The Solenoid Valve Standard

If AVENTICS handles the air that *moves* things, **ASCO** handles the valves that *switch and control fluids* — air, gas, water, coolant, and process media. ASCO is one of the most recognised names in solenoid valves, and on a car line it controls everything from the compressed air feeding a clamp to the water and fluids in utilities and processes.

ASCO Solenoid & Process Valves

Emerson · fluid control valves

Switching and controlling fluids. ASCO's solenoid valves electrically open, close, or direct the flow of air, gas, water, and other media — the on/off and directional control of fluid systems across the plant. The range spans general-purpose to specialised process and safety-rated valves, with multiple mounting options. Where AVENTICS moves cylinders, ASCO governs the fluids that feed and cool the whole operation.

CONTROLS

Air, gas, water, fluids

FUNCTION

On/off & directional

RANGE

General to process

TESCOM Regulators & TopWorx Sensing

Emerson · pressure & position

Precise pressure, and knowing valve state. **TESCOM** provides high-precision pressure regulators for demanding gas and fluid applications — controlling pressure where accuracy is critical. **TopWorx** provides valve position monitors and switches that report whether a valve is actually open or closed (the fluid-world cousin of the cylinder switches in Issue 09), feeding that state back to the control system.

TESCOM

Precision regulators

TOPWORX

Valve position sensing

BOTH

Fluid-control support

Rosemount: Quantifying the Process

Emerson's measurement brand is **Rosemount**, and it answers the questions a discrete sensor (Issue 09) cannot: not just "is the part there?" but "what is the exact pressure, temperature, level, or flow?" These are the instruments of *process* measurement — vital in a car plant's paint, e-coat, battery, coolant, and utility systems, where a precise analog value, not a yes/no, is what matters.

Rosemount 3051 Pressure Transmitter

Emerson · pressure measurement

The industry-standard pressure instrument. The Rosemount 3051 measures pressure, differential pressure, level, and flow with high accuracy, now with task-based menus, quick service buttons, Bluetooth connectivity, and built-in diagnostics. It is one of the most widely deployed process transmitters in the world — the instrument that turns a fluid pressure into a trustworthy number for the control system.

MEASURES

Pressure, DP, level, flow

FEATURES

Bluetooth, diagnostics

OUTPUT

to control system

The rest of the measured world. Beyond pressure, Rosemount covers **temperature** (transmitters with RTD/thermocouple sensors, Issue 09), **level** (radar, guided-wave, switches like the 2140 for overfill prevention), and **flow** (including Micro Motion Coriolis meters for mass flow). Plus wireless instrument versions for hard-to-wire points — secure, real-time data from anywhere on the plant.

TEMPERATURE

RTD / thermocouple

LEVEL

Radar, guided-wave

FLOW

Coriolis (Micro Motion)

DISCRETE VS PROCESS MEASUREMENT

Two different questions, two sensor worlds

Issue 09's sensors mostly answer *discrete* questions — present/absent, position, a torque reading. Rosemount instruments answer *continuous process* questions — the exact pressure in a line, the level in a tank, the flow of a coolant, the temperature of an oven. A car plant needs both: discrete sensing for assembly logic, process measurement for the paint, battery, and utility systems that behave like a process plant. Emerson is strong in the second world that Issue 09 only touched.

PART 5 · CONTROL & EDGE

PACSystems: The Controllers

All that motion and measurement needs a brain, and Emerson's control brand is **PACSystems** — programmable controllers (PLCs/PACs), edge controllers, and industrial PCs, the Emerson counterpart to a Siemens S7-1500 or a Rockwell ControlLogix (Issues 02/04). The emphasis is on openness and pushing intelligence to the machine edge.

PACSystems RSTi-EP CPE200 & RX3i

Emerson · programmable controllers

From compact edge to rack-scale control. The **RSTi-EP CPE200** family delivers large-PLC capability in a compact, rugged design with **security-by-design, open programming, and open communications** built in. The **RX3i** is the rack-based controller for larger systems (with a CPS400 safety variant). Programmed in PAC Machine Edition (PME) — the Emerson counterpart to TIA Portal or Studio 5000.

CPE200

Compact, rugged PLC

RX3i

Rack-scale + safety

SOFTWARE

PAC Machine Edition

PACSystems Edge: IPCs & PACEdge

Emerson · edge computing

Solving problems at the machine edge. The PACSystems IPC line (6010/7010/8010 industrial computers) and **PACEdge** software bring analytics and connectivity to the machine itself — the most advanced part of Emerson's portfolio for edge computing and control. This is the hardware that makes "floor to cloud" real: collecting, analysing, and forwarding machine data without shipping everything to a distant server first.

IPCS

6010 / 7010 / 8010

PACEEDGE

Edge analytics software

ENABLES

Floor to cloud

9+

major industrial brands in
the federation

5

fieldbuses the Series XV
speaks

3051

the world-standard pressure
transmitter

F→C

floor-to-cloud, the unifying
idea

PART 6 · VALIDATE & JOIN

NI Test & Branson Joining

Two more brands extend Emerson into corners no other vendor in this series reaches: validating that a finished module *works*, and joining materials without a fastener at all. Both are increasingly central to building an electric vehicle.

NI (National Instruments) Test & Measurement

Emerson · automated test

Proving the product works. NI — now an Emerson business — builds the platform-based, often **LabVIEW**-programmed automated test and measurement systems that validate electronics, battery modules, and components. In an EV plant, end-of-line and in-process electrical test is critical: every battery module and controller must be verified before it ships. NI is the brand behind those test rigs, increasingly with AI-assisted tools.

DOMAIN

Automated test

SOFTWARE

LabVIEW

VALIDATES

Electronics, batteries

Branson Welding & Joining

Emerson · ultrasonic & laser joining

Joining without a bolt. Branson provides **ultrasonic and laser welding** of plastics and metals — bonding components with vibration or light instead of a fastener or adhesive. In a car, that means welding plastic assemblies, battery components, and wire/foil connections. It is the joining counterpart to Atlas Copco's bolted fastening (Issue 11): a different way to make two parts one.

METHODS

Ultrasonic & laser

JOINS

Plastics, metals, foils

USE

Battery, plastics, wiring

WHY THIS MATTERS FOR EVS SPECIFICALLY

Batteries change what "assembly" means

A combustion car is mostly mechanical fastening; an electric car adds vast electrical and battery assembly — thousands of cell connections, modules to validate, busbars to weld. That shift makes **test** (NI, proving each module) and **non-bolt joining** (Branson, welding cells and connections) disproportionately important. Emerson's federation happens to hold both, alongside the pneumatics and measurement of a conventional line — a portfolio shaped well for the EV era.

REFERENCE

The Whole Federation, on One Page

Trace a single battery module through a plant and you can touch the whole Emerson federation: ASCO valves control the coolant and air; AVENTICS cylinders clamp the module in its fixture; Rosemount instruments watch the pressures and temperatures of the process

around it; a PACSystems controller runs the cell with PACEdge analysing it at the edge; Branson welds the cell connections; and an NI test rig validates the finished module before it ships. One company, many names, the whole journey.

Layer / job	Emerson brand	Counterpart in this series
Pneumatic motion	AVENTICS	Festo (Issue 12)
Fluid control valves	ASCO	(fluid-control specialty)
Pressure regulation	TESCOM	(air prep, Issue 12)
Process measurement	Rosemount	beyond discrete sensors (Issue 09)
Valve position sensing	TopWorx	cylinder switches (Issue 09)
Control (PLC/PAC)	PACSystems	S7-1500 / ControlLogix (02/04)
Edge & SCADA	PACEdge / Movicon	edge gateways (Issue 01/03)
Bolted fastening	(see Atlas Copco)	Atlas Copco (Issue 11)
Non-bolt joining	Branson	(welding & ultrasonic)
Test & validation	NI	(new to the series)

The arc that runs through every issue bends a little differently here. Most vendors win by *coherence* — one engine, one network, one tool. Emerson wins by *breadth* — assembling best-in-class brands across the widest span of the factory, from the air cylinder to the battery test rig, and betting that "floor to cloud" software can knit them into one data story. It is a different philosophy of how to be an automation giant: not a single tightly-built system, but a federation deep in every niche. For a plant that needs the best of many worlds — discrete *and* process *and* test — that breadth is the whole appeal. The names stay; the data flows. ■

FIELD GLOSSARY

The Emerson vocabulary in this issue

- **Emerson** — an industrial-automation powerhouse structured as a federation of acquired brands.
- **AVENTICS** — Emerson's pneumatics brand (valves, cylinders, actuators, vacuum).
- **Series XV / AV / SPRA** — AVENTICS's high-flow valve platform, modular valve system, and electric rod actuator.
- **ASCO** — Emerson's solenoid-valve brand for controlling air, gas, water, and fluids.
- **TESCOM** — high-precision pressure regulators.
- **TopWorx** — valve position monitors and switches.
- **Rosemount** — Emerson's process-measurement brand (pressure, temperature, level, flow).
- **Rosemount 3051** — the industry-standard pressure/DP/level/flow transmitter.
- **Micro Motion** — Coriolis mass-flow measurement.
- **PACSystems** — Emerson's controller brand (PLC/PAC, RSTi-EP CPE200, RX3i, IPCs).
- **PAC Machine Edition (PME)** — the programming software for PACSystems controllers.
- **PACEdge / Movicon** — Emerson's edge software and SCADA/HMI.
- **Branson** — Emerson's ultrasonic and laser welding/joining brand.
- **NI (National Instruments)** — Emerson's test-and-measurement business (LabVIEW, automated test).
- **Floor to cloud** — Emerson's framing for connecting machine data from the floor up to cloud analytics.

WHERE TO SEE THE REAL CATALOG

Every brand here has official Emerson product pages with specs and configurators. Start at the Emerson automation hub (emerson.com/en-us/automation); the AVENTICS pneumatics, ASCO valves, Rosemount measurement, and PACSystems control sections carry full detail, and NI is at (ni.com). This issue connects to Issue 03 (the fieldbuses & OPC UA the Series XV speaks), 09 (discrete vs process measurement), 11 (Atlas Copco — bolted joining, vs Branson's welding), and 12 (Festo — the pneumatics parallel to AVENTICS). Tesla's specific components are proprietary; models and figures here are representative of automotive-class plants.

THE INTEGRATION LAYER · Issue 13 · A technical magazine · Source: Emerson public product documentation

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 14

COVER STORY · MOVING & STORING — THE LOGISTICS OF PRODUCTION

The River of Parts: Material Handling, Flow & Inventory

The deep, three-layer discipline behind a factory that never starves and never drowns — the equipment that moves material, the flow that orchestrates it, and the inventory mathematics that decides how much to hold and when to order.

HANDLING IS WASTE

Why every metre a part travels is a cost, never value.

THE MILK RUN

How a tugger train on a timetable feeds the whole line.

THE EOQ FORMULA

The 110-year-old equation behind every order quantity.

THE BULLWHIP

How a small demand wobble becomes an upstream flood.

The Factory as a River

THE INTEGRATION LAYER · Feature · A comprehensive explainer grounded in material-handling & inventory engineering

A factory is best understood not as a building but as a river. Material enters as raw stock, flows through hundreds of operations, and leaves as finished product — and like any river, it can run dry or it can flood. Starve a line and it stops; drown it in inventory and you freeze cash, consume space, and hide every problem behind a wall of buffer stock. The whole discipline of production logistics exists to keep that river flowing at exactly the right rate: this issue is a deep look at how, across three tightly-coupled layers — the **physical** handling of material, the **design** of its flow, and the **mathematical control** of how much is held.

These three layers answer three different questions. *How is material physically moved and stored?* — the equipment: trucks, conveyors, cranes, racking, AGVs. *How is that movement organised?* — the flow: line-side delivery, milk runs, supermarkets, kitting. *How much should exist, and when do we order more?* — the control: the inventory mathematics of EOQ, reorder points, and safety stock. Master one and you have a piece; master all three and the river runs smooth.

EDITOR'S NOTE · ON SCOPE & HOW THIS DIFFERS FROM ISSUES 08 & 10

Issue 08 told the *story* of just-in-time and the kanban; Issue 10 covered the *conveyors* that move the car body. This issue is the **comprehensive engineering reference** beneath both: the full material-handling equipment taxonomy, the principles of flow design, and the quantitative inventory-control toolkit (with formulas). It is deliberately deeper and more technical. As always, Tesla's internal logistics are proprietary; the methods, equipment classes, and figures here are the **universal standards of production logistics**, representative of a plant of this class, not confirmed internal Tesla configuration. Diagrams are conceptual.

The one idea: material handling adds cost, never value

The thesis that unifies all three layers is uncomfortable but clarifying: *moving and storing material adds cost, not value*. A customer pays for a finished car, not for the forklift trips, the warehouse rent, or the weeks a part spent waiting in a bin. Every metre travelled, every day stored, every time a part is picked up and put down is pure cost. So the entire discipline is, at heart, a war on unnecessary handling and unnecessary stock — move less, store less, but never starve the line. Everything that follows is a tactic in that war.

Moving and storing material adds **cost, not **value**. The whole game is to do less of both — without ever starving the line.**

Layer 3 — CONTROL: how much, and when to order

EOQ · reorder point · safety stock · ABC · turns · the bullwhip

▼ governs

Layer 2 — FLOW: how movement is organised

line-side delivery · milk runs · supermarkets · kitting · FIFO

▼ directs

Layer 1 — HANDLING: how material physically moves & is stored

trucks · conveyors · cranes · AS/RS · racking · AGVs

Figure 1. The three layers of production logistics. Information (the control layer) governs the design (flow), which directs the physical equipment (handling). (*Conceptual.*)

In this issue

1. **Material handling** — the ten principles
2. **The equipment** — the handling & storage taxonomy
3. **Material flow** — designing how it moves
4. **Inventory control I** — the two costs & EOQ
5. **Inventory control II** — reorder, safety stock, ABC
6. **The bullwhip** — and how the layers interlock
7. **Synthesis** & the field glossary

PART 1 · THE DISCIPLINE

Material Handling & Its Ten Principles

Material handling is the movement, protection, storage, and control of material throughout manufacturing — and it has a formal body of engineering behind it. The widely-taught **Ten Principles of Material Handling** codify decades of practice into a checklist every plant designer works from. They are not arbitrary; each is a lever for cutting the cost that handling adds.

Principle	What it means in practice
1. Planning	Define the material, moves, and method before building anything — handling is designed, not improvised
2. Standardization	Standardise equipment, containers, and methods to cut variety, cost, and training
3. Work	Minimise the <i>work</i> = flow × distance; the best move is the one you eliminate
4. Ergonomic	Fit the task to the human — reduce strain, reaching, and lifting (recall the skillet's lift, Issue 10)
5. Unit load	Handle material in unit loads (pallets, totes) — move many pieces at once, not one by one
6. Space utilization	Use the cube of the building — go vertical; floor space is the costliest space
7. System	Treat handling as one integrated system from receiving to shipping, not isolated moves
8. Automation	Automate to improve consistency, throughput, and safety where it pays
9. Environmental	Design for energy, reuse, and minimal waste across the handling system

10. Life-cycle cost

Decide on total cost of ownership over the equipment's life, not purchase price alone

THE WORK PRINCIPLE, MADE CONCRETE

Work = flow × distance

The third principle is the one a logistics engineer lives by, because it is almost a formula: the handling work in a plant is proportional to *how much* material moves multiplied by *how far* it travels. That gives two levers — move less, or move it shorter. It is why factory layout (Part 3) matters so much: shortening the distance between stations cuts handling work directly, before a single forklift is bought. The cheapest move is the one designed out of existence.

PART 2 · THE PHYSICAL LAYER

The Handling & Storage Taxonomy

The equipment of material handling sorts into a clear taxonomy by what it does: vehicles that *transport*, systems that *store*, and machines that *lift and position*. A plant blends many of them, choosing each for a given material, distance, and rate. Here are the major families.

Industrial Trucks & Lift Trucks

transport · the flexible workhorse

Forklifts, reach trucks, pallet jacks. The flexible, go-anywhere movers of unit loads: counterbalance forklifts for general duty, reach trucks for high narrow-aisle racking, and manual or powered **pallet jacks** for short moves. Maximum flexibility, but each trip carries one load and needs a driver — which is exactly why lean plants try to replace forklift "taxi" trips with milk runs (Part 3).

ROLE

Flexible transport

LOADS

Pallets, unit loads

LIMIT

One load per trip

Tuggers & Tugger Trains

transport · the milk-run engine

A tractor towing a train of carts. A tugger pulls several wheeled carts in a train along a fixed route, dropping and collecting material at many stops in one loop — the physical engine of the **milk run**. One driver delivers to a dozen stations per trip instead of one, slashing traffic and enabling small, frequent, just-in-time deliveries.

ROLE

Routed delivery

CARRIES

Many carts/stops

ENABLES

Milk runs

AGVs, AMRs & Conveyors

transport · automated movement

Movement without a driver. Automated guided vehicles and autonomous mobile robots carry loads on software-defined routes; **conveyors** (Issue 10) provide fixed high-volume flow. These automate the transport layer — reroutable in software (AMR) or relentless and fast (conveyor) — and report every move back to the control system. Covered in depth in Issues 08 and 10.

AGV/AMR

Driverless, routed

CONVEYOR

Fixed high-volume

DATA

Reports moves

AS/RS & Automated Storage

storage · dense, automatic

Automated Storage & Retrieval Systems. Robotic cranes or shuttles store and fetch totes and pallets in dense, tall racking with no human walking an aisle — unit-load AS/RS for pallets, mini-load and shuttle systems for totes, and **vertical lift modules** and carousels that bring goods to the picker ("goods-to-person"). Maximum density and accuracy; the storage counterpart to the conveyor.

ROLE

Dense auto storage

FORMS

Unit-load / shuttle / VLM

WINS

Density + accuracy

Cranes, Hoists & Monorails

lifting · the heavy & the overhead

Lifting the big and awkward. Overhead bridge cranes, jib cranes, gantry cranes, and hoists move heavy or bulky loads through the air, freeing floor space below — dies, motors, large sub-assemblies, coils. Where a load is too heavy or too large for a truck, it goes overhead. (Overhead *conveyors* like Power & Free, Issue 10, are the continuous cousin.)

ROLE

Heavy lifting

FORMS

Bridge / jib / gantry

FREES

Floor space

Racking, Storage Systems & Unit Loads

storage · the passive backbone

Where material waits. Selective pallet racking (flexible), drive-in (dense), **flow racks** (gravity-fed FIFO, ideal for line-side and supermarkets), push-back, and pallet-shuttle systems — each trading density against access. And the humble **unit-load devices** they hold: standardised pallets, totes, and containers that let material be handled in efficient, countable batches (the Unit Load principle, made physical).

RACKING

Selective / drive-in / flow

TRADE

Density vs access

UNITS

Pallets, totes

THE SELECTION LOGIC

Rate and distance choose the equipment

Picking handling equipment echoes the conveyor logic of Issue 10: low volume and high variety favour **flexible** manual trucks; high, steady volume justifies **fixed** conveyors or **automated** AS/RS and AGVs. Long distances favour tuggers and conveyors over one-load forklift trips; dense storage favours going vertical with racking and AS/RS. The constant question is the Work Principle: what moves the most material the shortest distance with the least handling?

PART 3 · THE FLOW LAYER

Designing How Material Flows

Equipment moves material; **flow design** decides where, when, and how much — and it is where the biggest gains hide, because it attacks handling work before any equipment runs. The goal is to present each part to each operator exactly when needed, in the smallest sensible quantity, with the least travel. A family of lean techniques makes that real.

Flow layout vs functional layout

The first decision is layout. A **functional (process) layout** groups like machines together (all presses here, all welders there) — flexible, but material zig-zags across the plant, piling up

handling work. A **flow (product) layout** arranges stations in the sequence the product needs, so material moves in a straight, short line. Lean plants push hard toward flow layout because it directly minimises the flow×distance of the Work Principle. A **spaghetti diagram** — literally tracing a part's path on the floor plan — exposes the tangled travel a functional layout creates.

The Milk Run (Tugger-Train Replenishment)

flow · routed, timed delivery

A bus route for parts. Instead of forklifts making individual "taxi" trips on demand, a **milk run** sends a tugger train (Part 2) on a *fixed route* at a *fixed timetable*, delivering small quantities to many line-side stops and collecting empties each loop. It cuts traffic and congestion, smooths labour, and — crucially — enables the small, frequent deliveries that just-in-time (Issue 08) depends on. The route and timing are engineered like a transit schedule.

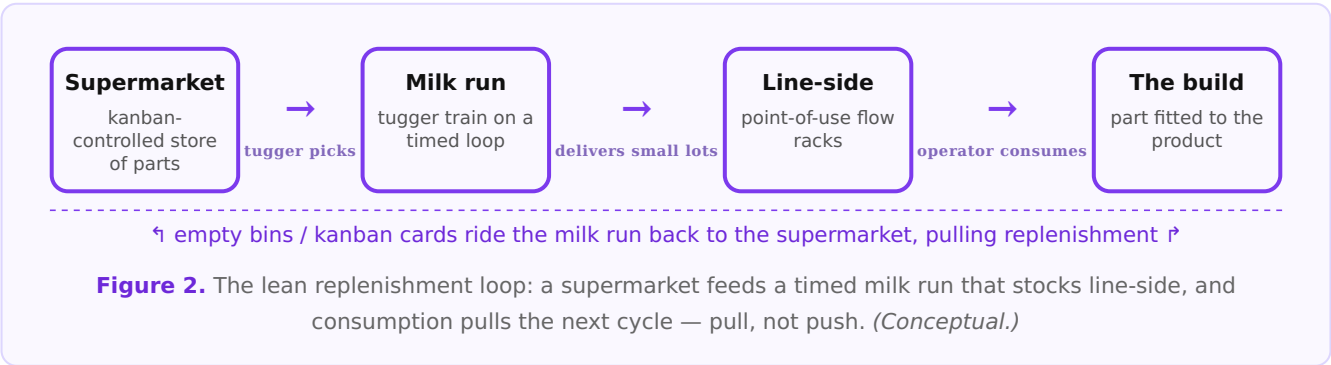
ROUTE Fixed loop	TIMING Fixed schedule	ENABLES Small-lot JIT
---------------------	--------------------------	--------------------------

Supermarkets & the Water Spider

flow · line-side replenishment

A controlled store, and the person who works it. A **supermarket** is a small, organised store of parts near the line, held in flow racks and replenished by kanban (Issue 08) — the line picks from it, and consumption pulls refills. The **water spider** (Japanese *mizusumashi*) is the dedicated, highly-trained material handler who follows a standardised route and timing to keep every line-side location stocked from the supermarket — so operators never leave their station to fetch parts.

SUPERMARKET Kanban-fed store	WATER SPIDER Routed replenisher	KEEPS Operators building
---------------------------------	------------------------------------	-----------------------------



Kitting, sequencing & FIFO

Two more flow tactics shape how parts arrive. **Kitting** gathers all the parts for one unit into a single tote or cart delivered together — the operator gets a complete "kit" rather than picking from many bins, cutting walking and errors. **Sequencing** (just-in-sequence, Issue 08) delivers parts in the exact order vehicles will consume them, so the right specific variant arrives as its car does. And a **FIFO lane** — a length-limited buffer between two processes — enforces first-in-first-out order and caps how much work-in-process can accumulate, linking two processes without fully coupling them.

WHY FLOW DESIGN BEATS EQUIPMENT

The cheapest handling is designed, not bought

A plant can spend millions automating a bad flow — or redesign the flow and need far less equipment. Shortening travel (flow layout), delivering in small timed loops (milk runs), presenting parts ready-to-use (kitting, supermarkets), and capping buffers (FIFO lanes) all cut handling work and inventory *before* any vehicle is purchased. This is why lean treats flow design as the highest-leverage layer: it changes the $\text{flow} \times \text{distance}$ equation itself.

PART 4 • THE CONTROL LAYER I

Inventory Control: The Two Costs & the EOQ

Now the mathematical layer that governs the whole river: how much material should exist, and when do we order more? It begins with a tension between two opposing costs. Order *too often* in small amounts and you pay a fortune in **ordering cost** (setups, paperwork, deliveries, receiving). Order *rarely* in huge amounts and you pay in **holding cost** (capital tied up, warehouse space, insurance, obsolescence). Somewhere between is the sweet spot — and a 110-year-old formula finds it.

THE COST OF HOLDING

Carrying inventory costs 20-30% of its value, every year

Holding stock is not free storage — its annual **carrying cost** typically runs **20-30% of the inventory's value**, once capital, space, insurance, shrinkage, and obsolescence are counted. A \$1M parts inventory can quietly cost \$200-300k a year simply to *hold*. That figure is why minimising inventory (the lean instinct of Issue 08) is worth so much, and why the math below matters.

The classic answer is the **Economic Order Quantity (EOQ)**, also called the Wilson formula — the order size that minimises the *total* of ordering plus holding cost over a year. Order in batches of the EOQ and the two costs balance at their combined minimum.

THE EOQ (WILSON) FORMULA

$$EOQ = \sqrt{(2 \cdot D \cdot S / H)}$$

D = annual demand (units/year) · **S** = ordering (or setup) cost per order · **H** = holding cost per unit per year.

Total cost **TC** = **PC** + **OC** + **HC** (purchase + ordering + holding). EOQ is the quantity where ordering cost and holding cost are equal — the minimum of the total-cost curve.

Figure 3. The total-cost curve. Ordering cost falls as order size grows; holding cost rises. Their sum is a U — and the EOQ sits at its bottom.

Order size →	Ordering cost / yr	Holding cost / yr	Total cost
Small (order often)	High (many orders)	Low (little on hand)	High
EOQ (the balance)	Moderate	Moderate (= ordering)	Minimum
Large (order rarely)	Low (few orders)	High (much on hand)	High

THE LEAN CRITIQUE OF EOQ

Don't accept the order cost — attack it

EOQ has a hidden assumption: that the ordering/setup cost **S** is fixed. Lean manufacturing refuses that. If a large setup cost forces a large EOQ (and thus large inventory), the lean answer is to *shrink S itself* — through quick changeovers (**SMED**) and streamlined ordering — which drives the EOQ down toward **one**. As $S \rightarrow 0$, the economic batch approaches single-piece flow, and the kanban pull system (Issue 08) becomes viable. So EOQ and kanban are not rivals: EOQ describes the optimum *given* your costs; lean changes the costs so the optimum becomes "small."

PART 5 • THE CONTROL LAYER II

Reorder Points, Safety Stock & ABC

EOQ answers *how much* to order; a second set of tools answers *when*, and *how much cushion* to keep against the unexpected. Together they turn inventory control from guesswork into a calculated system.

REORDER POINT & SAFETY STOCK

$$ROP = (d \cdot L) + SS$$

$$SS = z \cdot \sigma_d \cdot \sqrt{L}$$

d = average demand per day · **L** = lead time (days) · **SS** = safety stock

z = service-level factor (95% → $z \approx 1.65$; 99% → $z \approx 2.33$) · **σ_d** = std. deviation of daily demand.

Reorder when stock falls to the ROP — enough to cover demand during the lead time, plus a buffer for variability.

The logic is intuitive. The **reorder point** is the stock level that triggers a new order, set so the remaining stock lasts through the supplier's **lead time**. Because both demand and lead time vary, you add **safety stock** — a deliberate buffer sized by how much variability (σ) you face and how high a **service level** you want (the z-factor). Chasing a higher service level costs disproportionately more safety stock, which is why 100% is rarely economic.

$$\sqrt{(2DS/H)}$$

EOQ — the optimal order size

$$20-30\%$$

annual carrying cost, % of value

$$z \cdot \sigma \cdot \sqrt{L}$$

safety stock for a service level

$$\text{turns}$$

COGS / avg inventory — higher = leaner

Measuring health: turns & days of supply

Two ratios score how lean the inventory is. **Inventory turns** = annual cost of goods sold ÷ average inventory value — how many times a year the stock fully cycles; higher means leaner, faster-flowing inventory. **Days of supply** = 365 ÷ turns — how many days the current stock would last. A plant pushing just-in-time drives turns up and days-of-supply down, holding only what the next short interval needs.

Focusing effort: ABC and XYZ

Not every part deserves equal attention, so inventory is classified before it is controlled. **ABC analysis** (a Pareto split) ranks items by annual value: a vital few **A** items (~20% of items, ~80% of value) get tight control and frequent counts; the trivial many **C** items get simple two-bin kanban. A second axis, **XYZ**, classifies by *demand variability* — X steady and predictable, Z erratic and hard to forecast. Crossed together (AX, AZ, CZ...), they tell you exactly which parts need careful math and which can run on autopilot.

Class	By value (ABC)	Control approach
A	~20% of items, ~80% of value	Tight control, frequent cycle counts, low buffers, close supplier links
B	moderate value	Moderate control, periodic review
C	~50% of items, ~5% of value	Simple two-bin kanban; minimal effort

PART 6 · THE SYSTEM EFFECT

The Bullwhip Effect & How the Layers Interlock

One more phenomenon shapes inventory across the whole supply chain, and it explains why control discipline matters beyond a single plant: the **bullwhip effect**. A small, gentle change in customer demand at the end of a chain amplifies into ever-larger swings of orders and inventory as it travels upstream — like the tip of a whip turning a small wrist motion into a violent crack at the far end.

THE BULLWHIP EFFECT

A small demand wobble becomes an upstream flood

Each link in a supply chain, reacting to the orders of the link below it — batching orders, padding safety stock, reacting to delays — amplifies the variation it sees and passes a bigger swing upstream. The result: a modest bump in consumer demand can produce wild over- and under-ordering at component suppliers, with alternating gluts and shortages. The cures are exactly this issue's themes: **share real demand data** across the chain (information!), **order in smaller, more frequent lots** (milk runs, kanban), and **shorten lead times** — all of which damp the whip.

Step back and the three layers reveal themselves as one system, governed top-down by information. The **control** layer — EOQ, reorder points, ABC, the fight against the bullwhip — decides how much material should exist and when to replenish. That decision shapes the **flow** layer — whether you run big forklift deliveries or small timed milk runs from a kanban supermarket. And the flow design dictates the **handling** layer — which trucks, conveyors,

racking, and AS/RS you actually need. Change the inventory policy and the equipment requirement changes with it.

Information decides how much to hold; that shapes the flow; the flow chooses the equipment. The math governs the steel.

Layer	Question	Tools	Governed by
Control	How much & when?	EOQ, ROP, safety stock, ABC/XYZ, turns	Demand data & cost
Flow	How organised?	Layout, milk runs, supermarkets, kitting, FIFO	The control policy
Handling	How physically moved?	Trucks, tuggers, AGVs, conveyors, AS/RS, cranes	The flow design

PART 7 · SYNTHESIS

The River, Running Smooth

Return to the river. A factory that has mastered production logistics is one where material enters, flows to each point of use in small timed pulses, gets fitted, and leaves — with as little pooling, travel, and waiting as physics allows. No station starves; no warehouse drowns. Achieving that is not one trick but the patient integration of all three layers: equipment chosen for the Work Principle, flow designed to shorten and smooth the path, and inventory mathematics tuned so the right amount is always arriving and almost none is ever waiting.

And the through-line is the same one that runs across this entire magazine: *information governs the physical*. A forklift, a tugger train, a rack of totes — these are just steel until a decision about how much to hold and when to order gives them a job. The EOQ that sizes an order, the kanban that signals a refill, the demand data that tames the bullwhip: these are what turn a pile of equipment into a flowing river of parts. Get the mathematics right and the steel knows what to do. Get it wrong, and the finest fleet of AGVs in the world will faithfully deliver the wrong thing, in the wrong quantity, at the wrong time.

That is the quiet discipline beneath every car built: not the welding or the wiring that customers imagine, but the unglamorous, rigorous logistics of moving and storing material — the art of keeping the river flowing at exactly the right rate, neither dry nor flooded, from raw stock to finished vehicle. ■

FIELD GLOSSARY

The production-logistics vocabulary in this issue

- **Material handling** — the movement, protection, storage, and control of material through manufacturing.
- **Ten Principles of Material Handling** — the standard design checklist (planning, standardization, work, ergonomic, unit load, space, system, automation, environmental, life-cycle cost).
- **Work Principle** — handling work = flow × distance; minimise both.
- **Unit load** — material handled in standardised batches (pallets, totes) rather than piece by piece.
- **AS/RS** — Automated Storage & Retrieval System: robotic dense storage and goods-to-person picking.
- **VLM / carousel** — vertical lift module and carousel "goods-to-person" storage units.
- **Tugger / milk run** — a tractor towing carts on a fixed route and timetable, delivering small lots to many stops.
- **Supermarket** — a kanban-controlled line-side store of parts the line pulls from.
- **Water spider (mizusumashi)** — a dedicated material handler who replenishes line-side on a standard route.
- **Kitting** — gathering all parts for one unit into a single delivered kit.
- **Sequencing (JIS)** — delivering parts in the exact order of consumption.
- **FIFO lane** — a length-capped buffer enforcing first-in-first-out between processes.
- **Flow vs functional layout** — arranging stations by product sequence vs grouping like machines.
- **EOQ (Wilson formula)** — $\sqrt{(2DS/H)}$; the order size minimising ordering + holding cost.
- **Carrying / holding cost** — the annual cost of holding inventory, typically 20–30% of its value.
- **Reorder point (ROP)** — the stock level that triggers an order: lead-time demand + safety stock.
- **Safety stock** — a buffer against variability, sized by service level (z) and demand deviation.
- **Service level** — the probability of not stocking out; higher levels cost disproportionate safety stock.
- **Inventory turns / days of supply** — $\text{COGS} \div \text{average inventory}$; and $365 \div \text{turns}$.
- **ABC / XYZ analysis** — classifying items by value (ABC) and demand variability (XYZ) to focus control.
- **Bullwhip effect** — demand variability amplifying upstream through a supply chain.
- **EPQ** — Economic Production Quantity: the EOQ variant for produced (not purchased) parts.

FURTHER READING & SOURCING

This issue draws on standard production-logistics and material-handling engineering — the Ten Principles of Material Handling, lean material-flow practice (milk runs, supermarkets, water spider, kitting, FIFO), and quantitative inventory control (EOQ/Wilson, reorder point, safety stock, ABC/XYZ, inventory turns, the bullwhip effect) — from operations-management and material-handling literature. It is the in-depth companion to Issue 08 (just-in-time, kanban, WMS, vertical integration) and Issue 10 (conveyor systems & AGVs). Tesla's specific logistics systems are proprietary; methods, equipment classes, and figures here are representative of automotive-class plants, and diagrams are conceptual.

THE INTEGRATION LAYER · Issue 14 · A technical magazine · Sources: production-logistics & inventory-control fundamentals

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 15

COVER STORY · DECIDING WHAT TO BUILD, AND WHEN

Production Planning: The Factory's Mind

From a yearly forecast down to the exact car built in the next sixty seconds — the cascade of plans that decides what to make, how much, and when, and the leveling that keeps the whole river of work flowing smoothly.

THE PLANNING CASCADE

How a yearly forecast becomes a minute-by-minute build sequence.

MRP

The engine that explodes a schedule into every part needed.

TAKT TIME

The customer's drumbeat that sets the factory's pace.

HEIJUNKA

Why a level plan beats a fast one — leveling the mix.

The Factory's Mind

THE INTEGRATION LAYER · Feature · An explainer grounded in production-planning fundamentals

Everything else in this magazine answers the question "how does the factory *do* a thing?" — how a controller reasons, a drive moves, a sensor sees, a conveyor carries, a part is ordered. This issue answers a question that sits above all of them: *what should the factory do at all?* Before a single robot moves, someone — or some system — must decide which products to build, in what quantity, in what order, and at what moment, balancing customer demand against the plant's finite capacity. That decision-making is production planning, and it is the mind that gives every other layer its marching orders.

The defining feature of production planning is that it works at many time horizons at once, each feeding the next. A plan made a year out is coarse and strategic; a plan made for the next hour is exact and binding. The art is the cascade that turns the first into the second — a forecast for "about half a million cars next year" becoming "build *this* red dual-motor car with a tow hitch, now." This issue follows that cascade down, layer by layer, then examines the leveling discipline that keeps it from tearing the factory apart.

EDITOR'S NOTE · ON SCOPE & SOURCING

Tesla does not publish its internal production-planning systems or numbers. This is an explainer of the **standard production-planning framework universal to high-volume manufacturing** — the aggregate-plan / master-schedule / MRP / shop-floor hierarchy, takt time, level-vs-chase strategy, and Toyota-style heijunka leveling — from operations-management and lean literature. It builds directly on Issue 01 (MOS executing the plan), Issue 08 (JIT & kanban pull), and Issue 14 (inventory math & the bullwhip). Methods and figures are representative of an automotive-class plant, not confirmed Tesla configuration; diagrams are conceptual.

The one idea: match demand to capacity, across time

Strip planning to its essence and it is a single balancing act repeated at every horizon: reconcile what customers *want* with what the plant *can make*. Demand is uneven, capacity is finite and costly to change, and the two rarely line up on their own. Every planning tool in this issue — the aggregate plan, the master schedule, MRP, heijunka — is a different mechanism for making demand and capacity meet, at a different distance into the future. Plan well and the factory is calm and fully used; plan badly and it lurches between idle and overwhelmed.

Planning is one balancing act at every horizon: what customers *want* against what the plant *can make*.

LONG range — Business / Sales & Operations Plan

1-3 years · capacity, facilities, model lineup · "about how many, of what families"

▼ **constrains**

MEDIUM range — Aggregate Production Plan

3-18 months · workforce, inventory, output rate by product family

▼ **constrains**

SHORT range — Master Production Schedule (MPS)

weeks · how many of each specific end-product, per week

▼ **drives**

EXECUTION — MRP, sequencing & shop-floor schedule

days→minutes · every part, the build order, the next car (→ MOS, Issue 01)

Figure 1. The production-planning hierarchy: each horizon is more detailed and more binding than the one above, and constrains the one below. (*Conceptual.*)

In this issue

1. **The planning hierarchy** — horizons of decision
2. **Aggregate planning** — level vs chase
3. **The master schedule** — MPS & the time fence
4. **MRP** — exploding the bill of materials
5. **Takt time** — the customer's drumbeat
6. **Heijunka** — leveling volume & mix
7. **Push, pull & the digital plan** & glossary

PART 1 · HORIZONS OF DECISION

The Planning Hierarchy

Production planning is not one activity but a stack of them, distinguished by how far into the future they reach and how much detail they carry. The principle is simple: the further out, the coarser and more changeable; the nearer, the finer and more fixed. Each level produces constraints that the level below must respect — you cannot schedule next week's cars beyond the capacity that this year's business plan put in place.

At the top, the **business plan** and sales-and-operations planning (S&OP) set the multi-year strategy: which models, how much total capacity, what facilities and major suppliers. Beneath it, the **aggregate production plan** works in the medium term — months to a year and a half — deciding output rates, workforce size, and inventory levels for whole *product families*, not individual variants. Below that, the **master production schedule (MPS)** commits to how many of each *specific end product* to build, week by week. And at the bottom, execution systems — MRP, sequencing, and the shop-floor schedule — turn that into every part required and the exact order of build, handing off to the MOS (Issue 01) that runs the line.

WHY A HIERARCHY AT ALL?

Different decisions need different lead times

You cannot hire a thousand workers or pour a new building this afternoon, but you *can* resequence today's cars. Decisions have different lead times, so they must be made at different distances out. The hierarchy matches each decision to the horizon where it can actually be acted on — strategic capacity far ahead, build sequence at the last moment — and passes each level's commitments down as boundaries for the next. It is how a plant plans the un-plannable: by deciding the slow things early and the fast things late.

PART 2 · THE MEDIUM TERM

Aggregate Planning: Level vs Chase

The aggregate plan faces the central dilemma of planning head-on: demand rises and falls over the year, but changing output is expensive and disruptive. How should a plant respond? Two pure strategies bound the choice, and every real plan is a blend of them.

Level strategy

Hold a constant output rate and workforce; absorb demand swings with **inventory** (build ahead in slow periods, draw down in busy ones). Calm, stable, efficient — but it builds stock (and stock costs, Issue 14).

Chase strategy

Vary output and workforce to **match demand** period by period. Minimal inventory — but costly and disruptive: hiring, firing, overtime, and idle capacity as the rate chases the curve.

The trade-off is starkly clean: **level** trades inventory cost for stability; **chase** trades stability for low inventory. A pure level plan over-produces in quiet months; a pure chase plan whipsaws the workforce. Real aggregate plans sit between — a mostly-level base rate with bounded flexibility (overtime, a flex workforce, modest inventory swings) to handle the rest. This single choice ripples through everything below: a level plan makes the leveling of Part 6 natural, while a chase plan fights it.

Dimension	Level strategy	Chase strategy
Output rate	Constant	Varies with demand
Workforce	Stable	Hired/released to match
Inventory	Builds & draws down (buffer)	Minimal
Best when	Demand fairly steady; stock is cheap to hold	Demand volatile; stock is costly/perishable

PART 3 · THE COMMITMENT

The Master Production Schedule

If the aggregate plan decides "how much, of what families," the **master production schedule (MPS)** sharpens that into a real commitment: exactly how many of each specific end product to build in each near-term period — this many of *this* configuration next week. It

is the pivot of the whole hierarchy, the line where a strategy becomes a buildable promise that drives both customer delivery dates and the part-ordering engine below.

Master Production Schedule (MPS)

the buildable commitment

From plan to promise. The MPS states the quantity of each end item to produce per period over a near-term horizon. It must be **realistic** — checked against actual capacity via rough-cut capacity planning — because everything downstream trusts it. It balances demand (orders + forecast) against capacity, and it is the direct input to MRP (Part 4). Too optimistic an MPS and the plant promises what it cannot build; too conservative and it leaves capacity idle.

STATES

End items per period

CHECKED BY

Rough-cut capacity

FEEDS

MRP

THE TIME FENCE

The near future is frozen for a reason

An MPS is not equally changeable across its horizon. Inside the **time fence** — the immediate near term — the schedule is "frozen": parts are already ordered, sequences set, and changing it would cause chaos and shortages. Further out, the schedule is "slushy" or "liquid," open to adjustment as orders firm up. The time fence is how a plant stays responsive to new demand *without* destabilising the work already in motion — freedom far out, commitment near in. It is the scheduling cousin of the inventory time-buffers in Issue 14.

PART 4 • THE ENGINE

MRP: Exploding the Bill of Materials

The master schedule says "build 800 of this car next week." But a car is tens of thousands of parts, each with its own supplier and lead time. Turning the schedule into a precise list of *what to order and make, and when*, is the job of **Material Requirements Planning (MRP)** — the calculation engine at the heart of production planning, and the original "killer app" of the factory computer.

Material Requirements Planning (MRP)

the part-ordering calculator

Three inputs, one explosion. MRP takes the **MPS** (what to build, when), the **bill of materials** or BOM (what each product is made of), and **inventory records** (what's already on hand and on order). It "explodes" the BOM — multiplying out every sub-assembly and component — then nets against stock and offsets by each part's lead time to produce timed **planned orders**: make/buy this quantity, by this date. It answers what, how much, and when, for every part at once.

INPUTS

MPS + BOM + inventory

DOES

Explode & time-phase

OUTPUTS

Planned orders

INPUTS: Master Schedule + Bill of Materials + Inventory on hand

what to build · what it's made of · what we already have

▼ MRP explosion & netting

OUTPUT: time-phased planned orders for every part

"make 1,200 brackets by Tuesday" · "order 5,000 bolts for Friday"

Figure 2. MRP turns one master-schedule line into thousands of timed part orders, by exploding the BOM and netting against inventory. (*Conceptual.*)

MRP's reach grew over time. Extended with capacity, finance, and feedback it became **MRP II** (Manufacturing Resource Planning); folded into a company-wide system it became part of **ERP** (Issue 08). And its modern successor, **Advanced Planning & Scheduling (APS)**, adds finite-capacity awareness — planning against real constraints and optimising the sequence, rather than assuming infinite capacity as classic MRP does.

THE CATCH WITH CLASSIC MRP

It assumes infinite capacity — and fixed lead times

Classic MRP has two famous blind spots. It plans as if capacity were **unlimited** — happily scheduling more than the plant can build in a day — which is why rough-cut capacity checks and APS exist. And it treats lead times as **fixed constants**, when in reality they stretch under load (recall the EOQ assumption critique, Issue 14). This is precisely why lean plants lean on pull and leveling (Parts 5–6) to *complement* MRP: MRP plans the materials, while kanban and heijunka govern the actual flow on the floor.

PART 5 · THE DRUMBEAT

Takt Time: The Customer's Pace

Planning quantities is half the picture; planning *pace* is the other. The single most important number in a flow line comes not from the factory but from the customer: **takt time**, the rhythm at which products must be completed to exactly meet demand. From German *Takt* (a beat), it is the heartbeat the whole line is tuned to.

THE DEFINITION

Takt = available time ÷ customer demand

If a plant has **27,000 working seconds** in a shift and the customer wants **450 cars**, the takt time is $27,000 \div 450 = \mathbf{60 \text{ seconds}}$: a finished car must roll off every minute to meet demand, no faster and no slower. Takt is set by demand, *not* by how fast the plant could go. It converts an abstract quantity ("450 a day") into a concrete pace the line can be designed and balanced around.

Takt time must be distinguished from its cousin, **cycle time** — the actual time a process takes to complete one unit. The cardinal rule of flow design is that *cycle time must be at or below takt time* at every station: if any step takes longer than takt, the line cannot keep pace and demand is missed. Planning thus includes **line balancing** — distributing work across stations

so each one's cycle time fits just under takt, and no station is a bottleneck. Takt is the target; balancing is how the line is built to hit it.

Takt available time ÷ demand = required pace	$\leq$ Takt cycle time must fit under takt	Balance spread work so no station exceeds takt	60 s a car a minute, in the worked example
---	---	---	---

PART 6 · LEVELING

Heijunka: Leveling Volume & Mix

Here is the deepest idea in production planning, and Toyota's great contribution to it. Even with a sound schedule and the right takt, a plant can still tear itself apart by building in lumps — all the dual-motor cars Monday, all the base cars Tuesday — sending violent swings down to every supplier and station. **Heijunka** (Japanese for "leveling") is the discipline of smoothing production, and it works in two dimensions.

Heijunka has two parts	What it smooths	Why it matters
Leveling volume	The total quantity built per period — a steady rate, not bursts	Stabilises takt, workforce, and supplier draw
Leveling mix	The sequence of <i>variants</i> — interleave types rather than batch them	Smooths the demand for every distinct part across the day

Leveling the **mix** is the subtle, powerful half. Instead of building AAAABBBBCCCC (big batches of each variant), a leveled line builds **ABCABCABC** — interleaving variants in a repeating pattern. The effect is profound: every component supplier now sees a *steady, predictable* trickle of demand instead of a feast-or-famine batch, which is the single most effective cure for the bullwhip effect (Issue 14). A leveled schedule is the stable foundation that makes just-in-time and kanban (Issue 08) actually work.

The heijunka box

The classic tool that makes leveling visible is the **heijunka box** — a physical grid of pigeonholes, columns for time intervals and rows for product types, loaded with kanban cards in the leveled sequence. A material handler (the water spider, Issue 14) withdraws cards in order, releasing a steady, mixed stream of work to the line. It is leveling you can see and touch — the schedule made physical.

Variant ↓ / Time →	8:00	8:10	8:20	8:30	8:40	8:50
Model A	■		■		■	
Model B		■				■
Model C				■		

Figure 3. A heijunka box. Kanban cards are placed in a leveled, repeating sequence (A·B·A·C·A·B...) so variants are interleaved, not batched — steady demand for every part. (*Conceptual.*)

Build **ABCABC**, not **AAABBBCCC**. A level mix gives every supplier a steady drumbeat — and kills the bullwhip.

PART 7 · SYNTHESIS

Push, Pull & the Digital Plan

Two philosophies run through everything above, and a mature plant uses both. **Push** planning — MRP and the schedule — computes in advance what to build and pushes it onto the floor. **Pull** execution — kanban and JIT (Issue 08) — lets actual consumption signal replenishment. They are not enemies: the plan (push) sets the level rate and orders the long-lead materials, while pull (kanban) governs the fine-grained flow within that frame. Heijunka is the bridge — a leveled plan is exactly what makes pull stable.

Aspect	Push (plan-driven)	Pull (consumption-driven)
Trigger	Forecast & schedule, computed ahead	Actual downstream consumption
Tool	MRP / MPS / APS	Kanban / JIT
Strength	Plans long-lead items & capacity	Low inventory, self-regulating flow
Best for	Procurement, capacity, the level rate	Short-lead, repetitive shop-floor flow

Modern planning runs this whole cascade in software. **ERP** systems hold the business and material plan; **APS** optimises finite-capacity schedules; **MES/MOS** (Issue 01) executes and reports actuals back up; and increasingly, real production data flows back to continuously re-plan — a closed loop where the plan learns from what the floor actually did. The dashboards of Issue 01 are the visible tip of this: a plan, an execution, and a comparison, cycling continuously.

Step back and production planning reveals itself as the brain atop the body this magazine has been describing. The controllers, drives, sensors, conveyors, and logistics are the muscles and nerves; planning is the intention that tells them what to do. It reaches from a yearly forecast down to the car built in the next sixty seconds, reconciling demand with capacity at every horizon, and — through the quiet genius of leveling — keeping the river of work flowing smooth and steady rather than crashing in waves. The factory can only build well what it has

first planned well. Everything downstream is the faithful execution of that one upstream act of foresight. ■

FIELD GLOSSARY

The production-planning vocabulary in this issue

- **Production planning** — deciding what to make, how much, and when, by matching demand to capacity across time horizons.
- **Planning hierarchy** — the cascade from business plan → aggregate plan → master schedule → execution, each more detailed and binding.
- **Sales & Operations Planning (S&OP)** — the long-range business-level reconciliation of demand and supply.
- **Aggregate production plan** — a medium-term plan of output rate, workforce, and inventory by product family.
- **Level strategy** — constant output, absorbing demand swings with inventory.
- **Chase strategy** — varying output and workforce to match demand, minimising inventory.
- **Master Production Schedule (MPS)** — the committed quantity of each end product per near-term period.
- **Time fence** — the frozen near-term portion of the schedule, protected from change.
- **Rough-cut capacity planning** — checking the MPS against available capacity.
- **MRP (Material Requirements Planning)** — exploding the MPS through the BOM and netting inventory to produce timed part orders.
- **Bill of Materials (BOM)** — the structured list of everything a product is made of.
- **MRP II / ERP / APS** — MRP extended with resources; the company-wide system; and finite-capacity advanced scheduling.
- **Takt time** — available time ÷ demand; the required pace of production set by the customer.
- **Cycle time** — the actual time a process takes per unit; must be $\leq$ takt.
- **Line balancing** — distributing work across stations so each fits under takt.
- **Heijunka** — production leveling, smoothing both volume and the mix of variants.
- **Heijunka box** — a grid of kanban cards releasing a leveled, mixed sequence to the line.
- **Mixed-model production** — building several variants on one line in an interleaved sequence.
- **Push vs pull** — building to a computed plan (push) vs to actual consumption (pull); used together.
- **MTO vs MTS** — make-to-order (build on demand) vs make-to-stock (build to forecast).

FURTHER READING & SOURCING

This issue draws on standard production-planning and control theory — the aggregate / master-schedule / MRP / shop-floor hierarchy, level-vs-chase aggregate strategy, MPS and time fences, MRP / MRP II / ERP / APS, takt time and line balancing, and Toyota-style heijunka leveling and the heijunka box — from operations-management and lean-manufacturing literature. It is the planning-layer companion to Issue 01 (MOS executing the plan), Issue 08 (JIT, kanban & ERP), and Issue 14 (inventory mathematics & the bullwhip effect). Tesla's specific planning systems are proprietary; the framework and figures here are representative of automotive-class plants, and diagrams are conceptual.

THE INTEGRATION LAYER · Issue 15 · A technical magazine · Sources: production-planning & lean-manufacturing fundamentals

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 16

COVER STORY · WARP, MES & MOS — THE SYSTEMS THAT RUN THE PLANT

Code Over Steel: The Software Tesla Built

Most carmakers buy their factory software from SAP. Tesla wrote its own — a homemade stack of Warp, MES and MOS that plans, executes, and records every car. A look at the real systems, from public reporting.

BUILD, DON'T BUY

Why Musk refused SAP and wrote the software in-house.

MES & MOS

The air-traffic-control and floor-tracking of every car.

WARP

The planning & supply-chain brain, built in four months.

THE CLOSED LOOP

How the plan, the floor, and the data feed each other.

The Company That Writes Its Own Software

THE INTEGRATION LAYER · Feature · Assembled from public reporting on Tesla's software

Across fifteen issues this magazine has described the *kinds* of systems a factory runs — a manufacturing operating system, a planning hierarchy, an execution layer — using Tesla as the running example while noting, each time, that its actual internals are proprietary. This issue does something different: it gathers what is *publicly known* about the real, named software Tesla actually built and runs, and maps those concrete systems onto the conceptual layers the series has been describing. The names are real; the deep internals remain Tesla's secret.

The single fact that explains all of it: where almost every other automaker buys its factory and business software from vendors like SAP, Tesla decided to **write its own**. That choice — software built in-house and bolted tightly to the machines — is the root of Tesla's whole digital character, and the reason its stack looks nothing like a traditional plant's. This issue follows that homemade stack from the planning brain down to the floor, and back up through the data.

EDITOR'S NOTE · ON SOURCING & WHAT'S REALLY KNOWN

Unlike most issues, this one names real Tesla systems — so its standard of evidence matters. Everything here is drawn from **public reporting**: news coverage (including a 2018 lawsuit that revealed how MOS works), interviews, employee accounts, and analyst write-ups. That makes the broad architecture reliable, but the fine internals — exact algorithms, data models, how modules are built — are **not disclosed by Tesla** and are not claimed here. Where reports conflict or a detail is contested, this issue says so. Systems evolve; this reflects what has been publicly described up to the series' writing.

The one idea: vertically integrate the software, too

Tesla is famous for vertical integration in hardware — making its own cells, motors, and castings. The under-appreciated half is that it vertically integrated the *software* in exactly the same spirit. Reporting describes Tesla designing and coding its own production software, tightly integrated with the physical machinery, rather than relying on third-party ERP or MES tools. The payoff is the same as in hardware: no vendor handoffs, the ability to change a process on the fly, and a factory that behaves like one living software system rather than a static line wired to a box bought from someone else.

Most automakers **buy their factory software. Tesla decided to **write** it — and bolt it straight to the machines.**

WARP — planning, supply chain & business

product planning · inventory · sales orders · purchasing · finance · HR
(the ERP layer)

▼ **hands the build plan to**

MES — the production "air traffic controller"

routes each item · tracks production orders & quality · collects measurements

▼ **runs the floor through**

MOS — the Manufacturing Operating System

tracks each car through build/repair/test · logs parts, torque, quality · material tracking (barcode/RFID)

▼ **writes everything to**

DATA & DASHBOARDS

Oracle / SQL Server · in-house MySQL · QuickBase · BI dashboards — and back up to re-plan

Figure 1. Tesla's homemade stack as publicly described: Warp plans, MES orchestrates, MOS tracks on the floor, and data flows up to dashboards and back into planning. (*Conceptual, from public reporting.*)

In this issue

1. **Build vs buy** — the in-house philosophy
2. **Warp** — the planning & supply-chain brain
3. **MES** — the production air traffic controller
4. **MOS** — the floor & material tracking
5. **The data layer** — databases & dashboards
6. **The closed loop** & the bigger picture

PART 1 · THE PHILOSOPHY

Build, Don't Buy

The founding decision dates to 2012. As Tesla prepared to launch the Model S, the conventional move was to implement a trusted enterprise system like SAP — the default for any manufacturer. Instead, reporting recounts, Elon Musk tasked his chief information officer to develop a custom, end-to-end system fit to Tesla's particular needs as an automaker that also runs its own stores and service centers. The reasoning was that an off-the-shelf ERP could not move at Tesla's speed or match its unusual, vertically-integrated business model.

That became a standing mandate. Reports describe Musk directing that Tesla build its own software wherever possible rather than rely on packaged business and factory programs — not just SAP, but the whole category of dealer, service, and plant-management software other carmakers stitch together. Tesla still uses *some* outside tools where they clearly fit — the developer tracker Atlassian JIRA and the procurement platform Coupa are reported examples — but the stated aim is to be as self-reliant in software as it is in hardware.

WHY "BUILD" CAN WIN

Tesla as a "chain of startups"

Musk has described Tesla as a chain of startups — treating each major product line and plant like its own venture. In-house software fits that culture: a small team can ship and iterate a tool in weeks, change it the moment a process changes, and wire it directly to the machines without waiting on a vendor's roadmap. The result, reporting suggests, is the ability to eliminate handoffs between design, production, and the systems that run them — the software equivalent of Tesla's hardware vertical integration.

THE HONEST TRADE-OFF

Owning the software means owning every bug

Building your own stack is not free. You take on the entire burden a vendor would otherwise carry: maintenance, security, scaling, and the risk that a hastily-built internal tool becomes fragile as the company explodes in size. The same 2018 reporting that revealed these systems also documented friction and workarounds inside them. Build-vs-buy is a genuine trade — speed and control in exchange for the responsibility of being your own software vendor, forever.

PART 2 · THE PLANNING BRAIN

Warp: The Supply-Chain & Planning System

If the planning hierarchy of Issue 15 has a real name at Tesla, it is **Warp** — the homegrown system that plays the role SAP would in another company. Originally called Warpdrive (after warp drive), it was, by reporting, built and implemented in about four months by a small team of roughly 25 engineers, and it became central to Tesla's operations and unusual business model.

Warp (formerly Warpdrive)

Tesla · the in-house ERP / planning system

The business and planning backbone. Reporting describes Warp as centralising and integrating Tesla's business processes across **supply chain, product planning, inventory, sales-order management, assets, finance, HR, and admin** — the functions a traditional ERP covers. It is also described as the company's **purchasing-order** system, and is even used to manage work orders in Tesla's service centers. In the language of this series, Warp is the planning & supply-chain layer (Issues 14–15) made into one Tesla-specific platform.

ROLE

ERP / planning

COVERS

Supply chain → finance

ORIGIN

~25 engineers, ~4 months

FOUR MONTHS INSTEAD OF YEARS

The speed argument, made concrete

The detail that captures the whole philosophy: a major ERP rollout at a manufacturer is typically a multi-year, eight-figure undertaking. Tesla's account has a small team standing up Warp's first version in roughly four months. Whether or not every later extension was as fast, the origin story is the case *for* building in-house — a tool shaped exactly to the business, shipped at startup speed, owned outright.

Warp sits at the top of the material story this series has told. It is where the demand-and-capacity planning of Issue 15 would live, where the inventory and purchasing decisions of Issue 14 are made, and the place that tells the factory, in effect, *what* to build and *what* to order. From there, the job passes to the systems that actually run the floor.

PART 3 · THE ORCHESTRATOR

MES: The Production Air Traffic Controller

Between the plan and the physical line sits Tesla's **Manufacturing Execution System (MES)**. One widely-cited description calls it the *air traffic controller* of the entire production process — the system that directs what goes where, and watches it happen in real time.

Tesla MES (Manufacturing Execution System)

Tesla · real-time production control

Directing and watching the line. By reporting, Tesla's MES can tell an item on the assembly line where it should head next, keep track of production orders and quality issues, and collect basic measurements as production runs. It is the real-time layer between Warp's plan and the physical equipment — the concrete counterpart to the execution layer described in Issue 01. It does the orchestration; MOS (Part 4) does the detailed per-car record.

ROLE	TRACKS	COLLECTS
Real-time orchestration	Orders, quality, routing	Measurements

Underneath MES is a real database stack, and here the public record is unusually specific. Reporting states the MES uses **Oracle / SQL Server** databases; that more in-depth, customised **test data** lives in a separate **MySQL** database built by Tesla's in-house developers; that the web-based **QuickBase** has been used to track process changes and ongoing quality issues for its flexibility; and that plain **Excel** spreadsheets still handle one-off manual reports. It is a candid reminder that even the most software-forward factory runs on a pragmatic mix of the custom and the commodore.

MES VS MOS — TWO SYSTEMS, TWO JOBS

Orchestration vs the per-car record

People often blur Tesla's MES and MOS, but reporting treats them as distinct, complementary systems. **MES** is the real-time traffic controller — routing items, holding production orders, watching quality across the line. **MOS** (next) is the detailed operating system workers touch to record what happened to *each individual car* — this part installed, this torque achieved, this test passed. One conducts the orchestra; the other writes down every note each instrument played.

PART 4 · THE FLOOR

MOS: The Manufacturing Operating System

The system this series opened with, in Issue 01, is real and has a name workers use every shift: **MOS**, the Manufacturing Operating System. Much of what is publicly known about it

surfaced, unusually, through a 2018 lawsuit — which laid out in plain terms what the software does on the floor.

Tesla MOS (Manufacturing Operating System)

Tesla · the floor-level operating system

Tracking every car, step by step. By reporting, MOS tracks where a vehicle is in the process of manufacturing, repair, and testing. Workers log into it whether a body is fully assembled, whether specific items are installed — steering wheels, airbags, rocker mold — and the **torque levels** used by factory equipment to put the car together. It records whether each car has passed the necessary repairs, quality, and safety testing before it leaves. (This is the lineage of the 46.2 N·m data journey of Issue 01.)

TRACKS	LOGS	GATES
Build / repair / test state	Parts, torque, quality	Cars before they ship

Two further public details matter for this series. First, **material tracking**: reporting describes MOS following the flow of raw materials and finished products in real time using **barcode and RFID** — the concrete Tesla instance of the material-tracking ideas in Issues 08 and 14. Second, **reach**: MOS has been described as running across Tesla's Fremont plant, Gigafactory Nevada, the Buffalo factory, and a site in Ireland, and is used not only for cars but in the production of the Powerwall and battery packs. It is a company-wide operating system, not a single-line tool.

A CONTESTED DETAIL · "SCHEDULE-BASED REPLENISHMENT"

The kanban-by-another-name story

One widely-repeated report claims that within MOS, digital **kanban** modules were labeled "schedule-based replenishment" to avoid a term Musk reportedly disliked. It is a memorable anecdote — but a contested one: a Tesla spokesperson pushed back, saying kanban methods are in fact used widely in its factories. The honest version: MOS clearly contains pull-style replenishment logic (the digital cousin of the two-bin kanban of Issue 08); what it is called internally, and the story behind the name, is reported but disputed. A good reminder of how much "known" external detail is really second-hand.

PART 5 · THE DATA

The Data & Dashboard Layer

All three systems above are, in the end, machines for producing *data* — and a factory is only as good as what it can see. The bottom of Tesla's stack is the databases that store every logged result and the dashboards that turn them into decisions, the visible tip of which appeared in Issue 01.

Component (as reported)	Role
Oracle / SQL Server	The main MES production databases — orders, status, measurements
In-house MySQL (test DB)	Deeper, customised test data, on a database built by Tesla's own developers
QuickBase (web)	Tracking process changes and ongoing quality issues, valued for flexibility
BI dashboards (e.g. Tableau-class)	KPI and yield dashboards for leadership & supply-chain visibility

This layer is what makes the whole stack a *system* rather than a set of disconnected tools. Production data from MOS and MES flows into the databases; analytics and dashboards turn it into the real-time picture leadership acts on; and that picture feeds back into Warp's planning. It is the closed loop of plan → execute → measure → re-plan that this series has returned to again and again — here, running on Tesla's own homemade plumbing.

2012 Warp built — instead of SAP	~4 mo to stand up Warp's first version	3 core systems: Warp · MES · MOS	4+ plants reported running MOS
--	--	--	--

PART 6 · THE WHOLE PICTURE

The Closed Loop, and Why It Matters

Assemble the pieces and Tesla's software stack maps cleanly onto the conceptual layers this magazine has built across sixteen issues — only now with real names attached. **Warp** is the planning and supply-chain brain (Issues 14–15). **MES** is the real-time orchestrator and **MOS** the floor-level operating system and per-car record (Issue 01). The **data layer** closes the loop back to planning. The whole thing is bolted to the physical equipment, networks, and sensors the rest of the series described.

The traditional automaker

Buys SAP (or similar) for planning, a vendor MES for the floor, more vendors for dealers and service — then spends years integrating the seams between them.

Tesla, as reported

Writes Warp, MES, and MOS in-house, on its own databases, wired straight to the machines — one integrated stack it can change on the fly, and must maintain forever.

Figure 2. The structural difference: buy-and-integrate versus build-and-own. (*Conceptual.*)

Why does this matter beyond Tesla? Because the approach is now influential in its own right. Reporting notes that Palantir explicitly invoked Tesla and SpaceX as inspiration for its "Warp Speed" platform — pitched as a modern American manufacturing operating system — aimed at bringing this build-your-own, tightly-integrated software model to manufacturers that lack Tesla's in-house army of engineers. The idea that a factory's software should be a living, integrated system rather than a patchwork of purchased boxes is spreading outward from where this issue began.

Warp plans, MES orchestrates, MOS records, data closes the loop — one homemade stack, **wired straight to the steel.**

Step back to where the series started. Issue 01 followed a single torque reading — 46.2 N·m — from a sensor up to a manager's dashboard, and called the software that carried it the Manufacturing Operating System. Sixteen issues later, we can name the real systems that

reading would have crossed: logged in **MOS** on the floor, orchestrated by **MES**, against a build planned in **Warp**, stored in an **Oracle or MySQL** database, and surfaced on a **dashboard** — every layer of it written, as far as the public record shows, by Tesla itself. That is the quiet, unglamorous truth beneath the robots and the castings: a car company that decided it was also, fundamentally, a software company — and built the code that runs the steel. ■

FIELD GLOSSARY

The Tesla-software vocabulary in this issue

- **Warp (formerly Warpdrive)** — Tesla's in-house ERP/planning system: supply chain, product planning, inventory, sales orders, purchasing, finance, HR.
- **MES (Manufacturing Execution System)** — Tesla's real-time production "air traffic controller": routing, production orders, quality, measurements.
- **MOS (Manufacturing Operating System)** — the floor-level system workers use to record each car's build, parts, torque, quality, and test state.
- **Material tracking** — MOS following raw materials and finished goods in real time via barcode and RFID.
- **Schedule-based replenishment** — a reported (and contested) internal name for digital kanban / pull logic within MOS.
- **Build vs buy** — the choice to write software in-house rather than license a vendor package like SAP.
- **Vertical software integration** — building and owning the software stack, wired directly to the machines.
- **The data layer** — Oracle/SQL Server, an in-house MySQL test database, QuickBase, BI dashboards, and Excel.
- **Chain of startups** — Musk's framing of Tesla as many venture-like units, which the in-house software culture supports.
- **Closed loop** — plan (Warp) → execute (MES/MOS) → measure (data) → re-plan.

SOURCING & A NOTE ON CERTAINTY

This issue is assembled entirely from **public sources**: news reporting (notably 2016 coverage of Warp and 2018 coverage of MOS, including details that emerged via a lawsuit), analyst and industry write-ups, and employee accounts. The broad architecture — Warp for planning, MES + MOS for execution, a mixed database-and-dashboard layer, and an in-house build philosophy — is consistently reported and reliable. **Specific internals are not disclosed by Tesla**, some details (such as the "schedule-based replenishment" naming) are contested, and systems evolve over time. Nothing here should be read as confirmed current internal configuration. This is the real-systems companion to Issue 01 (MOS), Issue 08 (kanban/JIT), Issue 14 (inventory), and Issue 15 (planning).

THE INTEGRATION LAYER · Issue 16 · A technical magazine · Sources: public reporting on Tesla's software (Warp, MES, MOS)

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 17

COVER STORY · MEASURING THE FACTORY — OEE, KPIs & THE NUMBERS THAT MATTER

The Score: OEE, KPIs & Measuring a Plant

You can't improve what you don't measure. From the single most important number in manufacturing — OEE — to the handful of KPIs that actually run a factory, and how that score reaches a Tesla dashboard.

OEE

The one metric: Availability × Performance × Quality.

BEYOND OEE

FPY, throughput, MTBF, takt — the rest of the scorecard.

THE SIX BIG LOSSES

Where every minute of lost productivity hides.

A GOOD KPI

Why a number is useless unless it triggers an action.

You Can't Improve What You Don't Measure

THE INTEGRATION LAYER · Feature · An explainer grounded in manufacturing-metrics standards

Every issue of this magazine has, in the end, produced data — a torque reading, a part count, a quality result, a machine state. This issue is about what that data becomes: the **numbers** a factory uses to know whether it is winning. Manufacturing runs on Peter Drucker's old maxim — you cannot manage what you do not measure — and the discipline of choosing, calculating, and acting on the right performance indicators is what turns a flood of floor data into decisions. The dashboards of Issue 01 are the visible surface; this issue is the metrics beneath them.

At the centre sits one number widely called the single most important metric in manufacturing: **Overall Equipment Effectiveness (OEE)**. It compresses three different kinds of loss into one honest percentage, and the bulk of this issue unpacks it — what it is, how it's calculated, where it misleads, and the framework of "six big losses" it points to. Around it sits a small constellation of other KPIs — yield, throughput, reliability, schedule adherence — that together form a factory's scorecard.

EDITOR'S NOTE · ON SCOPE & SOURCING

This issue explains the **standard, universal manufacturing-metrics toolkit** — OEE and its three factors, the Six Big Losses, TEEP, first-pass yield, scrap, throughput, cycle time, MTBF/MTTR, and schedule adherence — from operations-management literature and the ISO 22400 KPI standard. It builds on Issue 01 (the dashboard), Issue 14–15 (flow & planning), and Issue 16 (the software that records the data). Tesla does not publish its internal KPI definitions or targets; what is said about Tesla here is limited to what is **publicly known** (its real-time data-driven dashboards and published production-rate targets), and is clearly flagged. Figures are illustrative; diagrams conceptual.

The one idea: a metric is only useful if it drives an action

Before any formula, the governing principle: a number is a **KPI** only if someone is accountable for it and it triggers a specific action when it moves — otherwise it is merely information. The best factories track *fewer* metrics, not more: eight to twelve well-defined, well-measured KPIs beat thirty half-measured ones. Every metric in this issue earns its place by pointing at a specific kind of loss and a specific lever to pull. Measurement is not the goal; *improvement* is.

A number is a KPI only if it has an owner and triggers an action. Otherwise it's just **information.**

PLANNED PRODUCTION TIME (the scheduled time to run)

all the time the asset was scheduled to produce

▼ – Availability loss (stops)

RUN TIME × ... → Availability

did it run when it was supposed to?

▼ – Performance loss (slow)

NET RUN TIME → Performance

did it run as fast as it should?

▼ – Quality loss (defects)

FULLY PRODUCTIVE TIME → Quality — this is OEE

only good parts, at full speed, with no stops

Figure 1. OEE as a funnel: planned time is eroded by three kinds of loss — stops, slowness, and defects — and what survives is fully productive time. (*Conceptual.*)

In this issue

1. **OEE** — the one metric, and its three factors
2. **The calculation** — a worked example
3. **The Six Big Losses** — and TEEP
4. **Quality metrics** — FPY, scrap, PPM
5. **Flow & reliability** — throughput, MTBF/MTTR
6. **Building a scorecard** & the Tesla picture

PART 1 • THE ONE METRIC

OEE: Availability × Performance × Quality

Overall Equipment Effectiveness is a composite metric that measures how productively equipment runs relative to its full potential. Its genius is compressing three completely different kinds of loss — stopping, running slow, and making junk — into one honest number. A score of 100% means a perfect state: only good parts, at full speed, with zero downtime. It is built from three factors, multiplied together.

THE OEE FORMULA

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

Availability = Run Time ÷ Planned Production Time (did it run when scheduled?)

Performance = (Ideal Cycle Time × Total Count) ÷ Run Time (did it run at full speed?)

Quality = Good Count ÷ Total Count (were the parts good first time?)

The crucial property is that the factors **multiply**. A line can look decent on each one alone yet score poorly overall, because the losses compound: 90% availability × 90% performance × 90% quality is only 73%, not 90%. One weak factor drags the whole score down — which is exactly the point. OEE refuses to let a plant hide a quality problem behind high uptime, or slow running behind good quality. All three must be high together.

What OEE rewards

Running when scheduled (no breakdowns or changeovers eating time), running at the ideal rate (no micro-stops or slow cycles), and making good parts the first time (no scrap or rework).

What OEE exposes

The gap between what a plant *could* produce and what it actually shipped — split into three diagnosable buckets so you know *which* kind of loss to attack first.

THE 85% BENCHMARK

What counts as a "good" OEE?

The widely-cited world-class benchmark for discrete manufacturing (cars, furniture, electronics) is **85%** — roughly 90% availability, 95% performance, 99% quality. Most plants start far lower: **60%** is typical, **40-65%** a common starting range, and below 60% signals real problems. An 85% score means converting about 85% of planned time into good output at speed — and very few achieve it consistently. The honest target, every source stresses, is *improvement over your own baseline*, not someone else's benchmark.

PART 2 · A WORKED EXAMPLE

Calculating OEE, Step by Step

The formula is clearest in a concrete shift. Take a line running one eight-hour shift with 420 minutes of planned production time (after scheduled breaks), an ideal cycle time of 1.0 second per unit, and these actual results — a standard worked example.

Step	Calculation	Result
Given	Planned time 420 min; 60 min unplanned downtime (2 breakdowns + 1 long changeover)	Run time = 360 min
Availability	$(420 - 60) \div 420 = 360 \div 420$	85.7%
Performance	$19,000 \text{ produced} \div 21,600 \text{ theoretical max (360 min at 1.0 s/unit)}$	88.0%
Quality	$18,050 \text{ good} \div 19,000 \text{ produced}$	95.0%
OEE	$0.857 \times 0.880 \times 0.950$	≈ 71.6%

THE SIMPLEST OEE FORMULA (ALGEBRAICALLY IDENTICAL)

$$\text{OEE} = (\text{Good Count} \times \text{Ideal Cycle Time}) \div \text{Planned Production Time}$$

If you substitute the three factor formulas and simplify, they collapse to this one expression — **good output, at ideal speed, as a fraction of all scheduled time**. The same number, reached directly. (*But computing the three factors separately is what tells you **where** the loss is — which is the whole value.*)

Notice the lesson hidden in the example: each factor looked reasonable — mid-80s to mid-90s — yet the line converted only about 72% of its planned time into good product. That ~28% gap is the opportunity, and because OEE split it into three buckets, the team knows the biggest single loss here is performance (speed), then availability (stops), then quality. The number does not just score; it *directs*.

The Six Big Losses & TEEP

OEE's three factors map onto a classic diagnostic framework: the **Six Big Losses**, two for each factor. This is what makes OEE actionable rather than merely descriptive — it tells you not just *that* you lost productivity but *which* of six specific ways, each with its own proven countermeasure.

OEE factor	The two losses	Typical countermeasure
Availability	1. Equipment failure (breakdowns) 2. Setup & adjustment (changeovers)	Top-loss analysis; SMED quick changeover (Issue 14)
Performance	3. Idling & minor stops 4. Reduced speed (slow cycles)	Short-interval control; root-cause the micro-stops
Quality	5. Process defects (scrap/rework) 6. Reduced yield (startup losses)	Standardized work; mistake-proofing

The discipline is Pareto: measure the six losses, find the biggest, attack it, repeat. A loss you cannot name you cannot fix — which is why splitting OEE into six causes matters more than the headline number. Improving the largest loss first compounds: because the factors multiply, lifting the weakest one lifts the whole score the most.

BEYOND OEE · TEEP

OEE measures planned time; TEEP measures all time

OEE scores effectiveness during *planned* production time. **TEEP** (Total Effective Equipment Performance) scores it against *all calendar time* — 24/7/365 — revealing hidden capacity in the hours you didn't schedule. The relationship: $TEEP = OEE \times \text{Utilization}$. An 85% OEE running two shifts (16 of 24 hours) is only ~57% TEEP — meaning ~43% of the asset's total capacity sits unused. That single comparison answers a CFO's question: *do we buy another machine, or just add a shift?*

THE DANGER OF OVER-INDEXING ON OEE

A metric chased blindly becomes a trap

Every serious source warns: OEE is one part of a balanced scorecard, not the whole truth. Chase it blindly and you invite perverse behaviour — **overproduction** (running to boost the number, building inventory nobody ordered, against everything in Issues 08 & 14), running equipment past safe parameters, or overstaffing. High OEE on the wrong product, or at the cost of safety, is a loss disguised as a win. The number serves the goal; it is not the goal.

PART 4 · THE QUALITY NUMBERS

First-Pass Yield, Scrap & PPM

OEE's quality factor is a blunt instrument; a real plant measures quality with sharper tools. The most important is **first-pass yield** — and the distinction it draws from ordinary yield is one of the most revealing numbers in manufacturing.

First-Pass Yield (FPY)

quality · right the first time

The percentage made correctly the first time, with no rework. $FPY = \text{good units with no rework} \div \text{total units started}$. It is harsher than simple yield because it *excludes* units that were eventually salvaged by rework — exposing the "invisible waste" hiding behind acceptable final output. A plant can ship 99% good units yet have a poor FPY if many needed fixing along the way. **Rolled throughput yield (RTY)** extends this across a whole sequence: multiply each step's FPY to get the realistic odds of a unit passing every step untouched. Targets: 95%+ excellent, 85-95% average.

FORMULA

Good first-time $\div$ total

HARSHER THAN

Simple yield

ACROSS STEPS

$RTY = \prod FPY$

Scrap Rate, Rework Rate & Defect PPM

quality · the cost of getting it wrong

What gets discarded, fixed, or fails. Scrap rate = scrapped $\div$ total; rework rate = reworked $\div$ total — both tie directly to margin (cutting scrap 1-2% can save a fortune in material). For low defect levels, quality is measured in **parts per million (PPM)**: the cross-industry median defect rate is around 4,500 PPM (~0.45%), while a Six Sigma world-class target is famously ~3.4 PPM. Breaking these down by shift, line, and cause is what turns a quality number into a root-cause hunt.

SCRAP

Discarded $\div$ total

MEDIAN

~4,500 PPM defects

SIX SIGMA

~3.4 PPM

PART 5 · FLOW & RELIABILITY

Throughput, Cycle Time & MTBF/MTTR

Two more families complete the core scorecard: the metrics of **flow** (how much, how fast) and of **reliability** (how dependable the equipment is). These are the numbers a manager checks first when output unexpectedly drops.

Throughput & Cycle Time

flow · how much, how fast

The most direct measures of output. **Throughput** is the volume of good units produced per hour/shift/day — the single clearest indicator of production capability and the first number to check when schedule attainment slips. **Cycle time** is the actual time to produce one unit start to finish. Compared against **takt time** (Issue 15: available time $\div$ demand), cycle time tells you whether the line can keep the customer's pace — the rule being cycle time $\leq$ takt at every station.

THROUGHPUT

Good units / time

CYCLE TIME

Actual time / unit

VS TAKT

must be $\leq$ takt

MTBF & MTTR

reliability · the maintenance pair

How reliable, and how fast to recover. **MTBF** (Mean Time Between Failures) measures how long equipment runs reliably between breakdowns — higher is better. **MTTR** (Mean Time To Repair) measures how quickly the team restores it after a failure — lower is better. Together they describe the availability story behind OEE: MTBF attacks *how often* you stop, MTTR attacks *how long* each stop lasts. They are the core of any maintenance scorecard and feed directly into availability.

MTBF

Uptime between fails $\uparrow$

MTTR

Time to repair $\downarrow$

DRIVES

Availability

85% world-class OEE (discrete)	A×P×Q the three factors, multiplied	6 big losses to diagnose	8-12 KPIs beat 30 half-measured ones
--	---	------------------------------------	--

PART 6 · THE SCORECARD & TESLA

Building a Scorecard — and the Tesla Picture

The canonical set is small and well-chosen: OEE with its three factors, FPY, scrap rate, MTBF, MTTR, changeover time, schedule adherence, on-time delivery, and energy per unit cover roughly 90% of what discrete manufacturing needs to manage. The art is the review cadence — OEE and downtime daily, maintenance KPIs weekly, cost and strategic metrics monthly — each with a named owner and a defined action.

Family	Key KPIs	Reviewed
Performance	OEE, TEEP, throughput, cycle time, capacity utilization	Daily
Quality	First-pass yield, scrap rate, rework rate, defect PPM	Daily / per shift
Reliability	MTBF, MTTR, availability, PM compliance	Weekly
Planning	Schedule adherence, on-time delivery (Issue 15)	Daily / weekly
Cost / energy	Cost per unit, energy per unit	Monthly

WHAT'S ACTUALLY KNOWN ABOUT TESLA'S METRICS

Public production rates, private KPI definitions

Tesla does **not** publish its internal OEE figures, KPI definitions, or line-level targets — those are proprietary. What *is* public is of two kinds. First, the **data infrastructure**: as Issue 16 covered, Tesla's MES/MOS capture production counts, quality results, and equipment state in real time, feeding the dashboards leadership acts on — exactly the plumbing a live OEE/yield scorecard needs. Second, the metric Tesla talks about openly is **production rate** — vehicles per week and annual run-rate targets (the famous Model 3 ramp from a handful of cars to thousands per week). That headline "units per week" is itself a throughput KPI; the OEE, FPY, and loss breakdowns beneath it are real but unpublished.

It is worth being precise about the difference. A public figure like "5,000 Model 3s per week" is a **throughput target** — the most visible KPI of all, and the one Musk has tied to existential stakes ("production hell"). But hitting it depends entirely on the unglamorous internal scorecard this issue describes: the availability of every robot, the first-pass yield of every weld, the MTTR when a line stops. Throughput is the number the world sees; OEE and its kin are the numbers that *produce* it. Tesla's openness about the former and silence about the latter is exactly what you'd expect — the output is a headline, the diagnostics are a trade secret.

The world sees **cars per week**. The factory runs on the **OEE**, **yield**, and **loss numbers** that **produce** it.

Step back and measurement closes the loop this whole series has traced. Planning (Issue 15) sets a target; the floor (Issues 02–13) executes it; the software (Issues 01, 16) records what happened; and metrics — OEE, yield, throughput, the six losses — turn that record into the truth about whether the plan was met, and where it failed. That truth flows back up to the next plan. A factory without good metrics is flying blind, however advanced its robots; a factory with them can see exactly where its potential leaks away, name the loss, and close the gap. The score is not vanity — it is the feedback that makes everything else improvable. ■

FIELD GLOSSARY

The performance-metrics vocabulary in this issue

- **KPI (Key Performance Indicator)** — a measured number with an owner and a triggered action; otherwise just information.
- **OEE (Overall Equipment Effectiveness)** — $\text{Availability} \times \text{Performance} \times \text{Quality}$; the share of planned time that is fully productive.
- **Availability** — $\text{run time} \div \text{planned production time}$ (the stops factor).
- **Performance** — $\text{actual vs ideal output rate}$ (the speed factor).
- **Quality** — $\text{good count} \div \text{total count}$ (the defect factor).
- **Six Big Losses** — breakdowns, setup/changeover, idling/minor stops, reduced speed, process defects, reduced yield.
- **TEEP** — Total Effective Equipment Performance: $\text{OEE} \times \text{utilization}$, against all calendar time.
- **First-Pass Yield (FPY)** — units made right the first time, excluding rework.
- **Rolled Throughput Yield (RTY)** — the product of FPYs across a sequence of steps.
- **Scrap / rework rate** — discarded or re-fixed units as a share of total.
- **Defect PPM** — defects per million; Six Sigma world-class ≈ 3.4 PPM.
- **Throughput** — good units produced per unit of time.
- **Cycle time** — actual time to produce one unit; compared to takt.
- **MTBF** — Mean Time Between Failures (reliability; higher is better).
- **MTTR** — Mean Time To Repair (recovery; lower is better).
- **Schedule adherence** — $\text{orders completed on schedule} \div \text{total scheduled}$.
- **ISO 22400** — the international standard defining manufacturing KPI calculations.

FURTHER READING & SOURCING

This issue draws on standard manufacturing-performance literature and the **ISO 22400** KPI standard — OEE and its three factors, the Six Big Losses, TEEP, first-pass and rolled-throughput yield, scrap/PPM, throughput, cycle time, MTBF/MTTR, and schedule adherence. The worked OEE example ($85.7\% \times 88.0\% \times 95.0\% \approx 71.6\%$) follows the canonical Vorne/Lean-Production formulation. It is the measurement-layer companion to Issue 01 (the dashboard), Issue 15 (planning & takt), and Issue 16 (the MES/MOS data that feeds these metrics). Tesla's internal KPI definitions and targets are proprietary; only its public data-driven approach and published production-rate targets are referenced, and clearly flagged as such. Figures are illustrative; diagrams conceptual.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 18

COVER STORY · THE SIGNAL THAT STARTS EVERYTHING

Demand: Orders, Forecasts & What Feeds Them

Every plan, every part order, every car begins with one question — how much will customers want? From the statistics of forecasting to Tesla's radical answer: don't forecast the order, take it directly.

FORECAST vs ORDER

The two ways a factory learns what to build.

THE INPUTS

History, trend, seasonality — what a forecast is built from.

EXPONENTIAL SMOOTHING

The workhorse formula that weights the recent past.

THE TESLA MODEL

Build-to-order: the configurator that skips the guess.

The Signal That Starts Everything

THE INTEGRATION LAYER · Feature · Forecasting theory through to Tesla's order-driven model

Trace any action in a factory back far enough — a robot moving, a bolt tightened, a part ordered, a plan drawn — and you arrive at a single origin: a belief about how much customers will want. Issue 15 showed how a production plan cascades down into a build sequence; but a plan is only an answer to a prior question, and that question is **demand**. Everything downstream is the faithful execution of a guess, or a fact, about what the market will buy. This issue is about how a factory gets that number — the demand signal that starts everything — and the two fundamentally different ways to get it: *forecast* it, or *take the order* directly.

The distinction is the spine of this issue. Most manufacturers must **forecast** — predict demand statistically and build to that prediction (make-to-stock), because customers expect to buy off the lot. A minority **build to order** — wait for a confirmed customer order, then build exactly that. Tesla is the famous modern example of the second model, and seeing both side by side reveals what forecasting is really for, where it fails, and why Tesla's order-driven approach is such a radical simplification of the oldest problem in manufacturing.

EDITOR'S NOTE · ON SCOPE & SOURCING

This issue explains the **standard demand-forecasting toolkit** (quantitative time-series methods, qualitative methods, forecast error/MAPE, and the make-to-stock vs build-to-order distinction) from operations-management literature, then describes **what is publicly known** about Tesla's order-driven model: its direct online configurator, build-to-order approach, and publicly-reported "order rate and backlog" language from its own filings. Tesla's internal forecasting algorithms and demand models are **proprietary and not disclosed**; statements about Tesla here are limited to its public sales model and reporting, and are flagged. This is the upstream companion to Issue 15 (planning) and Issue 14 (inventory). Figures are illustrative; diagrams conceptual.

The one idea: every plan is downstream of a demand number

The governing truth: *nothing* in a factory can be decided until someone commits to a demand figure. Safety stock is sized to forecast *error*; the reorder point uses forecast *demand* × lead time; the master schedule is a forecast made concrete; capacity is built years ahead against a forecast. As one source puts it, every inventory parameter is derived from a demand estimate, and the quality of those parameters is bounded by the quality of the forecast. Get demand right and everything downstream has a chance; get it wrong and the most sophisticated plant faithfully builds the wrong thing.

Every plan is downstream of a demand number. The whole factory is the execution of a **guess — or a **fact** — about what the market wants.**

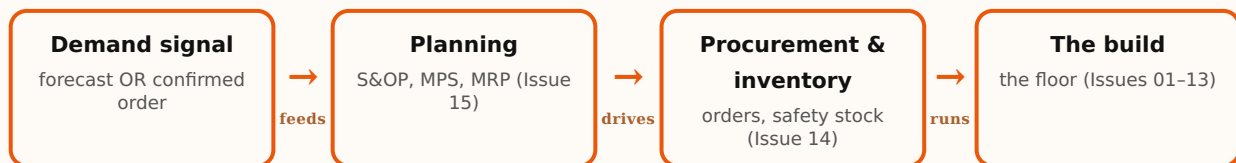


Figure 1. Demand is the headwater of the whole river. Everything this magazine has described flows downstream of the demand signal. (*Conceptual.*)

In this issue

1. **Forecast vs order** — the two models
2. **The inputs** — what a forecast is built from
3. **Quantitative methods** — averages & smoothing
4. **Forecast error** — why it drives safety stock
5. **Build-to-order** — and the Tesla model
6. **Orders, backlog & MOS** & the glossary

PART 1 • TWO MODELS

Forecast vs Order: How a Factory Learns Demand

There are only two ways to know what to build: predict it, or be told it. That choice — **make-to-stock** versus **build-to-order** — shapes everything about how a company forecasts, holds inventory, and takes risk. Neither is universally right; each fits a different product and market.

Make-to-stock (MTS)

Forecast demand, build ahead, sell from finished stock. Fast delivery off the shelf — but you bet capital on a prediction, risking unsold inventory if you guess high or stockouts if you guess low. The model for most consumer goods.

Build-to-order (BTO / MTO)

Wait for a confirmed customer order, then build exactly that. No finished-goods guess, minimal obsolescence — but the customer waits, and the plant must be fast and flexible. The model for expensive, highly-configurable products.

The trade is risk versus speed. MTS gives instant availability but carries the forecast risk on its balance sheet — every unsold unit is a bad bet made visible. BTO eliminates the finished-goods bet (you only build what's sold) but transfers the cost to *customer wait time* and demands a responsive supply chain. A combustion-era dealership lot is MTS in its purest form: thousands of pre-built cars guessing at demand. Tesla's website is BTO in its purest form: nothing built until you click "order."

THE HYBRID MOST PLANTS ACTUALLY RUN

The 80/20 compromise

Pure models are rare; most manufacturers blend. A common pattern (well illustrated by Toyota) is the **80/20 rule**: build the most popular ~20% of configurations to stock — covering ~80% of demand for fast delivery — while offering the remaining configurations build-to-order with a longer lead time. It captures most of MTS's speed and most of BTO's flexibility. Even order-driven makers keep some popular-spec inventory; even stock-driven makers take special orders. The question is rarely "which?" but "what mix?"

PART 2 · THE INPUTS

What a Forecast Is Built From

A forecast is only as good as what goes into it — and demand forecasting draws on two families of input: hard **quantitative** data and softer **qualitative** judgement. The best forecasts blend both, because numbers capture the past while judgement anticipates what the past cannot.

Quantitative Inputs (the hard data)

inputs · what the numbers say

The measurable history. The foundation is **historical demand** — past sales by period — from which three signals are extracted: the underlying **level** (the baseline rate), the **trend** (steady rise or fall over time), and **seasonality** (recurring patterns — holidays, back-to-school, weather). Add external quantitative inputs: economic indicators, price, promotions, web traffic, and order book. These feed the statistical methods of Part 3.

CORE

Historical demand

SIGNALS

Level, trend, season

EXTERNAL

Price, economy, traffic

Qualitative Inputs (informed judgement)

inputs · what the numbers can't see

Expert and market judgement. When history is missing or about to break — a new product, a market shift, a disruption — numbers fail and judgement fills the gap. Methods include the **Delphi method** (structured, anonymous rounds of expert opinion converging on consensus), sales-force estimates, market research, and customer/supplier input. Essential for new launches, where there is no past to extrapolate from.

WHEN

No/breaking history

METHOD

Delphi, expert panels

FOR

New products, shifts

WHY STATISTICS ALONE ISN'T ENOUGH

The past is a guide, not a guarantee

Purely statistical forecasting from historical consumption works beautifully *if* you sell roughly the same amount each period. The moment demand shifts — a competitor's move, a viral moment, a recession, a new model — history mis-predicts, producing stockouts or excess. That is exactly why qualitative judgement and fresh signals matter, and why the holy grail is replacing the forecast with *actual demand* wherever possible — the move Part 5 explores.

Quantitative Forecasting: Averages & Smoothing

The everyday engines of demand forecasting are **time-series** methods — they assume the future resembles the patterned past, and extract that pattern from history. Two dominate practice, and a family of refinements handles trend and seasonality.

Moving Average

method · the simple smoother

Average the recent past. A moving average takes the mean of demand over the last n periods and uses it as the next forecast — smoothing out short-term noise to reveal the underlying level. Simple, intuitive, and often the first method tried. Its weakness: it weights all n periods equally and **lags** behind any real trend, systematically under- or over-forecasting when demand is steadily moving.

DOESMean of last n periods**GOOD FOR**

No-trend, noisy data

WEAKNESS

Lags a trend

Exponential Smoothing

method · the weighted workhorse

Weight the recent past more heavily. The most widely-used method blends the most recent actual demand with the previous forecast, weighting recent data more via a smoothing constant α (alpha). A high α reacts fast to change; a low α smooths more. It adapts quickly, needs little data, and is the workhorse of short-term forecasting — introduced by Robert G. Brown in 1956.

DOES

Weighted recent demand

KNOB α (0–1)**WINS**

Adapts fast, low data

THE EXPONENTIAL SMOOTHING FORMULA

$$F_{\text{next}} = \alpha \cdot (\text{actual demand now}) + (1 - \alpha) \cdot (\text{forecast now})$$

α = smoothing constant between 0 and 1 — the weight on the latest actual demand.

High α (e.g. 0.5) → forecast reacts quickly to recent change. Low α (e.g. 0.1) → forecast stays smooth and stable. The new forecast is just a weighted average of *what just happened* and *what you last predicted*.

When demand has a **trend** or **seasonality**, simple methods lag, so they are extended. **Holt's method** (double exponential smoothing, 1957) adds a trend term; **Holt-Winters** (1960) adds a seasonal term too — the standard for SKU-level forecasts with recurring patterns. The practical rule: use a moving average or simple smoothing for flat, noisy demand; switch to Holt or Holt-Winters the moment a consistent trend or season appears, or the forecast will systematically lag reality.

Method	Handles	Use when
Moving average	Level (noise)	Demand is flat and choppy, no trend
Exponential smoothing	Level, adapts fast	Short-term, limited data, mild change
Holt (double)	Level + trend	Demand steadily rising or falling
Holt-Winters	Level + trend + season	Recurring seasonal patterns (per SKU)

Forecast Error: Why It Sizes Your Safety Stock

No forecast is exactly right — so the crucial second question is not just "what's the forecast?" but "how wrong is it usually?" That error is not a footnote; it directly determines how much safety stock a plant must hold, tying this issue straight back to the inventory math of Issue 14.

FORECAST ACCURACY • MAPE

$$\text{MAPE} = \text{average of } | \text{actual} - \text{forecast} | / \text{actual} \quad (\times 100\%)$$

MAPE = Mean Absolute Percentage Error — the average size of the forecast miss, as a percentage. Lower is better. It is the standard scorecard for "how good is our forecasting?" — tracked over time the way OEE (Issue 17) tracks the floor.

Here is the chain that makes error matter. **Safety stock is sized to cover forecast error** — specifically, the standard deviation (σ) of that error (recall $SS = z \cdot \sigma \cdot \sqrt{L}$ from Issue 14). So a *better* forecast — a lower MAPE, a smaller error σ — means you can hold **less safety stock** for the same service level, freeing working capital without raising stockout risk. Forecasting is not an academic exercise: every point of accuracy converts directly into less inventory or fewer stockouts. A biased forecast, meanwhile, triggers reorders systematically too early or too late.

level

the baseline demand rate

trend

steady rise or fall

season

recurring patterns

σ error

drives safety stock

THE BULLWHIP CONNECTION

Bad forecasts amplify upstream

Forecast error does not stay local. As Issue 14's bullwhip effect showed, each supply-chain link reacting to the orders below it — padding for forecast uncertainty, batching, second-guessing — amplifies demand swings upstream. Poor forecasting at the customer end becomes violent over- and under-ordering at component suppliers. This is the deepest argument for the order-driven model of Part 5: if you can replace the forecast with *real demand*, you cut the bullwhip off at its source.

Build-to-Order: Skipping the Guess

Now the radical alternative this issue has been building toward. What if, instead of forecasting demand and building to the prediction, you simply *waited for the order* and built exactly that? That is build-to-order — the oldest fulfillment model, reborn at scale by Tesla's direct, online, configure-to-order sales.

The order IS the demand signal. By public reporting, Tesla sells **direct to consumer** — no dealership lot of pre-built cars. A customer configures a car on Tesla's website or app (model, color, wheels, options) at a fixed, non-negotiable price, places a deposit, and that confirmed order flows into production. Reporting describes the configurator generating a bill of materials automatically and sending it to the production system — build-to-order, with the customer's exact spec as the trigger. No guessing which colors to pre-build; no lot of unsold inventory.

CHANNEL

Direct online / app

TRIGGER

Confirmed order → BOM

INVENTORY

Minimal pre-built stock

The advantages map exactly onto the failures of forecasting that Parts 3–4 described. No finished-goods forecast means no risk of building the wrong configurations; minimal finished inventory means little obsolescence and freed capital; and because demand arrives as *real orders*, not predictions, the forecast error and much of the bullwhip simply vanish at the source. Tesla's famed vertical integration of software (Issue 16) is what makes it work — the online order, the configurator, and the production system are one connected pipeline, so a click becomes a build order without a human re-keying anything.

BUT FORECASTING DOESN'T DISAPPEAR**You still forecast the parts, the capacity, and the aggregate**

Build-to-order removes the *finished-goods* forecast, but not forecasting altogether — a crucial honesty. Tesla must still forecast **aggregate demand** to plan capacity and factories years ahead (you cannot build a Gigafactory to order). It must forecast **component** demand to secure long-lead parts — cells, chips — well before specific orders arrive. And it forecasts to set production *rate*. Reporting and Tesla's own filings describe analyzing customer data, market trends, and historical sales for demand planning, and explicitly cite "current order rate trends and backlog" as supporting planned production levels. BTO moves the forecast *upstream* — from "which cars to build" to "how much capacity and how many parts" — rather than abolishing it.

PART 6 · ORDERS, BACKLOG & THE FLOOR**From Order to Backlog to the Line**

In an order-driven model, two concepts replace the forecast as the things managers watch: the **order rate** (how fast new orders arrive) and the **backlog** (confirmed orders not yet built). Together they are the real-time demand signal — and Tesla's own quarterly reporting has used exactly this language, stating that order-rate trends and backlog supported its planned production rates.

Customer order (configurator + deposit)

the confirmed demand signal — exact spec, fixed price

▼ enters

Order backlog & sequencing

confirmed orders queued; production rate set against order rate

▼ released to

Production schedule & BOM explosion (Issue 15)

the order's spec becomes a build sequence & part orders

▼ executed & tracked by

MES / MOS on the floor (Issues 01, 16)

each car built & tracked to its order — back to the buyer

Figure 2. The order-driven pipeline: a configured order becomes backlog, then a build sequence, then a tracked car — demand flows as fact, not forecast. (*Conceptual.*)

This is where demand rejoins the rest of the magazine. The confirmed order — not a forecast — becomes the master-schedule input (Issue 15); MRP explodes it into part orders (Issue 15); inventory policy stocks the components (Issue 14); the floor builds it (Issues 02–13); MES and MOS (Issues 01, 16) track that specific car against that specific order; and metrics (Issue 17) score how well it all ran. In a make-to-stock plant, that same pipeline runs on a *forecast* at the top instead of an order — the only difference is whether the headwater is a guess or a fact.

Forecast or order, the river is the same. The only question is whether the headwater is a **guess or a **fact**.**

Step back to where this series began and ends. Issue 01 followed a torque reading up from the floor; this issue follows the demand signal *down* into it. They are the two ends of one loop: demand at the top tells the factory what to build; data from the floor tells it what actually happened; and the gap between the two — forecast versus actual, planned versus built — is the entire art of running a plant. Tesla's contribution to that ancient problem is disarmingly simple: wherever you can, stop guessing what customers want and let them tell you directly, then wire that order straight through to the steel. Forecasting will never vanish — capacity and long-lead parts will always need a prediction — but the closer a factory can move its demand signal from *guess* to *fact*, the less it builds in vain. The whole river, from the first click to the finished car, is the answer to a single question asked well: *how much, of what, do they actually want?* ■

FIELD GLOSSARY

The demand & forecasting vocabulary in this issue

- **Demand forecasting** — predicting future demand to drive planning, procurement, and inventory.
- **Make-to-stock (MTS)** — forecasting demand and building to finished stock ahead of orders.
- **Build-to-order (BTO / MTO)** — building only after a confirmed customer order arrives.
- **80/20 (hybrid) model** — stocking the popular minority of configurations, building the rest to order.
- **Quantitative forecasting** — statistical methods using historical data (time series).
- **Qualitative forecasting** — judgement-based methods (Delphi, expert/market input) when data is thin.
- **Level, trend, seasonality** — the three signals extracted from demand history.
- **Moving average** — forecasting with the mean of the last n periods; lags trends.
- **Exponential smoothing** — a weighted average favouring recent demand, tuned by α .
- **Smoothing constant (α)** — the 0-1 weight on the latest actual demand.
- **Holt / Holt-Winters** — exponential smoothing extended for trend (Holt) and season (Holt-Winters).
- **MAPE** — Mean Absolute Percentage Error; the standard forecast-accuracy score.
- **Forecast error (σ)** — the variability of misses; it sizes safety stock (Issue 14).
- **Configurator** — the online tool where a customer specifies a product, generating its BOM.
- **Order rate** — how fast new confirmed orders arrive.
- **Backlog** — confirmed orders not yet built; a real-time demand signal.
- **S&OP** — Sales & Operations Planning; reconciling demand forecast with supply (Issue 15).

FURTHER READING & SOURCING

This issue draws on standard demand-forecasting and operations literature — quantitative time-series methods (moving average; exponential smoothing; Holt and Holt-Winters for trend/seasonality), qualitative methods (Delphi), forecast accuracy (MAPE), the make-to-stock vs build-to-order distinction, and the link between forecast error and safety stock. Statements about Tesla reflect **public reporting** on its direct-to-consumer, build-to-order model and online configurator, and its own filings' language on "order rate trends and backlog"; Tesla's internal forecasting models are proprietary and not disclosed. It is the upstream-demand companion to Issue 14 (inventory & the bullwhip), Issue 15 (planning, S&OP, MPS), and Issue 16 (the software pipeline). Figures are illustrative; diagrams conceptual.

THE INTEGRATION LAYER · Issue 18 · A technical magazine · Sources: demand-forecasting fundamentals & public reporting on Tesla's order model

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION · ISSUE 19

COVER STORY · WHERE THE THEORY MEETS THE PLANNER'S HANDS

The Spreadsheet: Excel for Inventory & Production Planning

The most-used planning tool in any factory isn't a robot or an ERP — it's a spreadsheet. How the inventory and planning mathematics of this series becomes real formulas in Excel: EOQ, safety stock, ABC, forecasts, and a master schedule.

THE CORE FUNCTIONS

The handful that do 90% of planning work.

FORECAST.ETS

Holt-Winters seasonality, built into Excel.

INVENTORY FORMULAS

EOQ, reorder point & safety stock, in real cells.

WHEN TO GRADUATE

Where the spreadsheet ends and MRP begins.

The Most Important Tool in the Plant Isn't a Robot

THE INTEGRATION LAYER · Feature · A hands-on guide putting the series' math into Excel

This series has spent eighteen issues on the theory and the machinery of a modern factory — controllers, drives, the mathematics of inventory and planning, the demand signal. This issue is different: it is about the unglamorous, universal tool where all of that theory actually meets a human being's hands for the first time. Not a robot, not a million-dollar MES — a **spreadsheet**. Before a plant runs MRP, before it buys an ERP, and very often *alongside* both forever, someone opens Excel and starts computing what to order and what to build. This issue shows exactly how.

It is a fitting tool to land on, because even the most software-forward factory in the world still reaches for it. Issue 16 noted, from public reporting, that Tesla — which wrote its own ERP rather than buy SAP — *still* uses plain Excel for one-off manual reports. The spreadsheet is the planner's pocket-knife: instantly available, endlessly flexible, and powerful enough to implement nearly every formula in Issues 14, 15, and 18 with a few built-in functions. This issue is the practical companion to those three — the place the equations become cells you can actually fill in.

EDITOR'S NOTE · ON SCOPE

This issue explains **standard Microsoft Excel functions** (available in current Excel; a few like XLOOKUP and FORECAST.ETS need Excel 2016 or later) applied to the inventory and planning mathematics established earlier in this series. The formulas shown are real and ready to type. The underlying math — EOQ, reorder point, safety stock, ABC, forecasting, takt, MRP — comes from Issues 14, 15, and 18, where it is derived and sourced. This issue also gives the honest limits of spreadsheets and when to move to MRP/ERP (Issues 15–16). Examples are illustrative; cell references assume a simple layout you can adapt.

The one idea: a formula is just a sentence in arithmetic

The thread of this issue is that every concept from the theory issues reduces to one short, typeable line. "Order the quantity that minimises cost" becomes $=\text{SQRT}(2 \cdot D \cdot S / H)$. "Hold enough buffer for 95% service" becomes $=\text{NORM.S.INV}(0.95) \cdot \dots$. "Forecast next month" becomes $=\text{FORECAST.ETS}(\dots)$. The spreadsheet is where abstract operations-management theory turns into a sentence in arithmetic that a planner writes once and reuses forever. Learn the dozen functions in this issue and you can build a working inventory-and-planning system before lunch.

Every equation in this series reduces to one typeable line. A formula is just a sentence in arithmetic.

1 — ORGANISE the data

tables of items, demand, costs, lead times (the raw material)

2 — CALCULATE the decisions

EOQ, reorder point, safety stock, takt (Issues 14–15)

3 — FORECAST the future

moving average, smoothing, FORECAST.ETS (Issue 18)

4 — ANALYSE & present

ABC/Pareto, PivotTables, conditional formatting, charts

Figure 1. The four jobs a spreadsheet does for a planner: organise, calculate, forecast, analyse — the practical form of this series' theory. (*Conceptual.*)

In this issue

1. **The toolkit** — the essential functions
2. **Inventory** — EOQ, ROP & safety stock
3. **ABC analysis** — Pareto in a spreadsheet
4. **Forecasting** — averages, smoothing, ETS
5. **Planning** — takt, MPS & a BOM explosion
6. **PivotTables & the limits** & the glossary

PART 1 • THE TOOLKIT

The Dozen Functions That Do the Work

You do not need to know hundreds of Excel functions to run inventory and planning — you need about a dozen, used well. They fall into four jobs: looking things up, doing conditional math, doing statistics, and summarising. Here is the working set this issue uses.

Function	What it does	Used here for
SUM / SUMIFS	Add up, optionally by criteria	Total demand, usage by category
AVERAGE / AVERAGEIFS	Mean, optionally by criteria	Average demand (moving average, ROP)
XLOOKUP (or VLOOKUP)	Find a row's value by a key	Pull a part's cost, lead time, BOM
IF / IFS	Branch on a condition	Classify A/B/C; flag low stock
SQRT	Square root	The EOQ formula
STDEV.S	Standard deviation (variability)	Demand σ for safety stock
NORM.S.INV	Z-value for a probability	Service-level factor (95% → 1.645)
FORECAST.ETS / TREND	Project future from history	Demand forecasting (Issue 18)
SUMPRODUCT	Multiply arrays, then add	BOM explosion (parts × quantities)
RANK / SORT	Order by value	ABC ranking (Pareto)
PivotTable	Summarise & cross-tab a table	Usage by part, demand by month
Conditional formatting	Colour cells by rule	Visual alerts (red = below ROP)

A NOTE ON VERSIONS & SYNTAX

XLOOKUP, dynamic arrays, and the dot-functions

Two small modernisations matter. **XLOOKUP** (Excel 2019/365) replaces the older **VLOOKUP** — same idea, simpler and more robust; if you are on an older version, **VLOOKUP**(key, table, col, FALSE) still works. And Microsoft renamed many statistics functions with a dot (**STDEV.S**, **NORM.S.INV**) — the legacy names (**STDEV**, **NORMSINV**) still function. This issue uses the modern names; substitute the legacy ones freely on older Excel.

PART 2 · INVENTORY CONTROL (ISSUE 14)

EOQ, Reorder Point & Safety Stock in Cells

The inventory mathematics of Issue 14 maps almost one-to-one onto Excel. Lay out your inputs in named cells — annual demand, order cost, holding cost, lead time, demand history — and four short formulas give you the complete reorder policy.

THE INVENTORY FORMULAS, IN EXCEL

EOQ — optimal order quantity

`=SQRT(2*D*S/H)`

Safety stock — buffer for a service level

`=NORM.S.INV(0.95)*STDEV.S(demand)*SQRT(L)`

Reorder point — when to reorder

`=AVERAGE(demand)*L + SafetyStock`

Inventory turns — how lean

`=COGS/AVERAGE(inventory)`

D = annual demand · S = order cost · H = holding cost per unit/yr · L = lead time (in the same period units as your demand history). "demand" / "inventory" are ranges of cells.

Read them against Issue 14 and the correspondence is exact. `=SQRT(2*D*S/H)` is the EOQ / Wilson formula verbatim. The safety-stock line is $z \cdot \sigma \cdot \sqrt{L}$ with `NORM.S.INV(0.95)` supplying the z-factor (it returns 1.645 for 95% service, 2.326 for 99% — the very numbers from Issue 14) and `STDEV.S` supplying σ from your demand history. The reorder point adds lead-time demand to that buffer. Four cells, and the whole reorder policy is live — change any input and every answer updates instantly.

WORKED NUMBERS

A quick sanity check

Say D = 12,000 units/yr, S = \$50/order, H = \$3/unit/yr. Then `=SQRT(2*12000*50/3)` returns **632** — order about 632 at a time. With average weekly demand 230, $\sigma = 40$, lead time L = 2 weeks and 95% service: safety stock `=NORM.S.INV(0.95)*40*SQRT(2)` $\approx$ **93**, and reorder point `=230*2+93` = **553**. Reorder when stock hits 553; order 632. The spreadsheet did in four cells what would take a page of arithmetic by hand.

Pareto Classification in a Spreadsheet

ABC analysis — the Pareto split that decides which parts deserve tight control — is one of the most satisfying things to build in Excel, because it is just five columns. Start with each item's annual usage and unit cost; the spreadsheet does the rest.

A: Item	B: Annual qty	C: Unit cost	D: Annual value	E: Cum. %	F: Class
Part-001	1,200	180.00	216,000	62%	A
Part-002	3,000	22.00	66,000	81%	A
Part-003	800	40.00	32,000	90%	B
Part-004	9,500	2.50	23,750	97%	B
Part-005	5,000	2.00	10,000	100%	C

BUILDING THE ABC COLUMNS

D — annual value (per row)

`=B2*C2`

Sort by column D, largest first

Data > Sort ↓ (or `SORT()`)

E — cumulative % of total value

`=SUM($D$2:D2)/SUM($D$2:$D$13)`

F — classify A / B / C

`=IFS(E2<=0.8, "A", E2<=0.95, "B", TRUE, "C")`

The mechanism is exactly Issue 14's Pareto logic. Compute each item's annual value (qty × cost), sort descending so the biggest-value items rise to the top, run a **cumulative percentage** down the list with a clever anchored `SUM($D$2:D2)` (the locked first reference and the moving second create a running total), then let `IFS` stamp each row A, B, or C at the 80% and 95% thresholds. The vital few "A" items now sit at the top, flagged for tight control and frequent counts; the trivial many "C" items fall to simple two-bin kanban. Add conditional formatting to colour the classes and the analysis is also a picture.

Averages, Smoothing & FORECAST.ETS

Issue 18's forecasting methods are all built into Excel — from a one-cell moving average to a function that performs full Holt-Winters seasonality automatically. Lay your demand history down a column with dates beside it, and you can forecast in a single formula.

THE FORECASTING FORMULAS, IN EXCEL

Moving average (last 3 periods)	<code>=AVERAGE(C2:C4)</code>
Exponential smoothing (α in \$G\$1)	<code>=\$G\$1*C2+(1-\$G\$1)*D1</code>
Linear trend projection	<code>=TREND(C2:C13,A2:A13,A14)</code>
Trend + seasonality (Holt-Winters)	<code>=FORECAST.ETS(A14,C2:C13,A2:A13)</code>
Forecast accuracy (MAPE)	<code>=AVERAGE(ABS(actual-forecast)/actual)</code>

The progression mirrors Issue 18 exactly. `=AVERAGE(C2:C4)` is the moving average. The smoothing line is the formula from Issue 18 verbatim — $\alpha \cdot \text{actual} + (1-\alpha) \cdot \text{previous forecast}$ — with the smoothing constant parked in a cell (\$G\$1) so you can tune it and watch the forecast react. And the jewel is **FORECAST.ETS** (Excel 2016+): it performs *exponential triple smoothing* — the Holt-Winters method of Issue 18, detecting trend *and* seasonality automatically — in one function. Excel even ships a "Forecast Sheet" button (Data > Forecast Sheet) that builds an ETS forecast with confidence bounds and a chart in two clicks.

MEASURING THE FORECAST

MAPE in one array formula

Track accuracy exactly as Issue 17-18 prescribe: put actuals and forecasts in two columns and compute `=AVERAGE(ABS(actual-forecast)/actual)` as the Mean Absolute Percentage Error. (In older Excel, enter it as an array with Ctrl+Shift+Enter; in current Excel it just works.) A lower MAPE means a tighter forecast — which, per Issue 14, lets you carry less safety stock for the same service level. The whole chain, live in one sheet.

PART 5 · PRODUCTION PLANNING (ISSUE 15)

Takt, a Master Schedule & a BOM Explosion

The planning layer of Issue 15 also lives comfortably in a spreadsheet — takt time is a single division, a master schedule is a grid, and even an MRP-style requirements calculation is a lookup and a multiply. This is where a planner often starts before any MRP system exists.

THE PLANNING FORMULAS, IN EXCEL

Takt time — the customer's pace	<code>=AvailableTime/Demand</code>
Net requirement (MRP netting)	<code>=MAX(0, Gross - OnHand - OnOrder)</code>
Component need (BOM explosion)	<code>=SUMPRODUCT(BuildQty, QtyPerCar)</code>
Lookup a part's lead time	<code>=XLOOKUP(PartID, Parts, LeadTimes)</code>

Takt time (Issue 15) is just available time ÷ demand — one cell. A **master production schedule** is a grid with weeks across the top and products down the side, summed and checked against a capacity row. And the heart of MRP — turning a build plan into part orders — is two

functions: SUMPRODUCT multiplies how many of each car you'll build by how many of a part each car needs (the BOM explosion of Issue 15), and MAX(0, gross - on-hand - on-order) nets that against stock to give the true quantity to order. A small MPS grid makes it concrete:

Product / Week	W1	W2	W3	W4	=SUM
Model A	120	120	140	140	520
Model B	60	80	80	90	310
Total build	180	200	220	230	830
Capacity	210	210	230	230	880
Load %	86%	95%	96%	100%	94%

The "Load %" row — =Total/Capacity — instantly shows W4 is at 100% capacity (no room for more), the rough-cut capacity check of Issue 15 in a single formula. Conditional formatting turns any cell over 100% red, and the planner sees the bottleneck at a glance. This is a genuine, if simple, planning system — thousands of real factories run exactly this.

PART 6 · PRESENTING, AND THE LIMITS

PivotTables, Dashboards — and When to Stop

Two features turn a calculating sheet into a management tool. **PivotTables** summarise a long transaction list into any cross-tab you want — usage by part, demand by month, scrap by line — by dragging fields, no formulas required. And **conditional formatting** plus charts make the numbers visual: stock below the reorder point glows red, a demand chart shows the trend, a Pareto chart draws the ABC curve. With these, a spreadsheet becomes the small, honest dashboard of Issue 17.

Why Excel wins

Instantly available, no setup, infinitely flexible, universally understood, and powerful enough for real inventory & planning math. The fastest path from a question to an answer. Perfect for one plant, one analyst, one model.

Where Excel breaks

Error-prone (one wrong reference, silent wrong answers), no real audit trail, fragile links, single-user, no live shop-floor data, and it does not scale to thousands of parts across sites. "Spreadsheet hell" is real.

WHEN TO GRADUATE FROM THE SPREADSHEET

The spreadsheet is the start, not the end

Excel's flexibility is also its danger: it has no guardrails, so at scale errors hide and links rot. The moment you have many parts, multiple sites, multiple users, or need *live* data from the floor, you have outgrown it — and that is precisely why MRP, ERP, and APS systems (Issue 15) and bespoke stacks like Tesla's Warp (Issue 16) exist: to do this same math with integrity, at scale, on real-time data, for many people at once. The honest rule: prototype and learn in Excel; run the enterprise on a real system. Tesla keeps Excel for one-off reports precisely *because* it has Warp for everything else.

And that placement is the real lesson of this issue, and a fitting near-final note for the series. Everything this magazine has described — the controllers, the networks, the inventory math, the planning hierarchy, the demand signal, the giant software stacks — can be *understood*, and often first *built*, in a spreadsheet. The EOQ that sizes an order, the safety stock that

prevents a stockout, the forecast that anticipates demand, the schedule that paces the line: all of it is a handful of cells before it is ever a system. The spreadsheet is where a planner thinks, where theory is tested cheaply, and where the math of this entire series becomes something you can touch, change, and watch recalculate. Master it, and you hold the whole factory in a grid — small enough to fit on one screen, and powerful enough to start running a plant. ■

FIELD GLOSSARY

The Excel-for-planning vocabulary in this issue

- **SUMIFS / AVERAGEIFS** — add or average cells that meet given criteria.
- **XLOOKUP (VLOOKUP)** — return a value from a table by matching a key (e.g. a part's cost).
- **IF / IFS** — choose an output based on conditions (e.g. ABC class).
- **SQRT** — square root; the core of the EOQ formula.
- **STDEV.S** — sample standard deviation; the demand σ for safety stock.
- **NORM.S.INV(p)** — the z-value for probability p; service-level factor (0.95 → 1.645).
- **AVERAGE** — arithmetic mean; moving-average forecast and mean demand.
- **TREND / FORECAST.LINEAR** — project a linear trend forward.
- **FORECAST.ETS** — exponential triple smoothing (Holt-Winters) with automatic seasonality.
- **Forecast Sheet** — Excel's one-click ETS forecast with confidence bounds and chart.
- **SUMPRODUCT** — multiply paired arrays and sum; the BOM-explosion engine.
- **MAX(0, ...)** — floor a value at zero; MRP net-requirement netting.
- **Cumulative % (anchored SUM)** — a running total via SUM(\$D\$2:D2) for Pareto/ABC.
- **MAPE** — Mean Absolute Percentage Error; forecast-accuracy score (Issue 18).
- **PivotTable** — drag-and-drop summary/cross-tab of a data table.
- **Conditional formatting** — colour cells by rule (e.g. red below reorder point).

FURTHER READING & SOURCING

This issue applies **standard Microsoft Excel functions** (current Excel; XLOOKUP and FORECAST.ETS require Excel 2016 or later) to the operations mathematics derived and sourced earlier in this series: inventory control (Issue 14 — EOQ, reorder point, safety stock, ABC), production planning (Issue 15 — takt, MPS, MRP/BOM explosion), and demand forecasting (Issue 18 — moving average, exponential smoothing, Holt-Winters, MAPE). All formulas shown are real Excel syntax. The limits-of-spreadsheets discussion and the move to MRP/ERP follow Issues 15–16 (including Tesla's reported use of Excel for one-off reports alongside its in-house Warp). Examples and cell references are illustrative and meant to be adapted.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION



APPENDIX & REFERENCE

Volume One: Issues 01-13

*A consolidated reference to the complete series — for every issue, a summary, its tools & product families, its terminology, and its links.
The whole factory, indexed.*

13 issues · 171 terminology entries · 92 tool & product families

Covering Tesla MOS, Siemens, communication protocols, Rockwell/Allen-Bradley, Beckhoff, motors & drives, logistics & inventory, sensors, conveyors, Atlas Copco, Festo, and Emerson.

The Whole Series, Indexed

This appendix consolidates all nineteen issues of *The Integration Layer* into one reference. Each issue below carries four sections: a **Summary** (what the issue covers), its **Tools & product families** (the products and systems profiled), its **Terminology** (the issue's Field Glossary), and its **Links** (the catalog and standards URLs cited). The colour chip beside each title matches that issue's theme. Concept/fundamentals issues (motors, logistics, sensors, conveyors) profile technologies rather than a single vendor's catalog, so they list fewer named products and, in some cases, no vendor URLs.

#	Issue title	Tools	Terms	Links
01	 Tesla MOS — The Operating System of the Factory	6	7	—
02	 The Siemens Stack Behind a Gigafactory	3	12	3
03	 The Languages of the Factory Floor	3	14	—
04	 The Rockwell Stack & the Language of CIP	6	15	2
05	 The Rockwell & Allen-Bradley Portfolio	16	12	2
06	 The Beckhoff Way — PC-based Control & EtherCAT	12	11	2
07	 Motors, Drives & the Art of Controlled Motion	3	13	3
08	 Logistics, Inventory & the Art of Not Holding Stock	2	12	—
09	 The Sensors That Watch Every Car Get Built	8	16	—
10	 The Lines That Move the Line	10	14	—
11	 Atlas Copco — Torque, Air & the Perfect Joint	9	15	2
12	 Festo — Pneumatics, Motion & the Grip	10	15	1
13	 Emerson — One Company, a Dozen Names	10	15	2
14	 Material Handling, Flow & Inventory Control	8	22	—
15	 Production Planning — Plan, Schedule, MRP & Heijunka	2	20	—
16	 Tesla's Real Software Stack — Warp, MES & MOS	3	10	—
17	 KPIs, OEE & Measuring a Plant	4	17	—
18	 Demand — Orders, Forecasting & What Feeds Them	5	17	—
19	 Excel for Inventory & Production Planning	10	16	1

Counts are the number of product/tool families, terminology entries, and catalog links captured per issue.

01 Tesla MOS — The Operating System of the Factory

SUMMARY

Tesla's in-house Manufacturing Operating System: the software that orchestrates equipment, captures every result, binds it to each vehicle's VIN, and gates progress. The cover story of the series, tracing why Tesla built rather than bought its factory OS.

TOOLS & PRODUCT FAMILIES

Manufacturing Operating System (MOS) — Tesla · in-house factory OS

MES / MOS execution layer — binds operations to each VIN

ERP integration — enterprise planning above the floor

Vehicle firmware / Tesla OS — the product as a networked computer

Executive factory dashboard — real-time plant-wide KPIs

Operator & Garage portals — station-level guidance & service data

TERMINOLOGY

MOS (Manufacturing Operating System) — Tesla's in-house software that runs the factory: orchestrating equipment, capturing results, binding them to each vehicle, and gating progress. Its exact internal design is not public.

MES (Manufacturing Execution System) — the general class of software that manages and records shop-floor production; MOS is Tesla's MES-class platform.

ERP (Enterprise Resource Planning) — business-wide planning software (e.g. SAP, Oracle); the kind of thing most manufacturers buy rather than build.

Tesla OS / vehicle firmware — the software *inside the car*; distinct from MOS, which is the software that *builds* the car.

As-built record / genealogy — the complete, linked history of how one specific vehicle was made, assembled from every captured result.

Factory mode — a testing approach that allows incremental change to the line without breaking the whole.

Vertical OS — the idea of treating factory, product, and cloud as one integrated system with a fast feedback loop.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

02 The Siemens Stack Behind a Gigafactory

SUMMARY

A family-by-family tour of the Siemens automation portfolio — the S7-1500 controller, SINAMICS drives and SIMOTICS motors, PROFINET/SCALANCE networking, and the TIA Portal that unifies them under 'Totally Integrated Automation'.

TOOLS & PRODUCT FAMILIES

SIMATIC S7-1500 — CPU range — SIMATIC · order prefix 6ES7

SINAMICS — the drive tiers — SINAMICS · order prefix 6SL3

SCALANCE — the network hardware — SCALANCE · industrial Ethernet

TERMINOLOGY

SIMATIC S7-1500 — Siemens' flagship modular PLC; the controller that runs a cell or line in real time.

ET 200 — the distributed I/O family, mounted near the machine and linked over PROFINET.

SINAMICS — the drive family (S120, S210, V90, G120) that controls motor speed, torque, and position.

SIMOTICS — the electric-motor family, matched to SINAMICS drives.

PROFINET — Siemens' industrial-Ethernet network for real-time control traffic.

SCALANCE — the network hardware: switches, routers, wireless, and firewalls.

SIRIUS — low-voltage controls: contactors, overload relays, safety relays.

SITOP — the power-supply family delivering stable 24 V DC.

SIMATIC Ident — RFID and optical code readers for traceability.

TIA Portal — the integrated engineering software that configures the whole stack from one project.

DRIVE-CLiQ — the digital interface linking Siemens drives, motors, and encoders.

OPC UA — a secure, structured-data standard, built into the S7-1500, bridging the floor to IT.

LINKS

[siemens.com/global/en/products/automation/systems/industrial/plc.html](https://www.siemens.com/global/en/products/automation/systems/industrial/plc.html)
[siemens.com/global/en/products/drives/sinamics.html](https://www.siemens.com/global/en/products/drives/sinamics.html)
mall.industry.siemens.com

03 The Languages of the Factory Floor

SUMMARY

A field guide to the communication protocols connecting a PLC to every device — determinism, PROFINET (RT/IRT), the fieldbus landscape (EtherNet/IP, EtherCAT, Modbus), IO-Link at the last metre, and OPC UA / MQTT bridging the floor to IT.

TOOLS & PRODUCT FAMILIES

PROFINET — Industrial Ethernet · the control backbone

IO-Link — IEC 61131-9 · the last-metre link

OPC UA — IEC 62541 · the structured bridge

TERMINOLOGY

Determinism — the guarantee that a message arrives within a known, bounded time, every time; the defining requirement of control networks.

Fieldbus — the general family of digital industrial networks connecting field devices to controllers.

PROFINET — Siemens' industrial-Ethernet control network, with real-time (RT) and isochronous (IRT) classes.

PROFIBUS — the older serial fieldbus that PROFINET succeeds; still common on existing lines.

PROFIsafe — a profile carrying safety signals over the same network as normal data.

PROFIdrive — a profile defining controller-to-drive communication for synchronised motion.

EtherNet/IP — the Rockwell-world industrial-Ethernet protocol, built on CIP.

EtherCAT — an ultra-low-latency Ethernet protocol favoured for high-performance motion.

Modbus TCP — a simple, near-universal protocol for basic and legacy devices.

IO-Link — a point-to-point digital link giving individual sensors and valves a two-way voice (value, health, config).

OPC UA — a secure, structured, self-describing standard bridging the floor to IT.

MQTT / Sparkplug B — a lightweight publish/subscribe protocol (and its industrial profile) for moving telemetry to the cloud.

Gateway — a device that translates between two incompatible protocols.

Unified Namespace — a single, live, structured picture of an operation, often built on MQTT/OPC UA.

LINKS

No vendor catalog URLs — governed by standards bodies: PI (PROFIBUS & PROFINET International), OPC Foundation (IEC 62541 / OPC UA), ODVA (EtherNet/IP & CIP), EtherCAT Technology Group, and OASIS + Eclipse (MQTT / Sparkplug).

04 The Rockwell Stack & the Language of CIP

SUMMARY

Allen-Bradley / Rockwell hardware and its defining idea: a single application protocol, CIP, carried over EtherNet/IP for control, motion, and safety alike — the Logix controllers, Device Level Ring, and the safety-in-the-same-chassis approach.

TOOLS & PRODUCT FAMILIES

ControlLogix 5580 — Allen-Bradley · 1756 · chassis-based
CompactLogix 5380 & 5480 — Allen-Bradley · 5069 · DIN-rail
EtherNet/IP + CIP — ODVA · the Rockwell house network
Kinetix — Allen-Bradley · servo drives
GuardLogix & Compact GuardLogix — Allen-Bradley · safety controllers
Compact 5000 & FLEX I/O — Allen-Bradley · distributed I/O

TERMINOLOGY

Allen-Bradley — Rockwell Automation's brand for its industrial control hardware.
Integrated Architecture — Rockwell's name for its unified portfolio, the parallel to Siemens TIA.
Logix engine — the common control engine shared across all modern Logix controllers.
ControlLogix (1756) — the large, chassis-based flagship controller.
CompactLogix (5069) — the compact, DIN-rail controller; the 5480 adds embedded Windows.
GuardLogix — the safety controller running safety logic in the same chassis, up to SIL 3 / PLe.
Studio 5000 Logix Designer — the single engineering software for all Logix controllers.
CIP (Common Industrial Protocol) — the one application protocol carrying control, motion, and safety as profiles.
EtherNet/IP — CIP carried over standard Ethernet; "IP" means Industrial Protocol.
CIP Motion / CIP Safety / CIP Sync — the motion, safety, and time-sync profiles of CIP.
Device Level Ring (DLR) — a ring topology that survives a single cable break without stopping.
Kinetix — the servo-drive family for integrated motion over EtherNet/IP.
Compact 5000 / FLEX I/O — the distributed I/O families.
FactoryTalk / PanelView — the SCADA/analytics suite and the panel-HMI family.
ODVA — the open standards body governing CIP and EtherNet/IP.

LINKS

rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html
odva.org

05 The Rockwell & Allen-Bradley Portfolio

SUMMARY

The full Allen-Bradley catalog, layer by layer: controllers, distributed I/O, PowerFlex VFDs, Kinetix motion, Stratix networking, GuardLogix safety, PanelView/FactoryTalk visualization, components, and the Studio 5000 software that ties it together.

TOOLS & PRODUCT FAMILIES

ControlLogix 5580 — Allen-Bradley · 1756 · flagship
CompactLogix 5380 & 5480 — Allen-Bradley · 5069 · compact
Micro800 & MicroLogix — Allen-Bradley · micro & nano control
Compact 5000 I/O — Allen-Bradley · 5069
FLEX 5000 & POINT I/O — Allen-Bradley · distributed I/O
PowerFlex 520-Series — Allen-Bradley · compact VFD
PowerFlex 753 / 755 / 755T — Allen-Bradley · high-performance VFD
Kinetix 5500 / 5700 — Allen-Bradley · servo drives
Kinetix 7000 & servo motors — Allen-Bradley · high-power servo & motors
Stratix Industrial Switches — Allen-Bradley · powered by Cisco
GuardLogix & Compact GuardLogix — Allen-Bradley · safety controllers
Guardmaster Safety Devices — Allen-Bradley · safety field devices
PanelView 5000 & 800 — Allen-Bradley · operator terminals (HMI)
FactoryTalk View — Rockwell · HMI / SCADA software
Studio 5000 Logix Designer — Rockwell · engineering software
FactoryTalk Suite — Rockwell · operations & information software

TERMINOLOGY

Integrated Architecture — Rockwell's name for its unified, connect-everything portfolio.
Logix engine — the shared control engine inside all modern Logix controllers.
ControlLogix / CompactLogix / GuardLogix / Micro800 — the large, compact, safety, and micro controllers.
Compact 5000 / FLEX 5000 / POINT I/O — the distributed I/O platforms.
PowerFlex — the VFD (variable frequency drive) family.
Kinetix — the servo-drive family for integrated motion over EtherNet/IP.
Stratix — the Cisco-built industrial Ethernet switch family.
EtherNet/IP & CIP — the network transport and the one application protocol it carries.
Guardmaster — the safety field-device line (relays, curtains, interlocks).
PanelView — the operator-terminal (HMI) family.
FactoryTalk — the visualization, historian, and analytics software suite.
Studio 5000 — the single engineering environment for the whole catalog.

LINKS

rockwellautomation.com/en-us/products/hardware/allen-bradley.html
odva.org

06 The Beckhoff Way — PC-based Control & EtherCAT

SUMMARY

Beckhoff's contrarian philosophy: the industrial PC *is* the controller, with PLC, motion, vision, and safety all running as software tasks in TwinCAT over EtherCAT — plus the unique XTS and XPlanar 'flying-magnet' motion systems.

TOOLS & PRODUCT FAMILIES

C60xx Ultra-Compact IPC — Beckhoff · cabinet Industrial PC
Embedded PC (CX series) — Beckhoff · DIN-rail controller
C7015 & Panel PCs — Beckhoff · on-machine & HMI
EtherCAT — EtherCAT Technology Group · the real-time bus
EtherCAT Terminals (EL series) — Beckhoff · in-cabinet I/O
EtherCAT Box (EP / EPP series) — Beckhoff · on-machine I/O (IP67)
AX8000 Multi-Axis Servo System — Beckhoff · high-performance servo
AMP8000 Distributed Servo System — Beckhoff · drive-in-motor
Servomotors & linear motors — Beckhoff · AM, AL, AA families
XTS — eXtended Transport System — Beckhoff · linear transport
XPlanar — Planar Motor System — Beckhoff · 2D levitating motion
TwinSAFE — Beckhoff · functional safety

TERMINOLOGY

PC-based control — Beckhoff's founding philosophy: a standard industrial PC *is* the controller, running control as software.
EtherCAT — the real-time industrial-Ethernet bus Beckhoff invented (2003); "processing on the fly" gives microsecond cycles.
Distributed Clocks — EtherCAT's mechanism for synchronising all devices to within nanoseconds.
EtherCAT G / EtherCAT P — the gigabit variant, and the variant carrying power + data on one cable.
Industrial PC (C60xx / CX / C7015) — the cabinet, DIN-rail embedded, and on-machine computers that run TwinCAT.
EtherCAT Terminals (EL) / EtherCAT Box — the slim in-cabinet I/O and the IP67 on-machine I/O.
AX8000 / AMP8000 — the multi-axis servo system, and the distributed drive-in-motor system.
XTS — the eXtended Transport System: independently controlled magnetic movers on a track.
XPlanar — the planar motor system: magnetic movers levitating and moving freely in 2D.
TwinCAT 3 — the engineering environment and real-time runtime, built in Visual Studio, running PLC, motion, vision, and more.
TwinSAFE — Beckhoff's EtherCAT-based functional-safety system.

LINKS

beckhoff.com/en-en/products/
ethercat.org

07 Motors, Drives & the Art of Controlled Motion

SUMMARY

A cross-vendor fundamentals issue built on one distinction — open loop (VFD) vs closed loop (servo): how a VFD makes adjustable AC, how a servo positions to the micron, and why the motor type (induction vs permanent-magnet) decides the drive type.

TOOLS & PRODUCT FAMILIES

The AC Induction Motor — the workhorse · squirrel-cage rotor
The Permanent-Magnet (PM) Servo Motor — the precision machine
The Closed-Loop Servo — controller · drive · motor · encoder

TERMINOLOGY

VFD (Variable Frequency Drive) — a drive that controls an AC motor's speed by synthesising AC at an adjustable frequency; also called an inverter or AC drive.

Servo drive / servo system — a closed-loop system (controller + drive + motor + feedback) for precise position, velocity, and torque control.

Open loop / closed loop — whether the system continuously checks the motor's actual output (closed) or not (open); the defining VFD-vs-servo distinction.

Stator / rotor — the stationary windings that create the rotating field, and the rotating part that follows it.

Induction motor — an AC motor whose cage rotor follows the field by induced current, always lagging by slip.

Permanent-magnet (PM) motor — a synchronous motor with magnets in the rotor, locking it to the field; the basis of servo motors.

Slip — the small speed gap by which an induction rotor lags the stator field.

Rectifier / DC bus / inverter — the three stages inside a VFD: AC→DC, smoothing, DC→adjustable AC.

PWM (Pulse-Width Modulation) — the technique of switching the DC bus into pulses that average to a sine wave at the desired frequency.

V/Hz & vector control — the ladder of VFD control methods, from a simple voltage/frequency rule to a full motor model.

Field weakening — running a motor above base speed by accepting reduced torque.

Encoder / resolver — the feedback device reporting a servo motor's actual position.

Constant- vs variable-torque load — loads needing the same torque at all speeds (conveyor) vs much less at low speed (pump/fan, the cube law).

LINKS

[siemens.com](https://www.siemens.com)

[rockwellautomation.com](https://www.rockwellautomation.com)

[beckhoff.com](https://www.beckhoff.com)

08 Logistics, Inventory & the Art of Not Holding Stock

SUMMARY

How a Gigafactory keeps parts flowing just-in-time: push vs pull, the two-bin kanban, the WMS, barcode/RFID tracking and cycle counting, ABC analysis, vertical integration, and warehouse automation (AGV/AMR/AS-RS).

TOOLS & PRODUCT FAMILIES

Warehouse Management System (WMS) — the material operating system

The Automated Warehouse Toolkit — material handling automation

TERMINOLOGY

Just-in-Time (JIT) — supplying parts precisely when needed to minimise stored inventory.

Just-in-Sequence (JIS) — JIT plus arriving in the exact order the line will use them.

Push vs pull — building to forecast (push) versus building only when downstream demand signals (pull).

Kanban — a visual signal authorising replenishment; the two-bin form reorders automatically when a bin empties.

WMS (Warehouse Management System) — software tracking every item by location and directing all warehouse moves.

ERP — enterprise planning software deciding what to order and make; sits above the WMS.

Cycle counting — continuously counting small subsets of stock to keep records accurate, instead of one annual count.

ABC analysis — sorting items by value so control effort focuses on the high-value few.

Reorder point / lead time / safety stock — the level that triggers an order, the time a supplier takes, and the deliberate buffer against variability.

Inventory turns — how many times a year stock is used and replaced; higher is leaner.

Vertical integration — making components in-house to control and shorten the supply chain.

AGV / AMR / AS/RS — automated guided vehicles, autonomous mobile robots, and automated storage/retrieval systems.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

09 The Sensors That Watch Every Car Get Built

SUMMARY

A cross-vendor tour of the factory's sense organs, grouped by the question each answers: presence/proximity (inductive, capacitive, magnetic), photoelectric & distance, machine vision, force/torque/pressure, position/temperature, and the IO-Link smart sensor.

TOOLS & PRODUCT FAMILIES

Inductive Proximity Sensor — detects metal · the floor's workhorse

Capacitive Proximity Sensor — detects almost anything

Magnetic (Hall / Reed) Sensor — detects a magnetic field

Machine Vision System — camera + lighting + processing

Torque Transducer (the nutrunner sensor) — strain gauges · the safety-critical bolt

Load Cell & Force Sensor — press force · assembly verification

Pressure Sensor — pneumatics · process · 4–20 mA

The IO-Link Smart Sensor — IEC 61131-9 · the last metre, revisited

TERMINOLOGY

Proximity sensor — detects presence without contact; inductive, capacitive, or magnetic.

Inductive sensor — detects metal via an electromagnetic field; the rugged floor workhorse.

Capacitive sensor — detects almost any material (incl. levels) via capacitance change.

Magnetic (Hall / reed) sensor — detects a magnetic field; standard for cylinder position.

Photoelectric sensor — detects objects with a light beam; through-beam, retro-reflective, or diffuse.

Laser distance sensor — measures precise distance by triangulation or time-of-flight.

Machine vision — a camera + processing system that reads codes, inspects, and guides.

Torque transducer — strain-gauge sensor measuring fastening torque (and, with an encoder, angle).

Load cell — a strain-gauge sensor measuring force or load.

Pressure sensor — measures pneumatic or fluid pressure, often as a 4–20 mA signal.

Encoder — reports exact position, angle, or speed; closes every servo loop.

LVDT — a precise linear-displacement sensor for dimensional gauging.

Ultrasonic sensor — measures distance to any material via sound time-of-flight.

RTD / thermocouple — temperature sensors for processes, motors, and fluids.

IO-Link / smart sensor — a two-way digital link giving a sensor value, health, and remote config.

IP rating — ingress protection (e.g. IP69K) defining dust/water resistance.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

10 The Lines That Move the Line

SUMMARY

The material-transport backbone of assembly, built on the synchronous-vs-asynchronous (paced vs accumulating) distinction: belt/roller/chain/slat conveyors, skids and skillets, overhead Power & Free and EMS, transfers, and the shift to AGVs and Tesla's 'Unboxed' idea.

TOOLS & PRODUCT FAMILIES

Belt Conveyor — continuous belt · parts & totes

Roller Conveyor — gravity or powered rollers

Chain & Slat Conveyor — heavy loads · mainline duty

The Skid — body-in-white · the body shop & paint

The Skillet — general assembly · trim & final line

Power & Free (P&F) — overhead · the automotive workhorse

Electrified Monorail System (EMS) — overhead · independently powered carriers

The Drive: Gearmotors & VFDs — motion, applied to a belt

Tracking & Identification — knowing which body is where

AGVs & AMRs — conveyance without a conveyor

TERMINOLOGY

Synchronous (paced) line — all units locked to one speed, moving as a single body matched to takt.

Asynchronous (accumulating) line — units that stop, queue, and release independently.

Accumulation — letting units queue on the conveyor as a physical buffer between stations.

Takt time — the line's target pace, the drumbeat the conveyor is set to (Issue 01).

Belt / roller / chain / slat conveyor — the floor-level families, by load and duty.

Skid — the rigid steel fixture carrying a bare body through body shop and paint.

Skillet — the general-assembly mainline carrier with an onboard scissor lift; often inductively powered.

Body marriage (decking) — the station where the upper body is joined to the chassis/powertrain.

Power & Free (P&F) — a two-track overhead conveyor whose carriers disengage and accumulate via pneumatic stops.

EMS (Electrified Monorail System) — an overhead system of individually powered, individually controlled carriers.

Shuttle / fork transfer / turntable — mechanisms that move carriers sideways, between conveyors, or rotate them.

Gearmotor — an induction motor plus gearbox, the typical conveyor drive (speed set by a VFD).

AGV / AMR — automated guided vehicles and autonomous mobile robots: conveyance with no fixed track.

"Unboxed" — Tesla's stated direction of building modules in parallel cells and joining them late, shrinking the linear line.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

11 Atlas Copco — Torque, Air & the Perfect Joint

SUMMARY

The company behind the fastening (and the plant's air): smart Tensor nutrunners that tighten a safety-critical bolt to exact torque and angle, Power Focus controllers that prove and log every joint, and oil-free (Class 0) compressors and vacuum. Answers 'what tightened Issue 01's 46.2 N·m?'

TOOLS & PRODUCT FAMILIES

Tensor STB / SRB (cordless) — Atlas Copco · battery nutrunner

Tensor ES / STR (cabled) — Atlas Copco · corded nutrunner

Tensor SB / fixtured spindles — Atlas Copco · handheld & automated

Power Focus & MTF6000 controllers — Atlas Copco · tightening controllers

Pneumatic Nutrunners (LTV, RTP) — Atlas Copco · air tools

Low-Torque Screwdrivers — Atlas Copco · small fasteners

ZR / ZT Oil-Free Rotary Compressors — Atlas Copco · oil-free, Class 0

GA Oil-Injected & ZH Centrifugal — Atlas Copco · general & high-flow air

Industrial Vacuum Pumps — Atlas Copco · vacuum technique

TERMINOLOGY

- Nutrunner** — a powered tool for tightening threaded fasteners to a controlled torque (and angle).
- Clamp load** — the actual clamping force a tightened bolt produces; the real goal, which torque only approximates.
- Torque & angle monitoring** — measuring both turning effort and rotation to verify a joint, catching faults torque alone misses.
- Transducerised tool** — a tool with a built-in torque transducer (Issue 09) for measured, verified tightening.
- Tensor** — Atlas Copco's flagship family of intelligent electric nutrunners.
- Tensor IxB** — the Tensor line with the controller built into the tool, removing the separate controller box.
- TurboTight / Tensor Pulse / TBP** — low-reaction tightening strategies for speed and operator ergonomics.
- TrueAngle** — angle compensation that guards against the operator moving during tightening.
- Power Focus / MTF6000** — Atlas Copco tightening controllers that program, judge, and log fastenings.
- Error-proofing** — features that prevent and detect mistakes (refusing bad joints, counting bolts, guiding the operator).
- Joint traceability** — recording every fastening result and binding it to the vehicle's VIN.
- Oil-free / Class 0** — compressed air certified to contain no oil — essential for paint and battery processes.
- VSD (Variable Speed Drive)** — matching a compressor's motor speed to air demand to save energy (Issue 07).
- ZR/ZT, GA, ZH** — oil-free rotary, oil-injected, and centrifugal compressor ranges.
- Industrial vacuum** — pumps that power suction grippers and hold parts; the inverse of compressed air.

LINKS

atlas-copco.com/en-us/itba
atlas-copco.com/en-us/compressors

12 Festo — Pneumatics, Motion & the Grip

SUMMARY

The pneumatics-and-motion house at the hinge between air and electric: cylinders, valve terminals (and the app-controlled VTEM Motion Terminal), the CPX/CPX-AP platform, CMMT servo drives, electric axes, grippers, and vacuum — and the steady migration from pneumatic to electric motion.

TOOLS & PRODUCT FAMILIES

- ISO & Compact Cylinders (DSBC, DNC, ADVU)** — Festo · piston-rod cylinders
- Guided Drives & Mini Slides (DGSL)** — Festo · guided linear motion
- Rodless Cylinders & Rotary (DGC, swivel)** — Festo · long-travel & rotation
- Valve Terminals (VTUG, VTSA, MPA, CPV)** — Festo · valve manifolds
- CPX / CPX-AP Automation Platform** — Festo · electrical backplane & remote I/O
- CMMT-AS / CMMT-ST Servo Drives** — Festo · multiprotocol servo drives
- Electric Axes & Actuators (EGC, mini slides)** — Festo · electromechanical motion
- Grippers (pneumatic & electric)** — Festo · gripping technology
- Vacuum Technology** — Festo · suction handling
- Air Preparation (FRL: MS, LFR)** — Festo · service units

TERMINOLOGY

Pneumatic cylinder — a piston in a tube pushed by compressed air; the basic unit of pneumatic motion.

ISO / compact cylinder (DSBC, ADVU) — standardised piston-rod cylinders for a straight push.

Guided drive / mini slide (DGSL) — a linear motion with guide rails so it cannot twist.

Rodless cylinder (DGC) — a cylinder whose load rides along the body, for long travels.

Solenoid valve — an electrically switched valve that directs air to a cylinder.

Valve terminal (VTUG/VTSA/MPA/CPV) — a manifold packing many valves into one networked block.

CPX / CPX-AP — Festo's electrical backplane and automation platform linking valves, I/O, and IO-Link to the network.

Motion Terminal (VTEM) — a valve terminal whose valve functions are defined by software apps ("digital pneumatics").

FRL / service unit (MS, LFR) — Filter-Regulator-Lubricator that conditions air before use.

PPS cushioning — Festo's self-adjusting end-of-stroke cushioning for smooth cylinder stops.

Simplified Motion Series — electric actuators designed to be as easy to set up as a pneumatic cylinder.

CMMT-AS / -ST — Festo's multiprotocol servo drives (EtherNet/IP, EtherCAT, PROFINET, Modbus TCP).

Electric axis (EGC) — an electromechanical linear actuator for programmable positioning.

Gripper — the pneumatic or electric "hand" that grips and orients a part.

Vacuum generator / gripper — venturi-based suction handling for flat or delicate parts.

LINKS

festo.com/us/en/c/products/industrial-automation-id_pim1/

13 Emerson — One Company, a Dozen Names

SUMMARY

Not one brand but a federation of famous names — AVENTICS pneumatics, ASCO valves, Rosemount measurement, PACSystems control, Branson joining, and NI test — united by the 'floor-to-cloud' data idea, spanning discrete automation, process measurement, and test.

TOOLS & PRODUCT FAMILIES

AVENTICS Series XV Valves — Emerson · next-gen valve platform

AVENTICS AV Valve System & SPRA actuator — Emerson · modular valves & electric

ASCO Solenoid & Process Valves — Emerson · fluid control valves

TESCOM Regulators & TopWorx Sensing — Emerson · pressure & position

Rosemount 3051 Pressure Transmitter — Emerson · pressure measurement

Rosemount Temperature, Level & Flow — Emerson · the instrument family

PACSystems RSTi-EP CPE200 & RX3i — Emerson · programmable controllers

PACSystems Edge: IPCs & PACEdge — Emerson · edge computing

NI (National Instruments) Test & Measurement — Emerson · automated test

Branson Welding & Joining — Emerson · ultrasonic & laser joining

TERMINOLOGY

Emerson — an industrial-automation powerhouse structured as a federation of acquired brands.

AVENTICS — Emerson's pneumatics brand (valves, cylinders, actuators, vacuum).

Series XV / AV / SPRA — AVENTICS's high-flow valve platform, modular valve system, and electric rod actuator.

ASCO — Emerson's solenoid-valve brand for controlling air, gas, water, and fluids.

TESCOM — high-precision pressure regulators.

TopWorx — valve position monitors and switches.

Rosemount — Emerson's process-measurement brand (pressure, temperature, level, flow).

Rosemount 3051 — the industry-standard pressure/DP/level/flow transmitter.

Micro Motion — Coriolis mass-flow measurement.

PACSystems — Emerson's controller brand (PLC/PAC, RSTi-EP CPE200, RX3i, IPCs).

PAC Machine Edition (PME) — the programming software for PACSystems controllers.

PACEdge / Movicon — Emerson's edge software and SCADA/HMI.

Branson — Emerson's ultrasonic and laser welding/joining brand.

NI (National Instruments) — Emerson's test-and-measurement business (LabVIEW, automated test).

Floor to cloud — Emerson's framing for connecting machine data from the floor up to cloud analytics.

LINKS

emerson.com/en-us/automation

ni.com

14 Material Handling, Flow & Inventory Control

SUMMARY

The deep three-layer discipline of production logistics: the equipment that physically moves and stores material (trucks, tuggers, AGVs, AS/RS, cranes, racking), the design of material flow (line-side delivery, milk runs, supermarkets, the water spider, kitting, FIFO), and the quantitative mathematics of inventory control (EOQ, reorder point, safety stock, ABC/XYZ, the bullwhip effect).

TOOLS & PRODUCT FAMILIES

Industrial Trucks & Lift Trucks — transport · the flexible workhorse

Tuggers & Tugger Trains — transport · the milk-run engine

AGVs, AMRs & Conveyors — transport · automated movement

AS/RS & Automated Storage — storage · dense, automatic

Cranes, Hoists & Monorails — lifting · the heavy & the overhead

Racking, Storage Systems & Unit Loads — storage · the passive backbone

The Milk Run (Tugger-Train Replenishment) — flow · routed, timed delivery

Supermarkets & the Water Spider — flow · line-side replenishment

TERMINOLOGY

Material handling — the movement, protection, storage, and control of material through manufacturing.

Ten Principles of Material Handling — the standard design checklist (planning, standardization, work, ergonomic, unit load, space, system, automation, environmental, life-cycle cost).

Work Principle — handling work = flow × distance; minimise both.

Unit load — material handled in standardised batches (pallets, totes) rather than piece by piece.

AS/RS — Automated Storage & Retrieval System: robotic dense storage and goods-to-person picking.

VLM / carousel — vertical lift module and carousel "goods-to-person" storage units.

Tugger / milk run — a tractor towing carts on a fixed route and timetable, delivering small lots to many stops.

Supermarket — a kanban-controlled line-side store of parts the line pulls from.

Water spider (mizusumashi) — a dedicated material handler who replenishes line-side on a standard route.

Kitting — gathering all parts for one unit into a single delivered kit.

Sequencing (JIS) — delivering parts in the exact order of consumption.

FIFO lane — a length-capped buffer enforcing first-in-first-out between processes.

Flow vs functional layout — arranging stations by product sequence vs grouping like machines.

EOQ (Wilson formula) — $\sqrt{(2DS/H)}$; the order size minimising ordering + holding cost.

Carrying / holding cost — the annual cost of holding inventory, typically 20–30% of its value.

Reorder point (ROP) — the stock level that triggers an order: lead-time demand + safety stock.

Safety stock — a buffer against variability, sized by service level (z) and demand deviation.

Service level — the probability of not stocking out; higher levels cost disproportionate safety stock.

Inventory turns / days of supply — $\text{COGS} \div \text{average inventory}$; and $365 \div \text{turns}$.

ABC / XYZ analysis — classifying items by value (ABC) and demand variability (XYZ) to focus control.

Bullwhip effect — demand variability amplifying upstream through a supply chain.

EPQ — Economic Production Quantity: the EOQ variant for produced (not purchased) parts.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

15 Production Planning — Plan, Schedule, MRP & Heijunka

SUMMARY

The factory's planning brain: the cascade from a yearly business plan down to the exact car built next — aggregate planning (level vs chase), the master production schedule and time fence, MRP exploding the bill of materials, takt time as the customer's drumbeat, and Toyota-style heijunka leveling of volume and mix.

TOOLS & PRODUCT FAMILIES

Master Production Schedule (MPS) — the buildable commitment

Material Requirements Planning (MRP) — the part-ordering calculator

TERMINOLOGY

- Production planning** — deciding what to make, how much, and when, by matching demand to capacity across time horizons.
- Planning hierarchy** — the cascade from business plan → aggregate plan → master schedule → execution, each more detailed and binding.
- Sales & Operations Planning (S&OP)** — the long-range business-level reconciliation of demand and supply.
- Aggregate production plan** — a medium-term plan of output rate, workforce, and inventory by product family.
- Level strategy** — constant output, absorbing demand swings with inventory.
- Chase strategy** — varying output and workforce to match demand, minimising inventory.
- Master Production Schedule (MPS)** — the committed quantity of each end product per near-term period.
- Time fence** — the frozen near-term portion of the schedule, protected from change.
- Rough-cut capacity planning** — checking the MPS against available capacity.
- MRP (Material Requirements Planning)** — exploding the MPS through the BOM and netting inventory to produce timed part orders.
- Bill of Materials (BOM)** — the structured list of everything a product is made of.
- MRP II / ERP / APS** — MRP extended with resources; the company-wide system; and finite-capacity advanced scheduling.
- Takt time** — available time ÷ demand; the required pace of production set by the customer.
- Cycle time** — the actual time a process takes per unit; must be ≤ takt.
- Line balancing** — distributing work across stations so each fits under takt.
- Heijunka** — production leveling, smoothing both volume and the mix of variants.
- Heijunka box** — a grid of kanban cards releasing a leveled, mixed sequence to the line.
- Mixed-model production** — building several variants on one line in an interleaved sequence.
- Push vs pull** — building to a computed plan (push) vs to actual consumption (pull); used together.
- MTO vs MTS** — make-to-order (build on demand) vs make-to-stock (build to forecast).

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

16 Tesla's Real Software Stack — Warp, MES & MOS

SUMMARY

Tesla's real, named software, assembled from public reporting: Warp (the in-house ERP/planning system, built in ~4 months instead of SAP), MES (the production 'air traffic controller'), MOS (the floor-level operating system tracking each car's build, parts, torque, quality and material flow via barcode/RFID), and the database-and-dashboard layer — all bolted to the machines under a build-don't-buy philosophy.

TOOLS & PRODUCT FAMILIES

- Warp (formerly Warpdrive)** — Tesla · the in-house ERP / planning system
- Tesla MES (Manufacturing Execution System)** — Tesla · real-time production control
- Tesla MOS (Manufacturing Operating System)** — Tesla · the floor-level operating system

TERMINOLOGY

Warp (formerly Warpdrive) — Tesla's in-house ERP/planning system: supply chain, product planning, inventory, sales orders, purchasing, finance, HR.

MES (Manufacturing Execution System) — Tesla's real-time production "air traffic controller": routing, production orders, quality, measurements.

MOS (Manufacturing Operating System) — the floor-level system workers use to record each car's build, parts, torque, quality, and test state.

Material tracking — MOS following raw materials and finished goods in real time via barcode and RFID.

Schedule-based replenishment — a reported (and contested) internal name for digital kanban / pull logic within MOS.

Build vs buy — the choice to write software in-house rather than license a vendor package like SAP.

Vertical software integration — building and owning the software stack, wired directly to the machines.

The data layer — Oracle/SQL Server, an in-house MySQL test database, QuickBase, BI dashboards, and Excel.

Chain of startups — Musk's framing of Tesla as many venture-like units, which the in-house software culture supports.

Closed loop — plan (Warp) → execute (MES/MOS) → measure (data) → re-plan.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

17 KPIs, OEE & Measuring a Plant

SUMMARY

Measuring the factory: OEE (Availability × Performance × Quality) as the single most important metric, the Six Big Losses framework and TEEP, and the wider scorecard — first-pass yield, scrap/PPM, throughput, cycle time, MTBF/MTTR, schedule adherence — plus what is publicly known about Tesla's data-driven dashboards and production-rate targets.

TOOLS & PRODUCT FAMILIES

First-Pass Yield (FPY) — quality · right the first time

Scrap Rate, Rework Rate & Defect PPM — quality · the cost of getting it wrong

Throughput & Cycle Time — flow · how much, how fast

MTBF & MTTR — reliability · the maintenance pair

TERMINOLOGY

KPI (Key Performance Indicator) — a measured number with an owner and a triggered action; otherwise just information.

OEE (Overall Equipment Effectiveness) — Availability × Performance × Quality; the share of planned time that is fully productive.

Availability — run time ÷ planned production time (the stops factor).

Performance — actual vs ideal output rate (the speed factor).

Quality — good count ÷ total count (the defect factor).

Six Big Losses — breakdowns, setup/changeover, idling/minor stops, reduced speed, process defects, reduced yield.

TEEP — Total Effective Equipment Performance: OEE × utilization, against all calendar time.

First-Pass Yield (FPY) — units made right the first time, excluding rework.

Rolled Throughput Yield (RTY) — the product of FPYs across a sequence of steps.

Scrap / rework rate — discarded or re-fixed units as a share of total.

Defect PPM — defects per million; Six Sigma world-class ≈ 3.4 PPM.

Throughput — good units produced per unit of time.

Cycle time — actual time to produce one unit; compared to takt.

MTBF — Mean Time Between Failures (reliability; higher is better).

MTTR — Mean Time To Repair (recovery; lower is better).

Schedule adherence — orders completed on schedule ÷ total scheduled.

ISO 22400 — the international standard defining manufacturing KPI calculations.

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

18 Demand — Orders, Forecasting & What Feeds Them

SUMMARY

The demand signal that starts everything: the make-to-stock (forecast) vs build-to-order distinction, the quantitative and qualitative inputs of a forecast, time-series methods (moving average, exponential smoothing, Holt-Winters), forecast error (MAPE) and how it sizes safety stock — and Tesla's order-driven model, where the online configurator turns a confirmed order into a build, moving the forecast upstream to capacity and parts.

TOOLS & PRODUCT FAMILIES

Quantitative Inputs (the hard data) — inputs · what the numbers say

Qualitative Inputs (informed judgement) — inputs · what the numbers can't see

Moving Average — method · the simple smoother

Exponential Smoothing — method · the weighted workhorse

Tesla's Direct Build-to-Order Model — Tesla · the configurator as demand signal

TERMINOLOGY

Demand forecasting — predicting future demand to drive planning, procurement, and inventory.

Make-to-stock (MTS) — forecasting demand and building to finished stock ahead of orders.

Build-to-order (BTO / MTO) — building only after a confirmed customer order arrives.

80/20 (hybrid) model — stocking the popular minority of configurations, building the rest to order.

Quantitative forecasting — statistical methods using historical data (time series).

Qualitative forecasting — judgement-based methods (Delphi, expert/market input) when data is thin.

Level, trend, seasonality — the three signals extracted from demand history.

Moving average — forecasting with the mean of the last n periods; lags trends.

Exponential smoothing — a weighted average favouring recent demand, tuned by α .

Smoothing constant (α) — the 0–1 weight on the latest actual demand.

Holt / Holt-Winters — exponential smoothing extended for trend (Holt) and season (Holt-Winters).

MAPE — Mean Absolute Percentage Error; the standard forecast-accuracy score.

Forecast error (σ) — the variability of misses; it sizes safety stock (Issue 14).

Configurator — the online tool where a customer specifies a product, generating its BOM.

Order rate — how fast new confirmed orders arrive.

Backlog — confirmed orders not yet built; a real-time demand signal.

S&OP — Sales & Operations Planning: reconciling demand forecast with supply (Issue 15).

LINKS

No vendor catalog URLs cited in this issue — a concepts/fundamentals issue; see the vendors' technical documentation.

19 Excel for Inventory & Production Planning

SUMMARY

The practical layer: how the inventory and planning mathematics of the series becomes real Excel formulas — EOQ, reorder point and safety stock (SQRT, NORM.S.INV, STDEV.S), ABC/Pareto classification, demand forecasting (FORECAST.ETS / Holt-Winters, MAPE), takt and an MRP-style BOM explosion (SUMPRODUCT) — plus PivotTables, the honest limits of spreadsheets, and when to graduate to MRP/ERP.

TOOLS & PRODUCT FAMILIES

SQRT — EOQ — Excel function · $=SQRT(2*D*S/H)$

NORM.S.INV — service-level z — Excel function · safety-stock factor

STDEV.S — demand variability — Excel function · the σ for safety stock

AVERAGE / SUMIFS — Excel functions · mean demand, usage by criteria

XLOOKUP (VLOOKUP) — Excel function · pull cost / lead time / BOM

IFS — ABC classification — Excel function · A/B/C thresholds

FORECAST.ETS — Holt-Winters — Excel function · trend + seasonality forecast

TREND / FORECAST.LINEAR — Excel functions · linear trend projection

SUMPRODUCT — BOM explosion — Excel function · parts $\times$ quantities

PivotTable & conditional formatting — Excel features · summarise & visual alerts

TERMINOLOGY

SUMIFS / AVERAGEIFS — add or average cells that meet given criteria.

XLOOKUP (VLOOKUP) — return a value from a table by matching a key (e.g. a part's cost).

IF / IFS — choose an output based on conditions (e.g. ABC class).

SQRT — square root; the core of the EOQ formula.

STDEV.S — sample standard deviation; the demand σ for safety stock.

NORM.S.INV(p) — the z-value for probability p; service-level factor (0.95 → 1.645).

AVERAGE — arithmetic mean; moving-average forecast and mean demand.

TREND / FORECAST.LINEAR — project a linear trend forward.

FORECAST.ETS — exponential triple smoothing (Holt-Winters) with automatic seasonality.

Forecast Sheet — Excel's one-click ETS forecast with confidence bounds and chart.

SUMPRODUCT — multiply paired arrays and sum; the BOM-explosion engine.

MAX(0, ...) — floor a value at zero; MRP net-requirement netting.

Cumulative % (anchored SUM) — a running total via `SUM($D$2:D2)` for Pareto/ABC.

MAPE — Mean Absolute Percentage Error; forecast-accuracy score (Issue 18).

PivotTable — drag-and-drop summary/cross-tab of a data table.

Conditional formatting — colour cells by rule (e.g. red below reorder point).

LINKS

microsoft.com/en-us/microsoft-365/excel

END OF APPENDIX

THE INTEGRATION LAYER · Appendix & Reference, Volume One (Issues 01–13) · A technical magazine on industrial automation.

Summaries, terminology, and links are drawn from the thirteen issues; vendor names and models are referenced descriptively from public product documentation. Tesla's internal systems are proprietary; examples are representative of automotive-class plants.

THE INTEGRATION LAYER

A TECHNICAL MAGAZINE ON INDUSTRIAL AUTOMATION



DEVICE & CATALOG-LINK INDEX

Every Device, Its Catalog Link

For each of the nineteen issues, every device and product family with a direct link to its official catalog page — organised journal by journal.

13 issues · 146 devices indexed · 97 catalog links

Links resolve to the official vendor catalog: a direct product page where verified, otherwise the product-family / category hub or the vendor's online catalog (e.g. Siemens Industry Mall) — the exact model/series name pinpoints the device. Concept & standards issues are marked, with standards-body or representative links where a single catalog does not exist.

Device & Catalog Links, Journal by Journal

This index lists, for every issue of *The Integration Layer*, each device or product family alongside a link to its official catalog. Where the exact product page was verified, it is given directly; otherwise the link points to the vendor's official product-family / category page or online catalog, and the **exact model/series name in the left column** pinpoints the device on that page. Eight issues profile specific vendor catalogs (Siemens, Rockwell ×2, Beckhoff, Atlas Copco, Festo, Emerson); the remaining eleven are concept / fundamentals issues whose "devices" are generic technologies, methods, communication standards, or proprietary systems — these are marked **CONCEPT / STANDARDS** and link to the governing standards body or a representative vendor where one applies, or are noted as having no single public catalog.

01

Tesla MOS — The Operating System of the Factory

CONCEPT / STANDARDS

Issue 01 profiles Tesla's *internal* manufacturing software and systems — proprietary and not sold from a public catalog.

Device / product	Catalog link
Manufacturing Operating System (MOS)	— no public catalog — Proprietary Tesla in-house system — no public catalog.
MES / execution layer	— no public catalog — Generic MES category (Tesla's is in-house).
ERP integration	— no public catalog — Enterprise software layer — vendor-dependent, not a floor device.
Vehicle firmware / Tesla OS	— no public catalog — Proprietary Tesla software.
Executive dashboard / Garage & Operator portals	— no public catalog — Proprietary Tesla applications.

02

The Siemens Stack Behind a Gigafactory

VENDOR CATALOG

Siemens catalog. Direct product page where verified; otherwise the SIMATIC hub or the Siemens **Industry Mall** online catalog (search the model).

Device / product	Catalog link
SIMATIC S7-1500 controller	siemens.com/en-us/products/simatic/s7-1500/ Verified product page.
SIMATIC S7-1200 / ET 200SP CPU	siemens.com/en-us/products/simatic/ SIMATIC product hub.
SIMATIC ET 200 distributed I/O	mall.industry.siemens.com Industry Mall — search 'ET 200'.
SINAMICS drives (S120 / G120 / S210)	siemens.com/global/en/products/drives/sinamics.html Verified family page.
SIMOTICS motors (1FK7 servo)	mall.industry.siemens.com Industry Mall — search 'SIMOTICS'.

SCALANCE network switches	mall.industry.siemens.com Industry Mall — search 'SCALANCE'.
SIRIUS switching / SITOP power	mall.industry.siemens.com Industry Mall — search 'SIRIUS' / 'SITOP'.
SIMATIC Ident (RFID/codes)	mall.industry.siemens.com Industry Mall — search 'SIMATIC Ident'.
TIA Portal (engineering software)	siemens.com/en-us/products/automation/industry-software/automation-software/tia-portal/ Software product page.

03 The Languages of the Factory Floor (protocols)

CONCEPT / STANDARDS

Issue 03 covers communication *standards*, not catalog products — each is governed by a standards body (the authoritative 'catalog' of the spec).

Device / product	Catalog link
PROFINET / PROFIBUS / PROFI-safe / PROFIdrive	profibus.com PI — PROFIBUS & PROFINET International.
EtherNet/IP & CIP (CIP Motion/ Safety/Sync)	odva.org ODVA — governs EtherNet/IP & CIP.
EtherCAT	ethernetcat.org EtherCAT Technology Group.
Modbus / Modbus TCP	modbus.org Modbus Organization.
IO-Link (IEC 61131-9)	io-link.com IO-Link Community / consortium.
OPC UA (IEC 62541)	opcfoundation.org OPC Foundation.
MQTT / Sparkplug	sparkplug.eclipse.org Eclipse Sparkplug (MQTT spec via OASIS).

04 The Rockwell Stack & the Language of CIP

VENDOR CATALOG

Rockwell/Allen-Bradley catalog. Direct page where verified; otherwise the programmable-controllers hub or the Allen-Bradley products hub (search the model).

Device / product	Catalog link
ControlLogix 5580 (1756)	rockwellautomation.com/en-us/products/hardware/programmable-controllers/1756controllogix5580.html Verified product page.
CompactLogix 5380 / 5480 (5069)	rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html Controllers hub — search '5380'.
EtherNet/IP & CIP (protocol)	odva.org ODVA — protocol governance.

Device Level Ring (DLR)	rockwellautomation.com/en-us/products/hardware/allen-bradley/networks.html Networks hub.
GuardLogix safety controller	rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html Controllers hub — search 'GuardLogix'.
Kinetix servo drives	rockwellautomation.com/en-us/products/hardware/allen-bradley/motion-control.html Motion-control hub.
Compact 5000 / FLEX I/O	rockwellautomation.com/en-us/products/hardware/allen-bradley/io.html I/O hub.
Studio 5000 / FactoryTalk software	rockwellautomation.com/en-us/products/software.html Software hub.

05 The Rockwell & Allen-Bradley Portfolio

VENDOR CATALOG

Rockwell/Allen-Bradley full catalog — links to each product's family hub on rockwellautomation.com (search the exact model).

Device / product	Catalog link
ControlLogix 5580	rockwellautomation.com/en-us/products/hardware/programmable-controllers/1756controllogix5580.html Verified product page.
CompactLogix 5380 / 5480	rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html Controllers hub.
Micro800 / MicroLogix	rockwellautomation.com/en-us/products/hardware/allen-bradley/programmable-controllers.html Controllers hub.
Compact 5000 / FLEX 5000 / POINT I/O	rockwellautomation.com/en-us/products/hardware/allen-bradley/io.html I/O hub.
PowerFlex 520 / 750 VFDs	rockwellautomation.com/en-us/products/hardware/allen-bradley/drives.html Drives hub — search 'PowerFlex'.
Kinetix 5500 / 5700 / 7000 servo	rockwellautomation.com/en-us/products/hardware/allen-bradley/motion-control.html Motion hub — search 'Kinetix'.
Stratix industrial switches	rockwellautomation.com/en-us/products/hardware/allen-bradley/networks.html Networks hub — search 'Stratix'.
GuardLogix / Guardmaster safety	rockwellautomation.com/en-us/products/hardware/allen-bradley/safety-products.html Safety-products hub.

PanelView HMI / FactoryTalk View	rockwellautomation.com/en-us/products/hardware/allen-bradley/human-machine-interface.html HMI hub.
Studio 5000 / FactoryTalk	rockwellautomation.com/en-us/products/software.html Software hub.

06 The Beckhoff Way — PC-based Control & EtherCAT

VENDOR CATALOG

Beckhoff catalog. Direct page where verified; otherwise the relevant products category on beckhoff.com (search the exact model).

Device / product	Catalog link
C6030 ultra-compact IPC	beckhoff.com/en-us/products/ipc/pcs/c60xx-ultra-compact-industrial-pcs/c6030.html Verified product page.
Embedded PC (CX series)	beckhoff.com/en-us/products/ipc/embedded-pcs/ Embedded-PC category.
C7015 on-machine IPC	beckhoff.com/en-us/products/ipc/pcs/ Industrial-PC category.
EtherCAT Terminals (EL series)	beckhoff.com/en-en/products/i-o/ethercat-terminals/ Verified category page.
EtherCAT Box (EP/EPP, IP67)	beckhoff.com/en-us/products/i-o/ethercat-box/ EtherCAT-Box category.
AX8000 multi-axis servo	beckhoff.com/en-us/products/motion/servo-drives/ Drive-technology category.
AMP8000 distributed servo	beckhoff.com/en-us/products/motion/ Motion category — search 'AMP8000'.
Servomotors (AM / AL)	beckhoff.com/en-us/products/motion/servomotors/ Servomotor category.
XTS linear transport	beckhoff.com/en-us/products/motion/xts-linear-product-line/ XTS product line.
XPlanar planar motors	beckhoff.com/en-us/products/motion/xplanar-planar-motor-system/ XPlanar product line.
TwinCAT 3 software	beckhoff.com/en-us/products/automation/twincat/ TwinCAT category.
TwinSAFE functional safety	beckhoff.com/en-us/products/i-o/twinsafe/ TwinSAFE category.
EtherCAT (protocol)	ethercat.org EtherCAT Technology Group.

07 Motors, Drives & the Art of Controlled Motion

CONCEPT / STANDARDS

Issue 07 explains generic motor/drive *technologies* (multi-vendor). Representative catalogs from vendors already in the series:

Device / product	Catalog link
VFD (induction-motor drive)	siemens.com/global/en/products/drives/sinamics.html Representative: SINAMICS G120 (also Rockwell PowerFlex).
Servo drive (closed-loop)	rockwellautomation.com/en-us/products/hardware/allen-bradley/motion-control.html Representative: Kinetix (also SINAMICS S120, Beckhoff AX8000).
AC induction motor	mall.industry.siemens.com Representative: SIMOTICS GP/SD (Industry Mall).
Permanent-magnet servo motor	beckhoff.com/en-us/products/motion/servomotors/ Representative: Beckhoff AM (also SIMOTICS S, Kinetix motors).
Encoder / resolver (feedback)	— no public catalog — Generic feedback device — multi-vendor (Heidenhain, SICK, etc.).
PWM / vector control (technique)	— no public catalog — Drive control method — not a product.

08 Logistics, Inventory & the Art of Not Holding Stock

CONCEPT / STANDARDS

Issue 08 covers logistics *methods and software*, not floor catalog devices.

Device / product	Catalog link
Just-in-Time / Kanban (method)	— no public catalog — Operating method, not a product.
Two-bin kanban system	— no public catalog — Physical method (bins + cards) — no single catalog.
Warehouse Management System (WMS)	— no public catalog — Software category — vendor-dependent.
Barcode / RFID tracking	mall.industry.siemens.com Representative readers: SIMATIC Ident (Industry Mall).
ABC analysis / reorder point (method)	— no public catalog — Inventory technique, not a product.
AGV / AMR / AS-RS	beckhoff.com/en-us/products/ Generic — built by intralogistics integrators (Daifuku, Dematic, etc.).

09 The Sensors That Watch Every Car Get Built

CONCEPT / STANDARDS

Issue 09 covers generic sensor *types* (multi-vendor — e.g. SICK, Balluff, Pepperl+Fuchs, Keyence, Siemens). No single catalog; representative entry points noted.

Device / product	Catalog link
Inductive / capacitive / magnetic proximity	— no public catalog — Multi-vendor (SICK, Balluff, Pepperl+Fuchs, IFM).
Photoelectric & laser distance	— no public catalog — Multi-vendor (SICK, Keyence, Banner).
Machine vision system	— no public catalog — Multi-vendor (Cognex, Keyence; Beckhoff TwinCAT Vision).

Torque transducer (nutrunner)	atlascope.com/en-us/itba Built into Atlas Copco tools (Issue 11).
Load cell / pressure sensor	— <i>no public catalog</i> — Multi-vendor (HBM, Festo SDE, Rosemount — Issue 13).
Encoder / LVDT / ultrasonic / RTD	— <i>no public catalog</i> — Multi-vendor measurement devices.
IO-Link smart sensor	io-link.com IO-Link standard (devices from many vendors).

10 The Lines That Move the Line (conveyors)

CONCEPT / STANDARDS

Issue 10 covers conveyor *types* built by specialist material-handling integrators (e.g. Daifuku, Dürr, Eisenmann, Bosch Rexroth) — not sold from the automation vendors' catalogs. No single product catalog.

Device / product	Catalog link
Belt / roller / chain / slat conveyor	— <i>no public catalog</i> — Specialist conveyor builders — engineered to order.
Skid conveyor (body-in-white)	— <i>no public catalog</i> — Body-shop integrator-built.
Skillet (assembly, scissor-lift)	— <i>no public catalog</i> — General-assembly integrator-built.
Power & Free (overhead)	— <i>no public catalog</i> — Overhead-conveyor specialists.
EMS (Electrified Monorail System)	— <i>no public catalog</i> — Overhead-conveyor specialists.
Shuttle / fork transfer / turntable	— <i>no public catalog</i> — Transfer mechanisms — integrator-built.
AGV / AMR (mobile transport)	— <i>no public catalog</i> — Intralogistics-robot vendors.

11 Atlas Copco — Torque, Air & the Perfect Joint

VENDOR CATALOG

Atlas Copco catalog. Direct page where verified; otherwise the industrial-tools (ITBA) or compressors hub (search the exact model).

Device / product	Catalog link
Tensor SRB cordless nutrunner	atlascope.com/en-us/itba/products/assembly-solutions/electric-assembly-tools/cordless-assembly-tools/tensor-srb-cordless-nutrunner Verified product page.
Tensor STB / ES / STR nutrunners	atlascope.com/en-us/itba Industrial-tools hub — search 'Tensor'.
Tensor SB / fixtured spindles	atlascope.com/en-us/itba Industrial-tools hub.
Tensor IxB (controller-in-tool)	atlascope.com/en-us/itba Industrial-tools hub — search 'Tensor IxB'.

Power Focus / MTF6000 controllers	atlascope.com/en-us/itba Industrial-tools hub — search 'Power Focus'.
Pneumatic nutrunners (LTV, RTP)	atlascope.com/en-us/itba Industrial-tools hub.
Low-torque screwdrivers	atlascope.com/en-us/itba Industrial-tools hub.
ZR/ZT oil-free & GA compressors	atlascope.com/en-us/compressors Verified compressors hub.
Industrial vacuum pumps	atlascope.com/en-us/vacuum-solutions Vacuum-solutions hub.

12 Festo — Pneumatics, Motion & the Grip

VENDOR CATALOG

Festo catalog. Direct page where verified; otherwise the industrial-automation products hub or Festo online catalog (search the exact model/series).

Device / product	Catalog link
Valve terminal VTUX (with CPX-AP)	festo.com/us/en/p/valve-terminal-vtux-with-cpx-ap-a-terminal-id_VTUX_APA/ Verified product page.
ISO/compact cylinders (DSBC/DNC/ADVU)	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'DSBC'.
Guided drives / mini slides (DGSL)	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'DGSL'.
Rodless cylinders / rotary (DGC)	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'DGC'.
Valve terminals (VTUG / VTSA / MPA)	festo.com/vtug Festo short-link / products hub.
CPX / CPX-AP automation platform	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'CPX-AP'.
Motion Terminal VTEM	festo.com/vtem Festo short-link (digital pneumatics).
CMMT-AS / -ST servo drives	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'CMMT'.
Electric axes (EGC) & grippers	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'EGC' / 'gripper'.
FRL air-prep (MS / LFR)	festo.com/us/en/c/products/industrial-automation-id_pim1/ Products hub — search 'MS series'.

13 Emerson — One Company, a Dozen Names

VENDOR CATALOG

Emerson catalog (a federation of brands). Verified hub per domain; search the exact model/series on the brand page.

Device / product	Catalog link
------------------	--------------

AVENTICS Series XV valves	emerson.com/en-us/automation/fluid-control-pneumatics/pneumatic-valves-valve-systems Verified pneumatic-valves page.
AVENTICS AV system / SPRA actuator	emerson.com/en-us/automation/fluid-control-pneumatics Verified pneumatics hub.
ASCO solenoid & process valves	emerson.com/en-us/automation/fluid-control-pneumatics Pneumatics/fluid-control hub.
TESCOM regulators	emerson.com/en-us/automation/valves-actuators-regulators Valves/regulators hub.
TopWorx valve position sensing	emerson.com/en-us/automation/valves-actuators-regulators Valves/regulators hub.
Rosemount 3051 & measurement	emerson.com/en-us/automation/measurement-instrumentation Measurement-instrumentation hub.
PACSystems controllers (CPE200/RX3i)	emerson.com/en-us/automation/control-and-safety-systems Control-systems hub.
PACSystems IPCs & PACEdge	emerson.com/en-us/automation/control-and-safety-systems Control-systems hub.
Branson ultrasonic/laser welding	emerson.com/en-us/automation/welding-assembly-cleaning Welding & assembly hub.
NI (LabVIEW) test & measurement	ni.com Verified NI site.

14 Material Handling, Flow & Inventory Control

CONCEPT / STANDARDS

Issue 14's devices are generic material-handling equipment classes, built to order by intralogistics integrators — no single product catalog. Representative OEMs noted.

Device / product	Catalog link
Industrial trucks / forklifts / pallet jacks	— <i>no public catalog</i> — Generic — OEMs: Toyota Material Handling, Jungheinrich, Crown, Hyster.
Tuggers & tugger trains (milk run)	— <i>no public catalog</i> — Generic — intralogistics OEMs; engineered to the route.
AGV / AMR (mobile transport)	— <i>no public catalog</i> — Driverless-vehicle vendors (Daifuku, Dematic, MiR, etc.).
AS/RS & automated storage (VLM, shuttle)	— <i>no public catalog</i> — Integrator-built — Dematic, Daifuku, SSI Schaefer, Honeywell Intelligrated.
Cranes, hoists & monorails	— <i>no public catalog</i> — Generic — Konecranes, Demag, and others.
Racking & storage systems (flow rack)	— <i>no public catalog</i> — Generic — SSI Schaefer, Interroll (flow), Dexion; engineered to layout.
Unit-load devices (pallets, totes)	— <i>no public catalog</i> — Commodity/standardised — many suppliers; no single catalog.

15

Production Planning — Plan, Schedule, MRP & Heijunka

CONCEPT / STANDARDS

Issue 15's 'devices' are planning methods and the software that runs them. Methods are not products; MRP/APS/ERP live in business software (representative vendors noted).

Device / product	Catalog link
Aggregate plan / MPS / takt / line balancing	— <i>no public catalog</i> — Planning methods — not products.
Heijunka leveling / heijunka box	— <i>no public catalog</i> — A lean method (the box is a shop-built tool) — no catalog.
MRP / MRP II	— <i>no public catalog</i> — A calculation embedded in ERP software (SAP, Oracle, Microsoft Dynamics, etc.).
APS (Advanced Planning & Scheduling)	— <i>no public catalog</i> — Finite-capacity planning software — multiple vendors.
ERP (the company-wide system)	— <i>no public catalog</i> — Representative: SAP, Oracle, Microsoft Dynamics (Tesla uses in-house Warp — Issue 16).

16

Tesla's Real Software Stack — Warp, MES & MOS

CONCEPT / STANDARDS

Issue 16 mixes Tesla's proprietary in-house systems (no public catalog) with the commodity third-party tools it is reported to use (real product pages).

Device / product	Catalog link
Warp (Tesla in-house ERP/planning)	— <i>no public catalog</i> — Proprietary Tesla system — no public catalog.
MES (Tesla, in-house)	— <i>no public catalog</i> — Proprietary Tesla system — no public catalog.
MOS (Tesla Manufacturing Operating System)	— <i>no public catalog</i> — Proprietary Tesla system — no public catalog.
Atlassian JIRA (issue tracking)	atlassian.com/software/jira Real product (reported outside tool).
Coupa (procurement)	coupa.com Real product (reported outside tool).
Oracle Database	oracle.com/database Real product — MES database (as reported).
Microsoft SQL Server	microsoft.com/en-us/sql-server Real product — MES database (as reported).
MySQL (test database)	mysql.com Real product — in-house test DB built on it.
Quickbase	quickbase.com Real product — process/quality tracking (as reported).

17 KPIs, OEE & Measuring a Plant

CONCEPT / STANDARDS

Issue 17's 'devices' are metrics and a standard, not products. Metrics are computed by MES/dashboard software; the methodology has an ISO standard.

Device / product	Catalog link
OEE / TEEP (and Availability, Performance, Quality)	— no public catalog — Metrics — computed by MES/OEE-dashboard software, not a product.
First-pass yield / scrap / PPM	— no public catalog — Quality metrics — not products.
Throughput / cycle time	— no public catalog — Flow metrics — measured, not bought.
MTBF / MTTR	— no public catalog — Reliability metrics — tracked in CMMS/maintenance software.
ISO 22400 (KPI standard)	iso.org ISO — the standard defining manufacturing KPI calculations.

18 Demand — Orders, Forecasting & What Feeds Them

CONCEPT / STANDARDS

Issue 18's 'devices' are forecasting methods plus Tesla's order model. Methods are not products; forecasting lives in ERP/planning or Excel; Tesla's configurator is proprietary.

Device / product	Catalog link
Make-to-stock vs build-to-order	— no public catalog — Fulfilment strategies — not products.
Moving average / exponential smoothing	— no public catalog — Forecasting methods — built into Excel & planning software (Issue 19).
Holt-Winters (trend + seasonality)	— no public catalog — A method — e.g. Excel's FORECAST.ETS (Issue 19).
MAPE (forecast accuracy)	— no public catalog — A metric — computed in any spreadsheet/planning tool.
Delphi / qualitative forecasting	— no public catalog — A judgement method — not a product.
Tesla online configurator / order model	— no public catalog — Proprietary Tesla direct-to-consumer system — no public catalog.
Demand-planning / APS software	— no public catalog — Representative: SAP IBP, Oracle, o9, Kinaxis (Tesla uses in-house Warp — Issue 16).

19 Excel for Inventory & Production Planning

VENDOR CATALOG

Issue 19's 'device' is Microsoft Excel itself — a real product. The items below are its built-in functions/features; the link is Excel's product page.

Device / product	Catalog link
Microsoft Excel (Microsoft 365)	microsoft.com/en-us/microsoft-365/excel Verified product page — the spreadsheet itself.

SQRT / NORM.S.INV / STDEV.S	microsoft.com/en-us/microsoft-365/excel Built-in functions — EOQ & safety stock (Issue 14).
XLOOKUP / VLOOKUP / IFS	microsoft.com/en-us/microsoft-365/excel Built-in functions — lookups & ABC classification.
FORECAST.ETS / TREND	microsoft.com/en-us/microsoft-365/excel Built-in functions — forecasting / Holt-Winters (Issue 18).
SUMPRODUCT / SUMIFS	microsoft.com/en-us/microsoft-365/excel Built-in functions — BOM explosion & usage totals.
PivotTable & Forecast Sheet	microsoft.com/en-us/microsoft-365/excel Built-in features — summarise & one-click ETS forecast.

END OF INDEX

THE INTEGRATION LAYER · Device & Catalog-Link Index (Issues 01–17).

Links are to official manufacturer catalogs and standards bodies, current at time of writing; vendors occasionally reorganise URLs, so the model/series name is given to locate each device. Vendor and brand names are referenced descriptively. Tesla's internal systems are proprietary.

FASTENING ENGINEERING · FROM PRELOAD TO QUALITY CONTROL

The bolt is a spring: Bolted-joint engineering & tightening in industry

Preload, the torque–tension link, and tightening control

A bolted joint works not through the bolt itself but through **clamping force**. The whole craft of fastening engineering is to create that force at the right level and – more importantly – to **guarantee** it. This guide shows why torque is only a small part of the story, and how modern shop-floor tools (such as **Atlas Copco Tensor / PF6000**) control preload and trace it in **MOS**.

Preload — the heart of the joint

The bolt stretches like a spring; its tension clamps the parts together.

Tightening strategies

Torque, torque+angle, yield control, and direct stretch.

Torque ≠ tension

~90% of torque is spent on friction; only ~10% becomes bolt tension.

Quality on the line

The torque–angle curve, error-proofing, and traceability into MOS.

A note on the numbers

The figures and worked examples in this guide are **realistic but educational**, chosen to make the mechanism concrete; they are not the official values of any plant. Three layers of knowledge are combined: (1) **standard joint-engineering fundamentals** – **ISO 898-1** for property class, the **VDI 2230** / Motosh torque–tension equation, and general industrial tightening practice; (2) **publicly known facts about industrial tooling** – transducerized DC nutrunners and their controllers; and (3) one **fully worked numeric example** on an **M10×1.5** class **10.9** bolt. Treat every number as a plausible illustration.

CONTENTS

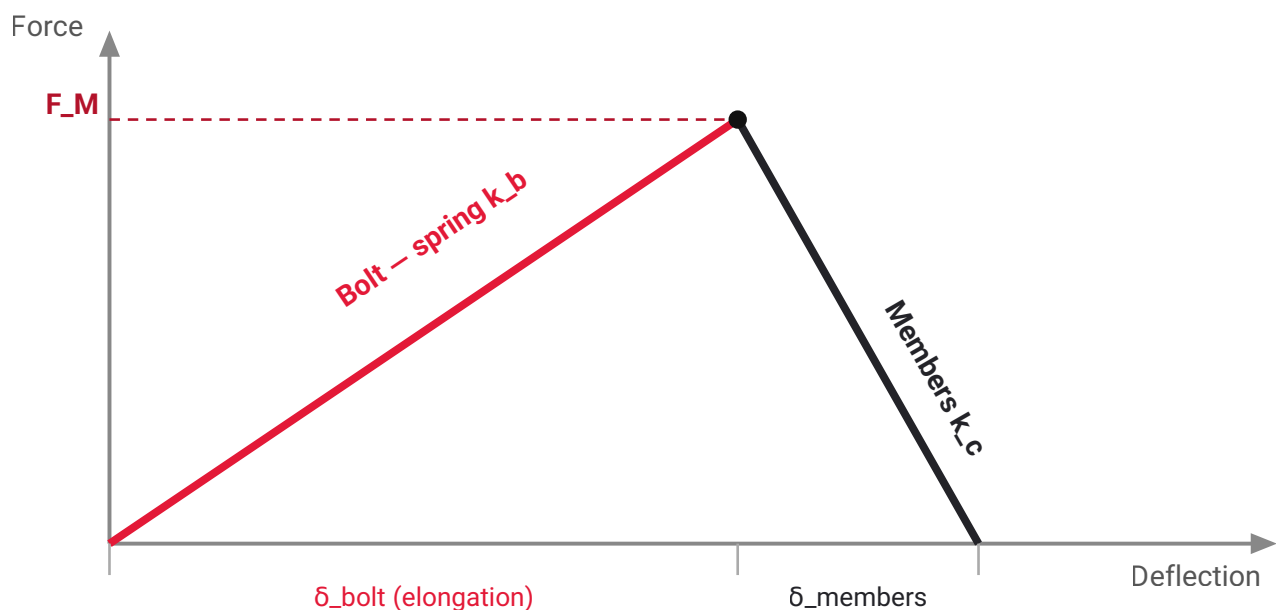
1	The bolt is a spring – fundamentals & preload	The bolted joint & preload
2	Thread geometry & property class	Threads & ISO 898-1
3	Torque–tension & why most torque is friction	Torque–tension
4	Tightening strategies & preload accuracy	Tightening strategies
5	Tools, the torque–angle curve & floor quality	Tools, curve & quality
6	Glossary & sources	Reference

1 The bolt is a spring – fundamentals of the bolted joint

Preload, clamp force, and the joint diagram

A common misconception is that “the bolt holds the parts together.” In reality, **clamp force** holds them. When we tighten a bolt, we stretch it like a **very stiff spring**; the tension inside the bolt – called **preload** – is the force that presses the members together. The entire craft of fastening engineering is to create that preload at the right level and then **guarantee** it.

The bolt and the members form a parallel-spring system: the bolt is a **tension** spring (it stretches, stiffness k_b) and the members are a **compression** spring (they are squeezed, stiffness k_c). At the moment of tightening, both reach a common force – the preload F_M . The “joint diagram” shows this balance:



Joint diagram: the bolt line has the gentler slope – it is the softer spring and stretches more; the stiffer members compress less. Both meet at the common preload F_M .

Why does preload matter so much? – what preload buys you

- **Prevents self-loosening:** high preload keeps the friction at the threads and under the head alive; without enough tension, vibration quickly backs the bolt out.
- **Carries external load:** as long as the external tensile load stays below the preload, the joint does not “separate” and sealing / rigidity is preserved.
- **Fatigue resistance:** this is the subtlest point. Because the bolt and members are parallel springs, a varying external load is shared between them in proportion to stiffness. Member stiffness is usually several times the bolt’s, so only a small fraction of the varying load (the load factor Φ) reaches the bolt; the rest simply reduces the clamp force. The result: **the bolt’s stress amplitude stays small and fatigue life is high** – provided preload is adequate.

One sentence that says it all

“The bolt doesn’t grip; the clamp does.”

The goal of tightening is to reach a defined **target tension** in the bolt, not merely to “make it tight.” Every method in Part 4 is an attempt to hit that target preload with the least scatter. And the core challenge — as Part 3 shows — is that what the tool measures (torque) does not have a direct, reliable relationship to what actually matters (tension).

2 Thread geometry & property class

Reading the bolt – dimensions and strength

To compute preload and torque we need two kinds of data: **thread geometry** (how much cross-section there is to carry tension) and **material / property class** (how far the bolt can be stressed). Both are fully defined in the metric standards.

2.1 The language of the metric thread – reading M10×1.5

The designation **M10×1.5** means a nominal diameter **d** = 10 mm and a pitch **P** = 1.5 mm (the distance between two adjacent threads). The metric thread profile angle is 60°. Three key diameters: the nominal diameter **d**, the pitch diameter **d₂** (where thread and groove widths are equal), and the minor / root diameter **d₃**. Tensile capacity is measured by the **tensile stress area** – a diameter slightly larger than the root:

$$A_s = (\pi/4) \cdot [(d_2 + d_3) / 2]^2$$

tensile stress area – the effective cross-section that carries the bolt tension

2.2 Property class (ISO 898-1) – what “10.9” means

Two numbers separated by a dot are stamped on every steel bolt head, such as **8.8**, **10.9** or **12.9**. Reading them is easy: the first number × 100 = the ultimate tensile strength **R_m** in MPa; and the product of the two numbers × 10 = the yield strength **R_{p0.2}**. For example, 10.9 means an ultimate of 1000 and a yield of 900 MPa. The **proof load** sits just below yield (~90% of it) and is the maximum load the bolt takes without permanent deformation.

Table 1 – Steel-bolt property classes ISO 898-1

Nominal strength values (MPa)

Property class	UTS R _m (MPa)	Yield R _{p0.2} (MPa)	Proof stress S _p (MPa)	Typical use
8.8	800	640	~580–600	General structural / chassis
10.9	1,000	900	~830	Powertrain, suspension, safety-critical
12.9	1,200	1,080	~970	High-load, compact joints

Rule of thumb: class **X.Y** → **R_m** = X×100 and **R_{p0.2}** = X×Y×10 MPa. Class 10.9 is the most common automotive grade.

2.3 From diameter to target preload – sizing the clamp

Given **A_s** and an allowable stress, the target preload follows. A common industrial rule is to stretch the bolt to about **70% of proof load** (leaving a safety margin while still providing ample clamp). The table below completes that chain for class 10.9 down to an approximate torque:

Table 2 – Stress area, target preload and approximate torque (class 10.9) worked sizing

Stress area → target preload (≈70% proof) → torque at nut factor K = 0.20

Bolt	A _s (mm ²)	Proof load (kN)	Target preload ≈0.7·proof (kN)	Torque @K=0.20 (N·m)
M6 × 1.0	20.1	16.7	11.7	14
M8 × 1.25	36.6	30.4	21.3	34

Bolt	A _s (mm ²)	Proof load (kN)	Target preload ≈0.7·proof (kN)	Torque @K=0.20 (N·m)
M10 × 1.5	58.0	48.1	33.7	67
M12 × 1.75	84.3	70.0	49.0	118
M16 × 2.0	157	130	91	292

The torque column comes from $T = K \cdot d \cdot F$ with $K = 0.20$ (dry / lightly oiled) – the subject of the next part. The highlighted bolt (M10×1.5) is followed throughout the guide.

3 Torque–tension — and why most torque is friction

The torque–tension relationship

The simplest industrial relationship between input torque and resulting preload is the “nut factor” formula:

$$T = K \cdot d \cdot F$$

T = applied torque · K = nut factor · d = nominal diameter · F = preload (clamp force)

This relationship is deceptively simple; all the physics is hidden inside the **nut factor K** . And that is exactly the problem: **K depends mostly on friction, and friction scatters**. To see why, let us follow where the input torque actually goes.

3.1 Where does the torque go? — the energy split

When we tighten a bolt, the input torque splits roughly three ways. Only a small part actually stretches the bolt; the rest is spent overcoming friction:

■ Bolt tension (preload) — ~10–15%
 ■ Thread friction — ~40%
 ■ Under-head friction — ~50%



In a typical tightening, only about **10–15%** of the input torque becomes useful bolt tension; over 85% is consumed by friction at the two contact surfaces. So any small change in lubrication, coating, rust or surface finish shifts K — and therefore preload — dramatically.

3.2 The full torque equation — the long-form (Motosh / VDI 2230) equation

The same three parts appear explicitly in the long-form equation. It shows where K comes from:

$$T = F \cdot [0.159 \cdot P + 0.577 \cdot \mu_{th} \cdot d_2 + 0.5 \cdot \mu_b \cdot D_{km}]$$

term 1 = pitch / lead (the only useful part) · term 2 = thread friction (μ_{th}) · term 3 = under-head friction (μ_b), D_{km} = effective bearing diameter

The first term (pitch) is the only part that would remain if friction were zero; the other two are friction. Dividing all three by **d** , their sum is exactly the **K** of the short formula. For dry steel, **$K \approx 0.20$** ; with good lubrication (wax / molybdenum) it drops to ~0.10–0.14, and with a rusty / rough surface it rises to ~0.25–0.30.

3.3 Worked example: why torque alone is not enough — the scatter problem on M10×1.5, class 10.9

Target bolt: **M10×1.5**, class **10.9**. From Table 2: stress area 58 mm², yield load 52.2 and proof load 48.1 kN. We take a target preload of 34 kN (~70% of proof). With **$K = 0.20$** :

$$T = 0.20 \times 0.010 \text{ m} \times 34,000 \text{ N} = \mathbf{68 \text{ N}\cdot\text{m}} \quad (\approx \text{target torque})$$

Now suppose we set the tool to that **fixed 67 N·m**, but friction varies from part to part. The actual preload comes out like this:

Table 3 — Effect of friction scatter on preload (at a fixed 67 N·m)

torque control · scatter

Same 67 N·m, varying nut factor K — resulting preload on M10 10.9

Surface condition	Nut factor K	Resulting preload (kN)	% of proof (48.1 kN)	Verdict
Well lubricated (wax / MoS ₂)	0.14	47.9	99%	At the proof limit — risk of yield
As-received (light oil)	0.20	33.5	70%	On target ✓
Dry / slightly rusted	0.25	26.8	56%	Under-clamped — loosening risk

The key takeaway: with a **single fixed torque**, friction scatter alone swings preload from 27 to 48 kN — from “far too loose” to “at the yield limit.” Torque control alone cannot guarantee preload; its typical scatter is **±25 to 35%**. This problem motivates the more accurate methods in the next part.

4 Tightening strategies & preload accuracy

Hitting the target with the least scatter

Every tightening method is an attempt to reach the target preload with the least scatter. The less a method depends on friction, the more accurate – and usually the more expensive and slower – it is. The four main families:

- **1) Torque control:** the tool tightens to a target torque and stops. The simplest and cheapest, but – as we saw – captive to friction scatter. Preload scatter $\sim\pm 30\%$.
- **2) Torque + angle:** first tighten to a “snug torque” so the joint pulls together, then turn a defined additional **angle**. Because beyond snug the rotation angle is roughly proportional to bolt tension (and independent of friction), accuracy improves sharply: $\sim\pm 12\text{--}15\%$. The dominant method in the automotive industry.
- **3) Yield / gradient control:** the tool monitors the **slope** of the torque–angle curve in real time and stops exactly where the slope drops (the onset of bolt yield). Because the yield point is a property of the bolt itself, friction is almost eliminated: $\sim\pm 8\%$. Used in powertrain and critical joints.
- **4) Bolt stretch / ultrasonic:** instead of torque, the bolt’s **elongation** is measured directly (with an ultrasonic sensor or strain gauge). The most accurate method ($\sim\pm 3\text{--}5\%$), but costly; reserved for the most critical joints such as cylinder-head bolts or key structural studs.

Table 4 – Comparison of tightening methods strategy comparison

Preload scatter, cost and typical application by method

Method	Controls into	Preload scatter	Friction sensitivity	Cost / complexity	Typical use
Torque	Elastic region	$\pm 25\text{--}35\%$	Very high	Low	General assembly, brackets
Torque + angle	Elastic region	$\pm 12\text{--}15\%$	Low	Medium	Most automotive joints
Yield / gradient	Onset of yield	$\pm \sim 8\%$	Very low	Medium–high	Powertrain, safety-critical
Bolt stretch / ultrasonic	Measured elongation	$\pm 3\text{--}5\%$	None (direct)	High	Cylinder head, structural studs

“Scatter” is the band of resulting preload around the target. Torque + angle is the industry sweet spot: good accuracy at reasonable cost – exactly what the tools in the next part deliver.

Hard vs soft joint

Why one fixed torque doesn’t suit every joint

Hard joint: metal on metal; beyond snug, torque rises very quickly with little rotation (steep slope). **Soft joint:** a rubber washer, gasket, or sheet metal; torque rises slowly over a large angle (gentle slope). It is this slope difference that the torque–angle and yield methods exploit to sense the state of the joint – the subject of the curve in the next part.

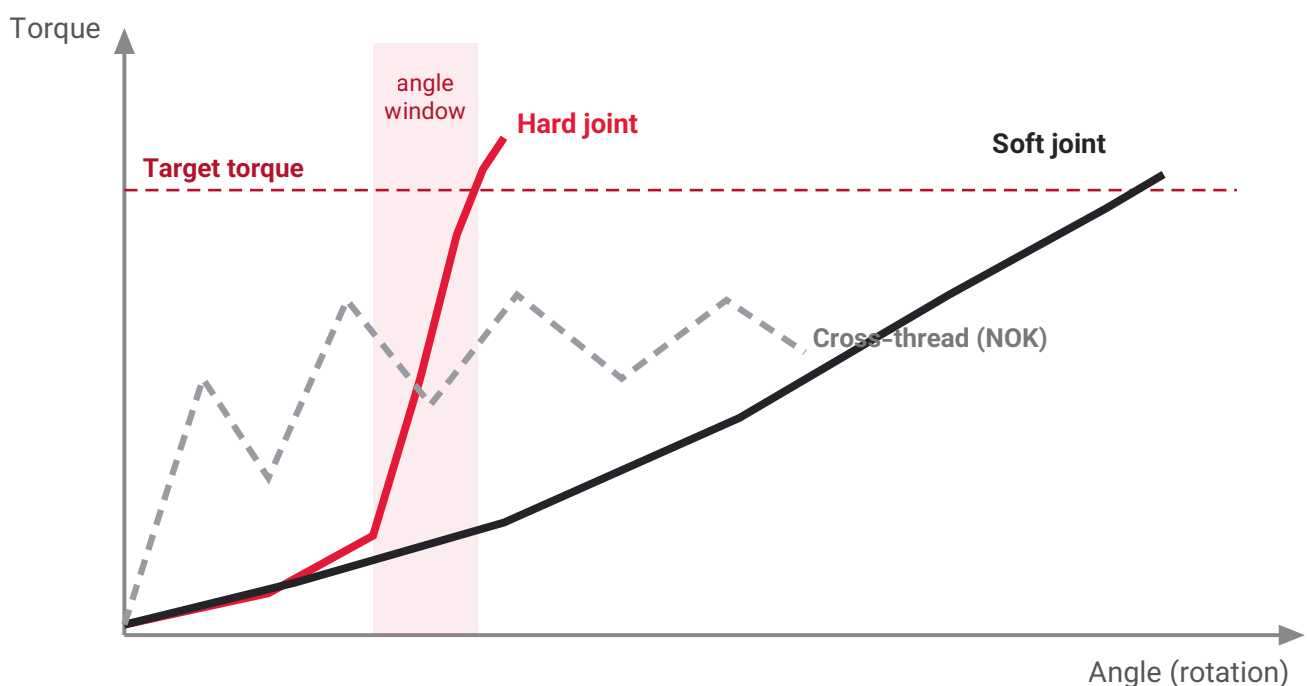
5 Tools, the torque–angle curve, and shop-floor quality

From a measured event to traceability

On a modern line, tightening is no longer “tighten until it feels tight”; it is a **measured and recorded** event. A transducerized DC nutrunner (such as the **Atlas Copco Tensor**) together with the **Power Focus 6000** controller controls torque and angle in a **closed loop** on every tightening, records the entire curve, and gives a final **OK / NOK** verdict.

5.1 The torque–angle curve: every joint’s signature – the torque–angle signature

The curve the tool records is the joint’s “fingerprint.” Its shape tells whether the joint was hard or soft, and if something is wrong – cross-threading, stripping, or abnormal friction – the curve gives it away immediately:



Three signatures: a hard joint (red) rises steeply after snug; a soft joint (black) climbs gradually over a large angle; a cross-thread (grey dashed) shows erratic prevailing torque and never reaches target cleanly – flagged NOK.

5.2 Error-proofing – error-proofing on the line

- **Sequence & count control:** the controller knows how many bolts and in what order they must be tightened (batch / sequence); the station will not pass until every bolt is OK.
- **Cross-thread & prevailing-torque detection:** if the torque before snug (prevailing torque) exceeds a limit, the controller rejects it as cross-threading / contamination.
- **Re-hit detection:** if an already-tightened bolt is torqued again, it is detected and flagged.
- **Socket position & Pset selection:** the tool runs only the correct program for that joint; a smart socket tray prevents using the wrong tool.

5.3 Traceability and the link to MOS – traceability into the Manufacturing Operating System

Every tightening produces a complete record: final torque, angle, Pset ID, time, tool, operator, and – most importantly – the vehicle identifier (**VIN**). This record is sent through industrial protocols (such as Open Protocol) to **MOS** (Tesla’s Manufacturing Operating System) to attach to that vehicle’s system of record. The result: for every critical joint on every vehicle, one can prove years later that “this bolt, at this torque and this angle, on this date, was tightened correctly” – and if it was NOK, the vehicle does not pass the quality gate until the problem is fixed.

Table 5 – Common faults and their signatures on the curve fault signatures

Common fastening faults and how they show on the torque–angle curve

Fault	Signature on the curve	Real preload	Action
Cross-thread	High, erratic prevailing torque before snug; noisy slope	Unreliable	Reject (NOK) – re-tap / scrap part
Stripped thread	Torque plateaus or drops, never reaches target	~Zero	Reject; replace nut/insert
False torque (high friction)	Reaches torque target with abnormally small angle	Too low	Flag; angle/yield method avoids it
Soft joint not seated	Very shallow slope; target hit at huge angle	Uncertain	Check gasket / re-slug
Re-hit (already tight)	Instant torque spike at ~zero angle	Unknown (no fresh stretch)	Detect & ignore / flag

This is precisely the power of the torque–angle method: because it sees both torque and angle, it can tell “false torque” (high friction) apart from a genuine clamp – something torque alone cannot do.

6 Glossary & sources

Reference

Term	Meaning
Preload / clamp force	The retained bolt tension that presses the parts together — the real goal of tightening
Nut factor (K)	The empirical coefficient in $T = K \cdot d \cdot F$; depends mostly on friction
Property class	An ISO 898-1 code such as 10.9; gives the bolt's ultimate and yield strength
Proof load	The maximum load the bolt takes without permanent deformation (~90% of yield)
Stress area (A_s)	The bolt's effective cross-section for carrying tension
Pitch (P) / lead	The distance between two adjacent threads
Thread friction (μ_{th})	Friction between the bolt and nut threads
Under-head friction (μ_b)	Friction between the bolt head bearing face and the part surface
Torque–angle	A tightening method: snug torque + a defined rotation angle
Yield / gradient control	Stops at the drop in curve slope, i.e. the onset of bolt yield
Snug torque	The initial torque that pulls the joint together before angle counting begins
Prevailing torque	Torque before snug; a high value signals cross-threading / contamination
Hard / soft joint	How fast torque rises with angle; steep (metal-metal) or gentle (gasket)
Scatter	The band of resulting preload around the target — the measure of a method's accuracy
Self-loosening	The bolt backing out under vibration when preload is inadequate

Sources & standards

- **Bolt standards:** ISO 898-1 (mechanical properties and property classes of steel bolts), ISO 724 (metric thread dimensions and stress area), and ISO 16047 (torque–tension test).
- **Joint design:** the VDI 2230 guideline for the systematic calculation of bolted joints under static and fatigue load; and the classic Motosh / Kellermann torque–tension equation.
- **Tightening technique:** general tooling-industry literature on tightening strategies and preload control (including Atlas Copco's published material on tightening technique and transducerized DC tools).
- **Shop-floor context:** public information about Tesla's MOS and its role in recording and tracing build / test, linked to the earlier guides in this series.
- **The numbers:** all illustrative and realistic, built solely to clarify the mechanism; they are not the official values of any plant. ■

FASTENER TYPES & STANDARDS · FROM CODE TO CHOICE

The language of fasteners: Types, technical specs & the standards system

Head & drive types, threads, property classes, nuts, washers & coatings

The fastener world is full of codes — **M10×1.5**, **10.9**, **ISO 4762**, **A2-70**. But they all come from a few orderly systems. This guide lays out head & drive types, thread forms, property classes & materials, nuts & washers, and coatings — so you can read a full fastener designation and choose the right one.

The standards system

Metric (ISO/DIN) vs imperial (ANSI/SAE) and the map of codes.

Head & drive

Hex, socket, countersunk, flange · Torx, Phillips, slotted.

Class & material

Steel class (8.8/10.9/12.9) and stainless (A2/A4).

Nuts, washers, coatings

Locking methods and how coating affects friction & corrosion.

A note

This document is a **compact educational reference**, not a substitute for the full text of the standards. The standard codes (ISO/DIN/ANSI) and strength values are real and follow common definitions; but for final design you should always consult the current edition of the standard itself and the manufacturer's datasheet. This guide complements the previous issue in this series (bolted-joint engineering): there we saw the physics of the joint; here, the catalog and language of fasteners.

CONTENTS

1	The standards system & reading a designation	Standards & designations
2	Head types & drive types	Head & drive types
3	Thread form, pitch & tolerance class	Threads
4	Property class, material & marking	Class & material
5	Nuts, washers & locking methods	Nuts, washers & locking
6	Coatings, corrosion & a selection guide	Coatings & selection
7	Glossary & sources	Reference

1 The standards system & reading a designation

Metric vs imperial, and the ISO / DIN map

Every code you see on a drawing or a fastener box belongs to one of the two great global systems. Knowing those two systems and their standards map is half the job of choosing a fastener.

1.1 The two great systems: metric and imperial – metric vs imperial

- **Metric (ISO / DIN / EN):** diameter and pitch in millimeters, with the prefix **M** (e.g. M10×1.5). The dominant system in automotive, Europe and Asia. Property class is stamped numerically (8.8, 10.9).
- **Imperial (ANSI / ASME, SAE):** diameter in fractional inches or a gauge number, and thread by **TPI** (threads per inch). Two main families: **UNC** (coarse) and **UNF** (fine). The dominant system in North America. Property class is shown by radial lines on the head (SAE Grade 5 / 8).

The thread flank angle is 60° in both systems, but the dimensions, pitch and tolerances differ; so metric and imperial fasteners are **absolutely not interchangeable** – even if they seem to fit, the thread is damaged.

1.2 The standards map (ISO ↔ DIN) – the standards map

Many legacy German standards (**DIN**) have today been superseded by their international **ISO** equivalent, but both names are still common in the market. The table below puts the most-used ones side by side:

Table 1 – Common fastener standards map **ISO ↔ DIN**

ISO designation and legacy DIN equivalent

Fastener	ISO	DIN (legacy)	Note
Hex bolt – partial thread	ISO 4014	DIN 931	shank + thread
Hex bolt – full thread	ISO 4017	DIN 933	threaded to head
Socket head cap screw	ISO 4762	DIN 912	internal hex (Allen)
Countersunk socket screw	ISO 10642	DIN 7991	90° flat head
Button head socket screw	ISO 7380	–	low dome head
Hex flange bolt	–	DIN 6921	integral washer flange
Set screw (grub)	ISO 4026–4029	DIN 913–916	headless
Hex nut (style 1)	ISO 4032	DIN 934	standard nut
Nylon-insert lock nut	ISO 10511	DIN 985	nylon ring
Plain washer	ISO 7089	DIN 125	flat
Mechanical properties – bolts	ISO 898-1	–	property classes
Stainless properties	ISO 3506-1	–	A2 / A4 grades

“DIN” today survives mostly as a familiar trade name; in modern ordering the ISO code is preferred, but both describe the same part.

1.3 Decoding a full designation – reading the code

A standard designation packs five things into one line: type/standard, thread and diameter, length, property class, and coating. For example:

ISO 4762 – M10×40 – 12.9 – zinc-plated

socket head cap screw · M10 (coarse) thread · 40 mm long · property class 12.9 · zinc coating

- **ISO 4762** = the standard and type (socket head cap screw, Allen drive).
- **M10** = metric, 10 mm nominal diameter (coarse pitch 1.5 implied; if fine, it would read M10×1.25).
- **×40** = 40 mm length.
- **12.9** = property class (Part 4).
- **zinc-plated** = surface coating / finish (Part 6; precise coding in ISO 4042).

2 Head types & drive types

The two visible features of every fastener

Every fastener's two visible features are the shape of the **head** (which spreads the clamp load and sets the look / space required) and the type of **drive** (the recess the tool grips to apply torque). The two are independent: a given head can come with different drives.

2.1 Head styles – head styles



Hex head
ISO 4014



Socket cap
ISO 4762



Countersunk
ISO 10642



Button head
ISO 7380



Hex flange
DIN 6921



Set screw
ISO 4026

Side view of six common heads. The countersunk head sits flush below the surface; the flange head has an integral washer; the headless set (grub) screw locks a part onto a shaft.

2.2 Drive types – drive types



Hex (ext.)



Hex socket



Torx



Phillips



Slotted



Pozidriv

Top view of six drives. Internal drives (hex socket, Torx) transmit more torque without slipping; thanks to its radial contact, Torx has the least “cam-out” (tool slip) and dominates the automotive assembly line.

Table 2 – Head/drive at a glance quick reference

Typical pairing and use

Head	Std	Common drive	Typical use
Hex head	ISO 4014/4017	External hex	Structural, high-torque, accessible joints
Socket cap	ISO 4762	Internal hex / Torx	Machinery, tight spaces, high strength (12.9)
Countersunk	ISO 10642	Internal hex / Torx	Flush surface required
Button head	ISO 7380	Internal hex	Low profile, light load, finish
Hex flange	DIN 6921	External hex	Spreads load, no separate washer (automotive)
Set screw	ISO 4026–29	Internal hex / slotted	Locking a hub/collar onto a shaft

3 Thread form, pitch & tolerance class

The helix that turns rotation into clamp

The thread is the heart of every fastener — the helix that converts rotation into axial force. Three thread features matter: the **profile form**, the **pitch** (coarse or fine), and the **tolerance class**.

3.1 Thread profile forms — thread forms

Most fasteners use a 60° triangular thread; but for transmitting power or sealing pipe, other forms exist:

Table 3 — Common thread profiles thread forms

Profiles and their use

Form	Flank angle	Use
Metric (M)	60°	General fastening — the most common worldwide
Unified (UNC / UNF)	60°	Imperial fastening (North America)
Whitworth (BSW)	55°	Legacy British equipment
Trapezoidal (Tr)	30°	Power / lead screws (motion, not clamping)
ACME	29°	Imperial power screws
Pipe — NPT / BSPT	60° / 55°	Tapered, sealing pipe threads

3.2 Coarse or fine pitch? — coarse vs fine

For a given diameter, two standard pitches exist: **coarse** (the default, not stated in the designation; e.g. M10) and **fine** (explicitly written; e.g. M10×1.25).

- **Coarse:** faster to run down, more tolerant of thread damage and corrosion, and less prone to stripping. The default choice for most applications.
- **Fine:** a slightly larger stress area (higher preload at the same diameter), better vibration resistance, and finer adjustment; suited to thin walls and tight spaces. But more sensitive to cross-threading and galling, especially in stainless.

3.3 Thread tolerance class — tolerance class

The fit of male and female threads is defined by a tolerance class: a lower-case letter for the external (male) thread, an upper-case letter for the internal (female). The standard industrial fit is **6g** (external) on **6H** (internal). The letter **g** gives a small allowance that leaves room for coating and lubrication; for thick coatings, **6e** (larger allowance) is sometimes chosen. The **h/H** class has zero allowance and is tighter.

3.4 Full thread & self-tapping — full thread & self-tapping

A bolt may be **fully threaded** (ISO 4017) or **partially threaded with a plain shank** (ISO 4014); the plain shank is better for precise location and carrying shear load. Another important family makes its own thread in the part: **thread-forming** screws (no chips — they displace soft material) and **thread-cutting** screws (a cutting edge that machines harder material). These eliminate the need for a nut or pre-tapped thread.

4 Property class, material & marking

How strong, and made of what

“How strong?” is answered by the property class, and “made of what?” by the material family. Both are marked on the head so they can be identified in the field.

4.1 Steel – ISO 898-1 classes – steel property classes

The rule (from the previous issue): the first number $\times 100$ = the ultimate strength, and the product of the two numbers $\times 10$ = the yield strength (MPa).

Table 4 – Steel property classes ISO 898-1

Steel bolt property classes (MPa) and material

Class	UTS R_m	Yield $R_{p0.2}$	Material	Typical use
4.6	400	240	Low-carbon steel	Low-load, general
5.8	500	400	Low-carbon steel	Light commercial
8.8	800	640	Medium-carbon, Q&T	General structural / chassis
10.9	1,000	900	Alloy steel, Q&T	Powertrain, suspension
12.9	1,200	1,080	Alloy steel, Q&T	High-load, compact

“Q&T” = quenched & tempered. From class 10.9 upward, the risk of hydrogen embrittlement during electroplating becomes serious (Part 6).

4.2 Stainless steel – ISO 3506 grades – stainless steel grades

The stainless code has two parts: a letter+digit for the material (**A2**, **A4** from the austenitic family) and a number after the dash for strength (equal to $R_m/10$). For example, **A4-80** means A4 material with an ultimate of 800 MPa.

Table 5 – Common austenitic stainless grades ISO 3506

Common austenitic stainless grades

Grade	~AISI	R_m (MPa)	$R_{p0.2}$ (MPa)	Corrosion / use
A2-70	304	700	450	General indoor/outdoor; food, architecture
A4-80	316	800	600	Marine / chemical (added molybdenum)

Thanks to molybdenum, A4 resists chloride corrosion (salt, sea) far better than A2. Note: austenitic stainless is usually non-magnetic and weaker than class 10.9 steel – it is chosen for corrosion, not maximum load.

4.3 Head marking – head marking

- **Metric:** the class number (e.g. **8.8** or **10.9**) plus the maker’s mark is stamped on the head.
- **Imperial (SAE):** by radial lines – **Grade 5** three lines, **Grade 8** six lines.
- **Stainless:** usually the material and class code such as **A2-70** on the head.

5 Nuts, washers & locking methods

Completing the joint and keeping it tight

A bolt does not work alone; the nut, washer and locking mechanism complete the joint. The most important job of these parts is to preserve the preload against vibration and loosening (the subject of the previous issue).

5.1 Nut types – nut types

Table 6 – Common nuts nuts

Common nut types and their feature

Nut	Std	Feature
Hex nut, style 1	ISO 4032 / DIN 934	Standard general-purpose nut
Hex nut, style 2	ISO 4033	Taller, for higher-strength joints
Nylon-insert lock nut	DIN 985	Nylon ring grips the thread (prevailing torque)
All-metal lock nut (Stover)	DIN 980	Deformed crown; reusable, high-temperature
Flange nut	DIN 6923	Integral washer face, often serrated
Castle nut	DIN 935	Slots for a cotter pin (positive lock)
Cap / dome nut	DIN 1587	Covers the thread end; finished look

5.2 Washer types – washer types

Table 7 – Common washers washers

Common washer types and their function

Washer	Std	Function
Plain (flat)	ISO 7089 / DIN 125	Spreads load, protects the surface
Spring (split)	DIN 127	Light spring; limited anti-loosening
Serrated / tooth	DIN 6797	Teeth bite into the surface
Belleville (disc spring)	DIN 2093	Stores spring force, helps maintain preload
Wedge-lock pair	Nord-Lock type	Cam ramps > thread pitch – vibration-proof

5.3 How joints are kept tight – locking methods

- **Prevailing torque:** a nylon-insert (Nyloc) or deformed all-metal nut that creates permanent friction against rotation.
- **Thread-locking adhesive:** an anaerobic adhesive (such as Loctite) that cures in the thread groove and locks rotation; available in removable to permanent grades.
- **Wedge-ramp washer:** a washer pair whose cam ramp is steeper than the thread pitch; any tendency to loosen makes the bolt tighter. The most effective against vibration.
- **Positive lock:** safety wire, a cotter pin in a castle nut, or a double nut – a mechanical lock independent of friction.

- **Adequate preload:** the most basic lock is a high, well-controlled preload itself — another reason the accurate tightening methods matter.

6 Coatings, corrosion & a selection guide

Protection, friction and the right pick

A coating does two jobs at once: protect against corrosion and — often forgotten — set the **friction**, and therefore the nut factor **K** in the torque–tension relationship. The wrong coating can both rust and upset the preload.

Table 8 — Common fastener coatings coatings

Corrosion, friction and notes

Coating	Corrosion protection	Friction / K effect	Note
Zinc electroplate (Zn)	Low–moderate	Variable, ~0.20	Cheapest; HE risk on ≥10.9
Zinc–nickel	High	Controlled	Automotive; far better than plain Zn
Hot-dip galvanized	High (thick)	High / rough	Outdoor steel; nut must be re-tapped
Zinc flake (Dacromet/Geomet)	High	Controlled, low	Thin, no hydrogen embrittlement — automotive standard
Black oxide	Very low	Low (with oil)	Appearance; needs oil film
Phosphate (Mn / Zn)	Low	Adhesive / paint base	Pre-paint, mild protection
PTFE / wax	—	Very low ~0.10–0.14	Used to control & stabilize friction

Warning — Hydrogen Embrittlement (HE)

High-strength bolts (class **10.9** and **12.9**, i.e. above 1000 MPa) can absorb hydrogen during acidic electroplating and suffer a **brittle, delayed fracture** — even hours after installation. Two preventions: **baking / de-embrittlement** at ~200°C for several hours immediately after plating, or using a **zinc-flake coating**, which is mechanically applied and essentially has no HE risk.

6.1 A quick selection guide — selection

Table 9 — Pick by requirement selection

Class / material / coating by need (illustrative)

Requirement	Class / material	Coating
General indoor structural	8.8 steel	Zinc plate
High load / compact (powertrain)	10.9 or 12.9 alloy steel	Zinc flake (avoid HE)
Outdoor / mild corrosion	8.8 + hot-dip, or A2 stainless	HDG / none
Light load + corrosion	A2 (304) stainless	Passive
Marine / chemical	A4 (316) stainless	Passive
Controlled, repeatable preload	any class	Wax / PTFE over zinc flake

The general rule: first **load and temperature** set the class and material, then the **corrosion environment** sets the coating; and finally, if preload accuracy matters, a friction-control layer is added.

7 Glossary & sources

Reference

Term	Meaning
Metric (M) thread	Diameter and pitch in millimeters, 60° profile (ISO)
UNC / UNF	Imperial coarse / fine thread, based on threads per inch (TPI)
Pitch (coarse/fine)	Spacing of the threads; fine = larger stress area
Tolerance class 6g/6H	The male/female thread fit; 6g/6H is the standard fit
Property class	An ISO 898-1 code such as 10.9 (ultimate / yield)
A2 / A4 (ISO 3506)	Austenitic stainless 304 / 316; A4 with molybdenum resists marine corrosion
Socket head cap screw	Cylindrical head with an internal hex (ISO 4762)
Countersunk	Conical head that sits flush with the surface (ISO 10642)
Hex flange bolt	Hex head with an integral washer (DIN 6921)
Set screw	Headless; locks a part onto a shaft
Torx	Six-lobe drive with the least slip (cam-out)
Prevailing-torque nut	Nyloc or all-metal; creates permanent friction
Wedge-lock washer	Cam ramp > thread pitch; anti-vibration
Zinc flake	Thin anti-corrosion coating with no hydrogen embrittlement
Hydrogen embrittlement	Brittle, delayed fracture of a plated high-strength bolt

Sources & standards

- **Dimensions & types:** the ISO standard series (4014, 4017, 4762, 10642, 7380, 4026–4029, 4032) and their DIN equivalents; and the ISO metric thread system ISO 261 / ISO 965 (tolerances).
- **Mechanical properties:** ISO 898-1 (steel bolts), ISO 898-2 (nuts), and ISO 3506 (stainless steel).
- **Coatings & surface:** ISO 4042 (electroplated coatings and their coding) and general industry literature on zinc flake, hot-dip galvanizing and hydrogen embrittlement.
- **Series link:** this reference complements the “bolted-joint engineering” issue; the concepts of preload, the K factor and torque control are explained there.
- The standard codes and strength values are real; the example numbers are illustrative, and for final design you should consult the current edition of the standard. ■

FASTENER CATALOG · SEE EVERY TYPE, FIND EVERY BRAND

The fastener catalog: Every type illustrated — and where to find the real thing

A visual gallery + a directory of major manufacturers, famous brands & their catalogs

This issue is a quick visual gallery of the common fastener types — bolts, screws, nuts, washers and more — each drawn with its standard, so you can recognize the shape at a glance. Then it points you to the great catalogs where you can see real product photos, dimensions and prices, and lists the major manufacturers and famous brands with direct links.

Visual gallery

Heads, drives, nuts, washers & more — illustrated with standards.

The great catalogs

McMaster-Carr, Bolt Depot, Bossard, Würth — with links.

Major manufacturers

Würth, ITW, LISI, Nucor, ARaymond, Howmet...

Famous brands

Nord-Lock, Huck, Loctite, SPAX, Hilti, Heli-Coil...

How to use this issue — and a note on images & trademarks

The drawings here are **original schematic illustrations** made to show the shape of each fastener type, not photographs of any specific product. For **real product photos**, exact dimensions, materials and prices, follow the catalog links in Part 2 — **McMaster-Carr** and **Bolt Depot** in particular have a clear picture for almost every type. All company, brand and standard names (ISO, DIN, Torx®, Nord-Lock®, Huck®, Loctite®, SPAX® and others) are trademarks of their respective owners and are listed for reference only.

CONTENTS

1	Visual gallery — bolts, screws, nuts, washers & more	The gallery
2	The great catalogs — where to see real product photos	Catalogs & links
3	Major manufacturers & famous brands — a directory	Brand directory

1 Visual gallery – bolts, screws, nuts, washers & more

Recognize the shape of every common type

Five quick panels cover the families you will meet most: bolt & screw **heads, drives, nuts, washers**, and a set of **other** fasteners. Each item is labeled with its name and the standard or family it belongs to.

1.1 Bolt & screw head styles



Hex head
ISO 4014



Socket cap
ISO 4762



Countersunk
ISO 10642



Button head
ISO 7380



Hex flange
DIN 6921



Set screw
ISO 4026

Side view. The countersunk head sits flush; the flange head has an integral washer; the headless set (grub) screw locks a part onto a shaft.
(Also common: pan, cheese and carriage heads.)

1.2 Drive types



Hex (ext.)



Hex socket



Torx



Phillips



Slotted



Pozidriv

Top view. Internal drives (hex socket, Torx) transmit more torque without slipping; Torx has the least “cam-out” and dominates the automotive line.

1.3 Nuts



Hex nut



Nylon lock



Flange nut



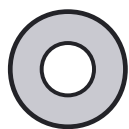
Castle nut



Wing nut

Hex nut (DIN 934), nylon-insert lock nut (the tan ring grips the thread), flange nut, castle nut (slots take a cotter pin), cap / dome nut (closed end), and wing nut (hand-tightened).

1.4 Washers



Plain (flat)



Square



Spring (split)



Ext. tooth



Belleville



Wedge-lock

Plain (DIN 125) and square load-spreading washers; split-spring and external-tooth lock washers; a Belleville disc spring (shown in section); and a wedge-lock pair whose cam ramps (red) resist vibration.

1.5 Other fasteners



Threaded rod
stud



Blind rivet
(pop)



Wood screw



Self-tapping



Wall anchor



Thread insert
Heli-Coil

Threaded rod / stud, blind (pop) rivet with its mandrel, wood screw, self-tapping screw, expansion wall anchor, and a helical thread insert (Heli-Coil) for repairing or reinforcing threads.

2 The great catalogs — where to see real product photos

Follow these links for photographs, dimensions & prices

The illustrations in Part 1 show the shape; for the actual part — real photos, every size, material, finish and price — these online catalogs are the best references. The first two are the quickest way to **see a clear picture of almost any fastener type**.

Table 1 — Online fastener catalogs catalogs

What each is best for, and the link

Catalog	Best for	Link
McMaster-Carr	The classic reference: a clean line-art image for almost every type, with full specs. ~595,000 products.	mcmaster.com
Bolt Depot	Simple, beginner-friendly “Fastener Type Chart” with pictures; buy single pieces; metric & imperial.	boltdepot.com
Bossard	200,000+ standard & branded parts; premium brands (Nord-Lock, Huck); strong engineering data.	bossard.com
Würth	125,000+ products, metric & inch; downloadable PDF catalogs; assembly materials & tools.	wuerth.com
Fastenal	Largest North-American distributor; OEM, MRO, construction & safety; broad stock.	fastenal.com
Grainger	Broad industrial / MRO supplier with fasteners and detailed data sheets.	grainger.com

Quickest way to “see the part”

Start with Bolt Depot’s type chart or McMaster-Carr

For a visual index, **Bolt Depot** groups everything into picture-led categories (bolts, screws, nuts, washers, anchors, rivets...), and **McMaster-Carr** shows a precise drawing plus dimensions for each variant. Between the two, you can match almost any fastener you have in hand to a name and a standard.

3 Major manufacturers & famous brands — a directory

Who makes what, grouped by field

Beyond the distributors above, these are the major manufacturers and the famous specialty brands, grouped by what they are best known for. Names are trademarks of their owners; links go to each company's official site.

Table 2 — Manufacturers & brands by category directory

Major fastener makers and famous brands (illustrative, not exhaustive)

Category	Brands — known for	Website
Distributors & catalogs	Würth (Europe's largest), Bossard , Fastenal , Grainger , McMaster-Carr , Bolt Depot — huge multi-brand stock + catalogs.	wuerth.com bossard.com
Industrial & OEM bolts	ITW (Illinois Tool Works), Nucor Fastener (US steel), Nederschroef (OEM cold-forming) — high-volume standard & engineered bolts.	itw.com nucor-fastener.com
Automotive fastening	ARaymond (clips & systems), Fontana Gruppo , Acument (the Torx family) — OEM assembly fasteners.	araymond.com acument.com
Aerospace fasteners	LISI Aerospace (Hi-Lok, Hi-Lite), Howmet Aerospace (ex-Alcoa) — high-strength, lightweight, flight-critical.	lisi-group.com howmet.com
Anti-loosening / locking	Nord-Lock (wedge-locking washers), HEICO (HEICO-LOCK), Loctite (Henkel thread-locking adhesives).	nord-lock.com loctite.com
Structural / blind / lockbolt	Huck (Howmet — lockbolts & structural blind rivets), Avdel & POP (Stanley Engineered Fastening).	hfsindustrial.com stanleyengineeredfastening.com
Anchors / construction	Hilti (anchors & systems), EJOT (building & industry), ITW Red Head (concrete anchors).	hilti.com ejot.com
Wood / self-drilling	SPAX (Altenloh, Brinck & Co. — Germany's oldest screw maker), Reisser — wood & construction screws.	spax.com
Thread inserts	Heli-Coil (helical wire inserts, Stanley), PEM (Penn Engineering self-clinching inserts).	heli-coil.com pemnet.com

Reference & sources

Catalog and brand information compiled from each company's official website and public industry directories (McMaster-Carr, Bolt Depot, Bossard, Würth, Fastenal, Grainger; and brand owners Nord-Lock, Howmet/Huck, Henkel/Loctite, LISI, ITW, Hilti, SPAX, Penn Engineering and others). The illustrations are original schematic drawings; product photographs and exact specifications live on the linked catalogs. Company, brand and standard names are trademarks of their respective owners, listed for reference only. ■

PRODUCTION PLANNING · FROM FORECAST TO THE LINE

Planning the machine: MPS, MRP & MRP II at the Tesla Gigafactory

And how the plan is handed to Tesla's MOS for execution

Three interlocking schedules decide how many vehicles are built each week, which materials are ordered and exactly when, and whether the plant can actually build it — and then how that plan is delivered to **MOS** (Tesla's Manufacturing Operating System) and run on the line. A worked walkthrough with real-shaped tables for a Model Y / Cybertruck plant.

MPS — what to build

The master build plan: the rate and sequence of finished vehicles.

MRP II — is it feasible?

Adding capacity, labor and money to the equation; the closed loop.

MRP — which parts, and when

Exploding the BOM, netting against inventory, offsetting by lead time.

MOS — execution on the line

Turning the plan into work orders, traceability and per-vehicle gating.

A note on the numbers

The tables in this guide are filled with **realistic but invented** figures, chosen to make the mechanism concrete; they are not leaked Tesla internal data. Three layers of knowledge are combined here: (1) **standard planning logic** (the **APICS** framework and the **ISA-95** pyramid), which is the same in any plant; (2) **publicly reported facts about Tesla** — its **MOS** and products such as **Model Y**, **Cybertruck** and the **4680** cell; and (3) one **fully worked numeric example** for teaching. Treat every specific number as a plausible illustration, not an official plant value.

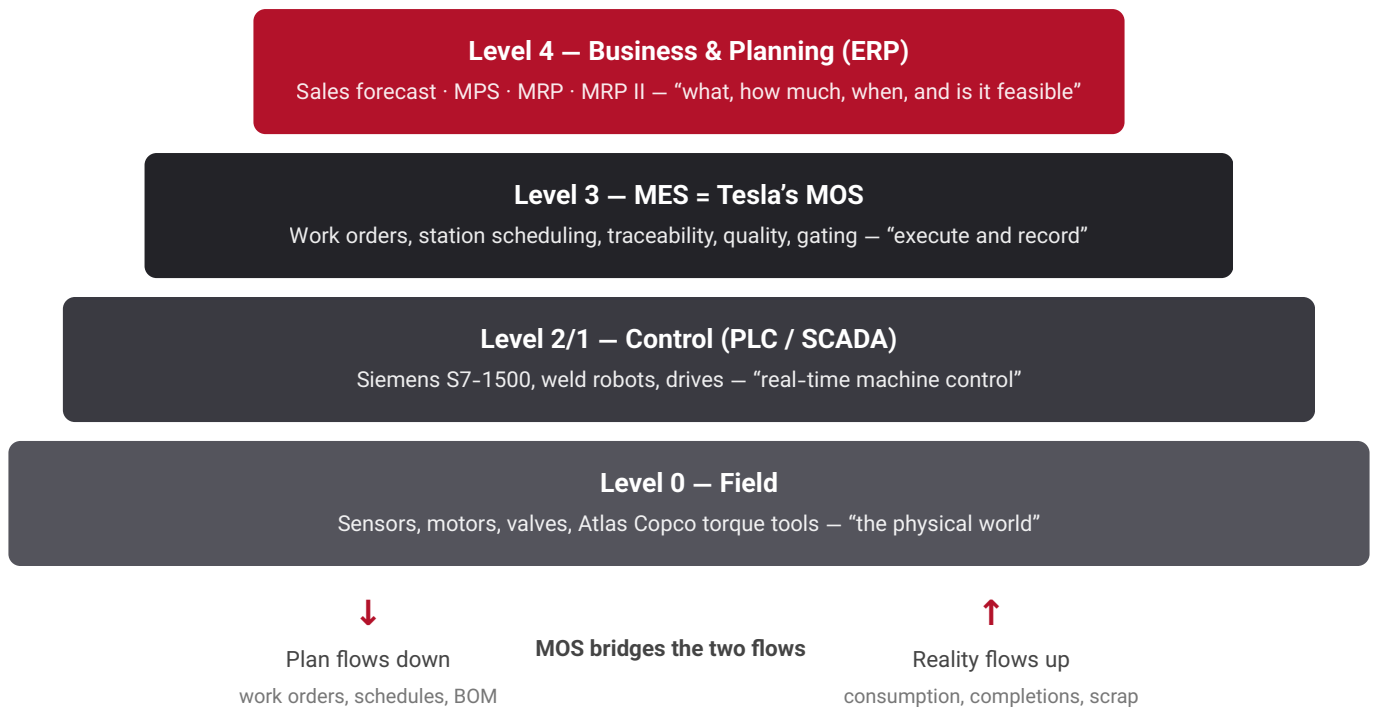
CONTENTS

1	Where planning sits relative to MOS — the automation pyramid	ISA-95
2	MPS — the master build plan (two tables)	Master Production Schedule
3	MRP — material planning (BOM + three tables)	Material Requirements Planning
4	MRP II — resource planning (capacity, labor, money)	Manufacturing Resource Planning
5	The closed loop and the link to every shop	Closed loop & the shop floor
6	Glossary & sources	Reference

1 Where does planning stand? The automation pyramid and its handshake with MOS

Where planning sits – the ISA-95 pyramid and its link to MOS

Before any table, one point has to be clear: **MPS**, **MRP** and **MRP II** are **planning** functions, and they live in the plant's upper layers (the **ERP** / business system). **MOS** – Tesla's own Manufacturing Operating System – sits in the **execution** layer. The plan flows top-down and MOS runs it on the line; then reality flows bottom-up and corrects the next plan. That round trip is the "closed loop."



Put simply: **planning decides what should happen; MOS makes it happen and records what actually did.** Three things pass from the planning layer to MOS – the vehicle build sequence (from MPS), the parts list for each vehicle and when to call them (from MRP), and the capacity ceiling of each work center (from MRP II). In return, MOS gives back three things: actual material consumption, actual vehicle completions, and scrap / rework. That feedback becomes the input to the next planning cycle.

MES vs MOS – a caveat

Why this separation matters

MRP / MPS / MRP II tables are **never** produced inside MOS itself; they are built in the ERP / planning system and their output is handed to MOS. In Tesla's public storytelling, MOS is the system of record for every vehicle and the conductor of the line – but the score was written by planning. Wherever you see a table in this guide, read it as a Level 4 output that is communicated down to Level 3 (MOS).

2 MPS – the master build plan

The Master Production Schedule

MPS answers a single question: “In each time bucket, how many of each **finished-product** configuration do we build?” It is the bridge between the **sales forecast** and the **shop floor**. Its inputs are the forecast and real customer orders; its output is a frozen build schedule that sets the rate and sequence of the line and becomes the root of every MRP calculation.

Key concepts in an MPS record – how to read it

- **Time buckets:** weekly in this example; in practice Tesla goes finer, down to day / shift level.
- **Time fences:** inside the **DTF** (demand fence) the plan is effectively frozen and only real orders count; outside it, the greater of forecast and orders is used. The **PTF** (planning fence) is the point out to which the system will auto-suggest production orders.
- **PAB** – projected available balance, the expected on-hand inventory; **ATP** – available-to-promise, the quantity not yet committed that can still be promised to a new customer.

Table 1 – Aggregate master schedule, vehicle final assembly (Giga Texas, weekly)

MPS · aggregate

Aggregate master schedule – vehicle final assembly, 8-week horizon (units)

Product configuration	W1	W2	W3	W4	W5	W6	W7	W8
Model Y Long Range AWD	3,200	3,200	3,400	3,400	3,400	3,600	3,600	3,600
Model Y Performance	800	800	850	850	850	900	900	900
Cybertruck AWD	600	650	700	700	750	750	800	800
Total vehicles / week	4,600	4,650	4,950	4,950	5,000	5,250	5,300	5,300

This “total” row is the same number that Part 4 converts into capacity and that Part 3 explodes into material requirements.

The table above is the high-level picture; but each MPS row has a calculation “engine” behind it. Table 2 opens that engine for the highest-volume configuration – **Model Y LR AWD** – so you can see where the build numbers come from.

Table 2 – Detailed MPS record for Model Y LR AWD

MPS · detail

Detailed MPS record · on-hand 450 · DTF = end of W2 (units)

Row	W1	W2	W3	W4
Forecast	3,000	3,000	3,200	3,200
Customer orders	3,150	2,800	1,900	900
Demand used	3,150	2,800	3,200	3,200
MPS (planned build)	3,200	3,200	3,400	3,400
PAB – projected balance	500	900	1,100	1,300
ATP – available to promise	500	400	1,500	2,500

Row logic: inside the DTF (W1–W2), “demand used” = real orders; outside the DTF, the greater of forecast / orders.

$PAB(t) = PAB(t-1) + MPS(t) - Demand(t)$ · $ATP(t) = MPS(t) - \sum \text{orders until next MPS}$ (W1 also counts opening on-hand).

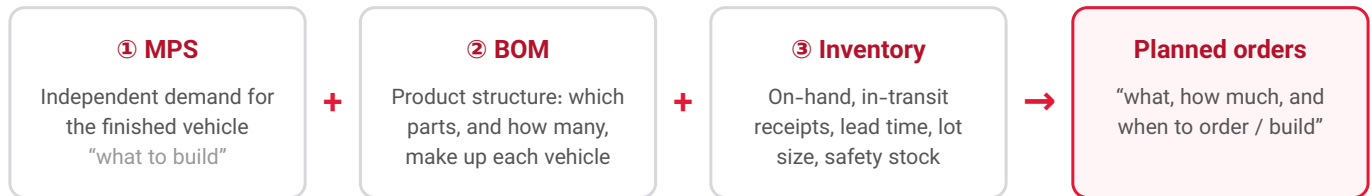
How does MPS connect to the floor? – what MPS hands over

The “total” figure and the configuration sequence are passed directly to **MOS** to set the order in which bodies enter the line and the takt time. It is the MPS that says this week the line rate should be set to ~5,000 vehicles and how the color / trim sequence should run. Every downstream shop – from stamping to final assembly – takes its rhythm from this plan. Crucially, MPS is the independent demand for the finished product; the next part shows how that demand is translated, through the BOM, into the dependent demand for thousands of parts.

3 MRP – Material Requirements Planning

From finished vehicles to time-phased part orders

If MPS says “4,000 vehicles next week,” then MRP asks “so how many stators, how many cells and how many megacastings do we need, and exactly **when** must we release the orders?” MRP fuses three inputs into one concrete output: **planned orders**.



3.1 The bill of materials (BOM)

A BOM is the product’s tree structure. The demand for one vehicle (the parent) becomes demand for its child parts by multiplying by the “quantity per”; this is called **dependent demand**. The tree below is a simplified illustration for Model Y (figures are illustrative):

Level	Component	Qty / vehicle	Source
0	Model Y LR AWD (vehicle)	1	MPS – final assembly
·1	Body-in-White	1	Body shop
··2	Front megacasting	1	Giga Press
··2	Rear megacasting	1	Giga Press
·1	Structural battery pack	1	Battery line
··2	4680 cell	830	Cell line / supplier
·1	Front drive unit	1	Drive-unit shop
·1	Rear drive unit (RDU)	1	Drive-unit shop
··2	Stator assembly	1	DU sub-assembly
··2	Rotor assembly	1	DU sub-assembly
··2	Inverter (power electronics)	1	Supplier / in-house
··2	Gearbox / reduction gear	1	DU sub-assembly
·1	Seat assembly	5	Supplier (JIS)
·1	Wheel & tire	4	Supplier (JIT)

The highlighted rows (RDU, Stator, 4680 cell) are tracked in the tables that follow, so you can watch the explosion and the lead-time offset in action.

3.2 Three MRP records, three levels of the tree – the explosion in action

The core of MRP is a time-phased grid. For each item we net the gross requirement against on-hand and in-transit receipts, then push the order back by the **lead time**. The golden rule: **each parent’s “planned order release” becomes, verbatim, its child’s “gross requirement.”**

Table 3 – MRP record: Rear Drive Unit (RDU) MRP · level 1

LT = 1 wk · lot-for-lot · safety stock 200 · on-hand 800 (units)

RDU	W1	W2	W3	W4
Gross requirements	4,000	4,000	4,250	4,250
Scheduled receipts	3,400	—	—	—
Projected on-hand	200	200	200	200
Net requirements	0	4,000	4,250	4,250
Planned order receipt	0	4,000	4,250	4,250
Planned order release	4,000	4,250	4,250	4,250

“Gross requirements” here = total Model Y from Table 1 (LR AWD + Performance). Note: because LT = 1 week, the W2 receipt comes from the W1 release — one column to the left.

Table 4 — MRP record: Stator assembly (child of RDU) MRP · level 2

LT = 2 wk · lot-for-lot · safety stock 300 · on-hand 1,000 (units)

Stator	W1	W2	W3	W4
Gross req. (= RDU release)	4,000	4,250	4,250	4,250
Scheduled receipts	3,300	4,250	—	—
Projected on-hand	300	300	300	300
Net requirements	0	0	4,250	4,250
Planned order receipt	0	0	4,250	4,250
Planned order release	4,250	4,250	—	—

The highlighted row is the RDU “planned order release” from Table 3, now serving as the stator’s gross requirement — this is the heart of MRP. Because LT = 2 weeks, releases fall two columns earlier than receipts.

Table 5 — MRP record: 4680 cell (supply-chain scale) MRP · cells

LT = 1 wk · lot-for-lot · safety stock 200,000 · on-hand 450,000 (cells)

4680 cell	W1	W2	W3	W4
Gross requirements	3,820,000	3,860,000	4,110,000	4,110,000
Scheduled receipts	3,570,000	—	—	—
Projected on-hand	200,000	200,000	200,000	200,000
Net requirements	0	3,860,000	4,110,000	4,110,000
Planned order release	3,860,000	4,110,000	4,110,000	—

Cell gross requirement ≈ (total Model Y + Cybertruck) × 830 cells per vehicle. This release row is the signal the cell line or supplier must meet — million-scale, weekly.

3.3 How does MRP connect to the floor and to supply? — releases become real actions

The “planned order releases” split into two branches: for purchased items (cells, inverters, seats, tires) they become a **purchase order / kanban / JIT call-off** to the supplier; for made items (RDU, stator, megacasting) they become an **internal work order** for the drive-unit, casting and battery shops. Those orders flow into **MOS**, which calls material to the line side and records consumption — and when MOS backflushes actual usage, the input inventory for the next MRP cycle is updated.

4 MRP II – Manufacturing Resource Planning

Capacity, labor and money – the closed loop

MRP assumes infinite capacity; a real plant is not. **MRP II** is MRP plus three things that make it “closed” and realistic: **capacity** (can the work centers actually take this load?), **labor**, and **money** (the same plan expressed in dollars). Its classic slogan: “one set of numbers for the whole company.”

4.1 Rough-cut capacity at the MPS level (RCCP) – Rough-Cut Capacity Planning

Before accepting an MPS, we test it against a few critical work centers. Using a “bill of resources” (standard hours per vehicle at each center) we compute the required load and compare it with available capacity:

Table 6 – RCCP: load vs capacity, by work center MRP II · capacity

Required vs available hours; peak week = W3/W4 at 4,950 vehicles

Work center	Std hr/veh	Req W1	Req W2	Req W3	Req W4	Capacity/wk	Peak util.	Status
Stamping / Giga Press	0.15	690	698	743	743	850	87%	OK
Body shop (welding)	0.30	1,380	1,395	1,485	1,485	1,600	93%	OK
Paint shop	0.40	1,840	1,860	1,980	1,980	2,000	99%	Monitor △
General Assembly	1.20	5,520	5,580	5,940	5,940	5,600	106%	OVER – action
Battery & Drive Unit	0.50	2,300	2,325	2,475	2,475	2,600	95%	OK

“Req” = (that week’s vehicles from Table 1) × “Std hr/veh.” Final assembly (GA) hits 106% in peak weeks: this is the plant’s bottleneck.

Closed loop

When capacity says no, the MPS changes

GA at 106% means the plan is “infeasible.” That feedback goes back up and one of these decisions is made: overtime / a Saturday shift, leveling the MPS (moving part of W3 into W2), or changing the product mix. This back-and-forth between plan and capacity is exactly what upgrades MRP into MRP II.

4.2 Detailed capacity for the bottleneck (CRP) – Capacity Requirements Planning, GA

RCCP is rough; CRP is finer and separates the load of **released (firm) orders** from **planned** orders, so we can see in which week the overload sits and what kind it is:

Table 7 – CRP for the General Assembly work center (GA) MRP II · CRP

Released vs planned load against capacity (hours)

GA – week	Released load	Planned load	Total load	Capacity	Over/(under)	Util.
W1	5,520	0	5,520	5,600	(80)	99%
W2	3,000	2,580	5,580	5,600	(20)	100%
W3	0	5,940	5,940	5,600	340	106%
W4	0	5,940	5,940	5,600	340	106%

Near-term weeks are mostly “released / firm”; far weeks are “planned.” The 340-hour W3/W4 overload is still only planned, so there is room to correct it.

4.3 Translating the plan into people and money – one set of numbers

MRP II's final distinction is that it ties the same physical plan to human and financial resources. The two views below translate Week 1 (4,600 vehicles) into headcount and cost (standard figures, illustrative):

Table 8a – Required crew from load MRP II · labor

W1 load → required crew (FTE $\approx$ load $\div$ 38 productive hr/wk)

Work center	Load W1 (hr)	Crew (FTE)
Stamping / Giga Press	690	18
Body shop	1,380	36
Paint shop	1,840	48
General Assembly	5,520	145
Battery & Drive Unit	2,300	61
Total direct (per shift basis)	11,730	308

Table 8b – Financial roll-up of the week MRP II · finance

Standard factory cost of the W1 plan (4,600 vehicles)

Cost element	Per vehicle (std)	× 4,600 vehicles
Direct material	\$27,500	\$126,500,000
Direct labor	\$1,450	\$6,670,000
Variable overhead	\$2,200	\$10,120,000
Total factory cost	\$31,150	\$143,290,000

This “one set of numbers” is what aligns finance, procurement and production on a single truth: capacity hours, required headcount and cost dollars all derive from the same MPS.

5 The closed loop and the link to every shop

The closed loop – how planning touches every shop

Now let us put it all side by side. For each key shop, the table below shows what MPS, MRP and MRP II contribute and what **MOS** does at the moment of execution. That last column is the contact point between planning and the floor.

Shop	MPS provides	MRP provides	MRP II provides	MOS executes (Level 3)
Stamping / Giga Press	Body rate & sequence	Aluminum ingot & insert orders; casting schedule	Press capacity & die availability; changeover time	Per-casting work order, serial / traceability, changeover logging
Body shop (BIW)	Body build sequence	Stamped panels + 2 megacastings per body	Weld-cell and robot capacity	Station dispatch, weld-data logging, fixture status
Paint shop	Color / trim mix	Paint, primer & sealer volumes	Oven & booth capacity; potential bottleneck	Vehicle tracking through booths, paint parameters, rework loop
General Assembly	Final sequence & takt time	Thousands of parts (seats, glass, harness, drive unit, pack) JIT/JIS	Labor & station capacity – the main bottleneck	Station-by-station work orders, torque logging, fit confirmation, gating
Drive-unit shop	Drive-unit demand (derived from MPS)	Stator, rotor, inverter, gearbox	Machining & test-bench capacity	Drive-unit build and end-of-line test record; serial-to-vehicle link
Battery / Cell	Cell & pack demand (derived from MPS)	4680 cell, foil, electrolyte, structural-pack parts	Cell-line & pack-line capacity	Cell / pack traceability, “marriage” of pack to vehicle
Inbound / suppliers	Forecast to the supplier	Order release → PO / kanban / JIT call-off	Supplier capacity & inventory dollars	Receiving, line-side replenishment signal, consumption backflush

In one glance

One sentence that stitches it all together

MPS says which vehicle and how many; **MRP** breaks that into parts and order timing; **MRP II** checks whether the labor, machines and money exist and corrects the MPS if not; and **MOS** runs that approved plan on the line in real time, records it, and feeds reality back to planning for the next cycle. ■

6 Glossary & sources

Reference

Term	Meaning
MPS	Master Production Schedule – the time-phased build plan for the finished product
MRP	Material Requirements Planning – explosion of the MPS through the BOM into time-phased orders
MRP II	Manufacturing Resource Planning – MRP + capacity + labor + finance, as a closed loop
BOM	Bill of materials – the tree structure of a product's parts and the quantity per
DTF / PTF	Demand / planning time fence – the time boundaries for freezing the plan and for auto-suggested orders
PAB	Projected available balance – expected on-hand inventory at the end of each bucket
ATP	Available to promise – quantity not yet committed to a customer and still promisable
GR / NR	Gross / net requirement – total need, and need after deducting on-hand and receipts
SR	Scheduled receipts – orders already released and in transit to be received
L4L / SS / LT	Lot-for-lot / safety stock / lead time – lot-sizing policy, inventory buffer, supply delay
RCCP / CRP	Rough-cut / detailed capacity – load-vs-capacity check at the MPS and MRP levels
Takt	Takt time – the heartbeat of the line: available time ÷ demand
MES / MOS	Manufacturing Execution System / Tesla's Manufacturing Operating System – the execution layer that runs the plan on the line and records it
ERP / ISA-95	Enterprise Resource Planning / the automation pyramid – the business layer and the reference model for plant layering

Sources & frameworks

- **Planning framework:** standard APICS / ASCM literature (MPS, MRP, MRP II, CRP) and the ANSI/ISA-95 reference model for Level 0–4 layering.
- **Publicly reported facts about Tesla's MOS:** general press coverage and plant tours (including CNBC and CleanTechnica reporting) that describe the role of MOS in build / repair / test traceability.
- **Product context:** public information about Tesla's Model Y, Cybertruck, the 4680 cell, the structural pack, and the Giga Press megacastings.
- **The numbers:** all illustrative and realistic, built solely to clarify the mechanism; they are not Tesla internal values.



EXPLAINED SIMPLY • NO TECH BACKGROUND NEEDED

The Factory Cloud

How a factory's machines quietly send their data to the internet —
and why it matters — told in plain, everyday language.

- What "**the cloud**" really means (no jargon)
- Why a factory can **predict a breakdown**

- How a machine sends its **temperature** to your screen
- How it's all **kept safe**

In This Issue

THE FACTORY CLOUD – EXPLAINED FOR EVERYONE

- 1 What "the cloud" really is
- 2 The journey of one number
- 3 How machines "talk" to each other
- 4 Why keep some thinking near the machine
- 5 What we actually ask the computer to do
- 6 Getting the number from a real machine
- 7 What happens up in the cloud
- 8 Turning numbers into decisions
- 9 Keeping it all safe
- ★ The whole story in one picture & a plain glossary

A Note Before We Begin

This guide is for the curious — not the technical. You don't need to know anything about computers, programming, or factories. We'll start from zero and explain every idea with **everyday comparisons**: post offices, filing cabinets, power grids, ID badges.

The big idea is simple. A modern factory is full of machines that constantly produce **numbers** — a temperature here, a speed there. If those numbers are gathered and read sensibly, they can warn you before something breaks. This is the story of how those numbers travel from a machine all the way to a screen — and back.

— The Industry 4.0 editorial team

ONE IDEA TO KEEP IN MIND

Picture a single number — say, the temperature of one machine — as a **letter** that has to travel from the factory floor to the manager's desk. Almost everything in this guide is just one step in that letter's journey. Keep the letter in mind and nothing here will feel complicated.

THE BASICS · NO. 1

1. What "the cloud" really is



The cloud" sounds mysterious, but it isn't. It simply means **using powerful computers that belong to someone else, over the internet** — instead of buying, housing, and maintaining your own. You pay only for what you use, and you can have more or less of it whenever you want.

THINK OF IT LIKE ELECTRICITY

You don't build a power plant in your backyard. You plug into the grid and pay for the electricity you actually use. "The cloud" is the same idea, but for computing: you plug into giant, professionally-run computer centers and pay for what you use — no plant to build, no upkeep.

From the cloud you essentially borrow three things: **brain-power** (to do calculations), **space** (to keep your information), and **ready-made tools** (useful services you'd otherwise have to build yourself).

1.1 Why would a factory want this?

A factory is full of machines that quietly produce numbers all day. Left inside the machine, those numbers are useless. Sending them to the cloud unlocks some very practical wins:

- ◆ **Watch everything from anywhere:** see how every machine is doing on a single screen — even from your phone, at home.
- ◆ **Get a warning before a breakdown:** just like a car that says "service the brakes soon," the data can flag a problem *before* it stops the line.
- ◆ **Grow without headaches:** one machine or a thousand across several sites — the cloud copes with both just as easily.
- ◆ **Spot patterns:** with so many numbers in one place, you can find waste, improve quality, and run the factory smarter.

A modern factory makes two things: products, and information. The cloud is where the information becomes useful.

A few words, in plain terms**GLOSSARY****The cloud**

Other people's computers that you rent over the internet, instead of owning your own.

Internet of Things

The simple idea of putting everyday objects — here, factory machines — online so they can send and receive information.

The machine's data (a "reading")

A single measurement a machine reports about itself, like a temperature or a speed.

The factory side vs. the office side

The world of physical machines on one hand, and the world of computers, screens, and the internet on the other. The cloud connects the two.

THE BIG PICTURE · THE JOURNEY

2. The journey of one number

Let's follow a single number. A machine — call it Machine 7 — has just measured a temperature of 82 degrees. What has to happen for that number to appear on the manager's screen? It makes a short journey, in stages. Each stage has one simple job and then passes the number along, like a relay race.



The journey of one number — from the machine (top) all the way to a screen (bottom)

THINK OF IT LIKE POSTING A LETTER

The sensor **writes** the letter. The nearby helper is the **local postbox**. "Sending it out" is the **mail truck**. The cloud's front desk is the **sorting office**. The helpers **act on** the letter, the filing cabinet **archives** it, and the screens turn a pile of letters into a tidy **report**. Same number, passed hand to hand.

It's a two-way street**IMPORTANT**

Numbers don't only travel *up* to the cloud. The cloud can also send instructions back *down* — like a remote control that can say "slow that machine down" or "switch it off." That's incredibly useful, but it's also why safety (Article 9) matters so much: a wrong instruction could affect a real, physical machine.

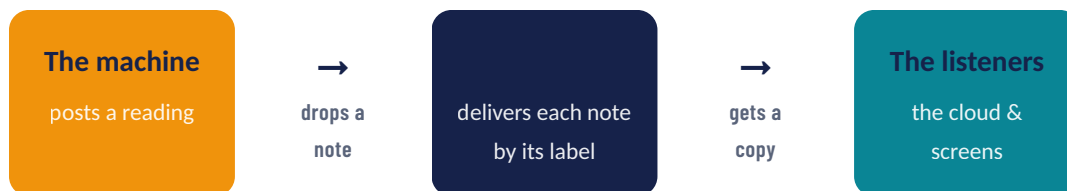
HOW IT WORKS · PASSING MESSAGES

3. How machines "talk" to each other

For a machine to share its readings, it needs an agreed way to pass messages around. The clever trick used almost everywhere is this: the machine doesn't phone each listener one by one. Instead, everything goes through a **middleman**.

THINK OF IT LIKE A POST OFFICE

The machine drops its note at a post office under a **label** — say, "Machine 7's temperature." Anyone who has signed up for that label automatically gets a copy. The wonderful part: the machine never has to know *who* is listening, or how many. It just posts; the post office delivers.



The machine posts; the post office delivers to whoever signed up

The "label" works like a folder-style address, going from general to specific. For example:

Line 1 › Machine 7 › temperature

This whole approach is deliberately **light**, so it works even on small devices and a weak internet connection — which is exactly what a busy factory needs. There are a handful of these "languages" in industry; this post-office style is the popular, lightweight one.

How hard do we try to deliver?

GOOD TO KNOW

You can choose how carefully each note is delivered: **send it and don't worry** if one slips through (fine for readings that repeat every second); **keep trying until it arrives** (the usual choice); or **make absolutely sure exactly one copy arrives** (for important messages that must never be lost or doubled).

SMART SHORTCUTS · THINKING NEARBY

4. Why keep some thinking near the machine

If the cloud is so capable, why not send every single number straight to it? Because that isn't always the smart choice. Often it's better to do a little thinking **right next to the machine** first, before anything is sent away.

THINK OF IT LIKE DOING THE MATH YOURSELF

If you can add up a quick bill in your head at your own desk, you do it right there — you don't mail the question to an accountant across the country and wait days for the answer. For fast, simple jobs, doing it on the spot is quicker, cheaper, and still works even when the post is closed.

4.1 Four reasons to think nearby

- ◆ **Speed:** for an instant safety reaction, you can't wait for a round trip to the cloud and back.
- ◆ **Less to send:** a machine can produce hundreds of readings a second; the helper nearby can summarize them and send only what matters.
- ◆ **Keeps working offline:** if the internet drops, the nearby helper holds the data, so the factory carries on.
- ◆ **Privacy:** sensitive details can stay on-site instead of leaving in full.

Rule of thumb: quick decisions happen right by the machine; the big-picture thinking happens in the cloud.

Who does what?

Done right by the machine	Done up in the cloud
Instant reactions and simple checks	Heavy number-crunching and learning from patterns
Trimming and summarizing the flood of readings	Keeping every reading for the long term
Carrying on during an internet outage	The big-picture view across every site

BEHIND THE SCENES · STEP 1

5. What we actually ask the computer to do

You might wonder what "telling a computer to do this" really involves. The good news: you don't need to read a single line of computer code to understand it. Here is the plain-English version of the instructions — think of each one as a simple recipe.

5.1 The machine's routine: take a reading, post it

In plain words**RECIPE 1**

What the "sensor" program is told to do, over and over:

- 1 Check in with the post office (so you're ready to send notes).
- 2 Every couple of seconds, read the machine's current temperature.
- 3 Write a short note with three facts: the machine's name, the reading, and the time.
- 4 Drop that note at the post office under the label "Machine 7's temperature."
- 5 Wait a moment, then do it all again — forever.

The little note it posts is nothing fancy — just three plain facts:

machine: Machine 7 · reading: 82.4° · time: 3:14:07 pm

5.2 The listener's routine: catch the note, show it

In plain words**RECIPE 2**

What a "listener" program is told to do:

- 1 Sign up at the post office for every label coming from Line 1.
- 2 Whenever a note arrives, open it and read the number.
- 3 Show that number (on a screen, a log, or a chart).

IN ONE SENTENCE

One small program keeps posting the machine's temperature to a noticeboard; another small program keeps watching that noticeboard and shows whatever appears. That's the whole idea — everything else is just a more capable version of these two simple habits.

BEHIND THE SCENES · STEP 2

6. Getting the number from a real machine

So far, we imagined the temperature reading appearing by magic. In a real factory, it comes from the machine's own little built-in controller — a small, rugged computer that runs the machine and knows its current readings. Think of it as **the machine's brain**.

THINK OF IT LIKE ASKING THE MACHINE DIRECTLY

It's as if you walked up to the machine and asked, "What's your temperature right now?" The machine's brain has the answer ready; you just need to know how to ask, and where it keeps that particular number.

6.1 The plain-English instructions

In plain words

RECIPE 3

How a program gets a reading straight from the machine's brain:

- 1 Connect to the machine's brain over the factory network (you need its address).
- 2 Ask it for the value kept in a specific spot inside it — the spot that holds the temperature.
- 3 Translate the brain's raw answer into an ordinary number we can read.

6.2 The simplest possible "helper nearby"

Now combine this with what we learned in Article 5. A tiny program can do just two things in a loop: **ask the machine for its reading**, then **post that reading to the post office**. That little "ask, then post" program is, in its simplest form, the **nearby helper** from the journey — the bridge between the machine and the wider world.

In plain words

RECIPE 4

The tiniest bridge between a machine and the cloud:

- 1 Ask the machine's brain for the current temperature.
- 2 Write the usual little note (machine, reading, time).
- 3 Post it to the post office.
- 4 Repeat, once a second.



These few simple steps close the old gap between the factory floor and the world of computers.

BEHIND THE SCENES · STEP 3

7. What happens up in the cloud

The note has been posted and has travelled to the cloud. What happens when it arrives? Three things, in quick succession — and remarkably, no one has to be sitting there waiting for it.

7.1 A worker who shows up only when needed

THINK OF IT LIKE HIRING A TEMP BY THE SECOND

Instead of keeping a worker on the payroll all day waiting for the rare moment a note arrives, the cloud hires a **temp worker who clocks in only for the two seconds of actual work, then clocks out**. You never pay for idle time — and if a thousand notes arrive at once, a thousand temps appear automatically, then vanish. This pay-only-for-the-moment style is what people mean by "serverless."

7.2 The plain-English instructions

In plain words

RECIPE 5

What happens the instant a reading reaches the cloud:

- 1 The note arrives at the cloud's front desk, which also checks the sender's ID.
- 2 A standing rule says: "Send any temperature note to a helper."
- 3 A tiny helper wakes up just for this note: it **files the reading away** in the filing cabinet, and if the number is too high, it **raises an alarm**.
- 4 The helper clocks out. You paid only for those couple of seconds.

Where do the readings get filed?

GOOD TO KNOW

A "smart filing cabinet"

A special store that's brilliant at organizing readings by the exact time they happened.

A "big cheap warehouse"

For keeping huge amounts of raw data inexpensively, in case it's useful later.

THE PAYOFF · FROM NUMBERS TO DECISIONS

8. Turning numbers into decisions

Filing a reading away is only the start. The real value comes when all those stored numbers are turned into something a person can act on.

8.1 A filing cabinet that's great with time

Factory readings are all about time: "Machine 7's temperature, second by second." The store that holds them is especially good at answering time-based questions — so you can instantly pull up "every temperature from last Tuesday afternoon" and see exactly what happened.

8.2 Dashboards, warnings, and predictions

From there, the numbers become a **dashboard** — like the dashboard of a car, but for the whole factory: gauges and warning lights you can read at a glance. And by studying the patterns, the system can even **predict trouble**: a small wobble today might mean a part will fail next week. That's the dream — fixing things *before* they break, not after.

THINK OF IT LIKE YOUR CAR'S WARNING LIGHT

Your car doesn't wait for the brakes to fail. It watches quietly and lights up "service soon" while there's still time. Predicting a machine's breakdown works the same way — a gentle heads-up before anything actually goes wrong.

The "digital twin"

A NEAT IDEA

A "digital twin" is a **living, on-screen copy of a real machine** — almost like a video-game version — that updates from the real machine's data in real time. With it, you can try out "what if we changed this?" safely on the copy, and see the likely result before touching the real thing.

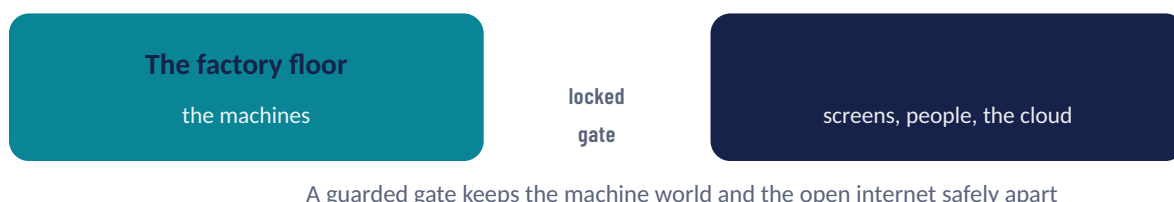
STAYING SAFE · THE LOCKS

9. Keeping it all safe

Because the cloud can send instructions back to real machines, safety here is not optional. A stranger who broke in could do more than peek at numbers — they could affect a physical machine. The good news is that a few common-sense habits stop most trouble.

9.1 Five simple safeguards

- ◆ **An ID badge for every device:** each machine and gadget must prove it really is itself — not just type a simple password.
- ◆ **Sealed, tamper-proof envelopes:** every message is locked while it travels, so no one can read it or secretly change it on the way.
- ◆ **Only the keys you need:** each device can open only the doors it genuinely needs — nothing more.
- ◆ **A locked fence:** the factory floor is kept separate from the open internet, connected only through carefully guarded gates.
- ◆ **Keep things up to date:** fix known weak spots as soon as repairs are available, like changing a lock that's known to be pickable.



A guarded gate keeps the machine world and the open internet safely apart

Do / Don't

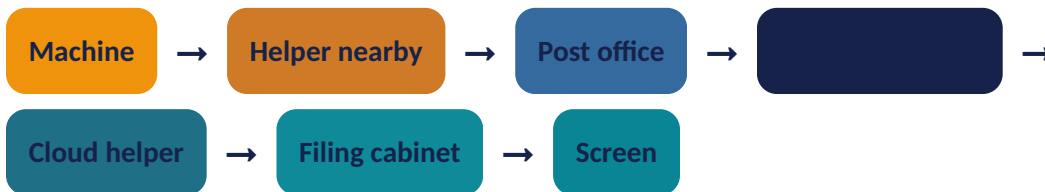
SAFETY

Do: give every device its own ID; keep the "sealed envelopes" on at all times; hand out the fewest keys possible; keep an eye on what's coming and going. **Don't:** leave the factory-default password in place; plug a machine straight into the open internet; or use one master key for everything.

THE WHOLE STORY · IN ONE PICTURE

★ Everything, at a glance

Now you've seen the whole journey. Here it is in a single line — the trip one number takes to become a useful decision:



From a machine on the floor, all the way to a decision on a screen

★ The one thing to remember

Strip away every fancy word and the whole "factory cloud" is just one simple chain:

Take a reading → send it → decide what to do → file it away → show it on a screen.

Every tool, service, and buzzword you might ever hear about this subject is simply one link in that chain. Once you can picture the chain, you understand the heart of it — and the rest is just detail.

In short: a modern factory quietly turns its machines' everyday numbers into early warnings and better decisions. It does this by passing each reading along a simple relay — from the machine, through a middleman, up to the cloud, and back to a human on a screen. No magic required.

REFERENCE

Plain-English glossary

The cloud

Other people's powerful computers, rented over the internet instead of owned.

Internet of Things

Putting everyday objects — like factory machines — online so they can share information.

Sensor

A device that measures something (like temperature) — a thermometer that can report its reading.

The machine's brain

The small built-in controller that runs a machine and knows its current readings.

A reading

One measurement a machine reports about itself.

The helper nearby (the "edge")

A small device beside the machine that tidies up data and can act fast on its own.

The post office (the middleman)

The go-between that takes each message and delivers a copy to everyone who signed up for it.

The label (or "address")

The folder-style tag on a message that decides who gets it.

The cloud's front desk

Where messages first arrive in the cloud, and where sender IDs are checked.

Serverless

A helper that's hired only for the seconds it's needed, then let go — you pay for nothing idle.

The filing cabinet (a database)

An organized store that keeps every reading and is great at finding things by time.

Dashboard

A car-style display of gauges and warning lights, but for the whole factory.

Predicting breakdowns

Spotting trouble early from the data — like a "service soon" light — so you fix it before it fails.

Digital twin

A live, on-screen copy of a real machine you can safely experiment on.

ID badge & sealed envelope

Proving who a device is, and locking messages so no one can read or change them in transit.

The locked fence

Keeping the factory floor separate from the open internet, with guarded gates between them.

INDUSTRY 4.0

A factory's numbers, finally explained.

No code, no jargon — just one number's journey from a machine to a screen, told the way you'd explain it to a friend.

NEXT ISSUE: How computers learn to predict a breakdown — explained just as simply.

INDUSTRY 4.0

TECHNICAL DEEP-DIVE

From Sensor to Cloud

The real journey of a single reading — traced end to end, from a hot bearing to a pixel on a dashboard. Every hop, with real values.

THE READING WE'LL FOLLOW

machine7 · bearing_temp = 82.4 °C

A companion to "The Factory Cloud" · concrete edition

THE SETUP

One reading, followed all the way

This companion takes **one concrete reading** and follows its *actual form* at every stage — as a current, as bytes, as a network packet, as a database row:

The bearing on Machine 7 is running at 82.4 °C. Watch what that number physically *is* at each step.

The full chain at a glance — the same value, transformed eleven times between the steel and the screen:



10

Time-series row

stored in the database

time, machine7, 82.4

↓
QUERY & RENDER

11

Dashboard pixel

plotted on a live chart

point at 82.4 °C

The same physical fact, represented eleven different ways between the bearing and the browser

HOW TO READ THE REST OF THIS DOCUMENT

Each section below zooms into one band of that trace and shows the **real mechanics**: the math, the protocol, the bytes, and the code or configuration that does the work. Values are representative of a typical Siemens-S7 + AWS-IoT stack.

BAND 1 · THE PHYSICAL WORLD

1. From heat to a number the PLC can read

Before any "data" exists, there is only heat. Turning that heat into a digital number takes three real conversions.

1.1 The sensor: heat → resistance

A common industrial temperature sensor is a **PT100 RTD** — a platinum element whose electrical resistance rises with temperature, from exactly $100\ \Omega$ at $0\ ^\circ\text{C}$. A close approximation is $R = 100 \times (1 + 0.00385 \times T)$, so at $82.4\ ^\circ\text{C}$ the element measures about **131.7 Ω** .

1.2 The transmitter: resistance → 4–20 mA

A transmitter converts that resistance into a **4–20 mA current loop**, linear across its configured range (here $0\text{--}150\ ^\circ\text{C}$). Current — not voltage — is used because it is immune to wire resistance and electrical noise over long factory cable runs. With $I = 4 + (T / 150) \times 16$, our $82.4\ ^\circ\text{C}$ becomes **12.79 mA**.

1.3 The analog input card: 4–20 mA → raw count

The PLC's analog input module digitizes the current. On Siemens hardware, the 4–20 mA span maps to an integer $0\ldots 27648$. With $\text{raw} = (I - 4) / 16 \times 27648$, 12.79 mA lands at a raw count of about **15189**.

The physical-to-digital chain, with real values

Stage	Conversion	Value at $82.4\ ^\circ\text{C}$
PT100 resistance	$R = 100 \cdot (1 + 0.00385 \cdot T)$	131.7 Ω
Loop current	$I = 4 + (T/150) \cdot 16$	12.79 mA
Raw ADC count	$(I-4)/16 \times 27648$	15189 / 27648

WHY 4–20 MA AND NOT 0–20?

The "live zero" is a safety feature: a healthy sensor never reads below 4 mA, so **0 mA means a broken wire**. The PLC can tell the difference between "0 $^\circ\text{C}$ " and "the cable is cut" — something a 0–20 mA or voltage signal can't do cleanly.

BAND 2 · INSIDE THE PLC

2. From raw counts to a number in memory

The raw count 15189 is meaningless to a human. The PLC program scales it back into engineering units and stores it as a proper floating-point number.

2.1 Scaling, in the PLC's own language

Siemens PLCs typically use `NORM_X` (normalize to 0.0–1.0) followed by `SCALE_X` (stretch to the real range). The result is written into a **Data Block** as a 32-bit REAL:

```
scale_temp.scl SIEMENS SCL

// "IW64" = analog input word, 0..27648 for 4..20 mA
norm  := NORM_X(MIN := 0,  VALUE := "IW64", MAX := 27648); // 0.0 .. 1.0
temp_c := SCALE_X(MIN := 0.0, VALUE := norm,  MAX := 150.0); // degC

// equivalent one-liner:
// temp_c := "IW64" * 150.0 / 27648.0; // = 82.4

"DB1".bearing_temp := temp_c; // stored as REAL at DB1.DBD0
```

2.2 What 82.4 actually looks like in memory

A REAL is a 32-bit **IEEE-754** float. The value 82.4 can't be stored exactly in binary, so it is kept as the nearest representable number, whose bit pattern is 0x42A4CCCD. Because S7 PLCs are **big-endian**, the four bytes sit in memory in this order — and this is exactly what the edge gateway will read back:

82.4 as a 32-bit REAL (IEEE-754, big-endian)

Field	Bits	Meaning
Sign	0	positive
Exponent	1000 0101	133 → multiply by 2^6
Mantissa	010 0100 1100 1100 1100 1101	the fractional part
Bytes	42 A4 CC CD (at DB1.DBD0)	

WHY THIS MATTERS DOWNSTREAM

Anything reading this value must agree on **type** (32-bit float), **byte order** (big-endian), and **address** (DB1.DBD0). Get any of the three wrong and 82.4 comes out as garbage. This is the most common real-world bug when first bridging a PLC.

BAND 3 · THE OT/IT BRIDGE

3. From the PLC to a message on the network

An edge gateway now reaches into the PLC, reads those four bytes, turns them back into a number, stamps the time, and packages everything as a small text message.

3.1 Reading the bytes back (S7comm)

read_and_build.py

PYTHON

```
import snap7, json, time
from snap7.util import get_real

plc = snap7.client.Client()
plc.connect("192.168.0.10", 0, 1)      # IP, rack, slot

raw = plc.db_read(1, 0, 4)           # DB1, start byte 0, 4 bytes
print(raw.hex(" "))                 # -> 42 a4 cc cd
temp_c = get_real(raw, 0)            # -> 82.4
plc.disconnect()
```

3.2 Wrapping it as a timestamped message

read_and_build.py (cont.)

PYTHON

```
payload = {
    "machine_id": "machine7",
    "metric":     "bearing_temp_c",
    "value":      round(temp_c, 1),      # 82.4
    "unit":       "degC",
    "quality":    "good",
    "ts":         int(time.time() * 1000), # epoch ms
}
msg = json.dumps(payload)               # ~120 bytes of text
```

The reading is now a self-describing, ~120-byte message: {"machine_id": "machine7", "metric": "bearing_temp_c", "value": 82.4, "unit": "degC", "quality": "good", "ts": 1718900047123}.

STAMP THE TIME AT THE EDGE, NOT IN THE CLOUD

The ts is added here, the moment the value is read. If you instead time-stamped it on arrival in the cloud, network jitter and retries would smear your time series — and "when did the bearing actually hit 82.4 °C?" would become unanswerable.

BAND 4 · ON THE WIRE

4. Crossing the internet, encrypted

The message leaves the factory over **MQTT**, but never in the clear. It rides inside a **TLS** tunnel on port 8883, and the device proves its identity with an **X.509 certificate** — the same "ID badge" idea, made concrete.

publish_to_cloud.py

PYTHON

```
import paho.mqtt.client as mqtt

ENDPOINT = "a1b2c3d4e5-ats.iot.eu-west-1.amazonaws.com"
client = mqtt.Client(client_id="edge-line1")

client.tls_set(ca_certs="AmazonRootCA1.pem",          # trust the cloud
               certfile="device.cert.pem",           # the device's badge
               keyfile="device.private.key")          # its private key
client.connect(ENDPOINT, 8883)                        # MQTT over TLS (mutual)

client.publish("factory/line1/machine7/bearing_temp",
              msg, qos=1)                             # at-least-once delivery
```

During the handshake, the device and the cloud each prove who they are (**mutual TLS**); from then on, the PUBLISH packet — topic plus our ~120-byte payload plus a few header bytes — is encrypted end to end. QoS 1 means the broker acknowledges receipt, and the edge re-sends if no acknowledgement comes back.

THE TOPIC IS DOING REAL WORK

factory/line1/machine7/bearing_temp isn't just a name — it's how the cloud will route, filter, and authorize this message. The device's certificate policy can even restrict it to only publish under factory/line1/#.

BAND 5 · IN THE CLOUD

5. Broker, rule, function

The packet arrives at **AWS IoT Core**. Three things happen in milliseconds: the message broker accepts it (after checking the certificate and its policy), the **rules engine** inspects it, and a matching rule fires an action.

5.1 A rule that listens for our topic

iot_rule.sql

IoT SQL

```
SELECT machine_id, value, ts
FROM 'factory/line1/+bearing_temp' -- + = any machine
WHERE value > 0
```

The + matches any machine on Line 1. When our message matches, the rule's action invokes a function and passes the parsed fields straight to it.

5.2 The function that files and checks it

process_telemetry.py

AWS LAMBDA

```
import boto3
ts = boto3.client("timestream-write")

def lambda_handler(event, context):
    machine = event["machine_id"]
    value = float(event["value"]) # 82.4

    ts.write_records(
        DatabaseName="FactoryDB", TableName="telemetry",
        Records=[{
            "Dimensions": [{"Name": "machine_id", "Value": machine}],
            "MeasureName": "bearing_temp_c",
            "MeasureValue": str(value),
            "MeasureValueType": "DOUBLE",
            "Time": str(event["ts"]), # the edge timestamp
        }],
    )
    if value > 80: # threshold alarm
        print(f"ALERT: {machine} bearing at {value} C")
    return {"status": "ok"}
```

IN PRACTICE, YOU MIGHT SKIP THE FUNCTION ENTIRELY

IoT Core can write **directly** to the database with a built-in Timestream action — no Lambda at all. The function is worth it only when you need real logic (custom alerts, enrichment, reformatting). Fewer moving parts means lower latency and cost.

BAND 6 · STORED, QUERIED, SHOWN

6. From a row to a pixel

Our reading is now a single **row** in a time-series database: a timestamp, the dimension `machine_id = machine7`, and the measure `bearing_temp_c = 82.4`. Recent rows live in a fast in-memory store; older ones roll off to cheaper long-term storage automatically.

6.1 The query a dashboard runs

A tool like Grafana repeatedly runs a query like this and draws the result as a line:

`dashboard_query.sql``SQL`

```
SELECT bin(time, 1m)           AS t,
       avg(measure_value::double) AS temp_c
FROM "FactoryDB"."telemetry"
WHERE machine_id = 'machine7'
      AND measure_name = 'bearing_temp_c'
      AND time > ago(1h)
GROUP BY bin(time, 1m)
ORDER BY t
```

The query returns one point per minute; the newest is our 82.4. The chart plots it, and an alert rule set at $> 80\text{ }^{\circ}\text{C}$ turns the panel red and notifies an engineer. Eleven transformations later, a hot bearing has become a red line on a screen.

Every layer simply re-encodes the same fact: 82.4 °C. Understand the encodings, and the whole system stops being mysterious.

THE NUMBERS · FLOOR TO GLASS

7. How long does the whole trip take?

"Floor-to-glass" is the time from the physical change to its appearance on a screen. Here is a realistic budget for our path — useful for judging whether a use case needs the edge (Band 3) or can live entirely in the cloud.

Representative end-to-end latency budget

Hop	Typical latency
Sensor + transmitter (thermal lag + filtering)	0.1 – 1 s
PLC scan cycle	1 – 10 ms
Edge poll interval (configurable)	0.1 – 1 s
MQTT publish over TLS to the cloud	20 – 150 ms
IoT rule + function	30 – 300 ms (cold start ~1 s)
Time-series write	5 – 50 ms
Dashboard refresh interval	1 – 10 s
Floor-to-glass (typical)	~1 – 3 s

THE LESSON HIDDEN IN THE TABLE

The slow steps are the **poll interval** and the **dashboard refresh**, not the cloud. If you need a sub-100 ms safety reaction (e.g. "trip the motor"), it must happen at the edge or in the PLC — the round trip to the cloud is simply too slow. The cloud is for visibility and intelligence, not reflexes.

The whole journey, in one breath: heat → resistance → current → raw count → a scaled **REAL** (bytes 42 A4 CC CD) → a JSON message → an encrypted MQTT packet → an authenticated cloud event → a rule → a function → a time-series row → a point on a chart. Same 82.4 °C, eleven costumes.